BIO303

BIOCHEMISTRY-II

VIRTUAL UNIVERSITY OF PAKISTAN

Sr. No.	Title	Page No.
1	BioEnergetics	
2	The Standard Free-Energy	
3	Eq constant and Standard free energy -I	
4	Eq constant and Standard free energy -2	
5	Chemical Logic and Common Biochemical Reactions-1	

6	Chemical Logic and Common Biochemical Reactions -2	
7	Chemical Logic and Common Biochemical Reactions- II-check final-3	
8	Phosphoryl Group Transfers and ATP-.pptx	
9	Phosphoryl Group Transfers in thioesters.pptx	
10	Bioenergetics -3-Hydrolysis of ATP- part 1	
11	Bioenergetics -3-Hydrolysis of ATP- part2.pptx	
12	Bioenergetics -3-ATP energizes active transport and muscle contraction.pptx	
13	Biological Oxidation -reduction reactions-2-4-18.pptx	
14	Reduction potential -	
15	Universal Electron Carriers	
16	GLYCOLYSIS-1	
17	Glycolysis- the pay off phase	
18	Preparatory phase of Glycolysis Requires ATP.pptx	
19	Preparatory phase of Glycolysis Requires ATP - cont-step 3	
20	Preparatory phase of Glycolysis Requires ATP-3	
21	Preparatory phase of Glycolysis -step 5	
22	Pay off phase-Last steps of glycolysis-6-7	
23	Last steps of glycolysis. -8-10	

24	Energy generated during Glycolysis- part 1 and 2.pptx	
25	Importance of Phosphorylated Intermediates.pptx	
26	Fates of Pyruvate.pptx	
27	Fates of Pyruvate -Lactic acid fermentation.pptx	
28	Ethanol Fermentation-.pptx	
29	Uses of Fermentation in Industry.pptx	
30	Industrial Fermentation-steps of fermentation.pptx	
31	Industrial Fermentation-cont-5	
32	Lecture - 32 - batch fermentation.pptx	
33	Continuous Fermentation-6.pptx	
34	Gluconeogenesis	
35	Gluconeogenesis -cont	
36	Gluconeogenesis - 1st by pass pathway-complete	
37	Gluconeogenesis - 2nd-3rd by pass pathways-	
38	Gluconeogenesis - cont - 4	
39	Gluconeogenesis - cont -5	
40	Pentose Phosphate Pathway - PPP -1	
41	Pentose Phosphate Pathway - PPP -2-.pptx	

42	Pentose Phosphate Pathway - PPP -3	
43	feeder pathways of glycolysis -1.	
44	feeder pathways of glycolysis -2	
45	feeder pathways of glycolysis-3 - galactosemia	
46	Denaturation of DNA	
47	Denaturation of DNA.. contd	
48	Renaturation of DNA	
49	RNA	
50	Difference b/w DNA and RNA	
51	Structure of RNA	
52	2ndry structure of RNA	
53	Function of mRNA	
54	Transfer RNA	
55	Transfer RNa..1	
56	Transfer RNa..2	
57	Ribosomal RNa	
58	snRNA	
59	Enzymes	

60	IUB Classification of enzymes-	
61	Lyases	
62	Ligases	
63	Ligand	
64	Mechanism of enzyme action	
65	Mechanism of enzyme action..cont -1	
66	Mechanism of enzyme action..cont -2	
67	Mechanism of enzyme action..cont -3	
68	Cofactor, coenzyme and prosthetic groups	
69	Cofactor, coenzyme and prosthetic groups-1	
70	Reaction rate and order of reaction	
71	Factors affecting enzyme activity	
72	Factors affecting enzyme activity (cont)	
73	Enzyme kinetics	
74	Enzyme kinetics..cont I	
75	Enzyme kinetics..cont II	
76	Enzymes...Michaelis-Mention Eq.	
77	Enzymes...Michaelis-Mention Eq.cont-1	

78	Enzymes...Michaelis-Mention Eq.cont-2	
79	Enzymes...Michaelis-Mention Eq.cont-3	
80	Enzymes... Order of Reaction	
81	Lineweaver-burk plot	
82	Inhibition of enzyme activity	
83	Competitive inhibition	
84	Competitive inhibition.. contd	
85	Non-competitive inhibition	
86	Enzymes.. effect on lineweaver-burk plot	
87	Water	
88	Ionization of water	
89	Buffer System.. working solutions	
90	Weak acids and bases	
91	Water, pH and Buffer Solutions.	
92	Buffer Solutions	
93	Citric acid cycle-1	
94	Citric acid cycle-2	
95	Citric acid cycle-3	

96	PDH complex-Citric acid cycle - 4	
97	PDH complex-2	
98	Steps of TCA-8 enzymes-	
99	TCA-step 1.	
100	TCA-step2	
101	TCA-step 4	
102	TCA-step 5	
103	TCA-ending.pptx	
104	fatty acid metabolism-1	
105	Anaplerotic reactions of TCA-	
106	Glyoxalate cycle -1	
107	Glyoxalate cycle -2	
108	Glyoxalate cycle -3	
109	Fatty acid-1.pptx	
110	fatty acid metabolism-2	
111	fatty acid metabolism-effect of hormones-	
112	carnitine shuttle	
113	fatty acid metabolism-fatty acid oxidation-3 stages	

114	beta oxidation-complete	
115	The four steps are repeated to yield acetyl CoA and ATP	
116	"Oxidation of unsaturated fatty acids requires 2 additional steps "	
117	Complete Oxidation of odd-numbered fatty acids requires 3 additional steps	
118	Coenzyme B12	
119	Transcription Factors Turn on the Synthesis of Proteins for Lipid Catabolism	
120	Oxidation in peroxisomes -	
121	Oxidation in peroxisomes -cont	
122	Omega Oxidation	
123	Oxidation of phytanic acid	
124	Ketone bodies	
125	Ketone bodies-continued	

Topic#1

BIOENERGETICS

Welcome to the course biochemistry-II- In this course you will learn the principles of biochemistry and their application on health and disease. In the beginning we will discuss the concepts of bioenergetics.. As the terminology is self explanatory.. BIO and nergetics.. Meaning that we will be dealing with the concepts of energy transfer in biological systems.

Generally speaking, we know that the Living cells have to perform certain work to stay alive, to grow, and to reproduce.

They should have the ability to trap energy and use it into biological work is a fundamental property of all living organisms, but they use it in different ways.

Modern organisms use the chemical energy in fuels to synthesize complex, highly ordered macromolecules

For example: Plants use energy in such a manner that causes them to release oxygen as a waste product, while animals (which includes humans, of course) require oxygen as a component to survive.

BIOENERGETICS:

Bioenergetics is the quantitative study of the energy transductions, that is.. changes of one form of energy into another—that occur in living systems, and of the nature and function of the chemical processes underlying these transductions.



fireflies convert energy into light

Bioenergetic processes:

Glycolysis is a biological process in which glucose is broken down into pyruvate thus generating two molecules of ATP (Adenosine triphosphate) per molecule of glucose

Gluconeogenesis is another process in which the cell synthesizes glucose from biomolecules such as proteins, amino acids and fats

Citric acid cycle generates molecules of ATP by oxidation of energy stored in food material

All animal and plant cells are powered by chemical energy stored in the chemical

Bonds of organic molecules. Animals convert the chemical energy of fuels into motion and heat and

In a few organisms such as fireflies chemical energy is converted into light…

Lets look at some Examples of major bioenergetic processes occurring in nature

ATP is the energy currency in biological systems

Citric acid cycle, that is the process of cellular respiration generates molecules of ATP by oxidation of energy stored in food material

These processes will be discussed in detail in coming lectures.

Laws of Thermodynamics:

Biological Energy Transformations Obey the Laws of Thermodynamics.There are two fundamental laws of thermodynamics.

1. First law

The first law is the principle of the conservation of energy: it states that

"for any physical or chemical change, the total amount of energy in the universe remains constant; energy may change form or it may be transported from one region to another, but it cannot be created or destroyed."

However, it looks that the direction of energy flow is from 1 direction to another.. To understand that we look at the 2nd law of thermodynamics for which should be familiar with the concept of entropy

Entropy:

The term "entropy," was first used in 1851 by Rudolf Clausius, which literally means "a change within" a system.. and it can be described as disorder..

Entropy = disorder

Entropy is a quantitative expression for the randomness or disorder in a system. When the products of a reaction are less complex and more disordered than the reactants, the reaction is said to proceed with a gain in entropy.

Denoted by S

°Delta S has a +ve sign when entropy increase and –ve when entropy decreases

units of entropy are joules/mole.Kelvin (J/mol.K) (joules/mole multiply by Kelvin). By convention, ° S has a positive sign when entropy increases and °H, as noted above, has a negative sign when heat is released by the system to its surroundings.

Second Law of Thermodynamics:

With this idea about entropy, we come to the 2nd law ..which states that..

"***the universe always tends towards increasing disorder and in all natural processes, the entropy of the universe increases… give example of hot cup.. Now the point is that entropy is not a measure of energy ..it only describes that how energy is distributed within a system."***

Enthalpy (H):

The term that describes the energy of a system is Enthalpy.Enthalpy is the heat content of the reacting system.

It reflects the number and kinds of chemical bonds in the reactants and products

When a chemical reaction releases heat, it is said to be exothermic and value of H is negative

When a chemical reaction takes up heat from surroundings, it is called as endothermic and have positive values of H
When a chemical reaction releases heat, it is said to be exothermic; the heat content of the products is less than that of the reactants and H has a negative value.

On the other hand, Reacting systems that takes up heat from their surroundings are endothermic and have positive values of H. Enthalpy and entropy of a system are related and help us to understand Gibb's free energy

Gibbs free energy (G):

Another terminology thermodynamic quantity that describe the energy changes occurring in a chemical reaction:

Gibbs free energy, G expresses the amount of energy capable of doing work during a reaction at constant temperature and pressure

When a reaction proceeds with the release of free energy the free-energy change, G, has a negative value and the reaction is said to be exergonic

In endergonic reactions, the system gains free energy and G is positive

The units of G and H are joules/mole or calories/mole

(1 cal is equal to 4.184 Joules)

Under the conditions existing in biological systems (including constant temperature and pressure), changes in free energy, enthalpy, and entropy are related to each other quantitatively by the equation

$$\Delta G = \Delta H - T\Delta S$$

°G: is the change in Gibbs free energy
°H: the change in enthalpy
T: the absolute temperature
° S: change in entropy

in which °G is the change in Gibbs free energy of the reacting system, °H is the change in enthalpy of the system, T is the absolute temperature, and °S is the change in entropy of the system.

Either of these conditions, which are typical of favorable processes, tend to make G negative. In fact, °G of a spontaneously reacting system is always negative.

Topic#2

The Standard Free-Energy

We already talked about Gibbs free energy.. Now we will show that why do we need to know about this terminology in biological systems…. We know that the living cells require some sources to provide Energy.. So that they can perform the functions of life..

will talk about Free Energy …that can be used by living cells…Cells Require Sources of Free Energy..

Cells are isothermal systems—they function at essentially constant temperature (and also function at constant pressure).

Heat flow is not a source of energy for cells, because heat can do work only as it passes to a zone or

object at a lower temperature. The energy that cells can and must use is free energy, is known as Gibbs freeenergy (G) in terms of thermodynamics.

.. This allows prediction of the direction of chemical reactions, their exact equilibrium position,

and the amount of work they can (in theory) perform at constant temperature and pressure. Both autotrophic

And Heterotrophic cells acquire it from absorbed solar radiation.

Both kinds of cells transform this free energy into

ATP and other energy-rich compounds that provide enrgy to carry out functions of life.

Cells require sources of Free Energy (G):

Cells work at constant temperature and pressure (isothermal)

Heat flow is NOT energy source

Change in Gibbs Free energy (ΔG)

Significance of Free-Energy Changes (°G):

There are some scientific facts that reveal the significance of Gibbs free energy..

Urdu.. Why is this so important for us.. Because of the reason that .. ..

Say all .. And we will discuss them while going through the present lesson

Actual free-Energy changes depend on Reactant
and Product Concentrations

Allows prediction of the direction of chemical reactions, their exact equilibrium position and the amount of work done

The free-energy change for a reaction is independent of the pathway by which the reaction occurs

Free-energy changes are additive; the net chemical reaction that results from successive reactions sharing a common intermediate has an overall free-energy change that is the sum of the °G values for the individual reactions

Standard free-energy G:

But before that we should know that what is standard free energy… Under standard conditions of temperature and pressure, meaning (298 K or 25C), when reactants and products are initially present at 1 M concentrations or, for gases, at partial pressures of 101.3 kilopascals (kPa), or 1 atm, the force driving the system toward equilibrium is defined as the standard free-energy change, ° G▼ where knot is the symbol to depict that the reaction occurred under standard conditions 0-0-

The free energy of reaction at under standard conditions

When Temperature: 298K or 25▼C, when reactants are initially present at 1 M concentrations or, for gases, at partial pressures of 101.3 kilopascals (kPa), or 1 atm, the force driving the system toward equilibrium is defined as the standard free-energy change, ° G▼

$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$$

Key conventions in biochemical reactions

However, most biochemical reactions, however, occur in well-buffered aqueous solutions near pH 7; both the pH and the concentration of water (55.5 M) are taken as constant

Most biochemical reactions occur near pH 7 (buffered)

Standard biochemical state

(H+) is 10-7 M

concentration of water is 55.5 M

Both the pH and the concentration of water are essentially constant

Under these conditions, °G is written as °G‘▼ (with prime)

Concentration of water, H+ and Mg+ are not included in equations for simplicity

Molar concentration of water:

Suppose we have 1 L of water. (1000 ml). The density of water is 1 g / ml, therefore in 1000 ml of water must be 1000g. Well, if the water is pure, it remains to be seen how many moles of water we have. 1 mol of water is 18 g, so 1000 g will therefore be 55.555 moles. Now if molarity is defined as moles / L, then we 55.555 moles / L is therefore 55.555 M. Similarly, concentrations are taken to solutions, that is, is perfect that you have to aproach what concentration of OH- or H + ions are in the water (supposedly 10'-7 M) that leaves the water

Autoprotolysis. I don´t want to go around the bush but be careful what you use this "concentration" for because if you're in a water balance is not good that you consider water as a species, as its "concentration" is contained somehow in constant balanc

Density of pure water = 1000 g/L

Molar mass of water = 18 g/mol (2H + 1 Oxygen=2+16=18)

Molar concentration of water = d/m

= 1000 / 18.02

= 55.5 mol/L

Topic#3

Eq constant and Standard free energy -I

The Standard Free-Energy is directly related to the Equilibrium constant

What is Equilibrium constant (K)?:

Now we are going to talk about the eq- constant K .. And its relationship with free energy.. The Standard Free-Energy Change Is Directly Related to the Equilibrium Constant

And how can we find the equilibrium constant of a reaction mixture..

Assume the reaction given here
An impotant point is … that.. The composition of a reacting system (a mixture of chemical reactants and products) tends to continue changing until it reaches equilibrium state

At the point of equilibrium concentration of reactants and products are exactly equal and also the rates of the forward and reverse reactions are exactly equal , because of which there will be no further net change occurs in the system.

AND.. At this point.. The concentrations of reactants and products at equilibrium define the equilibrium constant, Keq.

To understand this in a better way.. Consider the following reaction .. Wher A and B are the reactants, C and D are the products and the small letters depict their molar concentration in the reaction

The composition of a reacting system tends to continue changing until equilibrium is reached

At the equilibrium concentration of reactants and products, the rates of the forward and reverse reactions are exactly equal and no further net change occurs in the system.

The number that expresses the relationship between the amounts of products and reactants present at equilibrium in a reversible chemical reaction is known as the equilibrium constant, Keq

In the general reaction



How to calculate Equilibrium constant?:

Lets talk how to get the equilibrium constant

Assume the reaction given here… Let's try to understand it with an example.. We have a reacting system where A and B are the reactants, with molar concentration denoted by small a and b

Whereas C and D are the products of this particular reaction and their molar concentartions are denoted by small c and d respectively.

When the reaction is at equilibrium.. The eq. constant for this reaction is (the prioduct of conc. Of the products divided by the product of conc. Of the reactants

At this point, the equilibrium constant is the ratio of concentration of reactants to that of products and can be written as this equation..

When a reacting system is not at equilibrium, the

tendency to move toward equilibrium represents a

driving force, the magnitude of which can be expressed

as the free-energy change for the reaction, DG

The equilibrium constant is given by following equation where [A], [B], [C], and [D] are the molar concentrations of the reaction components at the point of equilibrium, equilibrium constant can be written as:

$$K_{eq} = \frac{[C]^c\,[D]^d}{[A]^a\,[B]^b}$$

Standard free-energy change:

ADD EQUATION

The standard free-energy change of a chemical reaction is simply an alternative mathematical way of expressing its equilibrium constant.

° G is characteristic for a given reaction and can be calculated from the equilibrium constant for the reaction:

Or we can say that delta G and eq constant are related b this equation .. Gibbs free energy (understandsrd conditions (rep by knot) = negative R (gas constant * temp _ln of eq constant

$$\Delta G'^{\circ} = -RT \ln K'eq$$

where

R is the gas constant (8.31 cal/Mol.K)

T is standard temperature (25 C or 298K)

K'eq is the equilibrium constant

And we should kn ow the value of delta G relevant to the value of eq constant

If the Keq is equal to 1.0

the standard free-energy change (°GO) of that reaction is 0.0 (the natural logarithm (ln) of 1.0 is zero)…EQUILIBRIUM

If Keq of a reaction is greater than 1.0

°G^O is negative, forward direction is favored

If Keq is less than 1.0

°G^O is positive, reverse direction is favored

WORKED EXAMPLE 13–1 **Calculation of $\Delta G'°$**

Calculate the standard free-energy change of the reaction catalyzed by the enzyme phosphoglucomutase

$$\text{Glucose 1-phosphate} \rightleftharpoons \text{glucose 6-phosphate}$$

given that, starting with 20 mM glucose 1-phosphate and no glucose 6-phosphate, the final equilibrium mixture at 25°C and pH 7.0 contains 1.0 mM glucose 1-phosphate and 19 mM glucose 6-phosphate. Does the reaction in the direction of glucose 6-phosphate formation proceed with a loss or a gain of free energy?

Solution: First we calculate the equilibrium constant:

$$K'_{eq} = \frac{[\text{glucose 6-phosphate}]}{[\text{glucose 1-phosphate}]} = \frac{19 \text{ mM}}{1.0 \text{ mM}} = 19$$

We can now calculate the standard free-energy change:

$$\begin{aligned} \Delta G'° &= -RT \ln K'_{eq} \\ &= -(8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K})(\ln 19) \\ &= -7.3 \text{ kJ/mol} \end{aligned}$$

Because the standard free-energy change is negative, the conversion of glucose 1-phosphate to glucose 6-phosphate proceeds with a loss (release) of free energy. (For the reverse reaction, $\Delta G'°$ has the same magnitude but the *opposite* sign.)

Topic#4

Eq constant and Standard free energy -2

The Standard Free-Energy is directly related to the Equilibrium constant.

° G and ° G'°are related:

The free energy change (DG) of a reaction determines its spontaneity. A reaction is spontaneous if DG is negative (if the free energy of products is less than that of reactants).

The actual free-energy change, °G is a variable that depends on °G'▼ and on the concentrations of reactants and products:

$$aA + bB \longleftrightarrow cC + dD$$

$$\Delta G = \Delta G'^{o} - RT \ln ([\text{products}]/[\text{reactants}])$$

$$\Delta G = \Delta G'^{o} + RT \ln \frac{[C]^c [D]^d}{[A]^a [B]^b}$$ (equation 13-4... according to Lehninger 6th Edition—chapter No 13)

The concentration terms in this equation express the effects commonly called mass action, and the term [C]c[D]d/[A]a[B]b is called the mass-action ratio, Q.

Thus the equation can be written a follows:

$$\Delta G = \Delta G'^{o} + RT \ln Q$$

Lets suppose that the reaction is under standard conditions..

$$aA + bB \longleftrightarrow cC + dD$$

But the conc. Of reactants and products is NEITHER equal to eachother… Nor the conc. Is 1M

non-standard concentrations

Actual conc. Of reactants and products but standard values of ° G'▼, R and T

Value of ° G at equilibrium (for spontaneous reaction)??

$$0 = \Delta G = \Delta G'^{\circ} + RT \ln \frac{[C]^{eq}[D]^{eq}}{[A]^{eq}[B]^{eq}}$$

$$\Delta G'^{\circ} = -RT \ln K'eq$$

Standard Free-Energy Changes Are Additive:

Consider two sequential chemical reactions

$$A \rightleftharpoons B \qquad \Delta G'^{0}_{1}$$

$$B \rightleftharpoons C \qquad \Delta G'^{0}_{2}$$

Since the two reactions are sequential, we can write the overall reaction as

$$A \rightleftharpoons C \qquad \Delta G'^{0}_{total}$$

The ΔG'0 values of sequential reactions are additive.

$$\Delta G'^{0}_{total} = \Delta G'^{0}_{1} + \Delta G'^{0}_{2}$$

This principle of bioenergetics explains how a thermodynamically unfavorable (endergonic) reaction can be driven in the forward direction by coupling it to a highly exergonic reaction through a common intermediate

Topic#5

Eq constant and Standard free energy -2

Chemical Logic and Common Biochemical Reactions

Most cells have the capacity to carry out thousands of specific, enzyme-catalyzed reactions:

for example, transformation of a simple nutrient such as glucose into amino acids, nucleotides, or lipids

extraction of energy from fuels by oxidation

polymerization of monomeric subunits into macromolecules

5 types of chemical reactions in cells:

The reactions going on in biological systems can be categorized into 5 types:

1. Reactions that make or break carbon-carbon bonds
2. Internal rearrangements, isomerizations, and eliminations
3. Free-radical reactions
4. Group transfer
5. Oxidation-reduction reactions

2 basic chemical principles:

And before we study the details of these chemical reactions we should revise 2 basic principles..1st principle is that..say on slide…

So what is a covalent bond.. Lets review the concept..A covalent bond, is a chemical bond that involves 2 atoms that share their electron pair. These electron pairs are known as shared pairs or bonding pairs, and the stable balance of attractive and repulsive forces between atoms, when they share electrons, is known as covalent bonding.

Atoms make covalent bond with other atoms in order to gain more stability, which is achieved by forming a full electron shell. By sharing their outer most (valence) electrons, atoms can fill up their outer electron shell and become more stable.

A covalent bond can be broken in two general ways

- Homolytic cleavage
- Heterolytic cleavage

A Covalent bond, also called a molecular bond, involves the sharing of electron pairs between atoms.

These electron pairs are known as shared pairs or bonding pairs, and the stable balance of attractive and repulsive forces between atoms, when they share electrons, is known as covalent bonding.

2 mechanisms for breaking a C-C or C-H bond:

Homolytic cleavage: each atom leaves the bond as a radical, carrying one unpaired electron

Heterolytic cleavage (more common): one atom retains both bonding electrons

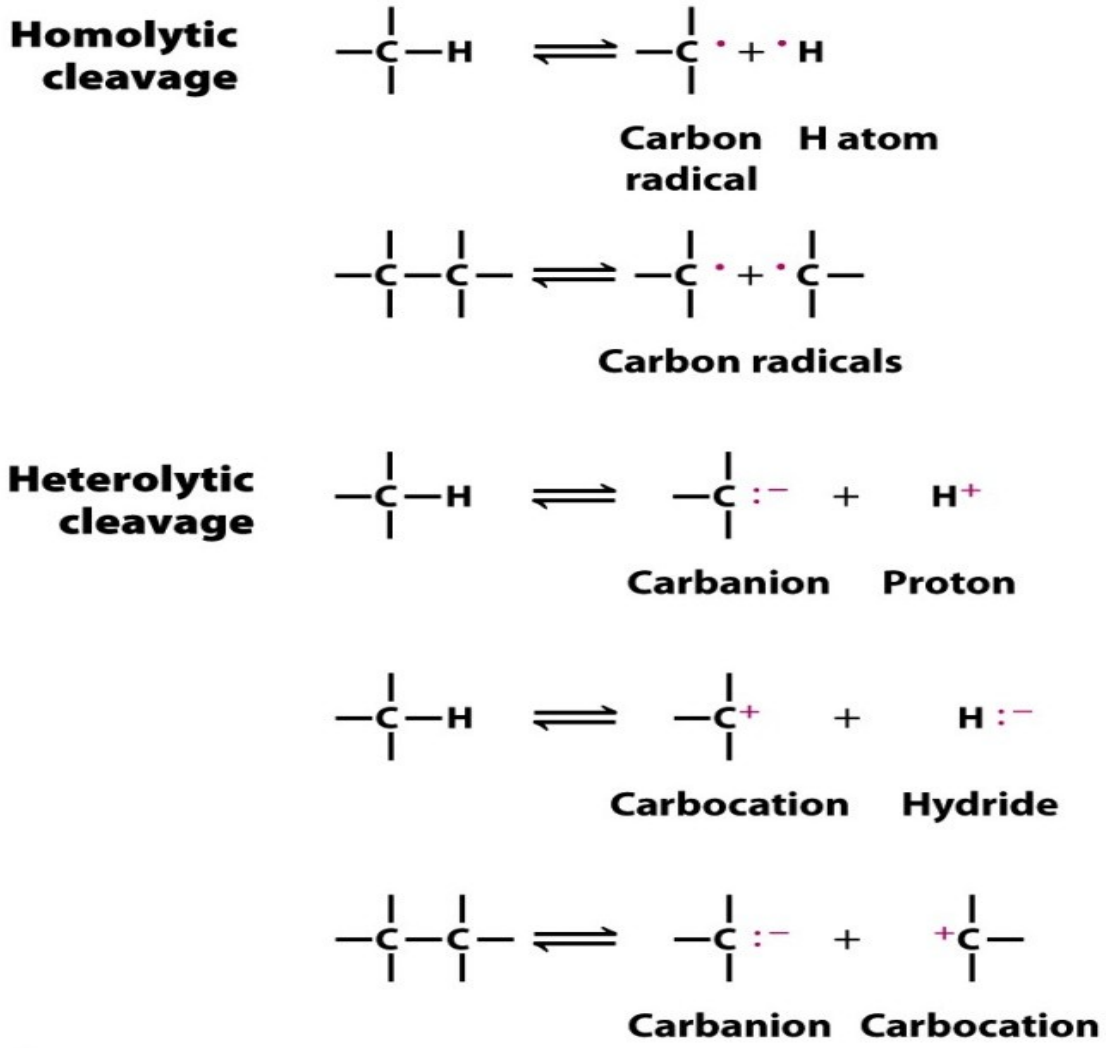


Figure 13-1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

<u>Second principle:</u>

Many biochemical reactions involve interactions between nucleophiles and electrophiles

- Nucleophiles: functional groups rich in and capable of donating electrons

(Nucleophiles are Lewis bases)

nucleophiles: meaning nucleus loving", or "positive-charge loving ..are the functional groups rich in electrons and can donate them to form a covalent bond

which is the exact definition of a Lewis base. In other words, nucleophiles are Lewis bases.

- Electrophiles: electron-deficient functional groups that seek electrons

Electrophile means.. electron-loving", or "negative-charge loving" these are the functional groups that are electron-deficient and seek electrons.. to form a covalent bond

Nucleophiles combine with Electrophiles and share electrons with them

There are some Common nucleophiles and electrophiles found in biochemical/biological reactions

Common nucleophiles and electrophiles in biochemical reactions:

There are some Common nucleophiles and electrophiles found in biological reactions

Examples of Nucleophiles include…. Name all of them and say.. As the picture shows, all these have –ve charge due to a pair of electrons that can be donated to make covalent bonds…And oxygen of hydroxide ion (Hydroxide is a diatomic anion with chemical formula OH^-. It consists of an oxygen and hydrogen atom held together by a covalent bond, and carries a negative electric charge. It is an important but usually minor constituent of water. It functions as a base, a ligand, a nucleophile and a catalyst.)

Protonated imine group.. Imine group is a functional group in a chemical compound that contains a carbon–nitrogen double bond. In protonated imine group, R group is activated and donates electrons to the carbon atom that can be given to the nitrogen that has +ve charge..

Another example is when the phosphorus atom in phosphate group

Or when a hydrogen atom having positive charge accepts an elctron pair..such kind of hydrogen atom is called a Brønsted acid, or just "acid".

the extent to which a species can accept a lone pair of electrons is called "electrophilicity".

The nitrogen atom can be attached to a hydrogen (H) or an organic group (R). If this group is not a hydrogen atom, then the compound can sometimes be referred to as a Schiff base.

Phosphorus (15P) - being the fifteenth element - has fifteen electrons, five valence electrons and the following electron configuration:

1s2
2s2 2p6
3s2 3p3

one might expect it to form 3 single covalent bonds and be happy. But it doesn't, it forms five bonds, presumably using each of its five electrons. My first question is - why would it do this? It seems to me that this would result in the Phosphorus atom having ten electrons in its outer shell (four from the double bond with the oxygen and two each from the single bonds with the other three oxygens).

Apart from the oxygen with a double-bond, we have three oxygens forming single bonds. Does this not leave them one electron short?

Finally, why are the oxygens shown as carrying a negative charge?

phosphorus has stable oxidation states in compounds, ranging from −III to +V.

First of all, let me state the obvious: Phosphorus is awesome. After we got that out of the way we can focus on why.

Apart from this it is possible to formulate many different cations and anions, that are derived from the above molecular structures. Just to name a few, there are P3+,P5+,P7+,P9+

But now for the most important part, phosphorus has stable oxidation states in compounds, ranging from −III

to +V. Here are a few examples:
P−IIIH3, P2−IIH4, [P−IH]n, P4±0, H3P+IO2, H4P2+IIO4, H3P+IIIO3, H2P2+IVO6, H3P+VO4

While dealing with these compounds it is usually completely unnecessary to describe bonding with hybrid orbitals.

In case of phosphane PH3

it would be wrong. Assuming sp3P one would expect ∠(H−P−H)≈109∘, while it is found to be ∠(H−P−H)=93.5∘, which is almost the same angle as the porbitals are having towards each other.

In general your assumption is correct, that it is possible to form only three covalent bonds to reach a stable configuration. And that will most likely be the case when phosphorus forms compounds with more electropositive elements.

Now dealing with oxygen, means dealing with a much more electronegative element, i.e. En(O)≈3.4

, En(P)≈2.2. That also means that bonds are much more polarised towards the oxygen.

Analysing the phosphate anion PO_4^{3-}

it is crucial to recognise its symmetry, which is tetrahedral Td. In this arrangement it is perfectly safe.

Very observant! What happens is that phosphorus is hybridized in the phosphate ion. Hybridization allows the central phosphate atom to form more bonds than you might expect from its ground state electron configuration.

Within the framework of hybridization, one may look at the bonds in the phosphate anion as having a great deal of ionic character.

The electronegativity of phosphorus is 2.19; the electronegativity of oxygen is 3.44. The difference is 1.25 - or rather sizable. As a result, electron density is largely withdrawn from the central phosphorus. Hence the negative charges you see on the three oxygen molecules in your picture.

Other depictions, which reject a large degree of d-orbital utilization in hybridization, might give a negative formal charge to all the oxygens and a positive formal charge to the central phosphorus by eschewing the P=O

double bond.

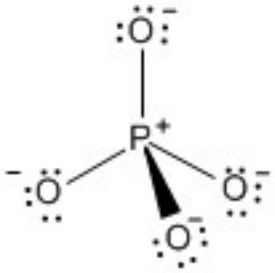
Apart from the oxygen with a double-bond, we have three oxygens forming single bonds. Does this not leave them one electron short?

Finally, why are the oxygens shown as carrying a negative charge?

You could draw double bonds for all the oxygens but there aren't that many valence electrons to be distributed. The structure shown above is a compromise between number of valence electrons and electronegativity considerations - we wouldn't want a surfeit of electrons around the central phosphorus since that's not as thermodynamically favorable as having the more electronegative oxygen support these electrons.

In addition, the sum of the formal charges on an atom must add up to the overall charge on the atom. Phosphate anion is triply negative charged; note how there are three oxygens bearing a single negative charge.

Electron pushing: The movement of a pair of electrons (a lone pair of electrons or a bond) from an electron rich site to an electron poor site.



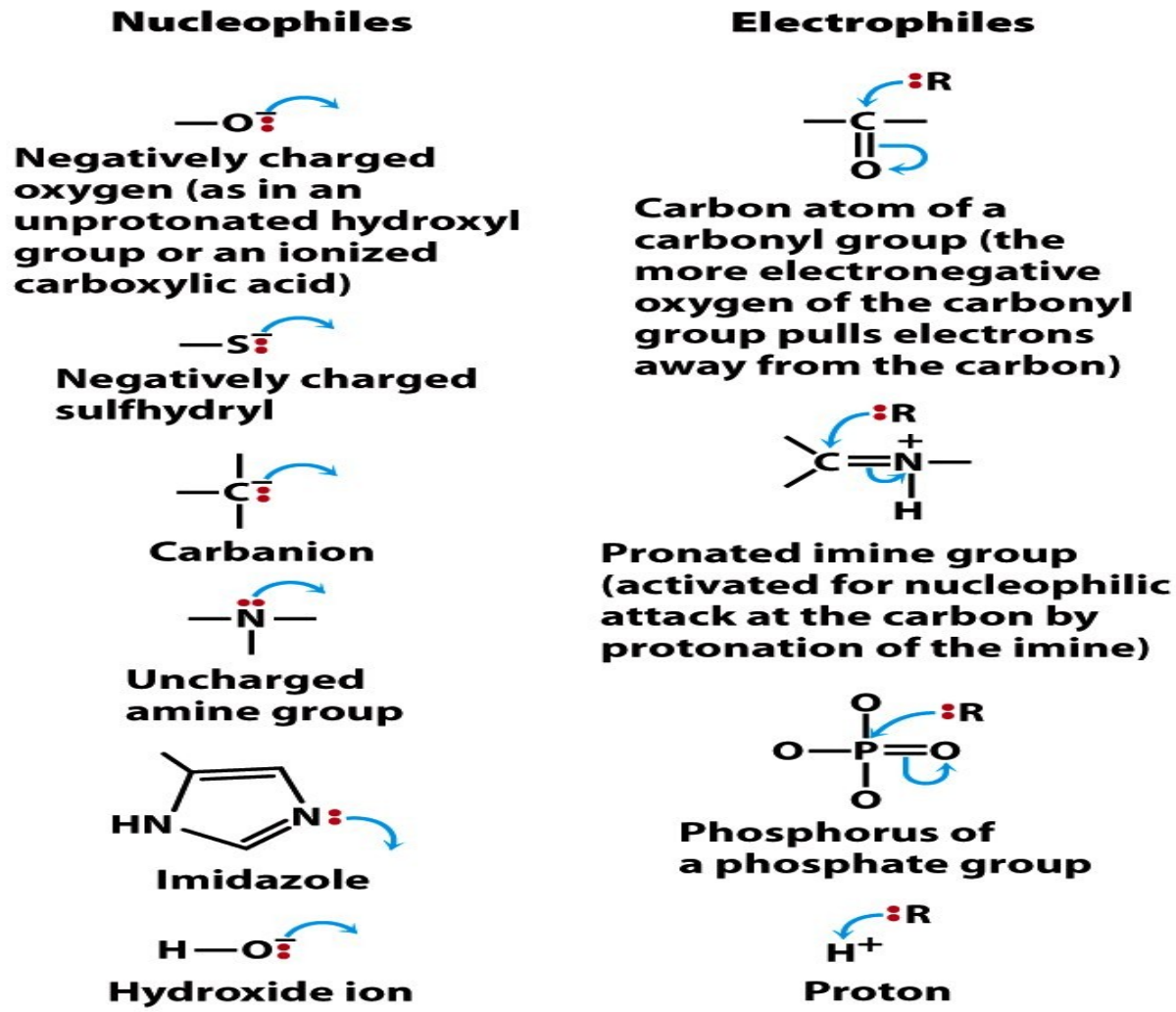


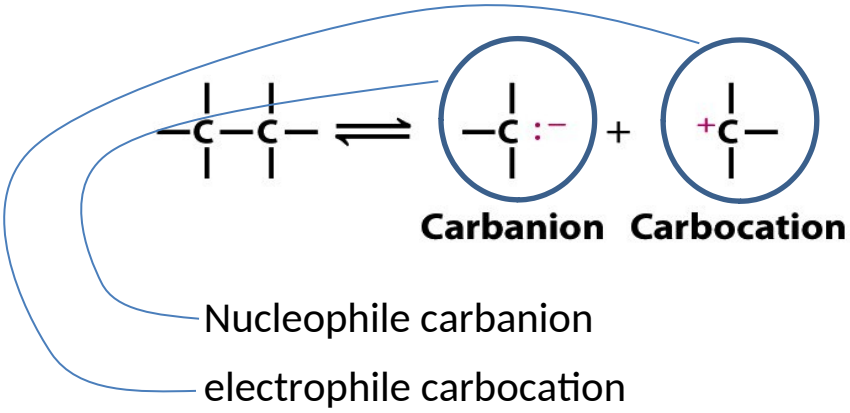
Figure 13-2
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Significance of Carbonyl group:

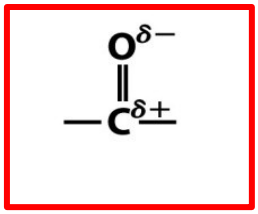
Carbonyl groups are particularly important in the chemical transformations of metabolic pathways…

- The carbon of a carbonyl group has a partial positive charge due to the electron-withdrawing property of the carbonyl oxygen✉ electrophilic carbon

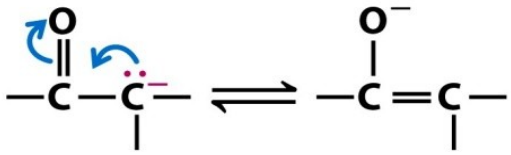
- Carbonyl group facilitate the formation of a carbanion's on an adjoining carbon by delocalizing the carbanion's negative charge
- An imine group can serve a similar function



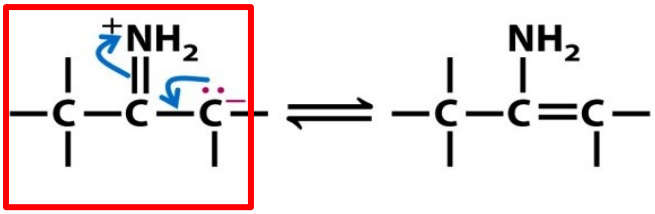
(a)



(b)



(c)



Topic#6

Chemical Logic and Common Biochemical Reactions -2

5 types of chemical reactions in cells:

The reactions going on in biological systems can be categorized into 5 types:

1. ***Reactions that make or break carbon-carbon bonds***

2. Internal rearrangements, isomerizations, and eliminations
3. Free-radical reactions
4. Group transfer
5. Oxidation-reduction reactions

Internal rearrangements, isomerizations, and eliminations:

Coming to the 2nd type of reactions taking place in biological systems, these include rearrangements within the molecule itself..this occurs by the: redistribution of electrons among the atoms and results in different types of alterations and distribution of charge .. But these rearrangements do not lead to any a change in the overall oxidation state of the molecule

different group in the molecule may undergo oxidation-reduction, with no net change in oxidation state of the molecule

groups at a double bond may undergo a cis-trans rearrangement MEANING…

the positions of double bonds may be transposed.. An d how does that happen..for example…

Isomerization

Enediol intermediate:The enediol is also an intermediate for the epimerization of an aldose or ketose. The reactions are usually base catalyzed, but can also take place under acid or neutral conditions. A typical rearrangement reaction is that between the aldose glyceraldehyde and the ketose dihydroxyacetone in a chemical equilibrium.

Reaction is also an example of elimination as .. Is eliminated ..

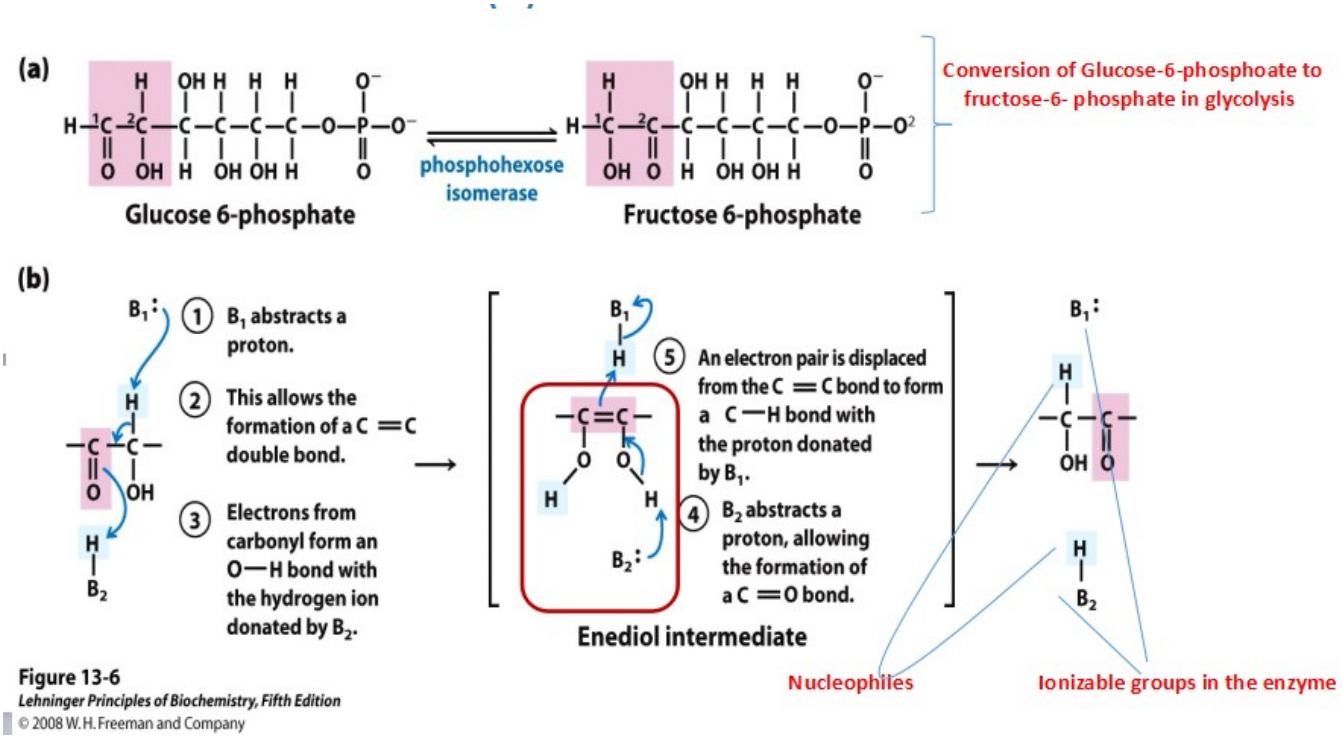


Figure 13-6
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Cis-trans rearrangement:

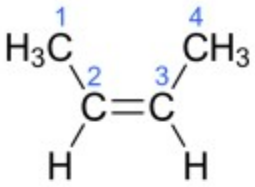
Cis–trans isomerism, also known as geometric isomerism or configurational isomerism. Cis is illustrated by... The presence of functional groups on the same side of the carbon chain while trans conveys that functional groups are on opposing sides of the carbon chain.

So while considering this kind of isomerism in living system, we can quote the example of proline amino acid. In general, Peptide bonds formed between amino acid residues are in trans-configuration… more than 99.95% have However, in case of proline, Peptide bonds formed are able to exist both as cis and trans isomers.. through peptidyl-prolyl (proline) isomerase enzyme

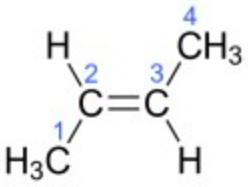
Prolyl isomerase ..this enzyme is (EC 5.2.1.8) found in both prokaryotes and eukaryotes and interconverts the cis and trans isomers of peptide bonds with the amino acid proline.

In general, Peptide bonds formed between amino acid residues with alpha carbons in trans-configuration… (more than 99.95% have trans configuration.)

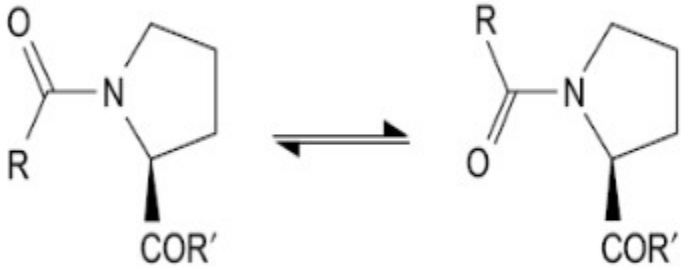
For peptide bonds involving the imino nitrogen of proline, peptide bonds are able to exist both in cis and trans configuration.. (6% are in the cis configuration) through peptidyl-prolyl (proline) isomerase enzyme



cis-butene



trans-butene



cis ***trans***

The *cis* and *trans* isomers of proline.

The process of the formation of *trans* and *cis* conformation is via isomerism, and the reaction is a reversible process.

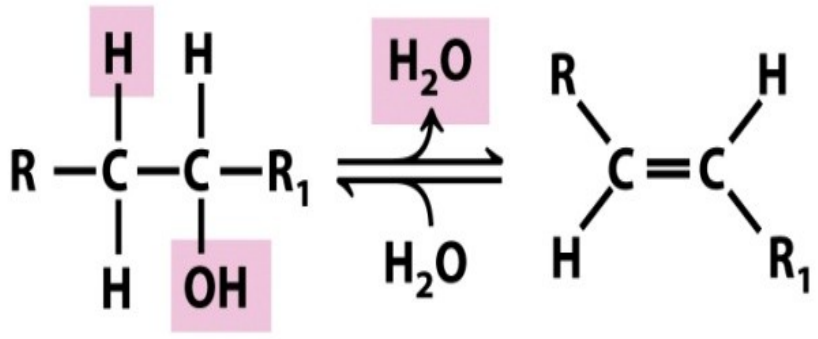
Proline

Elimination:

A water molecule is lost.. And a double bond is formed. overall oxidation state is the loss of water from analcohol, resulting in the introduction of a CP C bond d b/w 2 adjacent carbon atoms

The loss of water from an alcohol, resulting in the introduction of a C=C bond b/w 2 adjacent carbon atoms

An elimination reaction with no changes in oxidation state



Unnumbered 13 p498
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Loss of water from an alcohol → introduction of a C=C bond

Free-radical reactions:

Coming to free radical reactions.. From where we get these free radicals.. Remember we talked about Homolytic cleavage of covalent bond in which electrons pair is equally distributed and radicals are generated.. So cells have got some free radicals and they use them in certain reactions like/..

There are certain biological reactions in which Vit.B12 is used a s cofactor by enzymes. Among them, methylmalonyl-CoA mutase (MCM) has been extensively studied. This enzyme catalyzes the reversible isomerization of L-methylmalonyl-CoA to succinyl-CoA using adenosylcobalamin (AdoCbl) as a cofactor participating in the generation of radicals that allow isomerization of the substrate.

Certain radical-initiated decarboxylation reaction

3) Some reduction reaction, such as that catalyzed by ribonucleotide reductase

4) Some rearrangement reactions, such as that catalyzed by DNA photolyase.. That is a DNA repair enzyme. It binds to UV-damaged DNA containing pyrimidine dimers and, breaks the cyclobutane ring joining the two pyrimidines of the dimer and helps to avoid possible mutations.

Vitamin B12 is an organometallic compound with important metabolic derivatives that act as cofactors of certain enzymes, which have been grouped into three subfamilies depending on their cofactors. Among them, methylmalonyl-CoA mutase (MCM) has been extensively studied. This enzyme catalyzes the reversible isomerization of L-methylmalonyl-CoA to succinyl-CoA using adenosylcobalamin (AdoCbl) as a cofactor participating in the generation of radicals that allow isomerization of the substrate. The crystal structure of MCM determined in Propionibacterium freudenreichii var. shermanii has helped to elucidate the role of this cofactor AdoCbl in the

reaction to specify the mechanism by which radicals are generated from the coenzyme and to clarify the interactions between the enzyme, coenzyme, and substrate.

Homolytic cleavage of covalent bond ✉ produce free radical

- Isomerizations that make use of adenosylcobalamin (vitamin B12) for methylmalonyl-CoA mutase (MCM) reactions
- Radical-initiated decarboxylation reaction
- Some reduction reaction, such as that catalyzed by ribonucleotide reductase (converts ribonucleotides to deoxyribonucleotides)
- Some rearrangement reactions, such as that catalyzed by DNA photolyase

Free-radical reactions:

Oxygen-independent coproporphysinogen III oxidase catalyzes decarboxylation

In bacteria this process occurs in an oxygen-independent fashion by the enzyme..coproporphyrinogen-III oxidase catalyzes the oxygen-independent conversion of coproporphyrinogen-III to protoporphyrinogen-IX.and the step includes decarboxylation through free radical mechanism.. As co2 is release along with an electron…The accptor fot the released electron is not known.

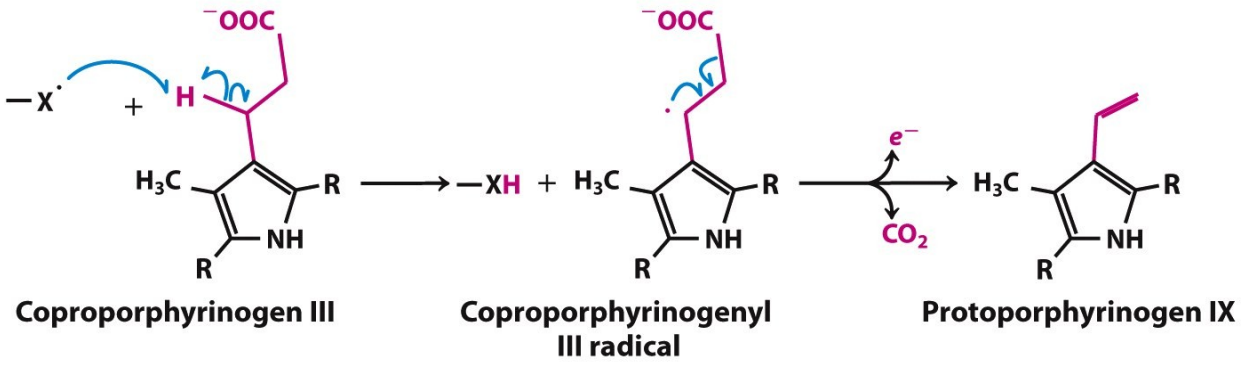


Figure 13-7
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

TOPIC#7

Chemical Logic and Common Biochemical Reactions- II-check final-3

Group Transfer Reactions:

In living cells, The transfer of acyl, glycosyl, and phosphoryl groups from one nucleophile to another is common

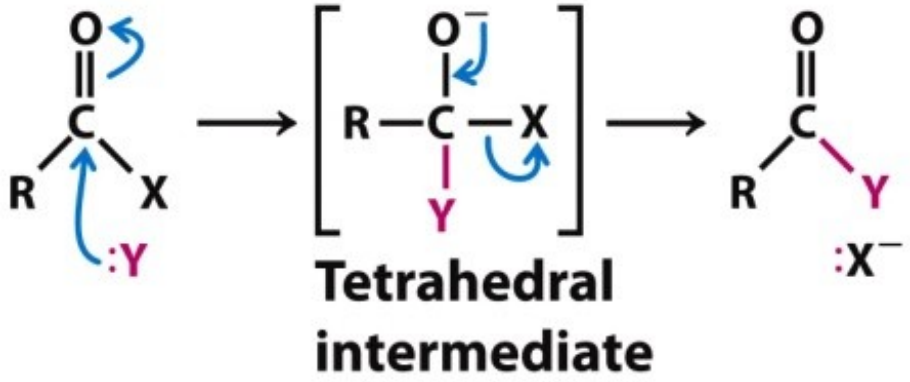
These include The transfer of certain acyl, glycosyl, and phosphoryl groups from one nucleophile carbonyl carbon of acyl group

to form a tetrahedral intermediate..and in nature, this takes place in Chymotrypsin reaction.. chymotrypsin is a proteolytic digestive enzyme of many organisms. breaks down proteins in the small intestine. It is secreted by the pancreas and converted into an active form by trypsin.

Glycosyl group transfer is enzyme catalyse reaction..occuring in … at C1 of sugar ring

In organic chemistry, the acryloyl group is form of enone with structure $H_2C{=}CH{-}C{-}$; it is the acyl group derived from acrylic acid. The preferred IUPAC name for the group is prop-2-enoyl, and it is also known as acrylyl or simply acryl

Acyl transfer: acyl group transfer generally involves the addition of a nucleophile to the carbonyl carbon of an acyl group to form a tetrahedral intermediate, eg. Chymotrypsin reaction



Unnumbered 13 p499
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

3) Transfer of phosphoryl group playes special roles in metabolic pathways

We can see that 3 o are single bonded while 4th ha 2 bonds with P giving a tetrahedral structure..

Nucleophilic substitution reactions requie Pi and PPI

- in several biochemical reactions where PI is required, inorganic orthophosphate (Pi): the ionized form of H3PO4 at neutral pH, a mixture of H2PO4- and HPO42-

- inorganic pyrophosphate (PPi): P2O74-

Nucleophilic substitution: attachment of a phosphoryl group to an poor leaving group (-OH): occur in hundreds of metabolic reactions

In chemistry, resonance or mesomerism is a way of describing delocalized electrons within certain molecules or polyatomic ions where the bonding cannot be expressed by one single Lewis structure.

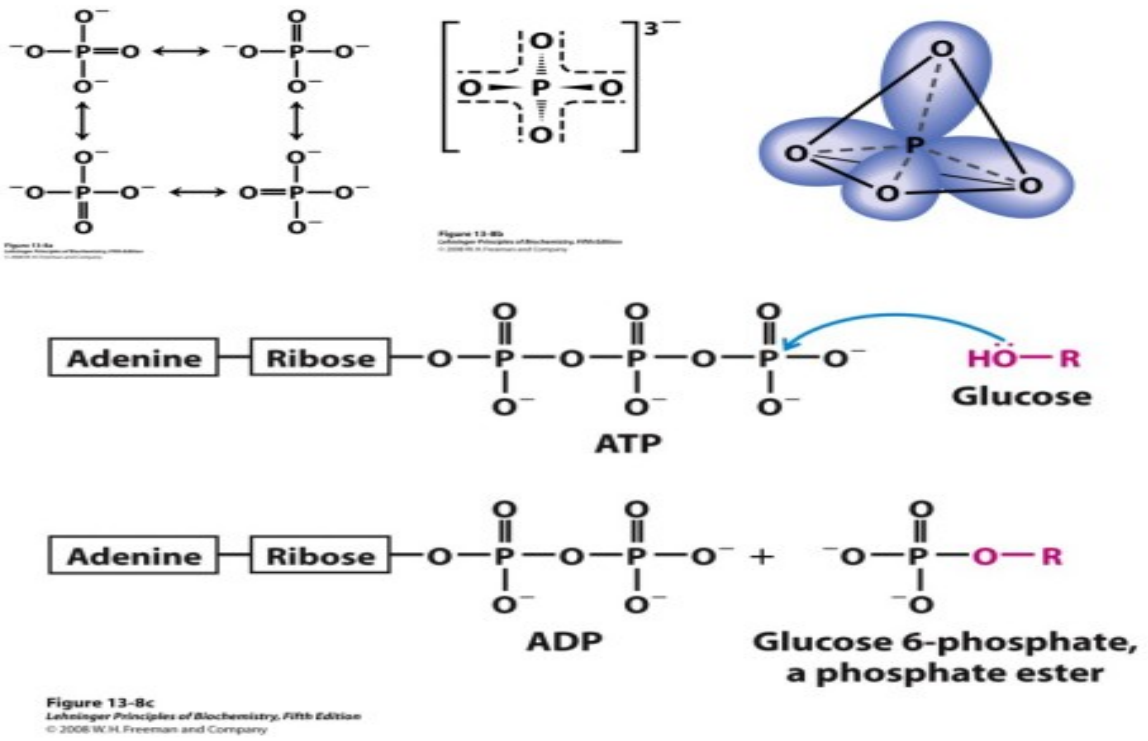


Figure 13-8c
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

Because oxygen is more electronegative than phosphorus, the sharing of electrons is unequal: the central phosphorus bears a partial positive charge and can therefore act as an electrophile

In many metabolic reactions, A phosphoryl group (-PO32-) can be transferred to an alcohol (✉ forming a phosphate ester), or to a carboxylic acid (✉mixed anhydride)

When a nucleophile attacks this Heletrophilic P in ATP, relatively stable pentacovalent structure forms as a reaction intermediate.

Reaction completes wiwhen ADP is released and so.. The trnsfer of

Kinases: enzymes that catalyze phosphoryl group transfer ..e g hexokinase transfers Phosphoryl group from ATP to glucose in glycolysis.

Thioalcohols (thiols): the oxygen atom of an alcohol is replaced with a sulfur atom

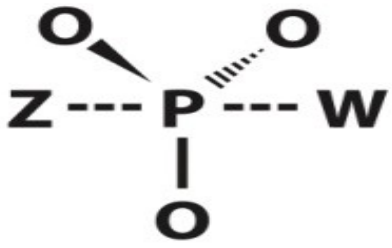


Figure 13-8d
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

Z = R — OH

W = ADP

<u>Oxidation-reduction reactions:</u>

Oxidation

- Addition of oxygen
- loss of electrons
- Loss of hydrogen

Reduction

- Removal of oxygen
- Gain of electrons
- Gain of hydrogen

Oxidation-reduction reactions:

The oxidation levels of carbon in biomolecules. Each compound is formed by oxidation of the red carbon in the compound shown immediately above. Carbon dioxide is the most highly oxidized form of carbon found in living systems.

Carbon atoms can exist in five oxidation states

The transition of these status are of crucial importance in metabolism

Biological reaction (commonly), a compound loses two electrons and two hydrogen ions: dehydrogenations (catalyzed by dehydrogenases)

In some biological oxidation, a carbon atom becomes covalently bonded to an oxygen atom (a enzyme catalyze these reaction calls oxidases)

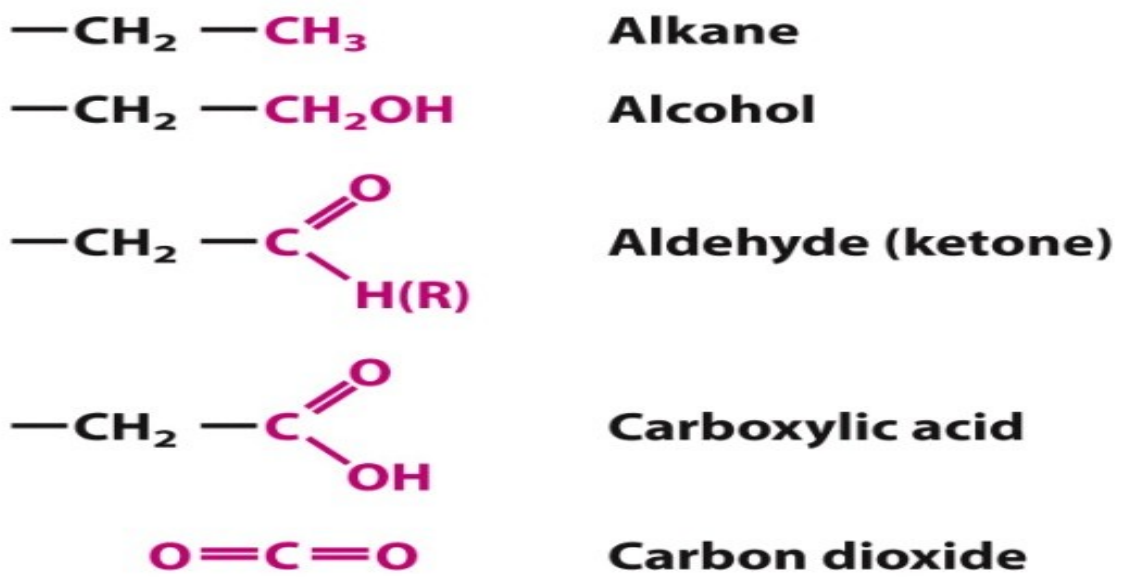


Figure 13-9
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Biological reaction (commonly), a compound loses two electrons and two hydrogen ions: dehydrogenations (catalyzed by dehydrogenases)

In some biological oxidation (but not all), a carbon atom becomes covalently bonded to an oxygen atom (, that is an oxidation reaction catatlysed by the nzyme oxidases)

Oxygenases: if oxygen is derived directly from molecular oxygen (O2).

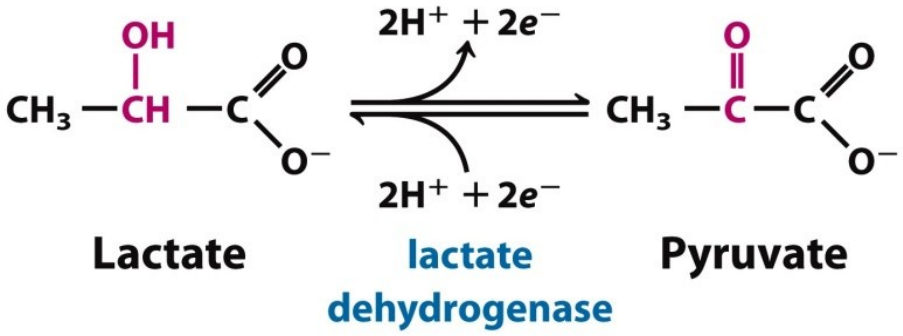
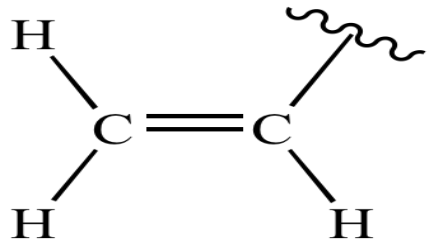


Figure 13-10
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

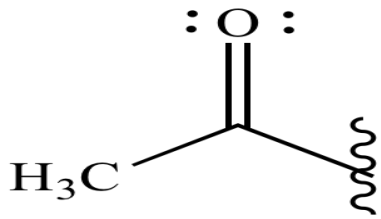
Vinyl group:

In chemistry, vinyl or ethenyl is the functional group with the formula −CH=CH2. It is the ethylene (IUPAC ethene) molecule ($H_2C=CH_2$) less one hydrogen atom.



Acyl group

It contains a double bonded oxygen atom and an alkyl group. In organic chemistry, the acyl group (IUPAC name: alkanoyl) is usually derived from a carboxylic acid. Acyl group is electron withdrawing due to the electronegativity of the oxygen atom.



The name is also used for any compound containing that group, namely R−CH=CH2 where R is any other group of atoms.

An oxoacid is an acid that contains oxygen. Specifically, it is a compound that contains hydrogen, oxygen, and at least one other element, with at least one hydrogen atom bond to oxygen that can dissociate to produce the H+ cation and the anion of the acid

In chemistry, alkyl refers to a group, a substituent, that is attached to other molecular fragments. For example, alkyl lithium reagents have the empirical formula Li(alkyl), where alkyl = methyl, ethyl, etc. A dialkyl ether is an ether with two alkyl groups, e.g., diethyl ether (O(C2H5)2).

Isopropyl group

Methyl group

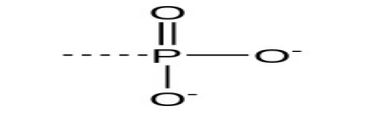
In organic chemistry, an alkyl substituent is an alkane missing one hydrogen

organic compound characterized by

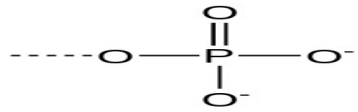
A hemiacetal or a hemiketal is a compound that results from the addition of an alcohol to an aldehyde or a ketone, respectively. The Greek word hèmi, meaning half, refers to the fact that a single alcohol has been added to the carbonyl group, in contrast to acetals or ketals, which are formed when a second alkoxy group has been added to the structure.[1]

Phosphoryl group

A phosphoryl group is the chemical entity PO_3^{2-}. The term is usually used for the compounds in which the phosphoryl group is attached to other atoms, e.g. phosphoryl chloride, or in the description of catalytic mechanism.



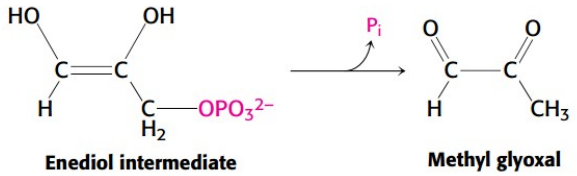
phosphoryl group



phosphate group

Enediol

An alkene enol with a hydroxyl group attached to both carbon atoms of the carbon double bond. A reducing sugar can form an enediol.

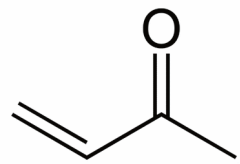


Enediol intermediate

Methyl glyoxal

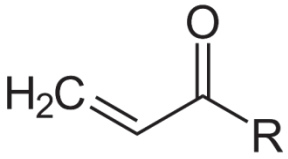
Enone:

An enone, also called an α, β-unsaturated carbonyl, is a type of organic compound consisting of an alkene conjugated to a ketone. The simplest enone is methyl vinyl ketone or $CH_2=CHCOCH_3$.



Acryloyl group

In organic chemistry, the acryloyl group is form of enone with structure $H_2C=CH–C(=O)–$. It is the acyl group derived from acrylic acid. The preferred IUPAC name for the group is prop-2-enoyl, and it is also known as acrylyl or simply acryl.



TOPIC#8

Phosphoryl Group Transfers and ATP

Phosphoryl Group Transfers and ATP

Energy currency of living cells

ATP is used for Phosphoryl transfer reactions

Heterotrophic organisms obtain energy from ATP by breakage of covalent bond, and can undergo one of the two following reactions

A) ATP ⟶ ADP + Pi OR

B) ATP ⟶ AMP + 2 Pi

What we have studied so far has told us that ATP is the Energy currency in biological systems

And how does it do this? .. By undergoing for Phosphoryl Transfer Reactions

So let take a look at the special role of ATP that links catabolism and

Anabolism and produces energy for functions of life.

Heterotrophic organisms obtain energy from ATP and this involves the breakage of covalent bond, and can undergo one of the two following reactions

A) ATP ⟶ ADP + Pi OR

B) ATP ⟶ AMP + 2 Pi

Living cells use that energy to make ATP from ADP and Pi

Living cells obtain free energy in a chemical form by the catabolism of nutrient molecules .They use that energy to make ATP from ADP and Pi .

ATP donates some of its chemical energy to

Endergonic processes such as the synthesis of metabolic intermediates and macromolecules from smaller precursors

The transport of substances across membranes against concentration gradients

Mechanical motion (muscle contraction), etc.

Energy donation involves group transfer and hydrolysis of ATP/energy rich phosphate compounds

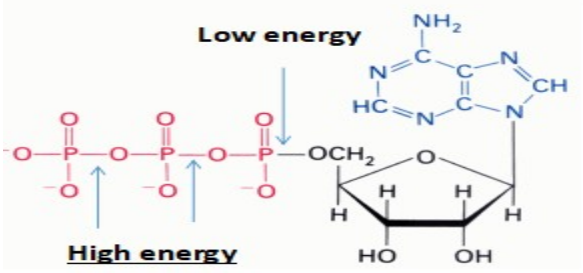
Energy donation does not involve only hydrolysis of ___ phosphoryl bond but also group transfer reactionsand cleaving of other energy rich phosphate compunds

The Free-Energy Change for ATP Hydrolysis Is Large and Negative

The hydrolytic cleavage of the terminal phosphoanhydride bond in ATP separates one of the three negatively charged phosphates and thus relieves some of the electrostatic repulsion in ATP

The terminal two phosphoanhydride bonds, are known as high energy bonds because their cleavage releases tremendous amount of free energy .Released Pi and ADP are highly soluble

The terminal two phosphoanhydride bonds, indicated in the red, are known as high energy bonds because when they are broken a tremendous amount of free energy is released.



Adenosine Triphosphate

Why large energy is released? This is depicted in this picture where we see

What we see in this picture is the hydrolytic cleavage of high energy bond.. That finally reduces the electrostatic repulsion among 4 negative charges on ATP

And the inorganic phosphate Pi is relaesed iis stabilized by the formation of several resonance forms not possible in ATP.. Now this mpolecule has got same degree of double bond character with all 4 oxygen atoms

3rdly ..Pi and ADP are highly soluble as compared to ATP that means that the products of reaction are in more stable condition as compared to the reactant

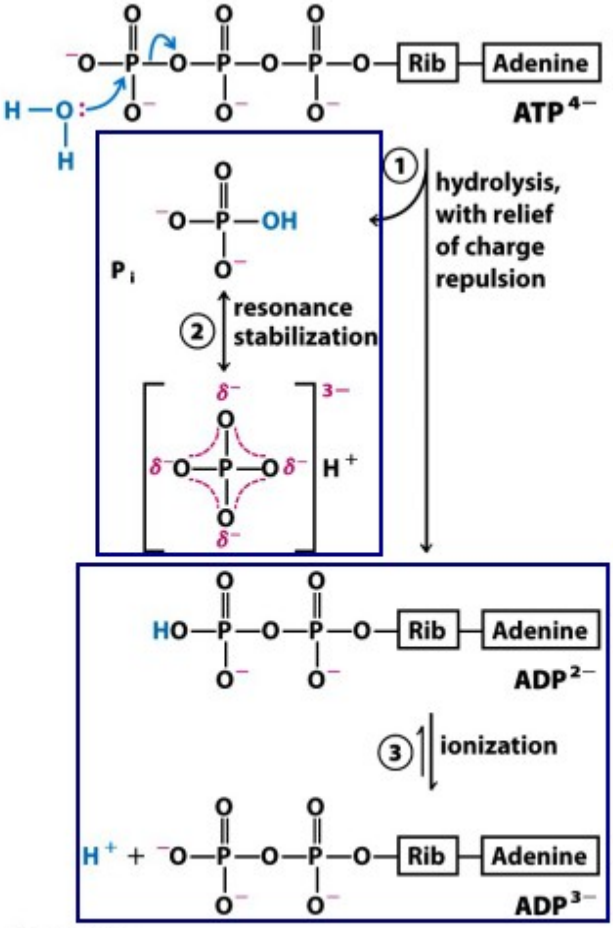


Figure 13-11
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

FIGURE 13–11 Chemical basis for the large free-energy change associated with ATP hydrolysis. ❶ The charge separation that results from hydrolysis relieves electrostatic repulsion among the four negative charges on ATP. ❷ The product inorganic phosphate (P_i) is stabilized by formation of a resonance hybrid, in which each of the four phosphorus–oxygen bonds has the same degree of double-bond character and the hydrogen ion is not permanently associated with any one of the oxygens. (Some degree of resonance stabilization also occurs in phosphates involved in ester or anhydride linkages, but fewer resonance forms are possible than for P_i.) A third factor (not shown) that favors ATP hydrolysis is the greater degree of solvation (hydration) of the products P_i and ADP relative to ATP, which further stabilizes the products relative to the reactants.

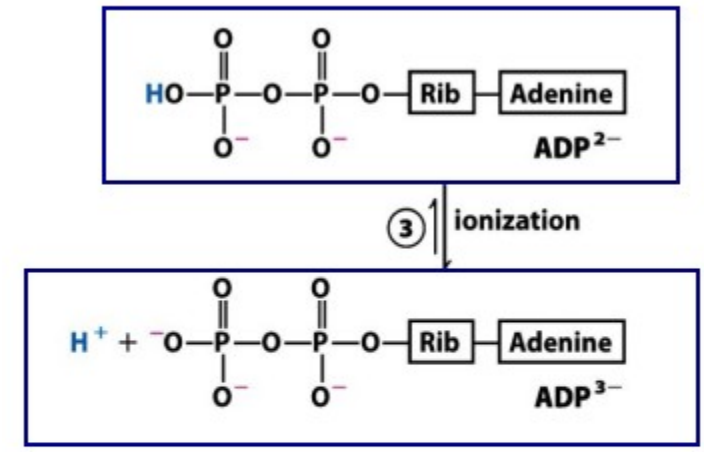
ADP2- is the other product of hydrolysis and it immediately ionizes, releasing H+ into a medium of very low [H+] (~10-7 M)

the hydrolysis of ATP is highly exergonic (ΔG'0 = -30.5 kj/mol), the ATP is stable at pH 7

Rapid hydrolysis of ATP occurs only when catalyzed by an enzyme

Meaning… that in living cells the standard conditions of molar concentration cant be followed..

Wha we studied about standard conditions of temp, pressure, and Atmospheric pressure that temp: 300K/25C, Molar con: 1M and P=1atm pressure.



$$ATP^{4-} + H_2O \longrightarrow ADP^{3-} + P_i^{2-} + H^+$$
$$\Delta G'^{\circ} = -30.5\ kJ/mol$$

Actual free energy of hydrolysis (ΔG) of ATP in living cells is very different because

- the cellular concentration of ATP, ADP, and Pi are not identical
- ATP, ADP, and Pi concentration are much lower than the standard condition of 1M
- Mg2+ in the cytosol binds to ATP and ADP

The free energy change for ATP hydrolysis is -30,5 kj/mol under standard conditions However,the actual free energy change (ΔG) of ATP hydrolysis in living cells is very different.

The cellular concentrations of ATP, ADP and Pi are not same and are much lower than the 1 M standard conditions.

Meaning… that in living cells the standard conditions of molar concentration cant be followed..

Wha we studied about standard conditions of temp, pressure, and Atmospheric pressure that temp: 300K/25C, Molar con: 1M and P=1atm pressure.

The cellular concentrations of ATP, ADP and Pi are not same and are much lower than the 1 M standard conditions.

In addition, Mg2+ in the cytosol binds to ATP and ADP and for most enzymatic reactions that involve ATP as phosphorly group donor, the true substrate is MgATP-2. The relevant ΔG'0 is therefore that for MgATP-2 hydrolysis.

In addition, Mg2+ in the cytosol binds to ATP and ADP and for most enzymatic reactions that involve ATP as phosphorly group donor, the true substrate is MgATP-2. The relevant ΔG'0 is therefore that for MgATP-2 hydrolysis.

TABLE 13–5 Adenine Nucleotide, Inorganic Phosphate, and Phosphocreatine Concentrations in Some Cells

	Concentration (mM)*				
	ATP	ADP†	AMP	P_i	PCr
Rat hepatocyte	3.38	1.32	0.29	4.8	0
Rat myocyte	8.05	0.93	0.04	8.05	28
Rat neuron	2.59	0.73	0.06	2.72	4.7
Human erythrocyte	2.25	0.25	0.02	1.65	0
E. coli cell	7.90	1.04	0.82	7.9	0

*For erythrocytes the concentrations are those of the cytosol (human erythrocytes lack a nucleus and mitochondria). In the other types of cells the data are for the entire cell contents, although the cytosol and the mitochondria have very different concentrations of ADP. PCr is phosphocreatine, discussed on p. 510.

†This value reflects total concentration; the true value for free ADP may be much lower (p. 503).

Table 13-5
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

For ATP as phosphoryl group donor, the true substrate is MgATP-2.

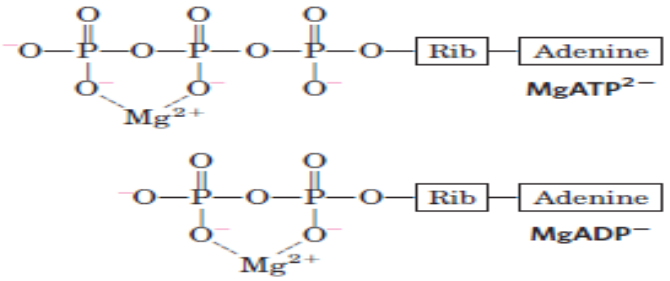


FIGURE 13–12 Mg^{2+} and ATP. Formation of Mg^{2+} complexes partially shields the negative charges and influences the conformation of the phosphate groups in nucleotides such as ATP and ADP.

Phosphorylation potential: The actual free energy of hydrolysis of ATP under intracellular conditions is often called its phosphorylation potential, °Gp.

WORKED EXAMPLE 13–2 Calculation of ΔG_p

Calculate the actual free energy of hydrolysis of ATP, ΔG_p, in human erythrocytes. The standard free energy of hydrolysis of ATP is −30.5 kJ/mol, and the concentrations of ATP, ADP, and P_i in erythrocytes are as shown in Table 13–5. Assume that the pH is 7.0 and the temperature is 37 °C (body temperature). What does this reveal about the amount of energy required to *synthesize* ATP under the same cellular conditions?

Solution: The concentrations of ATP, ADP, and P_i in human erythrocytes are 2.25, 0.25, and 1.65 mM, respectively. The actual free energy of hydrolysis of ATP under these conditions is given by the relationship (see Eqn 13–4)

$$\Delta G_p = \Delta G'^{\circ} + RT \ln \frac{[ADP][P_i]}{[ATP]}$$

Substituting the appropriate values we get

$$\begin{aligned}\Delta G_p &= -30.5\ \text{kJ/mol} + \left[(8.315\ \text{J/mol}\cdot\text{K})(310\ \text{K}) \ln \frac{(0.25 \times 10^{-3})(1.65 \times 10^{-3})}{(2.25 \times 10^{-3})}\right] \\ &= -30.5\ \text{kJ/mol} + (2.58\ \text{kJ/mol}) \ln 1.8 \times 10^{-4} \\ &= -30.5\ \text{kJ/mol} + (2.58\ \text{kJ/mol})(-8.6) \\ &= -30.5\ \text{kJ/mol} - 22\ \text{kJ/mol} \\ &= -52\ \text{kJ/mol}\end{aligned}$$

(Note that the final answer has been rounded to the correct number of significant figures (52.5 rounded to 52), following rules for rounding a number that ends in a 5 to the nearest even number.) Thus ΔG_p, the actual free-energy change for ATP hydrolysis in the intact erythrocyte (−52 kJ/mol), is much larger than the standard free-energy change (−30.5 kJ/mol). By the same token, the free energy required to *synthesize* ATP from ADP and P_i under the conditions prevailing in the erythrocyte would be 52 kJ/mol.

Phosphorylated compounds

Here are few compounds tht are phosphorylated inn nature and play important roles in appropriate functioning of living cells

- Phosphoenolpyruvate
- 1,3-bisphosphoglycerate
- Phosphocreatine

- ADP
- ATP
- AMP
- PPi
- Glucose 1-phosphate
- Fructose 6-phosphate
- Glucose 6-phosphate

Large Free Energies of Hydrolysis Phosphorylated Compounds and Thioesters

Orbital overlap between the O and C atoms allows resonance stabilization in oxygen esters; orbital overlap between S and C atoms is poorer and provides little resonance stabilization.

Other compounds that have large free energy of Hydrolysis

- Phosphorylated compounds
- Thioesters (Acetyl-CoA)

TOPIC#9

Phosphoryl Group Transfers in thioesters

Phosphoryl Group Transfers and ATP

Other Phosphorylated Compounds Have Large Free Energies of Hydrolysis

Orbital overlap between the O and C atoms allows resonance stabilization in oxygen esters; orbital overlap between S and C atoms is poorer and provides little resonance stabilization.

The products of both types of hydrolysis reaction have about the same free-energy content (G), but the thioester has a higher free-energy content than the oxygen ester.

Other compounds that have large free energy change

- Phosphorylated compounds

- Thioesters (Acetyl-CoA)

Phosphoenolpyruvate (PEP)

A phosphodiester bond occurs when exactly two of the hydroxyl groups in phosphoric acid react with hydroxyl groups on other molecules to form two ester bonds.[1]

Phosphodiester bonds are te basics of life on Earth,[fn 1] as they make up the backbone of the strands of nucleic acid. In DNA and RNA, the phosphodiester bond is the linkage between the 3' carbon atom of one sugar molecule and the 5' carbon atom of another, deoxyribose in DNA and ribose in RNA. Strong covalent bonds form between the phosphate group and two 5-carbon ring carbohydrates (pentoses) over two esterbonds.

The phosphate groups in the phosphodiester bond are negatively charged.

Phosphoenolpyruvate contains a phosphate ester bond that undergoes hydrolysis to yield to enol form of pyruvate.

The enol form of pyruvate can immediately tautomerize to the more stable keto form of pyruvate

Because phosphoenolpyruvate has only one form (enol) and the product, pyruvate, has two possible forms, the product is more stabilized relative to the reactant.

This is the greatest contributing factor to the high standard free energy change of hydrolysis of phosphoenolpyruvate (ΔG'0 = -61.9 kj/mol)

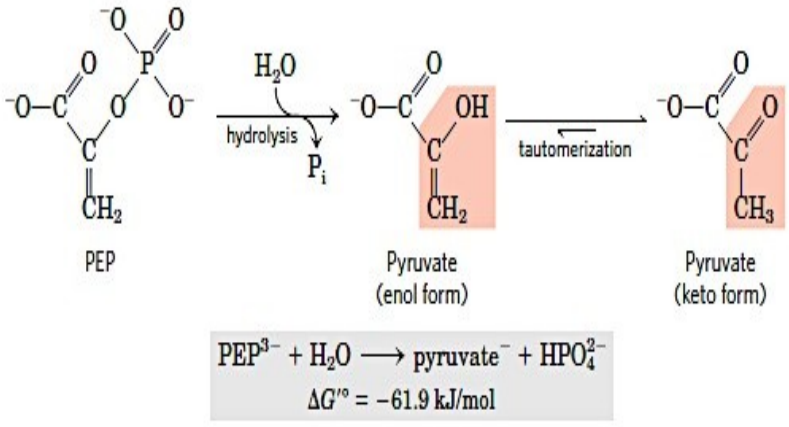


FIGURE 13-13 Hydrolysis of phosphoenolpyruvate (PEP). Catalyzed by pyruvate kinase, this reaction is followed by spontaneous tautomerization of the product, pyruvate. Tautomerization is not possible in PEP, and thus the products of hydrolysis are stabilized relative to the reactants. Resonance stabilization of P_i also occurs, as shown in Figure 13-11.

$$PEP^{3-} + H_2O \longrightarrow pyruvate^- + HPO_4^{2-}$$
$$\Delta G'^{\circ} = -61.9\ kJ/mol$$

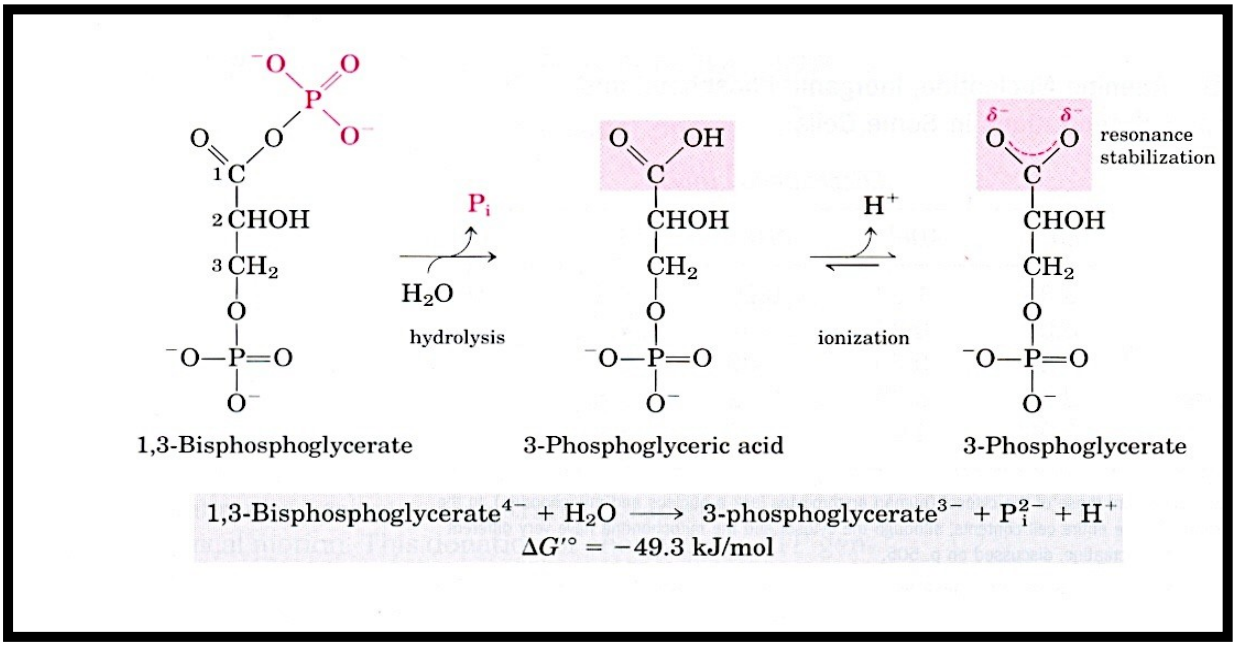
1,3-bisphosphoglycerate

1,3-bisphosphoglycerate contains an anhydride bond between the carboxyl group at C-1 and phosphoric acid.

Hydrolysis of this acyl phosphate is accompanied by a large, negative, standard free energy change ($\Delta G'0 = -49{,}3$ kj/mol) .This large, negative $\Delta G'0$ can, again, be explained in terms of the structure of reactants and products

Addition of water to anhyhride bond of 1,3-bisphosphoglycerate, one of the direct products, 3-phosphoglyceric acid, immediately leads to the lose a proton to give the carboxylate ion, 3-phosphoglycerate, which has two equally probable resonance forms

Removal of a direct product, 3-phospho-glyceric acid, and formation of resonance-stabilized ion favor the forward reaction.



Hydrolysis of 1,3-bisphosphoglycerate.

The direct product of hydrolysis is 3-phosphoglyceric acid, with an undissociated carboxylic acid. Its dissociation allows resonance structures that stabilize the product relative to the reactants.

Resonance stabilization of Pi further contributes to the negative free-energy change.

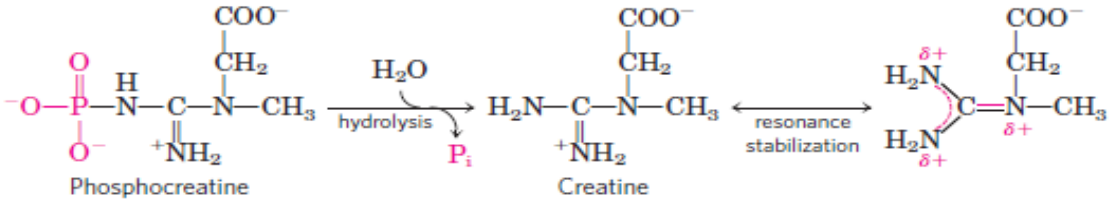
Phosphocreatine (Pcr)

In the phosphocreatine, the P-N bond can be hydrolyzed to generate free creatine and Pi.

The release of Pi and the resonance stabilization of creatine favor the forward reaction.

Pi is also resonance stabilized.

The standard free energy change of phosphocreatine is large and negative (ΔG'0 = -43 kj/mol).



$$\text{Phosphocreatine}^{2-} + H_2O \longrightarrow \text{creatine} + HPO_4^{2-}$$
$$\Delta G'^{\circ} = -43.0 \text{ kJ/mol}$$

Thioesters

Thioesters contain a sulfur atom in the position occupied

by an oxygen

Thioesters have large, negative standard free energy change

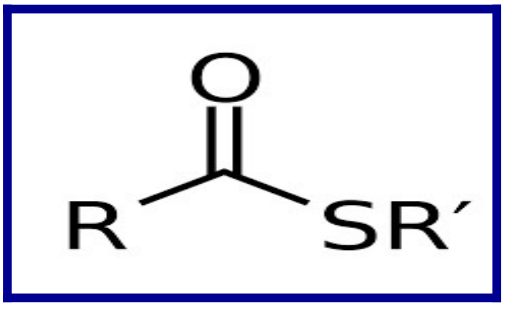
of hydrolysis.

Hydrolysis of acetyl-coenzyme A.. important in metabolism and has a large, negative, standard free energy of hydrolysis.

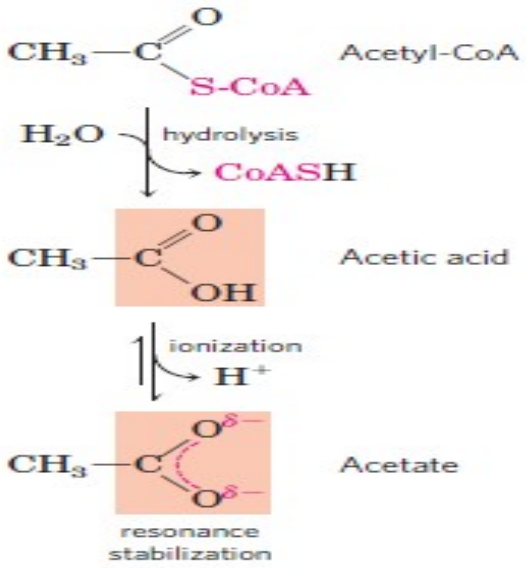
The acyl group in these compounds is activated for trans-acylation, condensation or oxidation-reduction reactions.Hydrolysis of the ester bond generates a carboxylic acid

ΔG'0 = -31,4 kj/mol for acetyl-CoA hydrolysis

Acetyl coenzyme A is one of many thioesters



Hydrolysis of acetyl-coenzyme A. Acetyl-CoA is a thioester with a large, negative, standard free energy of hydrolysis.



$$\text{Acetyl-CoA} + H_2O \longrightarrow \text{acetate}^- + \text{CoA} + H^+$$
$$\Delta G'^\circ = -31.4 \text{ kJ/mol}$$

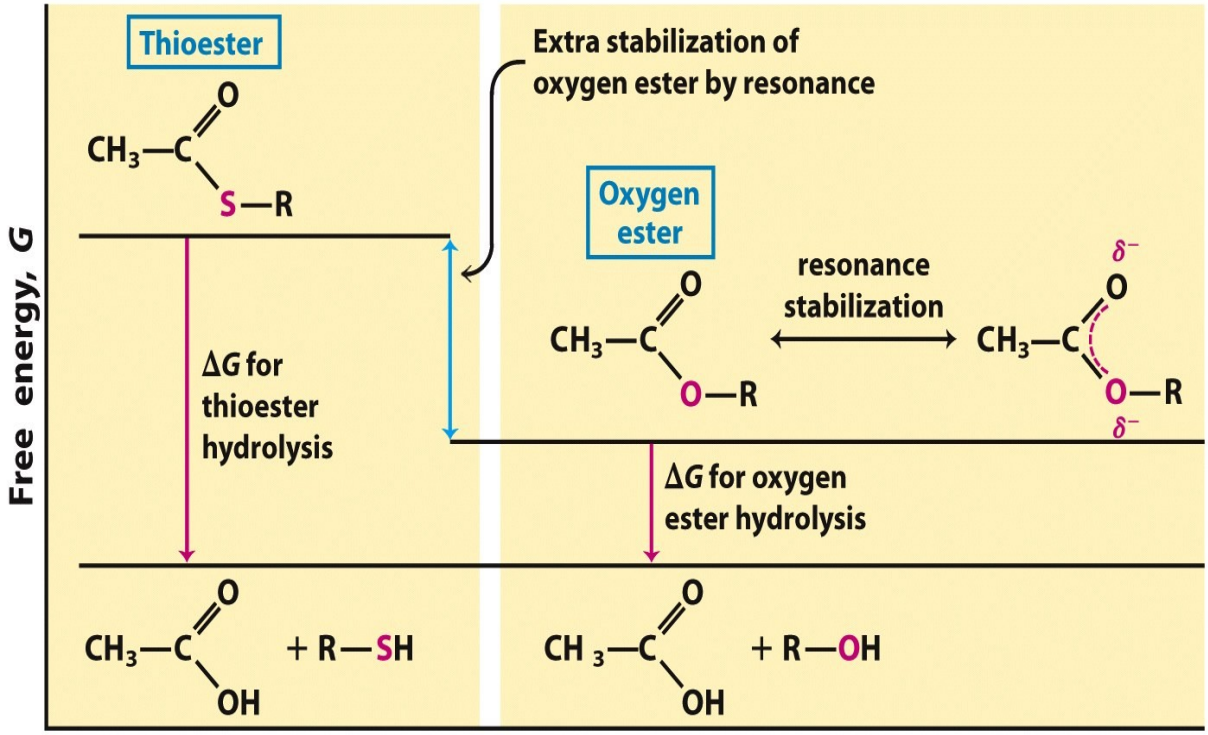


Figure 13-17
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Free energy released by hydrolysis of THIOESTERS is is more than OXYGEN esters

Summary for hydrolysis reactions

For hydrolysis reactions with large, negative standard free energy changes, the products are more stable than the reactants for one or more of the following reasons:

1. The bond strain in reactants due to electrostatic repulsion is relieved by charge separation, as for ATP

2. The products are stabilized by ionization, as for ATP, acyl phosphates, thioesters

3. The products are stabilized by isomerization (tautomerization) as for phosphoenolpyruvate

4. The products are stabilized by resonance as for creatine released from phosphocreatine, carboxylate ion released from acyl phosphates and thioesters and phosphate released from anhydride or ester linkages

The term 'high-energy phosphate bond' was used by biochemists to describe P-O bond broken in hydrolysis reactions for a long time. But it is incorrect and misleading as it wrongly suggests that the bond itself contains the energy.

In fact, the breaking of all chemical bonds requires an input of energy. The free energy released by hydrolysis of phosphate compounds does not come from the specific bond that is broken

results from the products of the reaction having a lower free energy content than the reactants

The phosphate compounds found in living organisms can be arbitrarily divided into two groups based on their standard free energy changes of hydrolysis.

- 'High-energy' compounds have a $\Delta G'^{0}$ of hydrolysis more negative than -25 kj/mol
- 'low-energy' compounds have a less negative $\Delta G'^{0}$

ATP with a ΔG'0 of hydrolysis of -30 kj/mol is a high-energy compound

glucose 6-phosphate is a low-energy compound (ΔG'0 = -13,8 kj/mol)

TABLE 13–6 Standard Free Energies of Hydrolysis of Some Phosphorylated Compounds and Acetyl-CoA (a Thioester)

	$\Delta G'^{\circ}$ (kJ/mol)	$\Delta G'^{\circ}$ (kcal/mol)
Phosphoenolpyruvate	−61.9	−14.8
1,3-bisphosphoglycerate ($\longrightarrow$ 3-phosphoglycerate + P_i)	−49.3	−11.8
Phosphocreatine	−43.0	−10.3
ADP ($\longrightarrow$ AMP + P_i)	−32.8	−7.8
ATP ($\longrightarrow$ ADP + P_i)	−30.5	−7.3
ATP ($\longrightarrow$ AMP + PP_i)	−45.6	−10.9
AMP ($\longrightarrow$ adenosine + P_i)	−14.2	−3.4
PP_i ($\longrightarrow$ 2P_i)	−19.2	−4.0
Glucose 3-phosphate	−20.9	−5.0
Fructose 6-phosphate	−15.9	−3.8
Glucose 6-phosphate	−13.8	−3.3
Glycerol 3-phosphate	−9.2	−2.2
Acetyl-CoA	−31.4	−7.5

Source: Data mostly from Jencks, W.P. (1976) in *Handbook of Biochemistry and Molecular Biology,* 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data*, Vol. 1, pp. 296–304, CRC Press, Boca Raton, FL. The value for the free energy of hydrolysis of PP_i is from Frey, P.A. & Arabshahi, A. (1995) Standard free-energy change for the hydrolysis of the α-β-phosphoanhydride bridge in ATP. *Biochemistry* **34**, 11,307–11,310.

Table 13-6
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

TOPIC#10

Bioenergetics -3-Hydrolysis of ATP

ATP Provides Energy by Group Transfers,Not by Simple Hydrolysis

ATP serves as the universal energy currency in all living cells.ATP can carry energy from high-energy phosphate compounds produced by catabolism to compounds such as glucose, converting them into more reactive species.

ATP Provides Energy by Group Transfers,Not by Simple Hydrolysis

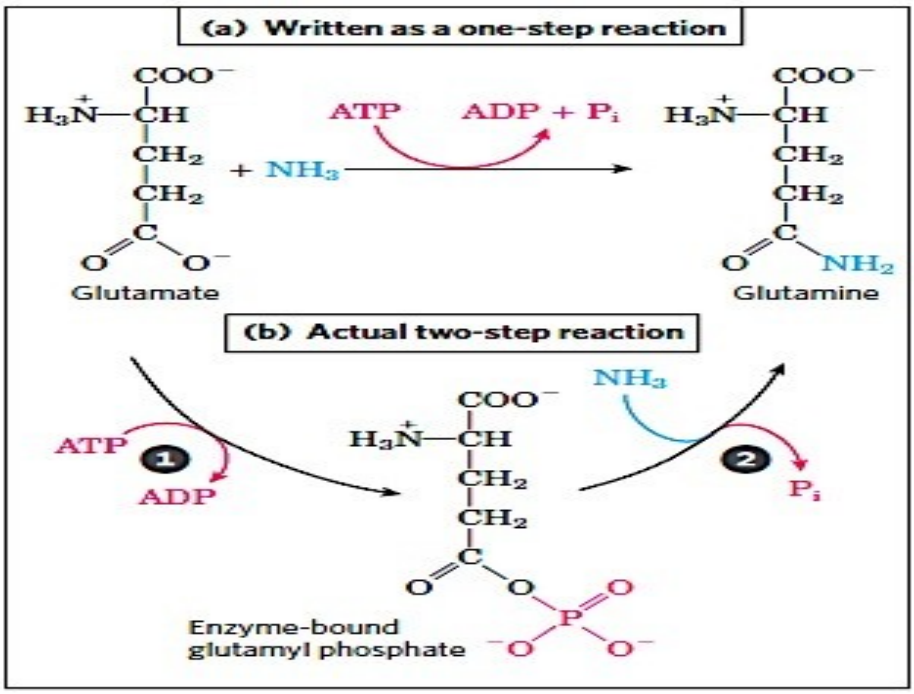
ATP hydrolysis in two steps.

The contribution of ATP to a reaction is often shown as a single step, but is almost always a two-step process.

Shown here is the reaction catalyzed by ATP-dependent glutamine synthetase.

A phosphoryl group is transferred from ATP to glutamate

the phosphoryl group is displaced by NH3 and released as Pi.



<u>Hydrolysis accompanied biological processes</u>

1. Noncovalent binding of ATP (or GTP), followed by its hydrolysis to ADP (or GDP) and Pi, (to cycle some proteins between two conformations), in muscle contraction
2. In the movement of enzymes along DNA
3. In the movement of ribosomes along messenger RNA
4. The energy-dependent reactions catalyzed by helicases and some topoisomerases GTP-binding proteins that act in signaling pathways to drive conformational changes

The phosphate compounds found in living organisms can be divided into two groups based on their standard free energy changes of hydrolysis.

- 'High-energy' compounds have a ΔG'0 of hydrolysis more negative than -25 kj/mol
- 'low-energy' compounds have a less negative ΔG'0

ATP with a ΔG'0 of hydrolysis of -30 kj/mol is a high-energy compound

glucose 6-phosphate is a low-energy compound (ΔG'0 = -13,8 kj/mol)

The term "high-energy phosphate bond," used to describe the P-O bond broken in hydrolysis reactions, is incorrect and misleading as it
wrongly suggests that the bond itself contains the energy.

the breaking of all chemical bonds requires an input of energy
As is evident from the additivity of free energy changes of sequential reactions, any phosphorylated compound can be synthesized by coupling the synthesis to the breakdown of another phosphorylated compound with a more negative standard free energy change of hydrolysis.

Example: $\Delta G'^0$

$PEP + H_2O \longrightarrow Pyruvate + P_i$ **-61,9**

$ADP + P_i \longrightarrow ATP + H_2O$ **+30,5**

$PEP + ADP \longrightarrow Pyruvate + ATP$ **-31,4**

Clevage of P_i from PEP releases more energy than is needed to drive to condensation of P_i with ADP, the direct donation of a phosphoryl group from PEP to ADP is thermodynamically feasible

Notice that while the energy of overall reaction is represented as the algebraic sum of first two reactions, the overall reaction (third) does not involve Pi ; PEP donates a phosphoryl group directly to ADP.

On the basis of their standard free energy changes of hydrolysis, phosphorylated compounds as having a high or low phosphoryl group transfer potential.

The transfer of a phosphoryl group to a compound effectively puts free energy into that compound that can be used in subsequent metabolic transformations.

Much of catabolism is directed toward the synthesis of high-energy phosphate compounds, but their formation is not an end in itself; they are the means of activating a wide variety of compounds for further chemical transformation.

TOPIC#11

Bioenergetics -3-Hydrolysis of ATP- part2.

ATP Provides Energy by Group Transfers,Not by Simple Hydrolysis.. cont

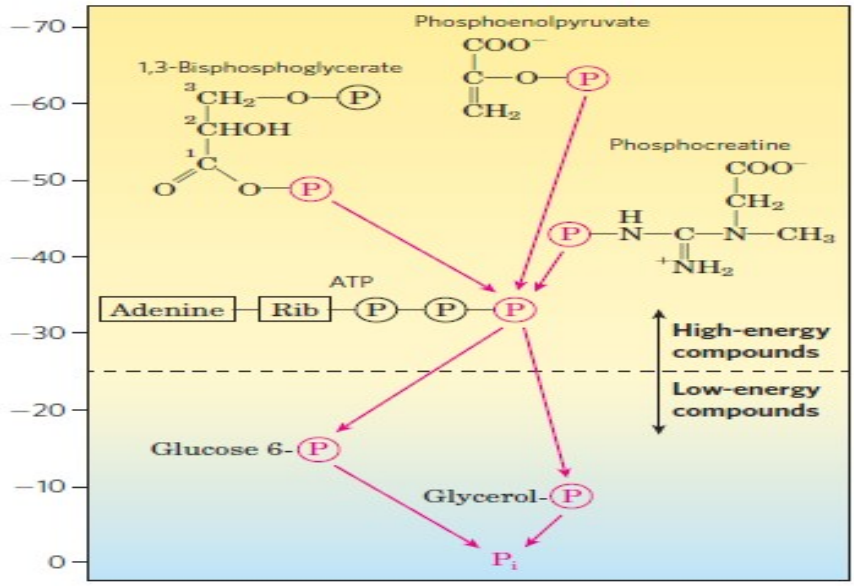
Ranking of biological phosphate compounds by standard free energies of hydrolysis

flow of phosphoryl groups, represented by P, from high-energy phosphoryl group donors via ATP to acceptor molecules (such as glucose and glycerol) to form their low-energy phosphate derivatives.

This flow of phosphoryl groups is catalyzed by kinases, and proceeds with an overall loss of free energy under intracellular conditions.

Hydrolysis of low-energy phosphate compounds releases Pi, which has an even lower phosphoryl group transfer potential

(The location of each compound's donor phosphoryl group along the scale approximately indicates the del G'°▼ of hydrolysis.)



One more chemical feature of ATP is crucial to its role in metabolism: although in aqueous solution ATP is thermo-dynamically unstable and is therefore a good phosphoryl group donor, it is kinetically stable.

Because of high activation energies required for uncatalyzed reaction ATP does not spontaneously donate phosphoryl groups to water or to the other potential acceptors in the cell.

ATP hydrolysis occurs only when specific enzymes which lower the energy of activation are present

In aqueous solution ATP is thermodynamically unstable and is therefore a good phosphoryl group donor, it is kinetically stable.

Because of high activation energies required for uncatalyzed reaction ATP does not spontaneously donate phosphoryl groups to water or to the other potential acceptors in the cell.

ATP hydrolysis occurs only when specific enzymes which lower the energy of activation are present

thermodynamically unstable: when energy level of products is lower than that of reactants

kinetically stable is when reaction has high activation energy due to some strong bonds.

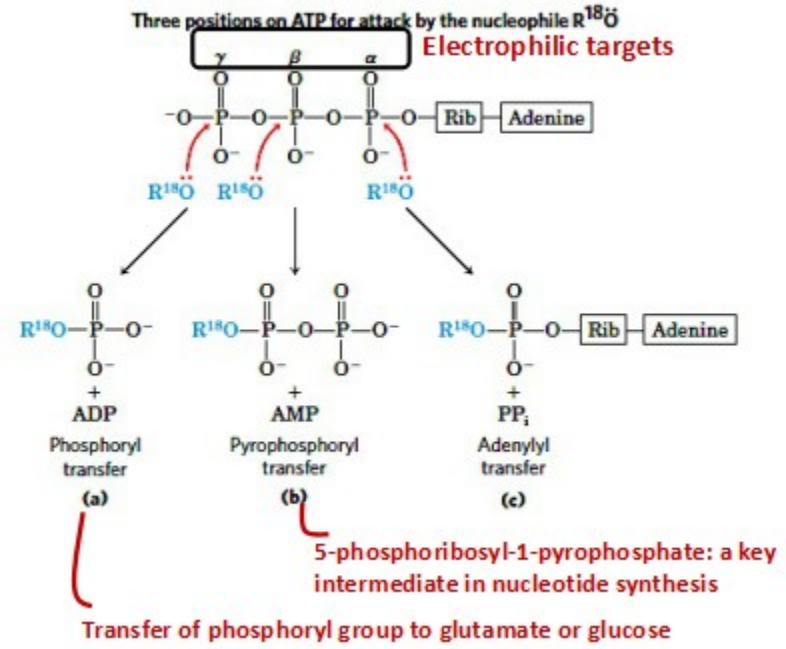
ATP Donates Phosphoryl, Pyrophosphoryl, and Adenylyl Groups

Nucleophilic displacement reactions of ATP.

Any of the three P atoms (➔, ↑, or ✚) may serve as the electrophilic target for nucleophilic attack.

By the labeled nucleophile R-18O: The nucleophile may be an alcohol (ROH), a carboxyl group (RCOO-), or a phosphoanhydride (a nucleoside mono- or diphosphate, for example).

When the oxygen of the nucleophile attacks the position, the bridge oxygen of the product is labeled, indicating that the group transferred from ATP is a phosphoryl (2PO3-2), not a phosphate (OPO3-2)



Each of the three phosphates of ATP is susceptible to nucleophilic attack and each position of attack yields a different type of product

Any of the three P atoms (➔, ↑, or ✚) may serve as the electrophilic target for nucleophilic attack. In this case, by the labeled nucleophile R-18O: The nucleophile may be an alcohol (ROH), a carboxyl group (RCOO-), or a phosphoanhydride (a nucleoside mono- or diphosphate, for example).

When the oxygen of the nucleophile attacks the position, the bridge oxygen of the product is labeled, indicating that the group transferred from ATP is a phosphoryl (2PO3-2), not a phosphate (OPO3-2)

Attack on the position displaces AMP and leads to the transfer of a pyrophosphoryl (not pyrophosphate) group to the nucleophile.

Attack on the a position displaces PPi and transfers the adenylyl group to the nucleophile.

Attack on the position displaces AMP and leads to the transfer of a pyrophosphoryl (not pyrophosphate) group to the nucleophile.

Attack on the a position displaces PPi and transfers the adenylyl group to the nucleophile.

5-phosphoribosyl-1-pyrophosphate is a key intermediate in nucleotide synthesis

- Hydrolysis of a-b phosphoanhydride bond: ~46 kJ/mol

- Hydrolysis of b-g phosphoanhydride bond: ~31 kJ/mol
- PPi → 2 Pi: -19.2 kJ/mol: total = -64.8 kJ/mol (Adenylation: produces Ppi hydrolysed to 2Pi by inorganic pyrophosphatase enzyme)

Activation of fatty acids

a) Energy yielding oxidation

b) Synthesis of more complex lipids (thiol esters)

The first step in the activation of a fatty acid— either for energy-yielding oxidation or for use in the synthesis of more complex lipids—is the formation of its thiol ester.

The direct condensation of a fatty acid with coenzyme A is endergonic, but the formation of fatty acyl–CoA is exergonic by stepwise removal of two phosphoryl groups from ATP.

First, adenylate (AMP) is transferred from ATP to the carboxyl group of the fatty acid, forming a mixed anhydride (fatty acyl adenylate) and liberating PPi.

The thiol group of coenzyme A then displaces the adenylyl group and forms a thioester with the fatty acid.

sum of these two reactions is energetically equivalent to the exergonic hydrolysis of ATP to AMP and Ppi

The formation of fatty acyl–CoA is endergonic but made energetically favorable (exergonic) by hydrolysis of the PPi by inorganic pyrophosphates.

Thus, in the activation of a fatty acid, both phosphoanhydride bonds of ATP are broken.

Luciferin

William McElroy and his colleagues at the Johns Hopkins University isolated the principal biochemical components: luciferin, a complex carboxylic acid, and luciferase, an enzyme.

The generation of a light flash requires activation of luciferin by an enzymatic reaction involving pyrophosphate cleavage of ATP to form luciferyl adenylate.

In the presence of molecular oxygen and luciferase, the luciferin undergoes a multistep oxidative decarboxylation to oxyluciferin.

This process is accompanied by emission of light.

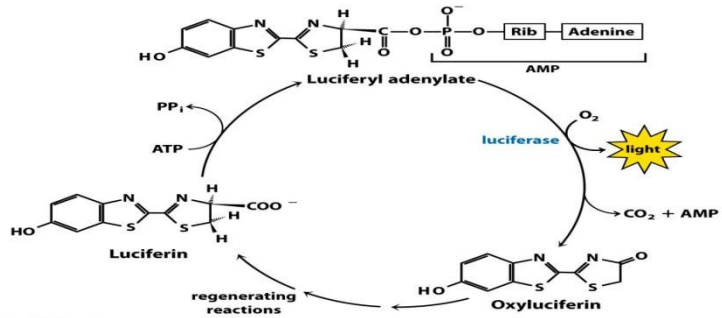
Luciferin is regenerated from oxyluciferin in a subsequent series of reactions.



Box 13-1 figure 1a
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

In the laboratory, pure firefly luciferin and luciferase are used to measure minute quantities of ATP by the intensity of the light flash produced. As little as a few picomoles (10-12 pico mol) of ATP can be measured in this way.

Next-gen pyrosequencing of DNA relies on flashes of light from the luciferin-luciferase reaction to detect the presence of ATP after addition of nucleotides to a growing strand of DNA



Box 13-1 figure 1b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Assembly of Informational Macromolecules Requires Energy

Next heading under this topic.. The precursors for DNA and RNA synthesis are nucleoside triphosphates, and polymerization is accompanied by cleavage of the phosphoanhydride linkage between ➔ and ➔ phosphates, with the release of PPi (Fig. 13–20).

The moieties transferred to the growing polymer in these reactions are adenylate (AMP), guanylate (GMP), cytidylate (CMP), or uridylate (UMP) for RNA synthesis, and their deoxy analogs (with TMP in place of UMP) for DNA synthesis.

Adenylyl groups from ATP

required for the activation of amino acids for protein synthesis, several steps on the ribosome are also accompanied by GTP hydrolysis.

TOPIC#12

Bioenergetics -3-ATP energizes active transport and muscle contraction

ATP energizes active transport and muscle contraction

Transport processes are major consumers of energy

In brain and kidney, two third of total is consumed to pump sodium and potassium ions across plasma membrane (Na + K+ ATPase)

Na+ dependent phosphorylation of Na+ K+ ATPase changes protein conformation, K+ dependent dephosphorylation brings back original conformation.

Phosphoryl group transfer: to enzyme, not to substrate

<u>ATP energizes active transport and muscle contraction</u>

Direct hydrolysis of ATP (ATP ADP+Pi) is the source of energy in the conformational changes that produce muscle contraction

ATP bind with head of myosin (not covalent) dissociate from actin ATP hydrolysis induces conformational change (still ADP and Pi bound tightly) Pi release conformational change of the head of myosin actin move

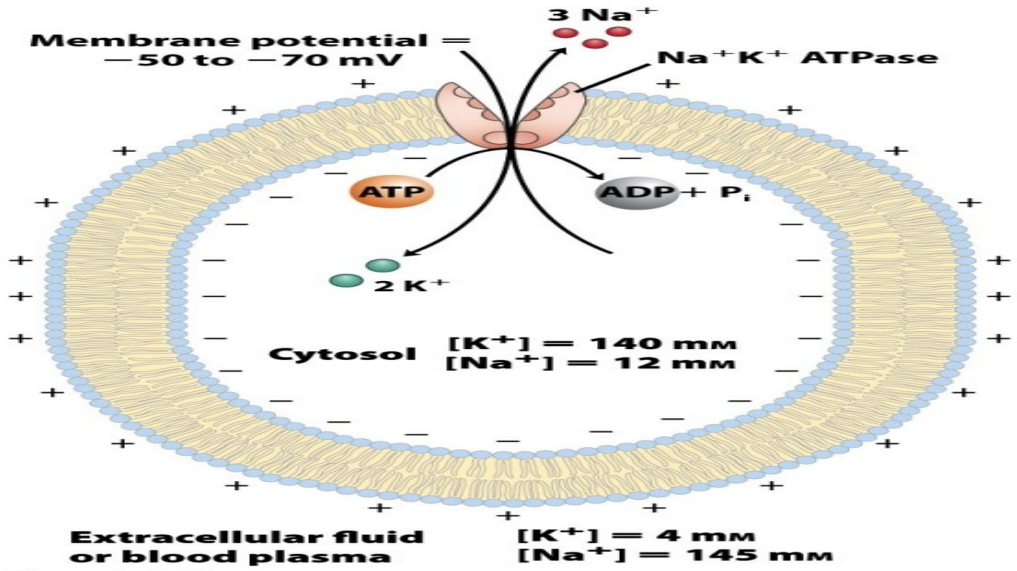


Figure 11-38
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

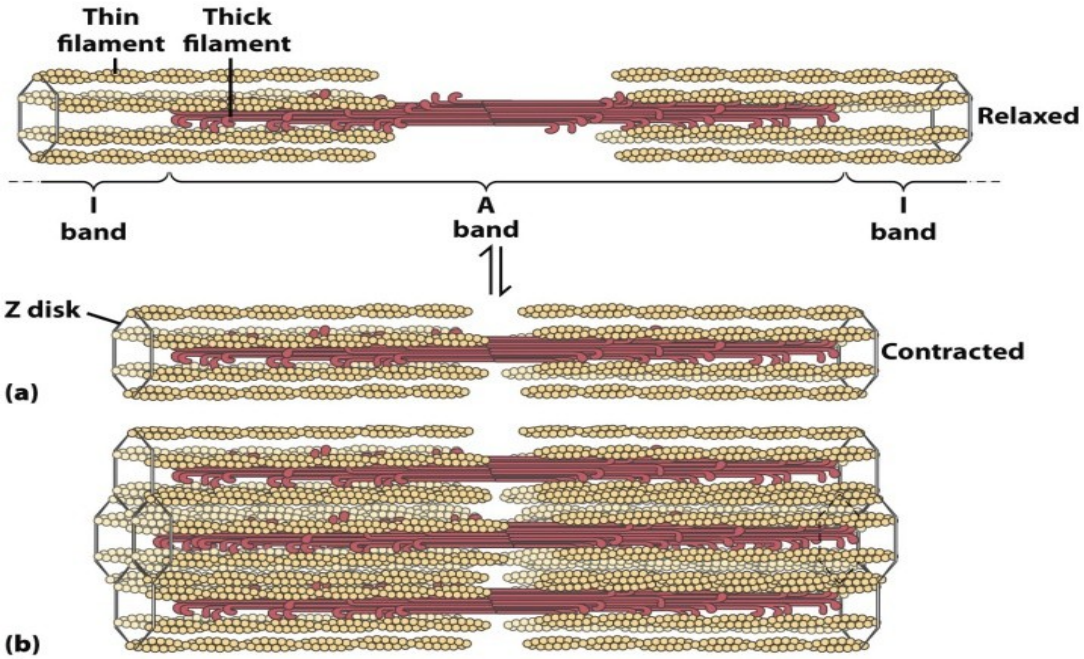


Figure 5-30
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

MUSCLE CONTRACTION

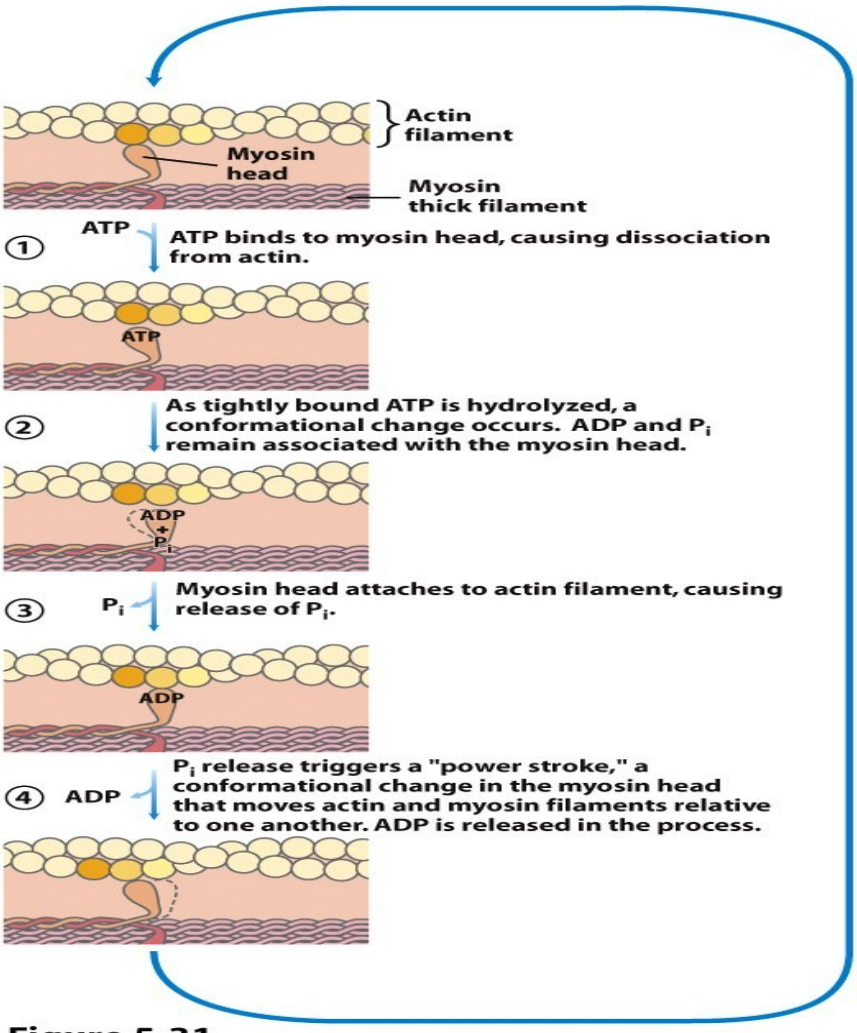


Figure 5-31
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

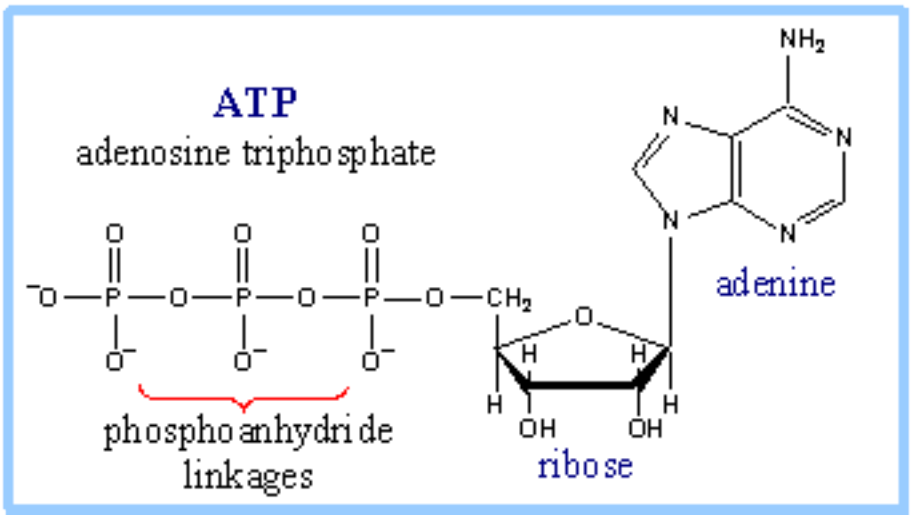
<u>Transphosphorylations between nucleotides occur in all cell types</u>

ATP is considered as the cell's energy currency and donor of phosphoryl groups, all other nucleoside triphosphates (GTP, UTP, CTP) and all the deoxynucleoside triphosphates (dATP, dGTP, dTTP and dCTP) are energetically equivalent to ATP

The free energy changes associated with hydrolysis of their phosphoanhydride linkages are very nearly identical with those for ATP.

In preparation for their various biological roles, other nucleotides are generated as the nucleoside triphosphate (NTP) forms by phosphoryl group transfer to the corresponding nucleoside diphosphates (NDPs) and monophosphates (NMPs)

ATP is the primary high-energy phosphate compound produced by catabolism in the processes of glycolysis, oxidative phosphorylation. Several enzymes carry phosphoryl groups from ATP to the other nucleotides



Nucleoside diphosphate kinases, found in all cells, catalyzes the reaction

$$\text{ATP+NDP(or dNDP)} \xrightleftharpoons{Mg^{2+}} \text{ADP+NTP(or dNDP)}$$

Although this reaction is fully reversible the relatively high ATP/ADP ratio in cells normally drives the reaction to the right, with the net formation of NTPs and dNTPs

Phosphoryl group transfers from ATP results in an accumulation of ADP. eg, when muscle is concracting vigorously ADP accumulates and interferes with ATP-dependent contraction.

During intense demand for ATP, the cell lowers the ADP concentration, and at the same time acquires ATP, by the action of adenylate kinase:

$$2\text{ADP} \xrightleftharpoons{Mg^{2+}} \text{ATP + AMP}$$

Ping-pong mechanism of nucleoside diphosphate kinase

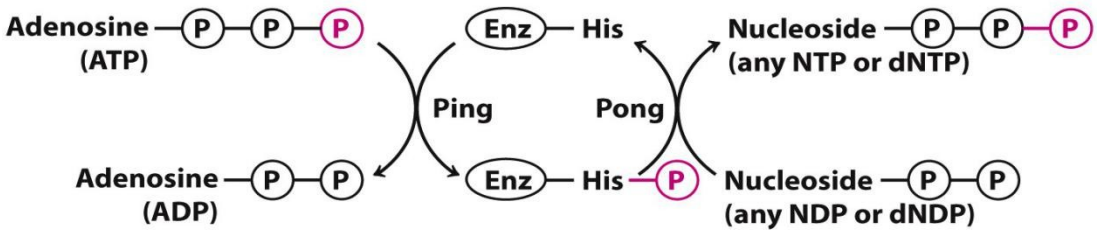


Figure 13-21
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

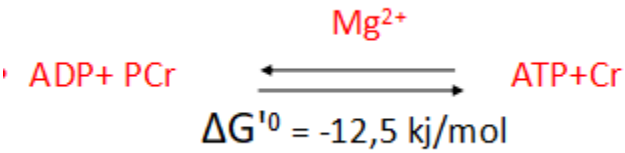
Ping-pong mechanism of nucleoside diphosphate kinase: The enzyme binds its first substrate (ATP in our example), and a phosphoryl group is transferred to the side chain of a His residue. ADP departs, and another nucleoside (or deoxynucleoside) diphosphate replaces it, and this is converted to the corresponding triphosphate by transfer of the phosphoryl group from the phosphohistidine residue.

This reaction is fully reversible, so after the intense demand for ATP ends, the enzyme can recycle AMP by converting it to ADP which can then be phosphorylated to ATP in mitochondria

A similar enzyme guanylate kinase, converts GMP to GDP at the expense of ATP.

By these pathways energy conserved in the catabolic production of ATP is used to supply the cell with all required NTPs and dNTPs

Phosphocreatine (PCr) serves as a ready source of phosphoryl groups for the quick synthesis of ATP from ADP. The phosphocreatine concentration in skeletal muscle is considerably higher than those in the other tissues. The enzyme creatine kinase catalyzes the reversible reaction.



When a sudden demand for energy depletes ATP, the PCr reservoir is used to replenish ATP at a rate faster than ATP can be synthesized by catabolic pathways

When the demand for energy slackens ATP produced by catabolism is used to replenish the PCr reservoir by reversal of the creatine kinase reaction.

Inorganic polyphosphate is a donates phosphoryl group

Inorganic polyphosphate ((polyP)n): linear polymer of Pi … Present in simple organisms of other phyla like yeast

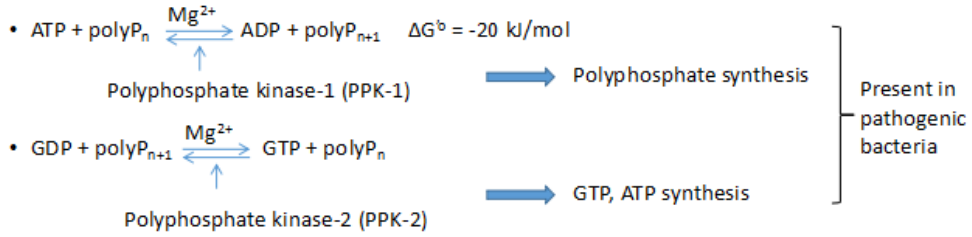
In yeast, polyP accumulated in vacuoles (= 200 mM)

Serve as a phosphagen (Reservoir of phosphoryl group).. Like PCr

The shortest polyP is PPi (n=2): can serve as the energy source for active transport of H+ in plant cells (across vacuolar membrane)

In bacteria, the enzyme polyphosphate kinase-1 (PPK-1) catalyzes the reversible reaction

polyphosphate kinase-2 (PPK-2), catalyzes the reversible synthesis of GTP (or ATP) from polyphosphate and GDP (or ADP):



TOPIC#13

Biological Oxidation -reduction reactions-2-4-18

Biological oxidation-reduction reactions

Central feature of metabolism

1) Phosphoryl group transfer

2) Electron transfer in oxidation-reduction

The flow of electrons in oxidation-reduction reactions is responsible for all work done by living organisms

Oxidation	**Reduction**
Addition of oxygen	Removal of oxygen
loss of electrons	Gain of electrons
Loss of hydrogen	Gain of hydrogen

Biological oxidation-reduction reactions

In nonphotosynthetic organism, source of electron: food

In photosythetic organism, source of electron: chemical compound excited by the absorption of light

Electrons move from various metabolic intermediates to specialized electron carriers in enzyme-catalyzed reaction ✉ energy release

1. The flow of electrons can do biological work

Electromotive force (emf): (measured in volts), is the voltage developed by any source of electrical energy such as a battery… In which two chemical species differ in their affinity for electrons, so the electrons flow spontaneously through the circuit, due to the difference in electron affinity

Living cells have an analogous biological "circuit", with relatively reduced compound such as glucose (the source of electrons)

Glucose ✉ enzymatically oxidized ✉ release electron ✉ spontaneously flow through electron-carrier intermediates to another chemical species, such as O2 this electron flow is exergonic reaction because O2 has a higher affinity for electrons than do electron-carrier intermediates

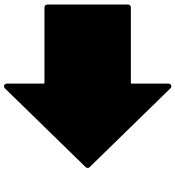
The resulting emf provides energy to a variety of molecular energy transducers (enzymes and other proteins) that do biological work

The flow of electrons can do biological work

In the mitochondrion, membrane-bound enzymes couple electron flow ✉ a production of transmembrane pH difference and a transmembrane electrical potential which helps to accomplish osmotic and electrical work

The proton gradient thus formed has potential energy, sometimes called the proton-motive force by analogy with electromotive force.

Another enzyme, ATP synthase in the inner mitochondrial membrane, uses the proton-motive force to do chemical work: synthesis of ATP from ADP and Pi as protons flow spontaneously across the membrane.



Bacterial flagellar motion: use proton-motive force

Thus, the proton potential has potential energy: electron-motive force

Oxidation-Reductions Can Be Described as Half-Reactions

Half-Reactions:

A half reaction is either the oxidation or reduction reaction component of a redox reaction.

A half reaction is obtained by considering the change in oxidation states of individual substances involved in the redox reaction

Although oxidation and reduction occur together, we consider the two halves of an oxidation-reduction reaction separately.

For example, the oxidation of ferrous ion by cupric ion is written as follows,

$$Fe^{+2} + Cu^{+2} \leftrightarrow Fe^{+3} + Cu^{+}$$

Same can be described as 2 half reactions in the following form:

1. $Fe^{+2} \leftrightarrow Fe^{+3} + e^-$

2. $Cu^{+2} + e^- \leftrightarrow Cu^{+}$

Conjugate redox pair

The electron-donating molecule in an oxidation-reduction reaction is called the reducing agent or reductant

the electron-accepting molecule is the oxidizing agent or oxidant.

A given agent, such as an iron cation existing in the ferrous (Fe+2) or ferric (Fe+3) state functions as a conjugate reductant-oxidant pair (redox pair), just as an acid and corresponding base function as a conjugate acid-base pair.

Acid-base reactions can be written as a general equation:

proton donor ↔ H+ + proton acceptor

In redox reactions we can write a similar general equation:

electron donor ↔ e - + electron acceptor

In the reversible half-reaction (1) above, Fe+2 is the electron donor and Fe+3 is the electron acceptor; together, Fe+2 and Fe+3 constitute a conjugate redox pair.

The oxidation of reducing sugar by cupric acid

Aldehydes and ketones with anomeric carbon can act reducing sugars.. Having capability to reduce copper ions in medium.. The reducing sugar reduces the copper(II) ions in these test solutions to copper(I), which forms a brick red copper(I) oxide precipitate

Basis of qualitative carbohydrate test (Benedict's test)

The reducing end of a disaccharide is the monosaccharide with a free anomeric carbon that is not involved in a glycosidic bond and is thus capable of converting to the open-chain form.

A nonreducing disaccharide has both anomeric carbons tied up in the glycosidic bond…

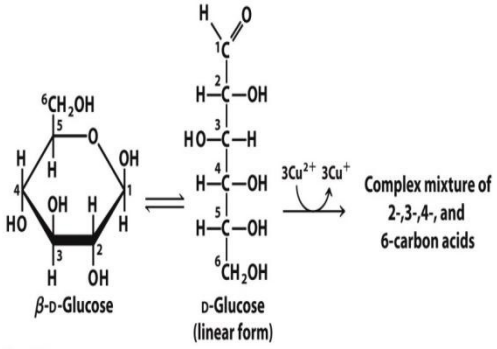
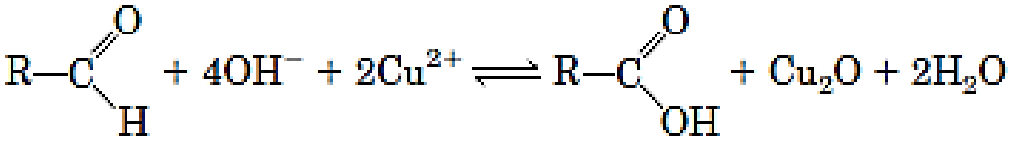
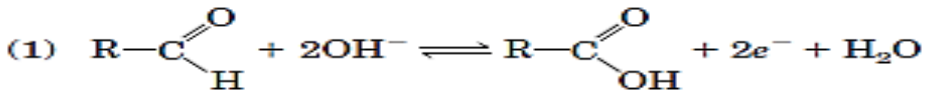


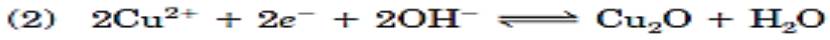
Figure 7-10
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

consider the oxidation of a reducing sugar by cupric ion:



This overall reaction can be expressed as two half reactions:





Because two electrons are removed from the aldehyde carbon, the second half-reaction (the one-electron reduction of cupric to cuprous ion) must be doubled to balance the overall equation.

The monosaccharides can be divided into two groups: the aldoses, which have an aldehyde group, and the ketoses, which have a ketone group. Ketoses must first tautomerize to aldoses before they can act asreducing sugars.. Having capability to reduce copper ions in medium.. The reducing sugar reduces the copper(II) ions in these test solutions to copper(I), which then forms a brick red copper(I) oxide precipitate has a free aldehyde group or a free ketone group

The reducing end of a disaccharide is the monosaccharide with a free anomeric carbon that is not involved in a glycosidic bond and is thus capable of converting to the open-chain form.

A nonreducing disaccharide has both anomeric carbons tied up in the glycosidic bond.The common dietary monosaccharides galactose, glucose and fructose are all reducing sugars.

<u>Biological Oxidations Often Involve Dehydrogenation</u>

In biological Systems oxidation (loss of electrons) is often synonymous with dehydrogenation and many enzymes that catalyze oxidation reactions are dehydrogenases.

The carbon in living cells exists in a range of oxidation status

Electronegativity: H<C<S<N<O

- for attached C atoms, i.e. C-C bonds electrons shared, \count 0
- for attached X atoms, i.e. C-X bonds (X more electronegative), \count -1 (per bond)

- for attached H atoms, i.e. C-H bonds (H is less electronegative than C), \ count +1

OXIDATION STATE

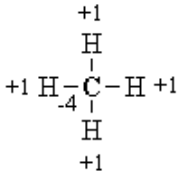
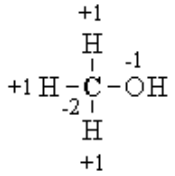
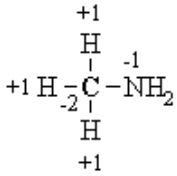
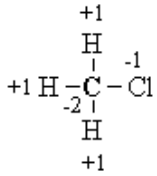
It is calculated by counting all the electrons bonded to carbon and add them in following way:

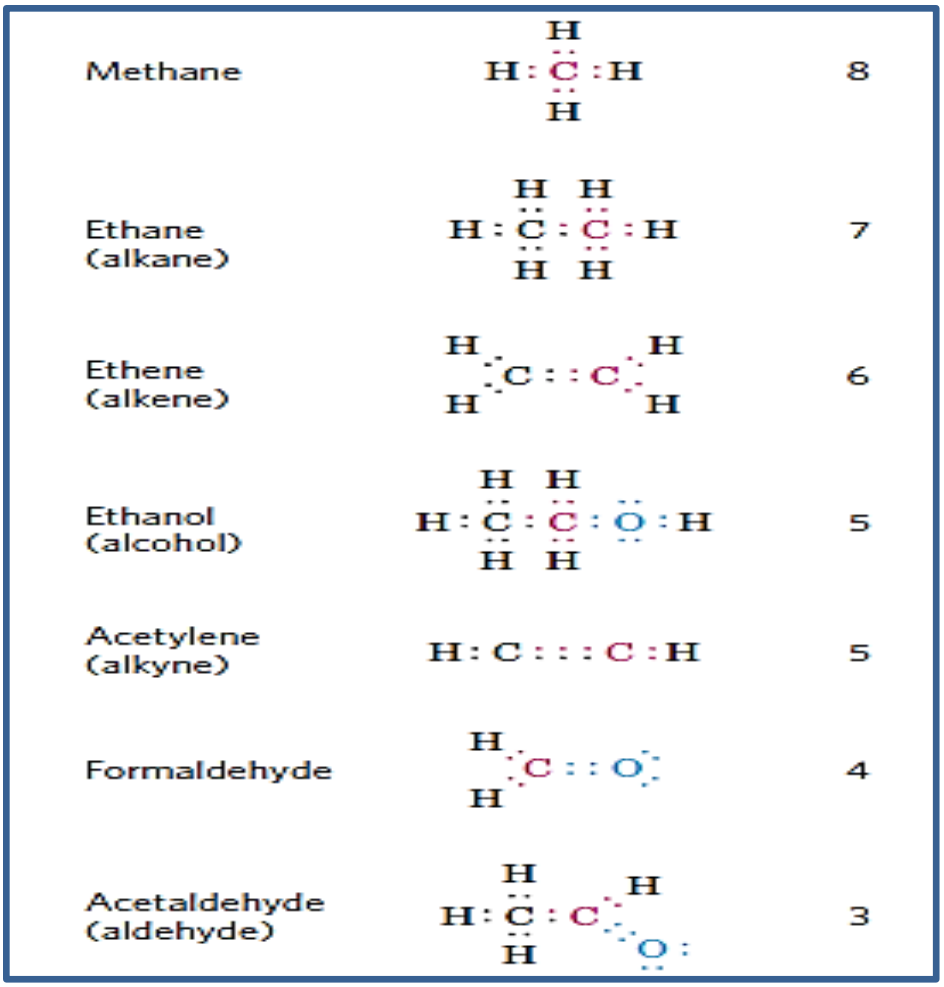
for attached C atoms, i.e. C-C bonds electrons shared, count 0

for attached X atoms, i.e. C-X bonds (X more electronegative), count -1 (per bond)

for attached H atoms, i.e. C-H bonds (H is less electronegative than C), count +1

Add the total for atoms attached to the C in question, then switch the sign.

Methane	Methanol	AminoMethane	Methayl chloride
			
C = - (+1 +1 +1 +1) = - 4	**C** = - (-1 +1 +1 +1) = - 2	**C** = - (-1 +1 +1 +1) = - 2	**C** = - (-1 +1 +1 +1) = - 2



For the figure given

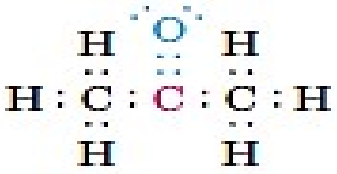
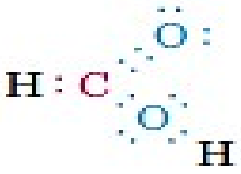
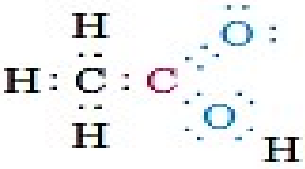
Different levels of oxidation of carbon compounds in the biosphere.

In order to estimate the level of oxidation of these compounds, focus on the red carbon atom and its bonding electrons.

When this carbon is bonded to the less electronegative H atom, both bonding electrons (red) are assigned to the carbon.

When carbon is bonded to another carbon, bonding electrons are shared equally, one of the two electrons is assigned to the red carbon. When the red carbon is bonded to the more electronegative O atom, the bonding electrons are assigned to the oxygen.

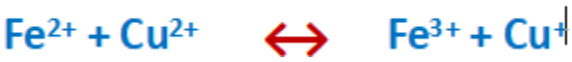
The number to the right of each compound is the number of electrons “owned” by the red carbon, a rough expression of the degree of oxidation of that compound. As the red carbon undergoes oxidation (loses electrons), the number gets smaller.

Acetone (ketone)		2
Formic acid (carboxylic acid)		2
Carbon monoxide		2
Acetic acid (carboxylic acid)		1
Carbon dioxide		0

Oxidation states of C reducing sugar (an aldehyde or ketone) by carbon in various compounds; from fully reduced (methane to fully oxidized (carbon dioxide)

Electrons are transferred from one molecule to another in 4 ways

1. Directly as electrons: Fe2+/Fe3+ redox pair can transfer an electron to the Cu/Cu2+ redox pair



2. As hydrogen atoms: hydrogen atom consists of a proton (H+) and a single electron (e-)



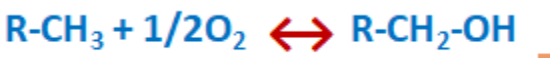
AH2/A: conjugate redox pair ✉ can reduce another compound B



3. As a hydride ion (:H-) .. has two electrons

This reaction occurs in NAD-linked dehydrogenase

4. Through direct combination with oxygen, where Oxygen covalently incorporated into product



TOPIC#14

Reduction potential

Reduction potential measures affinity for electrons

Reduction potential:

Reduction potential (redox potential, oxidation/reduction potential), is a measure of the tendency of a chemical species to acquire electrons and thereby get reduced. Units: Volts (V), or millivolts (mV).. denoted by E.

Standard reduction potential:

The tendency for a chemical species to be reduced, at standard conditions. The more positive the potential is the more likely it will be reduced… denoted by EO.

They help to measure the …. Redox reaction.. In terms of volts (V), or millivolts (mV and like we said.. That .. measure of the tendency of a chemical species to acquire electrons .. Aquiring e is to undergo reduction…

And The standard reduction potential is the tendency for a chemical species to be reduced, and is measured in volts at standard conditions. The more positive the potential is the more likely it ...

concentrations of 1 mol/L and at 1 atm (101.3kp) and 25°C temp.

The cell potential, EcellEcell, is the measure of the potential difference between two half cells in an electrochemical cell.

he potential difference is caused by the ability of electrons to flow from one half cell to the other. Electrons are able to move between electrodes because the chemical reaction is a redox reaction. A redox reaction occurs when a certain substance is oxidized, while another is reduced. During oxidation, the substance loses one or more electrons, and thus becomes positively charged. Conversely, during reduction, the substance gains electrons and becomes negatively charged.

so

And how is that helpful for us? And how it can be measured ..

To measure the Reduction potential.. Electrochemists use half cell.. A half cell is one of the two electrodes in a cell or simple battery. separated from other. Half cell.. . two half cells make An electrochemical cell in which redox reaction can take place.

.. So that they work only in the presence of…? To initiate e flow and conduct electricity.. Placing a piece of reactant in an electrolyte solution makes a half cell.

A Half-cell

A half cell is one of the two electrodes in a galvanic cell or simple battery submerged in an aqueous solution of the same metal ions.

Until or unless they are separately kept in their solutions.. There would be no redox reaction.

. For example, in the Zn-Cu battery, the two half cells make an oxidizing-reducing couple. The electrode is connected to the other half cell, which contains an electrode of some other metal submerged in an aqueous solution of subsequent metal ions. An example of this would be a zinc electrode, dipped in zn solution. in which the zn atoms in the electrode lose two electrons and becomes Zn2+ (oxidized(.. The electrons lost by the zn atoms in the electrode are then transferred to the second half cell,.

.In order to balance the charge on both sides of the cell, the half cells are connected by a salt bridge..

. As the anode half cell becomes overwhelmed with zn2+ ions, the negative anion of the salt will enter the solution and stabilized the charge. Similarly, in the cathode half cell, as the solution becomes more negatively charged, cations from the salt bridge will stabilize the charge.

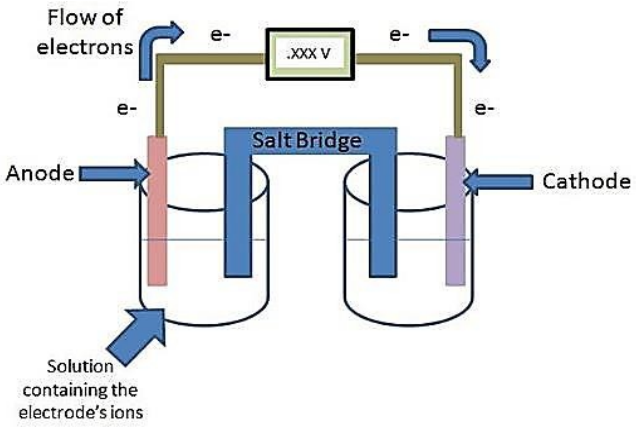
If no salt bridge were present, the solution in one half cell would accumulate negative charge and the solution in the other half cell would accumulate positive charge as the reaction proceeded, quickly preventing further reaction, and hence production of electricity

. Reduction potential of a half-cell depends on the activity of reduced and oxidized species which is approximated by their concentrations

On the anode, oxidation takes place.
Zn Zn2+ + 2 e-
Zn is the reducing agent, and Zn2+ the oxidizing agent.

On the cathode, reduction takes place.

Cu2+ + 2 e Cu
Cu2+ is the oxidizing agent and Cu the reducing agent.



A standard Half-cell

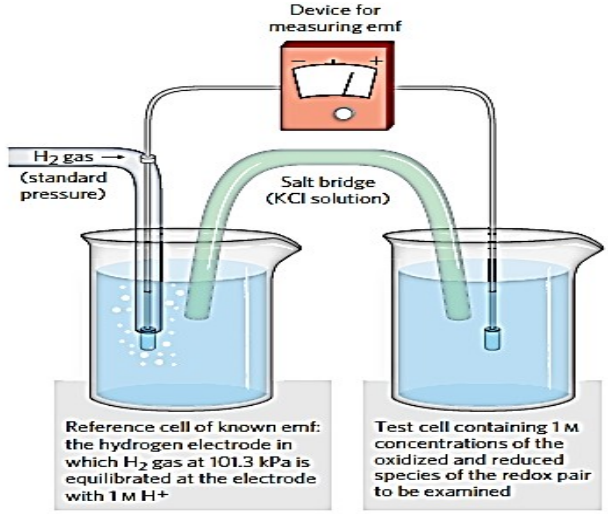
Standard Hydrogen electrode..(SHE) is the primary reference electrode with platinum electrode dipped in solution with H ions under standard conditions

The electromotive force (emf) of this electrode is designated 0.00 V. (At pH 7, 25 OC)

Using observed emf (of test solution) and the known emf of the reference cell, the emf of the test cell with the redox pair can be calculated.

The cell that gains electrons has, by convention, the more positive reduction potential.

The electrode at which this half-reaction occurs (called a half-cell) is arbitrarily assigned an EO of 0.00 V.



But the fact is that only the difference of reduction potential can help us to calculate it.. That is why we need standard or a reference elctrode..The half consists of a metal electrode of a certain metal submerged in an aqueous solution of the same metal ions.

.. A half cell consists of an electrode and the species to be oxidized or reduced. A standard half cell conists of

Unless it is connected to another half cell via an electric conductor and salt bridge, no reaction will take place in a half cell.

Standard reduction potentials can be used to calculate free-energy change

Electrons tend to flow to the half-cell with the more positive E, and the strength of that tendency is proportional to Δ E, the difference in reduction potential

We can then predict the direction in which electrons will tend to flow when the two half-cells are connected through an external circuit

Electrons tend to flow to the half-cell with the more positive E ✉ the strength of flow depends on ΔE (difference in reduction potential)

The energy made available by this spontaneous electron flow (the free-energy change, Δ G, for the oxidation-reduction reaction) is proportional to Δ E:

$$\Delta G = -n\mathcal{F}\Delta E \quad \text{or} \quad \Delta G'^{o} = -n\mathcal{F}\Delta E'^{o}$$

Why are reduction potentials so useful to the biochemist?When E values have been determined for any two halfcells,relative to the standard hydrogen electrode, we

also know their reduction potentials relative to eachother. We can then predict the direction in which electronswill tend to flow when the two half-cells are connected

through an external circuit or when component of both half-cells are present in the same solution.

where n is the number of electrons transferred in the reaction

F is Faraday (unit of electricity), used in the study of electrochemical reactions and equal to the amount of electric charge that liberates one gram equivalent of any ion from an electrolytic solution

With this equation we can calculate the actual free-energy change for any oxidation-reduction reaction from the values of E ’o (reduction potentials) and the concentrations of reacting species

When acetaldehyde is reduced by the biological electron carrier NADH, ΔG = -35.3 kJ/mol.

TABLE 13–7 Standard Reduction Potentials of Some Biologically Important Half-Reactions

Half-reaction	E'° (V)
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.816
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	0.771
$NO_3^- + 2H^+ + 2e^- \longrightarrow NO_2^- + H_2O$	0.421
Cytochrome *f* $(Fe^{3+}) + e^- \longrightarrow$ cytochrome *f* (Fe^{2+})	0.365
$Fe(CN)_6^{3-}$ (ferricyanide) $+ e^- \longrightarrow Fe(CN)_6^{4-}$	0.36
Cytochrome a_3 $(Fe^{3+}) + e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$O_2 + 2H^+ + 2e^- \longrightarrow H_2O_2$	0.295
Cytochrome *a* $(Fe^{3+}) + e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	0.29
Cytochrome c $(Fe^{3+}) + e^- \longrightarrow$ cytochrome *c* (Fe^{2+})	0.254
Cytochrome c_1 $(Fe^{3+}) + e^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome *b* $(Fe^{3+}) + e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	0.077
Ubiquinone $+ 2H^+ + 2e^- \longrightarrow$ ubiquinol $+ H_2$	0.045
$Fumarate^{2-} + 2H^+ + 2e^- \longrightarrow succinate^{2-}$	0.031
$2H^+ + 2e^- \longrightarrow H_2$ (at standard conditions, pH 0)	0.000
Crotonyl-CoA $+ 2H^+ + 2e^- \longrightarrow$ butyryl-CoA	−0.015
$Oxaloacetate^{2-} + 2H^+ + 2e^- \longrightarrow malate^{2-}$	−0.166
$Pyruvate^- + 2H^+ + 2e^- \longrightarrow lactate^-$	−0.185
Acetaldehyde $+ 2H^+ + 2e^- \longrightarrow$ ethanol	−0.197
FAD $+ 2H^+ + 2e^- \longrightarrow FADH_2$	−0.219*
Glutathione $+ 2H^+ + 2e^- \longrightarrow$ 2 reduced glutathione	−0.23
$S + 2H^+ + 2e^- \longrightarrow H_2S$	−0.243
Lipoic acid $+ 2H^+ + 2e^- \longrightarrow$ dihydrolipoic acid	−0.29
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
Acetoacetate $+ 2H^+ + 2e^- \longrightarrow$ β-hydroxybutyrate	−0.346
α-Ketoglutarate $+ CO_2 + 2H^+ + 2e^- \longrightarrow$ isocitrate	−0.38
$2H^+ + 2e^- \longrightarrow H_2$ (at pH 7)	−0.414
Ferredoxin $(Fe^{3+}) + e^- \longrightarrow$ ferredoxin (Fe^{2+})	−0.432

Source: Data mostly from Loach, R.A. (1976) In *Handbook of Biochemistry and Molecular Biology*, 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data*, Vol. 1, pp. 122–130, CRC Press, Boca Raton, FL.
* This is the value for free FAD; FAD bound to a specific flavoprotein (e.g., succinate dehydrogenase) has a different E'° that depends on its protein environment.

Table 13-7
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Conjugate redox pair: An electron donor and its corresponding electron acceptor form; for example, Cu+ (donor) and Cu2+ (acceptor), or NADH (donor) and NAD+ (acceptor) form Conjugate redox pair.

Half cell: A half cell is one of the two electrodes in a galvanic cell or simple battery. For example, in the Zn-Cu battery, the two half cells make an oxidizing-reducing couple.

Reduction potential (also known as redox potential, oxidation/reduction potential, or Eh) measuresthe tendency of a chemical species to acquire electrons and thereby be reduced.

TOPIC#15

Universal Electron Carriers

So.. We had been talking about redox reactions.. And half cell in which redox reaction takes place.. And we mentioned several times that reduction is the addition of electrons. . What happens in biological system is.. That many of the enzymes that catalyze cellular oxidations reactions.. channel electrons from their substrates through some types of universal electron carriers. Some of these proteins are water-soluble, but others are peripheral or integral membrane proteins

During the catabolic processes these carriers are reduced, substrate is oxidized and this results in the conservation of free energy

Enrgy genration is not only by glucose and ATP, but there are some other Behind the scene contributors too.. These are electron carrier molecules.. Glucose

Cellular oxidation of glucose to carbon dioxide requires specialized electron carriers

The principles of oxidation-reduction energetics ✉ can apply to the many metabolic reactions that involve electron transfers

Glucose oxidation:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O,$$

$$\Delta G'^{o} = -2{,}840 \text{ kJ/mol} > \text{ATP synthesis in cells: } 50\text{-}60 \text{ kJ/mol}$$

Cells does not convert to CO2 in a single, high energy releasing reaction but rather in a series of controlled reactions, some of which are oxidation for the ATP synthesis from ADP through NAD+ and FAD

A Few Types of Coenzymes and Proteins Serve as Universal Electron Carriers

1) The nucleotides NAD and NADP move readily from one enzyme to another
2) The flavin nucleotides FMN and FAD are usually very tightly bound to the enzymes, called flavoproteins, for which they serve as prosthetic groups
3) Lipid-soluble quinones: ubiquinone and plastoquinone act as electron carriers and proton donors in the non-aqueous environment of membranes

4) Iron-sulfur proteins and cytochromes, which have tightly bound prosthetic groups that undergo reversible oxidation and reduction, serve as electron carriers in many oxidation-reduction reactions

NAD, NADP, FMN, and FAD are water-soluble coenzymes and undergo reversible oxidation and reduction during many of the electron-transfer reactions of metabolism.

Enrgy genration is not only by glucose and ATP, but there are some other Behind the scene contributors too.. These are electron carrier molecules.. Glucose

. Some of these proteins are water-soluble, but others are peripheral or integral membrane proteins

The reduction of these carriers in catabolic processes results in the conservation of free energy released by substrate oxidation.

NAD, NADP, FMN, and FAD are water-soluble coenzymes and undergo reversible oxidation and reduction in many of the electron-transfer reactions of metabolism.

The nucleotides NAD and NAD move readily from one enzyme to another

The flavin nucleotides FMN and FAD are usually very tightly bound to the enzymes, called flavoproteins, for which they serve as prosthetic groups.

Lipid-soluble quinones: ubiquinone and plastoquinone act as electron carriers and proton donors in the non-aqueous environment of membranes.

Iron-sulfur proteins and cytochromes, which have tightly bound prosthetic groups that undergo reversible oxidation and reduction, also serve as electron carriers in many oxidation-reduction reactions.

<u>NAD and NADP are soluble electron carriers that act with dehydrogenase enzymes</u>

Activated electron carriers, involved in oxidation-reduction and coupling reactions

NAD+ (nicotinamide adenine dinucleotide)

NADP+ (nicotinamide adenine dinucleotide phosphate)

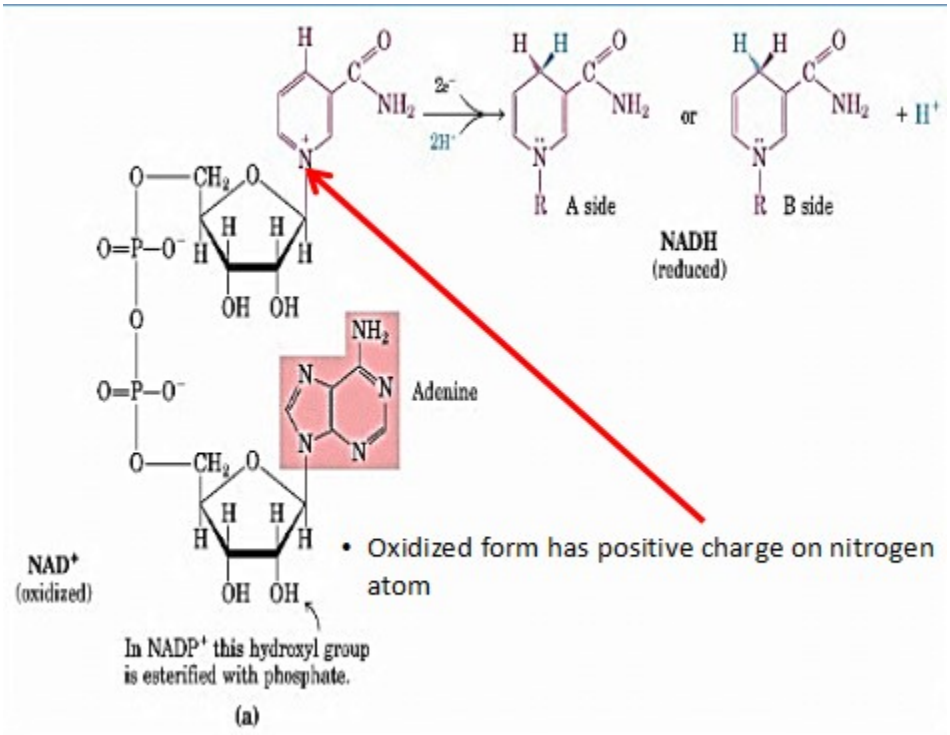
For NAD: ✉ NAD+: oxidized form

For NADP:✉ NADP+: oxidized form

Joined by phosphoanhydride bond

Pyridine nucleotides

Nicotinamide moiety comes from NIACIN Vitamin



And it's the flow 0f e in etc that is used to produce ATP ..

There are other important activated carriers involved in biosynthesis of molecules. These activated carriers are involved in oxidation-reduction reactions and involved in coupling reactions.

The most important of these carriers are NAD+ (nicotinamide adenine dinucleotide) and closely related molecule NADP+ (nicotinamide adenine dinucleotide phosphate).

A conjugate acid, within the Brønsted–Lowry acid–base theory, is a species formed by the reception of a proton (H+) by a base—in other words, it is a base with a hydrogen ion added to it. On the other hand, a conjugate base is merely what is left over after an acid has donated a proton during a chemical reaction

Mechanism. The reaction mechanism generally begins with the oxidative addition of an organic halide to the catalyst. ... The final step is reductive elimination of the two coupling fragments to regenerate the catalyst and give the organic product.

Both coenzymes undergo reversible reduction of the nicotinamide ring

As a substrate molecule undergoes oxidation (dehydrogenation), giving up two hydrogen atoms, the oxidized form of the nucleotide (NAD+ or NADP+) accepts a hydride ion (:H2, the equivalent of a proton and two electrons) and is reduced (to NADH or NADPH)

The second proton removed from the substrate is released to the aqueous solvent. The half-reactions for these nucleotide cofactors are:

NAD+ + 2e _ + 2H + + NADH NADH + H+

NADP + + 2e _ + 2H + + NADH NADPH + H +

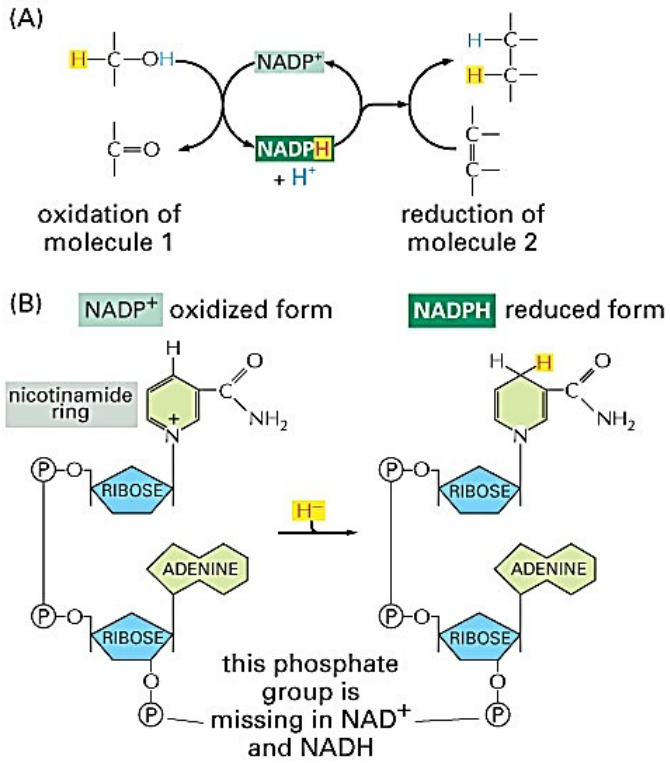


Figure 3-35 Essential Cell Biology, 2/e. (© 2004 Garland Science)

Reduction of NAD or NADP converts the benzenoid ring of the nicotinamide moiety (with a fixed positive charge on the ring nitrogen) to the quinonoid form (with no charge on the nitrogen).

The reduced nucleotides absorb light at 340 nm; the oxidized forms do not (used for detection)

Plus sign with NAD and NADP does not indicate the net charge (both are negatively charged) rather, it indicates that the nicotinamide ring is in its oxidized form, with a positive charge on the Nitrogen atom

"H" in NADH and NADPH, the denotes the added hydride ion

Note that the plus sign in the abbreviations NAD1 and NADP1 does not indicate the net charge on these molecules in fact, both are negatively charged); rather, it

indicates that the nicotinamide ring is in its oxidized form, with a positive charge on the nitrogen atom. In the abbreviations NADH and NADPH, the "H" denotes the

added hydride ion. To refer to these nucleotides without specifying their oxidation state, we use NAD and NADP.

Total concentration of NAD+ + NADH in most tissues: 10-5 M

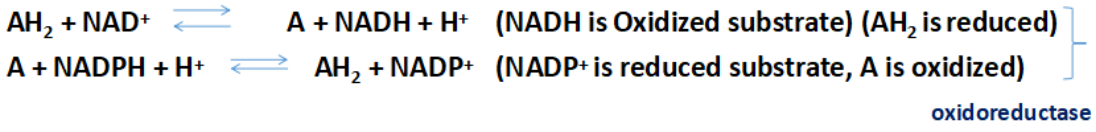
Total concentration of NADP+ + NADPH in most tissues: 10-6 M

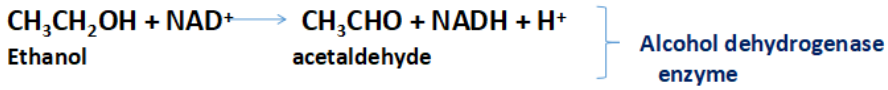
Concentration ratio: NAD+ > NADH ➡ favoring hydride transfer from a substrate to NAD+ to form NADH ➡ function in oxidation, occur in mitochondrial matrix

Concentration ratio: NADP+ < NADPH ➡ favoring hydride transfer from NADPH to a substrate ➡ function in reduction ➡ occur in cytosol

More than 200 enzymes are known to catalyze reactions in which NAD+ (or NADP+) accepts a hydride ion from a reduced substrate or NADPH (or NADH) donates a hydride ion to an oxidized substrate

General reaction





NAD and NADP:

NAD and its phosphorylated analog NADP undergo reduction to NADH and NADPH, accepting a hydride ion from an oxidizable substrate.

The hydride ion is added to either the front (the A side) or the back (the B side) of the planar nicotinamide ring

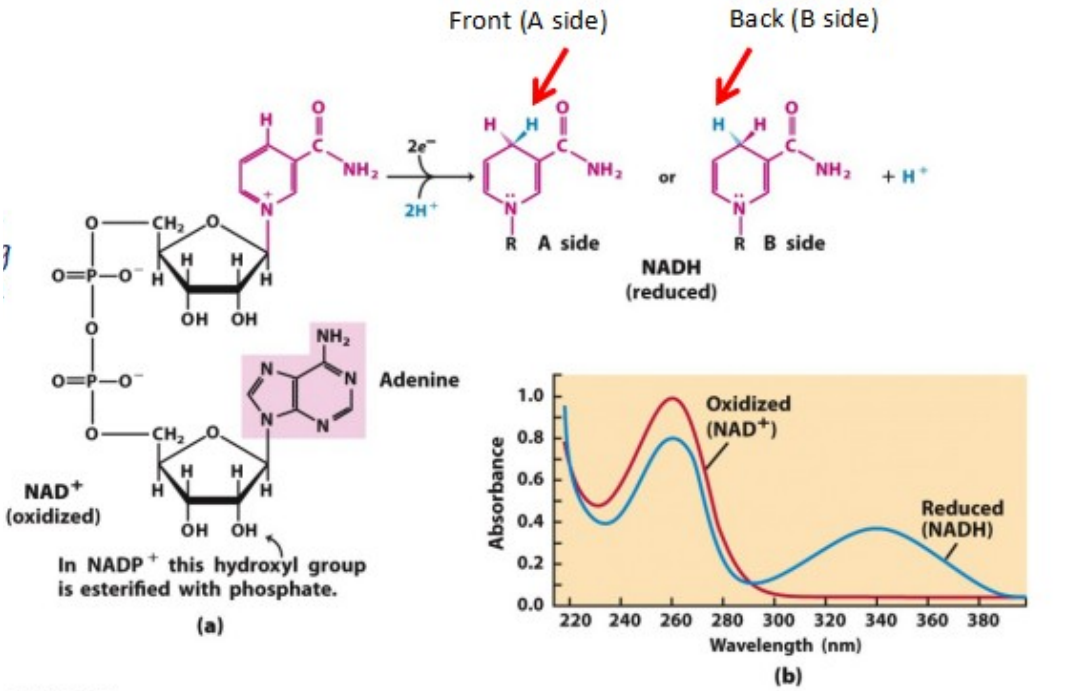


Figure 13-24
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

The UV absorption spectra of NAD and NADH

TABLE 13-8 Stereospecificity of Dehydrogenases That Employ NAD^+ or $NADP^+$ as Coenzymes

Enzyme	Coenzyme	Stereochemical specificity for nicotinamide ring (A or B)	Text page
Isocitrate dehydrogenase	NAD^+	A	643
α-Ketoglutarate dehydrogenase	NAD^+	B	644
Glucose 6-phosphate dehydrogenase	$NADP^+$	B	577
Malate dehydrogenase	NAD^+	A	647
Glutamate dehydrogenase	NAD^+ or $NADP^+$	B	702
Glyceraldehyde 3-phosphate dehydrogenase	NAD^+	B	553
Lactate dehydrogenase	NAD^+	A	563
Alcohol dehydrogenase	NAD^+	A	565

TABLE 11–7 Some Diseases Resulting from Ion Channel Defects

Ion channel	Affected gene	Disease
Na^+ (voltage-gated, skeletal muscle)	*SCN4A*	Hyperkalemic periodic paralysis (or paramyotonia congenita)
Na^+ (voltage-gated, neuronal)	*SCN1A*	Generalized epilepsy with febrile seizures
Na^+ (voltage-gated, cardiac muscle)	*SCN5A*	Long QT syndrome 3
Ca^{2+} (neuronal)	*CACNA1A*	Familial hemiplegic migraine
Ca^{2+} (voltage-gated, retina)	*CACNA1F*	Congenital stationary night blindness
Ca^{2+} (polycystin-1)	*PKD1*	Polycystic kidney disease
K^+ (neuronal)	*KCNQ4*	Dominant deafness
K^+ (voltage-gated, neuronal)	*KCNQ2*	Benign familial neonatal convulsions
Nonspecific cation (cGMP-gated, retinal)	*CNCG1*	Retinitis pigmentosa
Acetylcholine receptor (skeletal muscle)	*CHRNA1*	Congenital myasthenic syndrome
Cl^-	*CFTR*	Cystic fibrosis

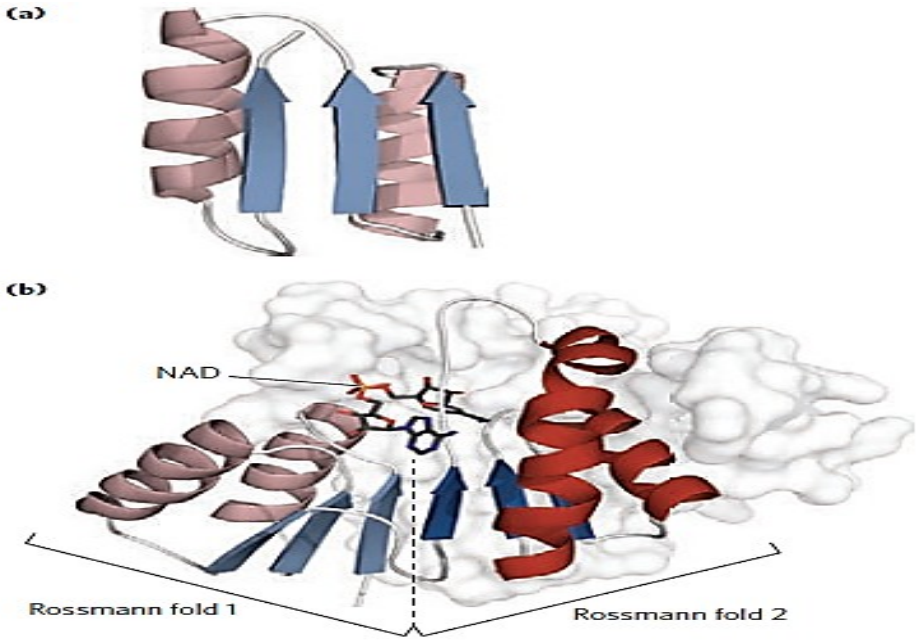
Rossmann fold

Most dehydrogenases that use NAD or NADH contain a conserved protein binding domain called the Rossmann fold

Rossmann fold: consists of a six-stranded parallel beta sheet and four associated alpha helices

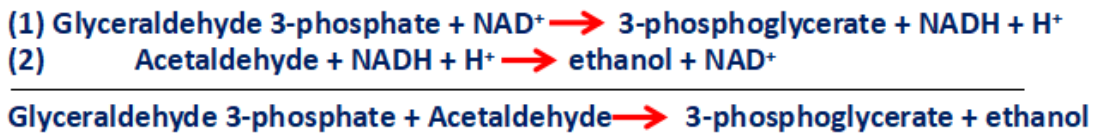
Nucleotide binding domain of lactate dehydrogenase with NAD is shown

NAD is bound to paired beta-alpha motifs of Rossmann fold



The association between dehydrogenases and cofactor is loose ✉ the coenzyme readily diffuse from one enzyme to another ✉ act as a water-soluble carrier of electron form one metabolite to another

In overall reaction, there is not net production or consumption of NAD+ or NADH ✉ no concentration change of NAD+ or NADH .. and NAD+ or NADH get recycled repeatedly



Dietary deficiency of Niacin causes Pellagra

Most coenzyme: derived from vitamins

NAD and NADP: derived from vitamin niacin (=nicotinic acid)

Human cannot synthesize sufficient quantities of niacin

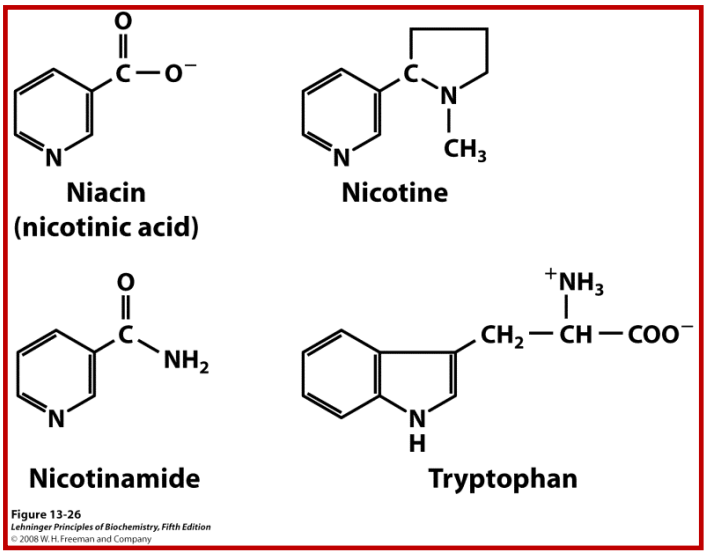
Niacin derived from tryptophan

In the laboratory, nicotinic acid was first produced by oxidation of the natural product.. nicotine —thus the name

Maize: content low tryptophan

Niacin deficiency: affects all the NAD(P)-dependent dehydrogenase ✉ causes the serious human disease pellagra (=rough skin): characterized by the Three Ds (dermatitis, diarrhea and dementia) ✉ death

Too much alcohol drinking ✉ reduce niacin absorption from the intestine



Niacin deficiency: affects all the NAD(P)-dependent dehydrogenase ✉ causes the serious human disease pellagra (=rough skin): characterized by the Three Ds (dermatitis, diarrhea and dementia)

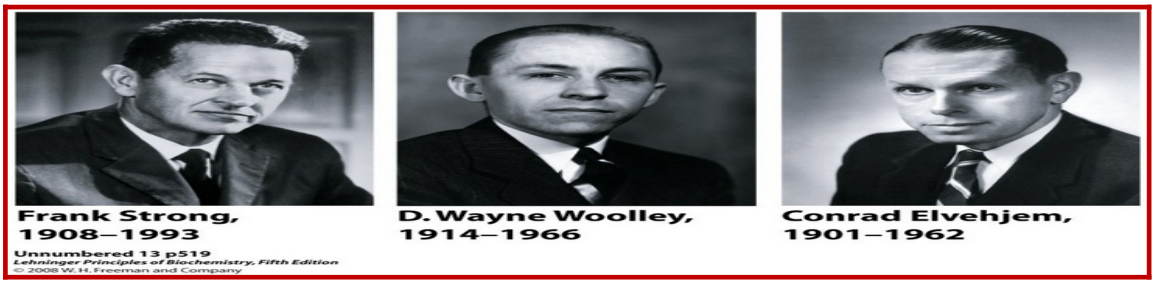
✉ death

similar disease black tongue in dogs

Niacin is curative agent (scientists)

Too much alcohol drinking ✉ reduce niacin absorption from the intestine

Both nicotinic acid and nicotinamide cure pellagra, but nicotine (from cigarettes or elsewhere) has no curative activity



Niacin (nicotinic acid) and its derivative nicotinamide

The biosynthetic precursor of these compounds is tryptophan.

In the laboratory, nicotinic acid was first produced by oxidation of the natural product.. nicotine —thus the name.

Both nicotinic acid and nicotinamide cure pellagra, but nicotine (from cigarettes or elsewhere) has no curative activity

Flavoproteins

Flavoproteins: enzymes that catalyze oxidation-reduction reactions using coenzyme

- Flavin mononucleotide (FMN)
- Flavin adenine dinucleotide (FAD)
- Flavin nucleotides ✉ derived from the vitamin riboflavin
- Flavoproteins act in a quite different role, as light receptors

The fused ring structure of the flavin nucleotide (isoalloxazine ring)

Isoalloxazine ring undergoes reversible reduction, accepting either one or two electrons in the form of one or two hydrogen atoms (each atom an electron plus a proton) from a reduced substrate

When a fully oxidized flavin nucleotide accepts only one electron (one hydrogen atom), the semiquinone form of the isoalloxazine ring is produced

The fully reduced forms are abbreviated FADH2 and FMNH2

Flavoproteins ✉ derived from the vitamin riboflavin

Flavoproteins: enzymes that catalyze oxidation-reduction reactions using coenzyme

Flavin mononucleotide

(FMN) or flavin adenine dinucleotide (FAD)

Certain flavoproteins act in a quite different role, as light receptors.

The fused ring structure of the flavin nucleotide (isoalloxazine ring)

Isoalloxazine ring undergoes reversible reduction, accepting either one or two electrons in the form of one or two hydrogen atoms (each atom an electron plus a proton) from a reduced substrate.

The fully reduced forms are abbreviated FADH2 and FMNH2

When a fully oxidized flavin nucleotide accepts only one electron (one hydrogen atom), the semiquinone form of the isoalloxazine ring is produced, FADH2 and FMNH2

NAD+ --> NADH
- Dehydrogenase always removes 2H+ (hydrogen protons) and 2e- from the metabolite. NAD can accept 2e- but only a single H+. Therefore, the second H+ is donated to the solution.
FAD --> FADH2
- When dehydrogenase removes 2H+ and 2e- from the glucose metabolite, FAD can accept 2e- AND 2H+. Therefore no extra hydrogen proton needs to be donated to solution

Oxidized and reduced forms of FAD and FMN

Some enzymes use flavin nucleotides for oxidation-reduction

Flavin nucleotides are tightly bound in flavoproteins

- Fully reduced: 360 nm
- Partially reduced: 450 nm
- Fully oxidized: 370 and 440 nm

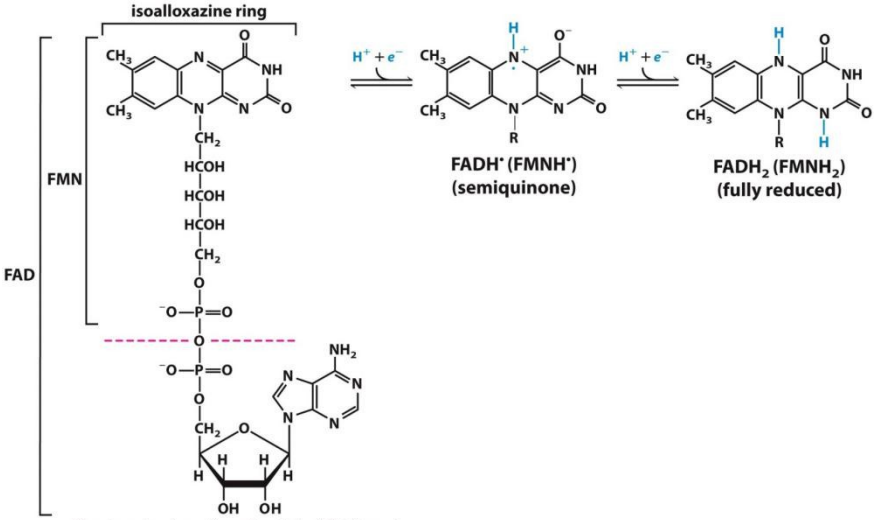


Figure 13-27
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

FMN consists of the structure above the dashed line on the FAD (oxidized form). The flavin nucleotides accept two hydrogen atoms (two electrons and two protons), both of which appear in the flavin ring system. When FAD or FMN accepts only one hydrogen atom, the semiquinone, a stable free radical, forms.

Enzymes that use FAD or FMN as coenzymes:

TABLE 13–9 Some Enzymes (Flavoproteins) That Employ Flavin Nucleotide Coenzymes

Enzyme	Flavin nucleotide	Text page(s)
Acyl-CoA dehydrogenase	FAD	673
Dihydrolipoyl dehydrogenase	FAD	637
Succinate dehydrogenase	FAD	646
Glycerol 3-phosphate dehydrogenase	FAD	759
Thioredoxin reductase	FAD	917
NADH dehydrogenase (Complex I)	FMN	738–739
Glycolate oxidase	FMN	813

Table 13-9
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Cryptochromes

Cryptochromes: Family of flavoproteins, widely distributed in the eukaryotic phyla, that mediate the effects of blue light on plant development and the effects of light on mammalian circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period).

The cryptochromes are homologs of another family of flavoproteins, the photolyases. Found in both bacteria and eukaryotes, photolyases use the energy of absorbed light to repair chemical defects in DNA

mediate the effects of blue light on plant development and the effects of light on mammalian circadian rhythms (oscillations in physiology and biochemistry, with a 24-hour period)

Homologs of Photolyases

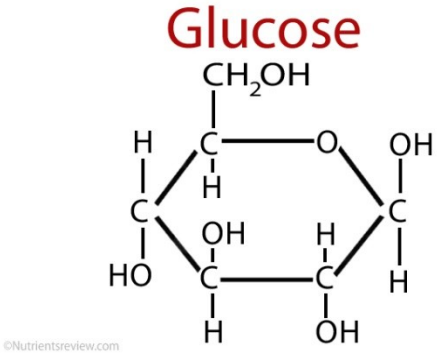
Photolyases: Use energy of light to repair defects in DNA

TOPIC#16

GLYCOLYSIS-1

GLUCOSE METABOLISM

Glucose is a simple sugar with the molecular formula $C_6H_{12}O_6$, which means that it is a molecule that is made of six carbon atoms, twelve hydrogen atoms, and six oxygen atoms. Glucose circulates in the blood of animals as blood sugar.



In animals and vascular plants, glucose has three major fates:

1. It may be stored (as a polysaccharide , i-e., glycogen or starch)

2. Oxidized to a three-carbon compound (pyruvate) via glycolysis to provide ATP and metabolic intermediates.

3. Oxidized via the pentose phosphate (phosphogluconate) pathway to yield ribose 5-phosphate for nucleic acid synthesis and NADPH for reductive biosynthetic processes.

Organisms that do not have access to glucose from other sources must make it

Photosynthetic organisms make glucose by first reducing atmospheric CO2 to trioses, then converting the trioses to glucose.

Non-photosynthetic cells make glucose from simpler three and four-carbon precursors by the process of gluconeogenesis, effectively reversing glycolysis in a pathway that uses many of the glycolytic enzymes.

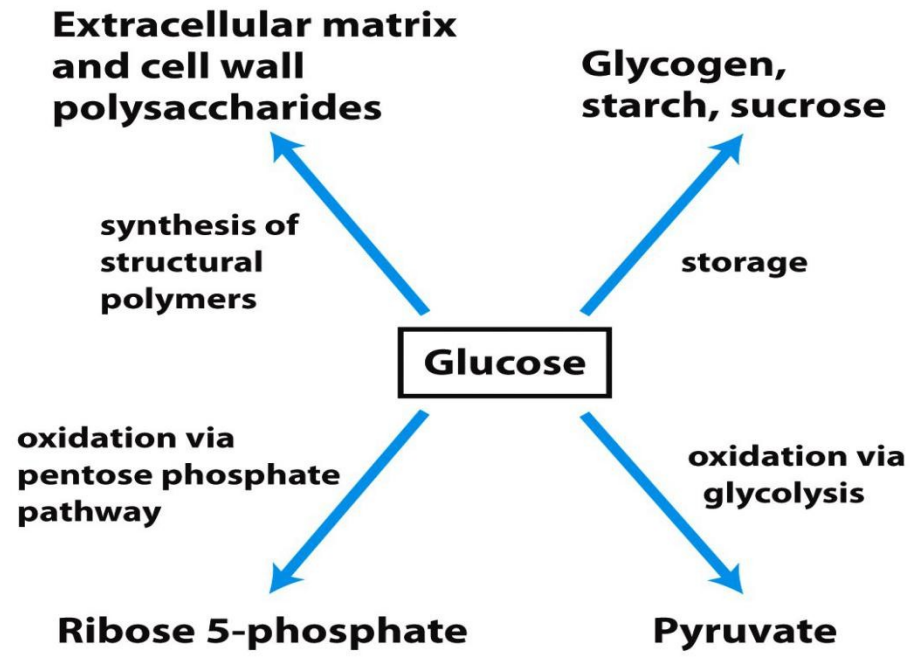


Figure 14-1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

GLYCOLYSIS

In glycolysis (from the Greek glykys, "sweet" or "sugar," and lysis, "splitting"), a molecule of glucose $C_6H_{12}O_6$is degraded in a series of enzyme-catalyzed reactions to yield two molecules of the three-carbon compound pyruvate $CH_3COCOO^- + H^+$.

During glycolysis, the free energy released is used to form the high-energy compounds ATP (adenosine triphosphate) and NADH (reduced nicotinamide adenine dinucleotide).

The pathway of glycolysis was explained by Otto Warburg and Hans von Euler-Chelpin in yeast and in muscle by Gustav Embden and Otto Meyerhof in1930s.

The most common type of glycolysis is the Embden–Meyerhof–Parnas (EMP pathway), which was discovered by Embden and Meyerhof and J.K. Parnas.

Glycolysis occurs in most organisms in the cytosol of the cell.

It is an oxygen independent metabolic pathway, meaning that it does not use molecular oxygen (i.e. atmospheric oxygen) for any of its reactions.

The products of glycolysis, that is, pyruvate and NADH + H^+) are sometimes disposed off using atmospheric oxygen.

Aerobic Glycolysis: When molecular oxygen is used in the disposal of the products of glycolysis the process is usually referred to as Aerobic Glycolysis.

Anaerobic Glycolysis: When no oxygen is used, the process is said to be Anaerobic.

2 phases of GLYCOLYSIS

The Preparatory Phase – in which ATP is consumed, hence also known as the investment phase

The Pay Off Phase – in which ATP is produced

Occurs in 10 steps, the first 5 of which constitute the preparatory phase.

The breakdown of the six-carbon glucose into two molecules of the three-carbon pyruvate

The Preparatory Phase

step 1: The first step in glycolysis is phosphorylation of glucose at the hydroxyl group on C-6 by a family of enzymes called hexokinases to form glucose-6-phosphate (G6P).

step 2: The D-glucose 6-phosphate thus formed is converted to D-fructose 6-phosphate,

step 3: which is again phosphorylated, this time at C-1, to yield D-fructose 1,6-bisphosphate

For both phosphorylations, ATP is the phosphoryl group donor.

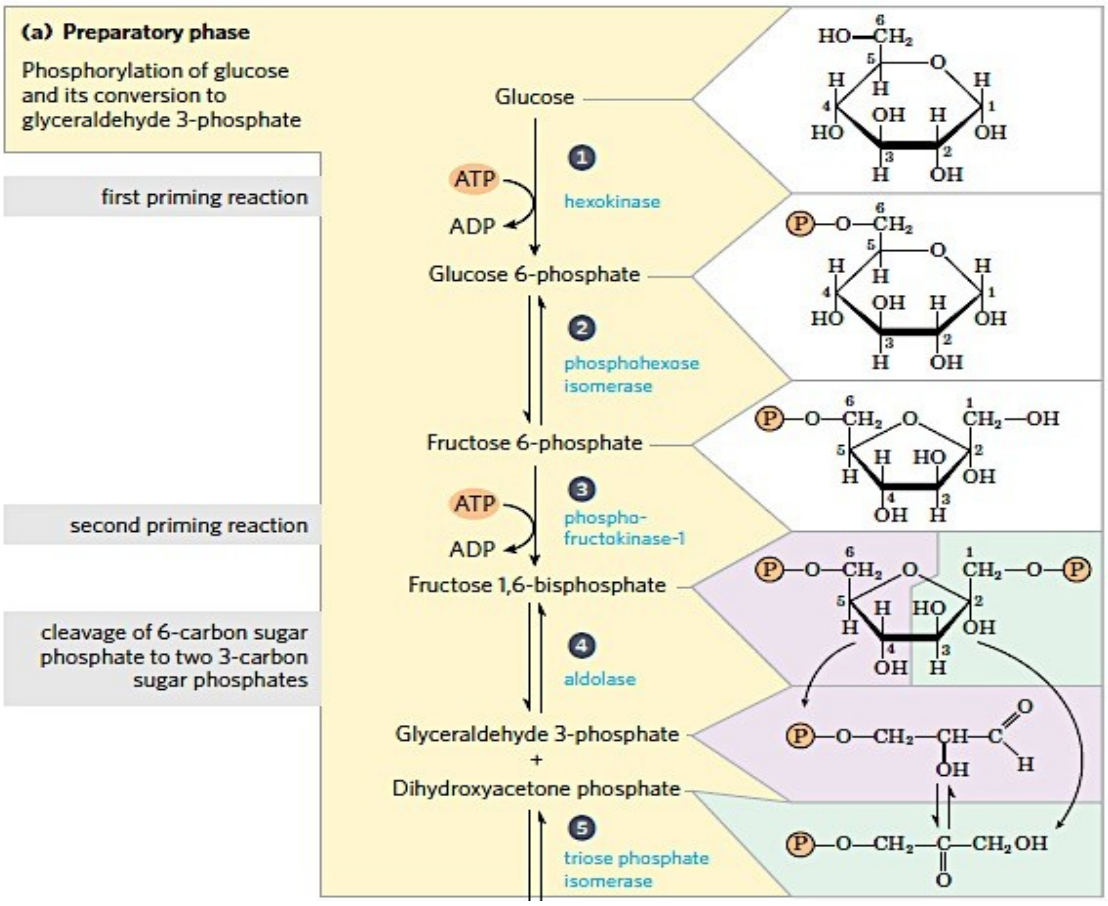
step 4: Fructose 1,6-bisphosphate is split to yield two three-carbon molecules, dihydroxyacetone phosphate and glyceraldehyde 3-phosphate (step 4); this is the "lysis" step that gives the pathway its name.

step 5: The dihydroxyacetone phosphate is isomerized to a second molecule of glyceraldehyde 3-phosphate (step 5), ending the first phase of glycolysis.

To summarize:

In the preparatory phase of glycolysis the energy of ATP is invested, raising the free-energy content of the intermediates, and the carbon chains of all the metabolized hexoses are converted to a common product, glyceraldehyde 3-phosphate.

Note that during preparatory phase, 2 molecules of ATP are invested before the cleavage of glucose into two three-carbon pieces.



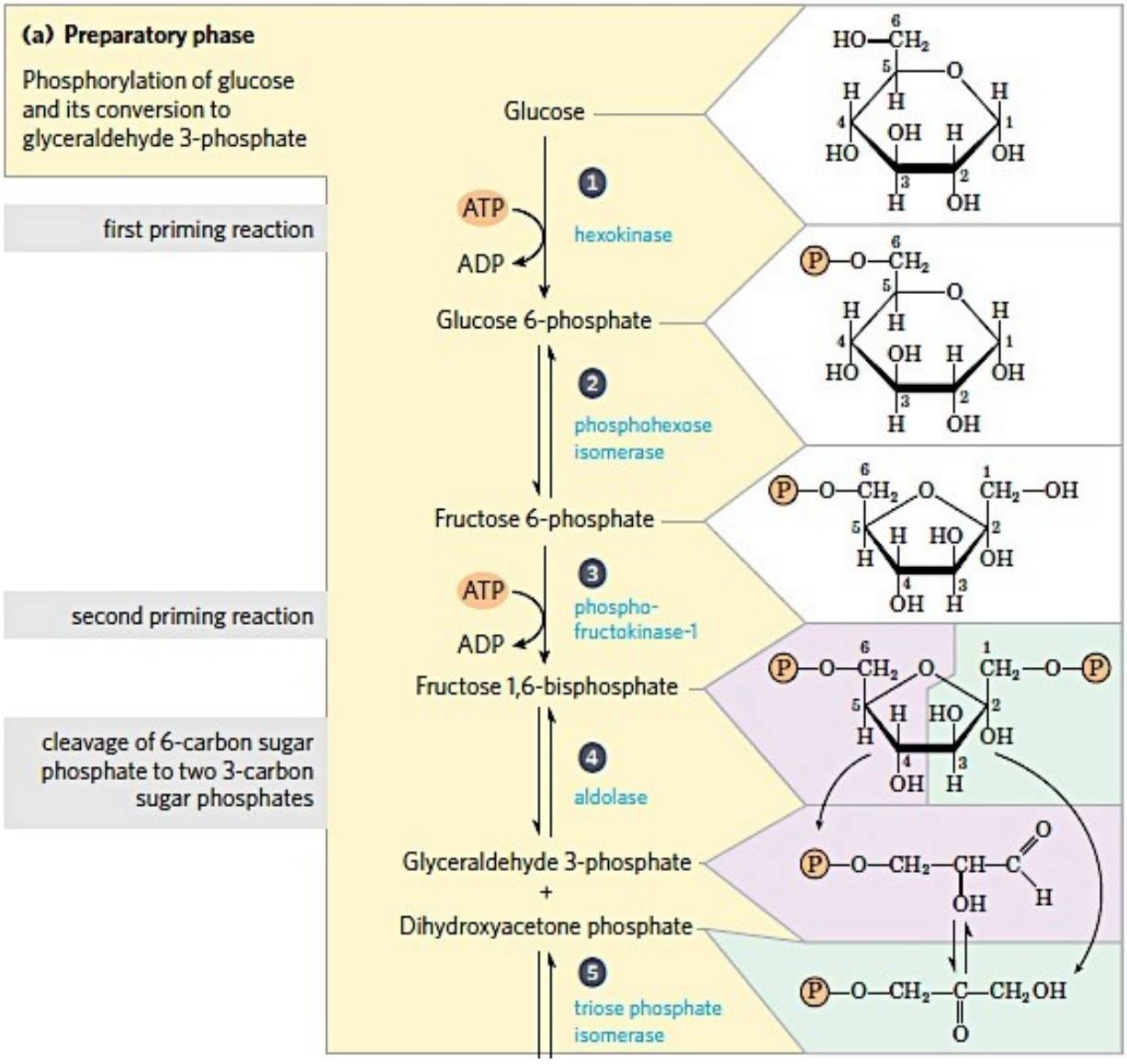
Preparatory phase of Glycolysis

For each molecule of glucose that passes through the preparatory phase (a), two molecules of glyceraldehyde 3-phosphate are formed; both pass through the payoff phase.

TOPIC#17

Glycolysis- the pay off phase

GLYCOLYSIS-The Pay-off Phase



The Pay-off Phase

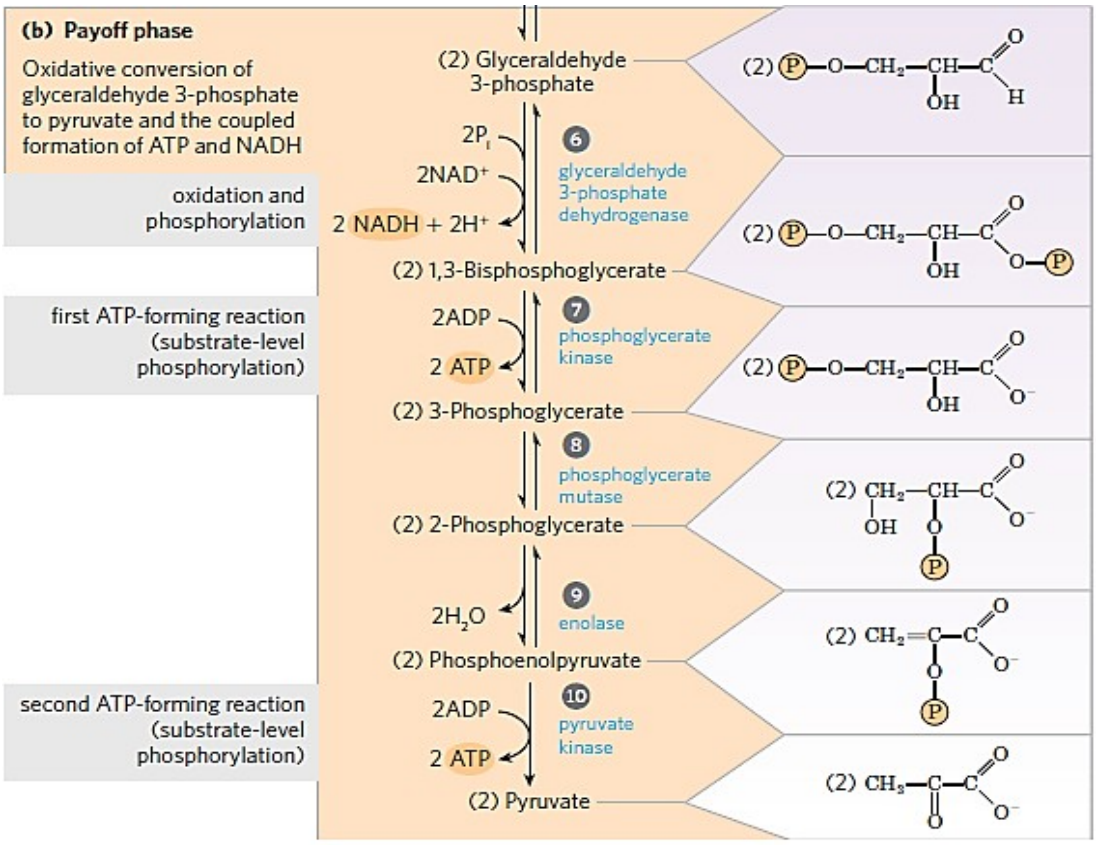
The energy gain comes in the payoff phase of glycolysis

step 6: Each molecule of glyceraldehyde 3-phosphate is oxidized and phosphorylated by inorganic phosphate (not by ATP) to form 1,3-bisphosphoglycerate

Energy is also conserved in the formation of two molecules of the electron carrier NADH per molecule of glucose.

steps 7 through 10: Much of this energy is conserved by the coupled phosphorylation of four molecules of ADP to ATP.

The net yield is two molecules of ATP per molecule of glucose, because two molecules of ATP were invested in the preparatory phase.



Pyruvate is the end product of the second phase of glycolysis.

For each glucose molecule, two ATP are consumed in the preparatory phase and four ATP are produced in the payoff phase,

This gives a net yield of two ATP per molecule of glucose converted to pyruvate.

each phosphoryl group, represented here as P, has two negative charges

Energy yield during Glycolysis

Two ATP are consumed in the preparatory phase and four ATP are produced in the payoff phase, giving a net yield of two ATP per molecule of glucose converted to pyruvate. Note that each phosphoryl group, represented as P, has two negative charges ($2PO_3^{-2}$)

Much of this energy is conserved by the coupled phosphorylation of four molecules of ADP to ATP.

Energy is also conserved in the payoff phase in the formation of two molecules of the electron carrier NADH per molecule of glucose.

In the sequential reactions of glycolysis, three types of chemical transformations are noteworthy:

(1) degradation of the carbon skeleton of glucose to yield pyruvate

(2) phosphorylation of ADP to ATP by compounds with high phosphoryl group transfer potential, formed during glycolysis.

(3) transfer of a hydride ion to NAD^+ forming NADH.

High Phosphoryl Transfer Potential Compounds. These compounds have a higher phosphoryl transfer potential than that of ATP and can be used to phosphorylate ADP to form ATP.

TOPIC#18

Lecture - 18 - Preparatory phase of Glycolysis Requires ATP

Preparatory phase of Glycolysis Requires ATP

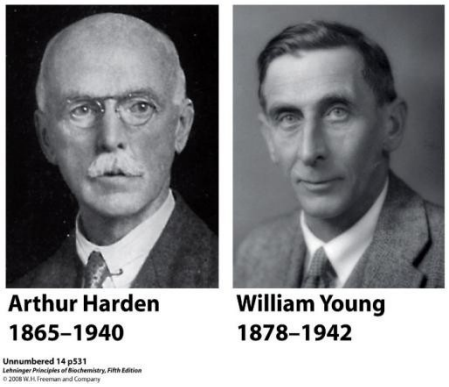
Experiment by A. Harden and W. Young

Preparatory phase of glycolysis, two molecules of ATP are invested and the hexose chain is cleaved into two triose phosphates

In 1906, Arthur Harden and William Young tested their hypothesis that inhibitors of proteolytic enzymes would stabilize the glucose fermenting enzymes in yeast extract.

They added blood serum (known to contain inhibitors of proteolytic enzymes) to yeast extracts and observed the predicted stimulation of glucose metabolism.

However, in a control experiment with the boiling serum, they discovered that boiled serum was just as effective at stimulating glycolysis



Arthur Harden
1865–1940

William Young
1878–1942

Unnumbered 14 p531
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

Harden and Young soon discovered that glucose added to their yeast extract was converted to a hexose bisphosphate (the "Harden- Young ester," eventually identified as fructose 1,6-bisphosphate.

Careful examination revealed that inorganic phosphate was responsible for the stimulation.

This was the beginning of investigations on the role of organic esters and anhydrides of phosphate in biochemistry, which has led to our current understanding of the central role of phosphoryl group transfer in biology.

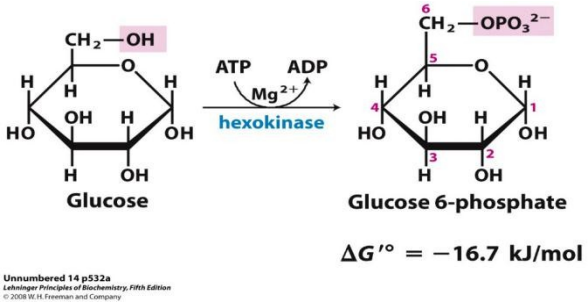
Preparatory phase of Glycolysis Requires ATP

1. Phosphorylation of Glucose

In the first step of glycolysis, glucose is activated for subsequent reactions by its phosphorylation at C-6 to yield glucose 6-phosphate, with ATP as the phosphoryl donor:

This reaction is irreversible under intracellular conditions, and catalyzed by hexokinase.

The acceptor in the case of hexokinase is a hexose, normally D-glucose, hexokinase also catalyzes the phosphorylation of other common hexoses, such as D-fructose and D-mannose, in some tissues.



Hexokinase requires Mg+2 for its activity, because the true substrate of the enzyme is not ATP-4 but the MgATP-2 complex

Hexokinase and other nine enzymes of are soluble cytosolic proteins

Note:

Isozymes

Two or more enzymes that catalyze the same reaction but are encoded by different genes are called isozymes

Kinase:

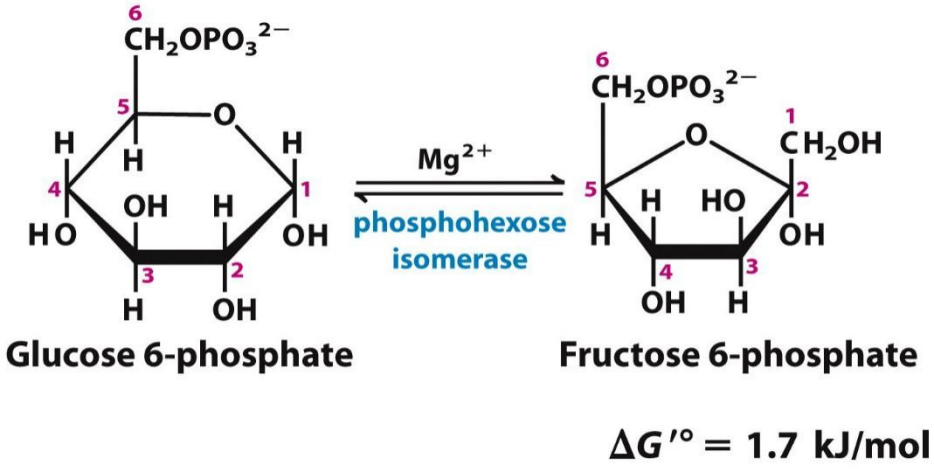
Kinases are a subclass of transferases...catalyse transfer of phosphate group from phosphate donating molecule to specific substrates

2. Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate

The enzyme phosphohexose isomerase (phosphoglucose isomerase) catalyzes the reversible isomerization of glucose 6-phosphate, an aldose, to fructose 6- phosphate, a ketose.

The mechanism for this reaction involves an enediol intermediate

Enediol: An enediol is an alkene enol with a hydroxyl group attached to both carbon atoms of the carbon double bond.



Unnumbered 14 p532b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

3 steps for Conversion of Glucose 6-Phosphate to Fructose 6-Phosphate (phosphohexose isomerase reaction)

1. opening of ring

2. Isomerization of glucose to fructose through formation of enediol intermediate

3. Closing of fructose ring

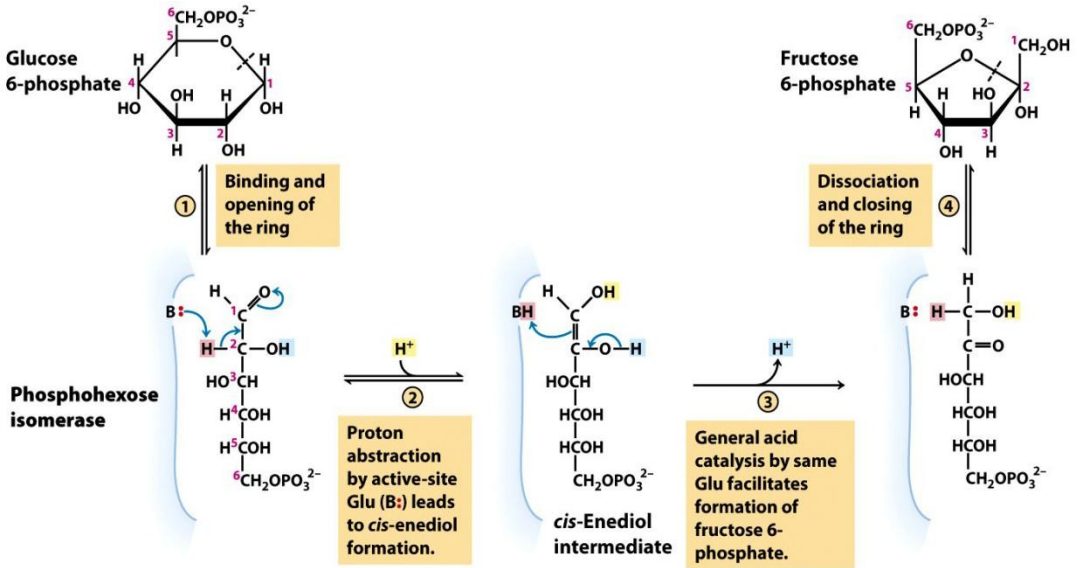


Figure 14-4
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Fig 14-5: The phosphohexose isomerase reaction.

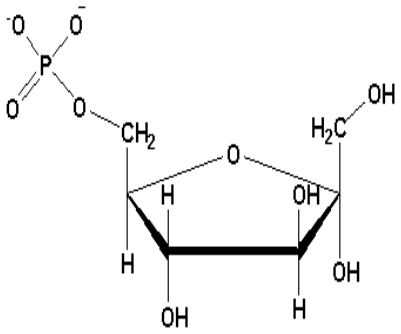
The ring opening and closing reactions (steps 1 and 4) are catalyzed by an active-site His residue, by mechanisms omitted here for simplicity.

The proton (light red) initially at C-2 is made more easily abstractable by electron withdrawal by the adjacent carbonyl and nearby hydroxyl groups. After its transfer from C-2 to the active-site Glu residue (a weak acid), the proton is freely exchanged with the surrounding solution; that is, the proton abstracted from C-2 in step 2 is not necessarily the same one that is added to C-1 in step 3.

TOPIC#19

Preparatory phase of Glycolysis Requires ATP

Preparatory phase of Glycolysis Requires ATP - 2



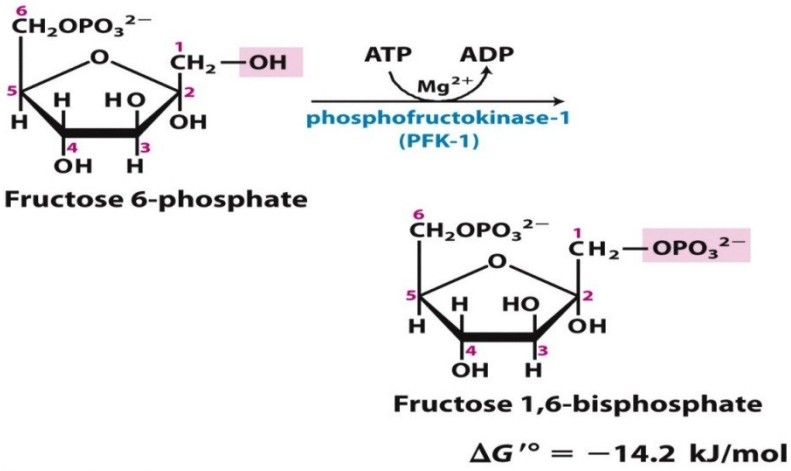
Fructose-6-phosphate (F6P)

3. Phosphorylation of Fructose 6-Phosphate to Fructose 1,6-Bisphosphate

Phosphofructokinase-1 (PFK-1) catalyzes the transfer of a phosphoryl group from ATP to fructose 6-phosphate to yield fructose 1,6-bisphosphate.

KEY CONVENTION:

Compounds that contain two phosphate or phosphoryl groups attached at different positions in the molecule are named bisphosphates (or bisphospho compounds); for example, fructose 1,6-bisphosphate and 1,3-bisphosphoglycerate. Compounds with two phosphates linked together as a pyrophosphoryl group are named diphosphates; for example, adenosine diphosphate (ADP). Similar rules apply for the naming of trisphosphates (such as inositol 1,4,5-trisphosphate;) and triphosphates (such as adenosine triphosphate, ATP)

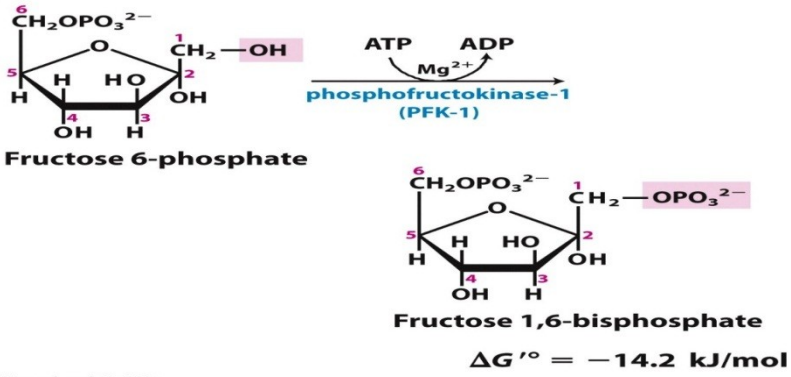


Unnumbered 14 p532c
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

The PFK-1 reaction is essentially irreversible under cellular conditions, and it is the first "committed" step in the glycolytic pathway

Glucose 6-phosphate and fructose 6-phosphate have other possible fates, but fructose 1,6-bisphosphate is targeted for glycolysis

Some bacteria and protists and perhaps all plants have a phosphofructokinase that uses pyrophosphate (PPi), not ATP, as the phosphoryl group donor in the synthesis of fructose 1,6-bisphosphate:



Unnumbered 14 p532c
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

In metabolic pathways, enzymes catalyzing essentially irreversible reactions are potential sites of control

Phosphofructokinase-1 is subject to complex allosteric regulation

The PFK1 has got binding sites for both, ATP-Mg2+ and F6P binding sites

The activity of PFK-1 enzyme and glycolysis is inhibited whenever the cell has high levels of ATP

In this way, the cell can increase or decrease the rate of glycolysis in response to the cell's energy requirements

In Eukaryotes, fructose 2,6-bisphosphate (not to be confused with the PFK-1 reaction product, fructose 1,6-bisphosphate) is a potent allosteric activator of PFK-1

Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway also activates phosphofructokinase indirectly.

Imine: An imine is a functional group or chemical compound containing a carbon–nitrogen double bond. The nitrogen atom can be attached to a hydrogen or an organic group.

Amine: an organic compound derived from ammonia by replacement of one or more hydrogen atoms by organic groups.

Schiff base intermediate:

They can be considered a sub-class of imines, being either secondary ketimines or secondary aldimines depending on their structure. (R-CH=NR' where R' ≠ H)

TOPIC#20

Preparatory phase of Glycolysis Requires ATP-3

Preparatory phase of Glycolysis Requires ATP - 2

In metabolic pathways, enzymes catalyzing essentially irreversible reactions are potential sites of control

Phosphofructokinase-1 is subject to complex allosteric regulation

The PFK1 has got binding sites for both, ATP-Mg2+ and F6P binding sites

The activity of PFK-1 enzyme and glycolysis is inhibited whenever the cell has high levels of ATP

In this way, the cell can increase or decrease the rate of glycolysis in response to the cell's energy requirements

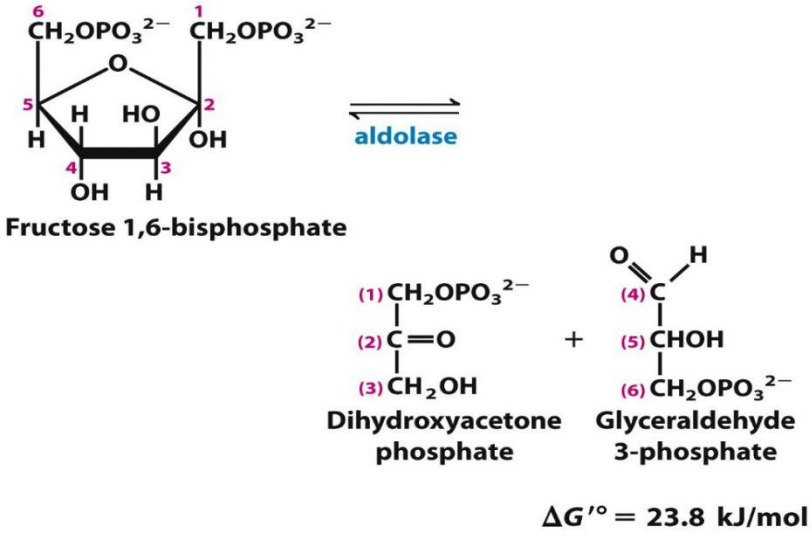
In Eukaryotes, fructose 2,6-bisphosphate (not to be confused with the PFK-1 reaction product, fructose 1,6-bisphosphate) is a potent allosteric activator of PFK-1

Ribulose 5-phosphate, an intermediate in the pentose phosphate pathway also activates phosphofructokinase indirectly.

4. Cleavage of Fructose 1,6-Bisphosphate

The enzyme fructose 1,6-bisphosphate aldolase, often called simply aldolase, catalyzes a reversible aldol condensation (see Fig. 13–4..Lehninger Edition 6).

Fructose 1,6-bisphosphate is cleaved to yield two different triose phosphates, glyceraldehyde 3-phosphate, an aldose, and dihydroxyacetone phosphate, a ketose.



Unnumbered 14 p533
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Two classes of aldolases

There are two classes of aldolases.

- Class I, found in animals and plants, form the Schiff base intermediate
- Class II, in fungi and bacteria, do not form the Schiff base intermediate

Imine: An imine is a functional group or chemical compound containing a carbon–nitrogen double bond. The nitrogen atom can be attached to a hydrogen or an organic group.

Schiff base intermediate:

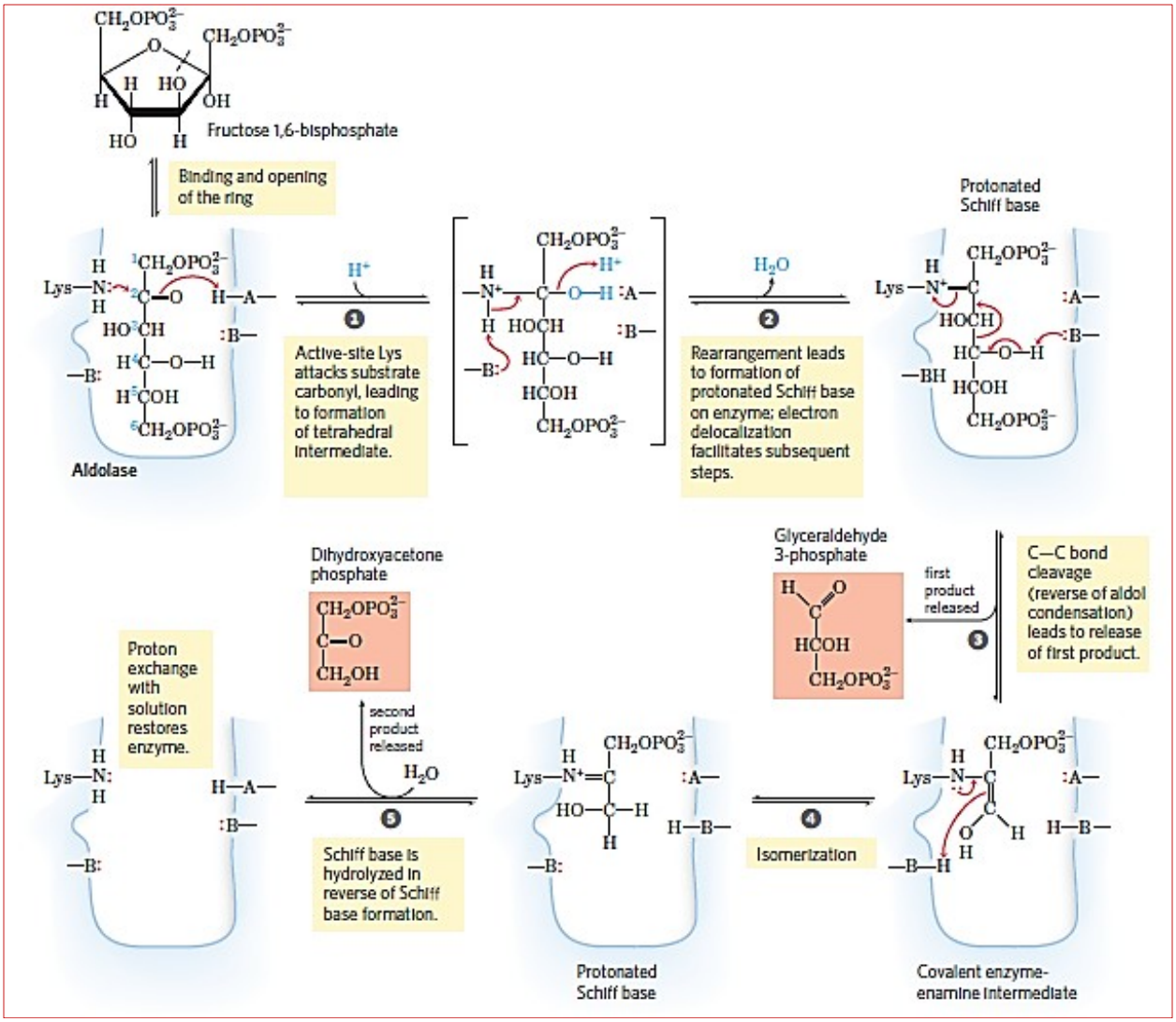
They can be considered a sub-class of imines, being either secondary ketimines or secondary aldimines depending on their structure. (R-CH=NR' where R' ≠ H)

MECHANISM

FBPA, class I aldolase reaction. The reaction shown here is the reverse of an aldol condensation.

Cleavage between C-3 and C-4 depends on the presence of the carbonyl group at C-2, which is converted to an imine on the enzyme.

A and B represent amino acid residues that serve as general acid (A) or base (B).



Imine: An imine is a functional group or chemical compound containing a carbon–nitrogen double bond. The nitrogen atom can be attached to a hydrogen or an organic group.

Schiff base intermediate:

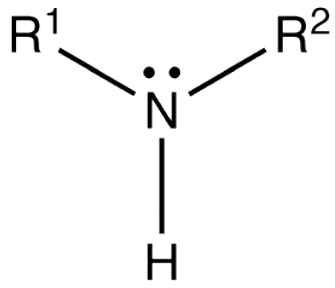
They can be considered a sub-class of imines, being either secondary ketimines or secondary aldimines depending on their structure. (R-CH=NR' where R' ≠ H)

Amine: an organic compound derived from ammonia by replacement of one or more hydrogen atoms by organic groups.

Secondary amine:

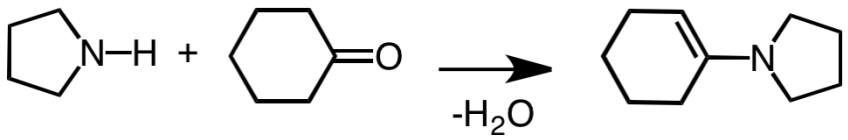
Secondary amines have two organic substituents (alkyl, aryl or both)

bound to the nitrogen together with one hydrogen.



Enamine:

An enamine is an unsaturated compound derived by the condensation of an aldehyde or ketone with a secondary amine.



TOPIC#21

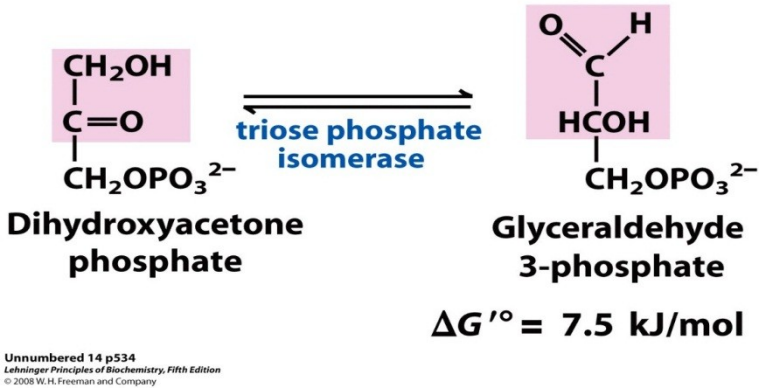
Preparatory phase of Glycolysis -step 5

1. **Interconversion of the Triose Phosphates**

Only one of the two triose phosphates formed by aldolase, glyceraldehyde 3-phosphate, can be directly degraded in

the subsequent steps of glycolysis.

The other product, dihydroxyacetone phosphate, is rapidly and reversibly converted to glyceraldehyde 3-phosphate by the fifth enzyme of the glycolytic sequence, triose phosphate isomerase:



Unnumbered 14 p534
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

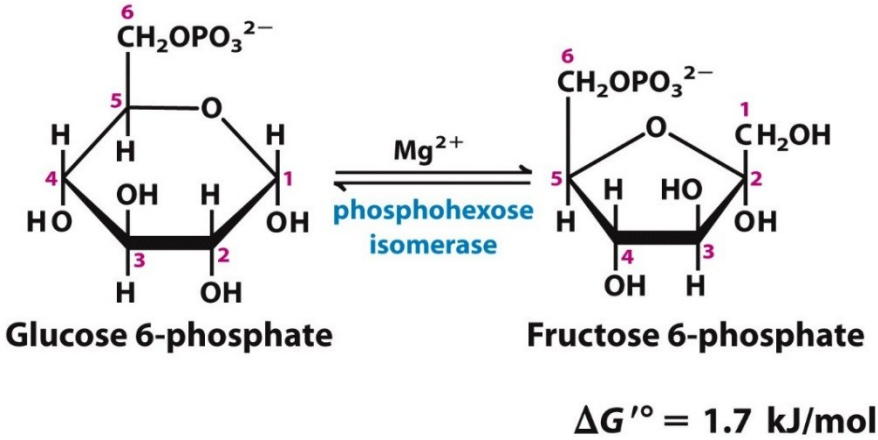
The reaction mechanism is similar to the reaction promoted by phosphohexose isomerase in step 2 of glycolysis (Fig. 14–5, Lehninger; edition 6).

After the triose phosphate isomerase reaction, the carbon atoms derived from C-1, C-2, and C-3 of the starting glucose are chemically indistinguishable

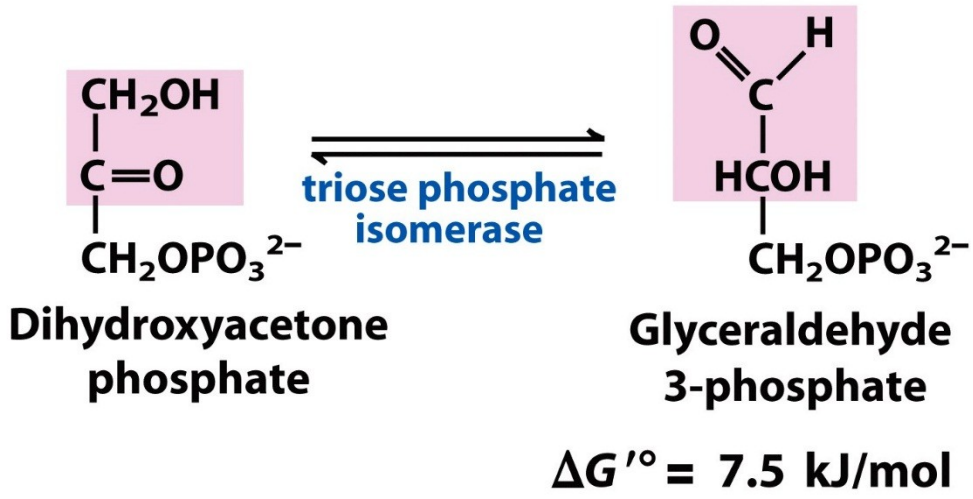
from C-6, C-5, and C-4, respectively, the two “halves” of glucose have both yielded glyceraldehyde 3-phosphate.

This reaction completes the preparatory phase of glycolysis.

The hexose molecule has been phosphorylated at C-1 and C-6 and then cleaved to form two molecules of glyceraldehyde 3-phosphate.



Unnumbered 14 p532b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

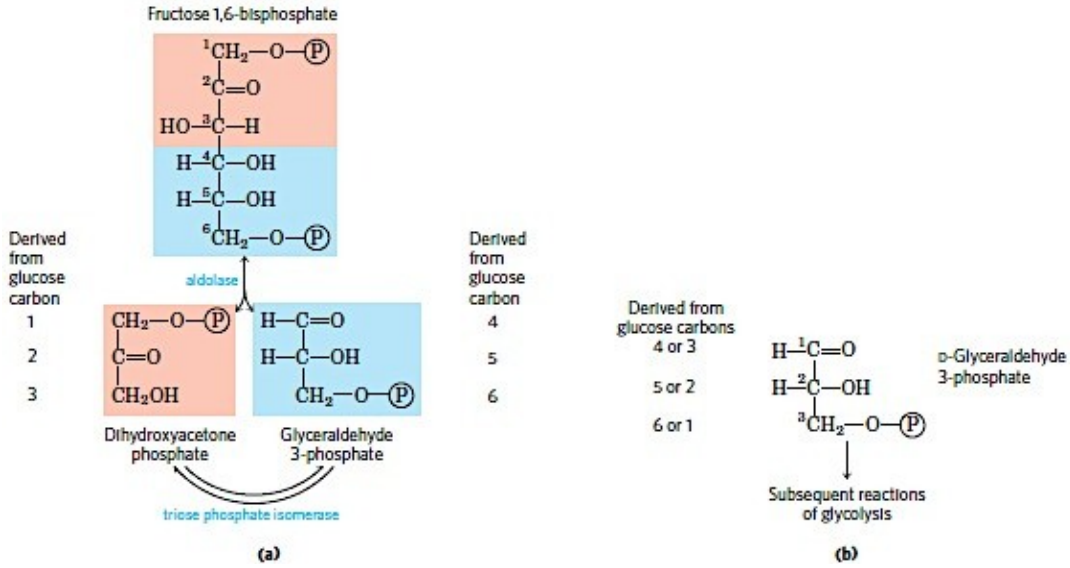


Unnumbered 14 p534
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Fate of the glucose carbons in the formation of glyceraldehyde 3-phosphate.

The origin of the carbons in the two 3-carbon products of the aldolase and triose phosphate isomerase reactions. The end product of the two reactions is glyceraldehyde 3-phosphate (two molecules).

Each carbon of glyceraldehyde 3-phosphate is derived from either of two specific carbons of glucose. Note that the numbering of the carbon atoms of glyceraldehyde 3-phosphate differs from that of the glucose from which it is derived. In glyceraldehyde 3-phosphate, the most complex functional group (the carbonyl) is specified as C-1. This numbering change is important for interpreting experiments with glucose in which a single carbon is labeled with a radioisotope.



Radioisotopes:

Atomic number (Z): no. of protons

Mass number (A): the total number of protons and neutrons in an atomic nucleus (together known as nucleons).

Isotopes are the atoms in an element that have the same atomic number but a different atomic mass; that is, the same number of protons and thus identical chemical properties, but different numbers of neutrons and consequently different physical properties.

Isotopes can be stable or unstable or radioisotopes.

In the latter, their nuclei have a special property: they emit energy in the form of ionizing radiation while searching for a more stable configuration.

TOPIC#22

Pay off phase-Last steps of glycolysis-6-7

The Payoff Phase of Glycolysis Yields ATP and NADH

The payoff phase of glycolysis includes the energy-conserving phosphorylation steps

The chemical energy of the glucose molecule is conserved in the form of ATP and NADH

The conversion of two molecules of glyceraldehyde 3-phosphate to two molecules of pyruvate is accompanied by the formation of four molecules of ATP from ADP.

the net yield of ATP per molecule of glucose degraded is only two, because two ATP were invested in the preparatory phase of glycolysis to phosphorylate the two ends of the hexose molecule**.**

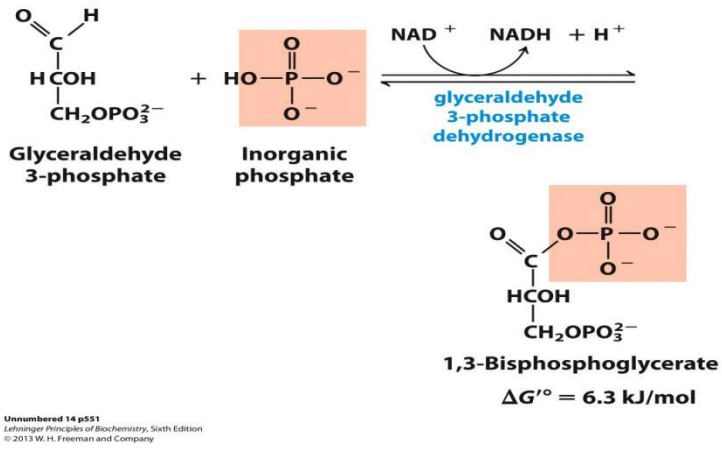
2. Oxidation of Glyceraldehyde 3-Phosphate to 1,3-Bisphosphoglycerate

The first step in the payoff phase is the oxidation of glyceraldehyde 3-phosphate to 1,3-bisphosphoglycerate, catalyzed by glyceraldehyde 3-phosphate dehydrogenase

This is the first of the two energy-conserving reactions of glycolysis that eventually lead to the formation of ATP.

The aldehyde group of glyceraldehyde 3-phosphate (G3P) is oxidized, not to a free carboxyl group but to a carboxylic acid anhydride with phosphoric acid.

This type of anhydride, called an acyl phosphate, has a very high standard free energy of hydrolysis (-49.3 kJ/mol)



Unnumbered 14 p551
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Much of the free energy of oxidation of the aldehyde group of glyceraldehyde 3-phosphate is conserved by formation of the acyl phosphate group at C-1 of 1,3-bisphosphoglycerate.

NAD+ is the e- acceptor (bound to Rossmann fold of dehydrogenase)

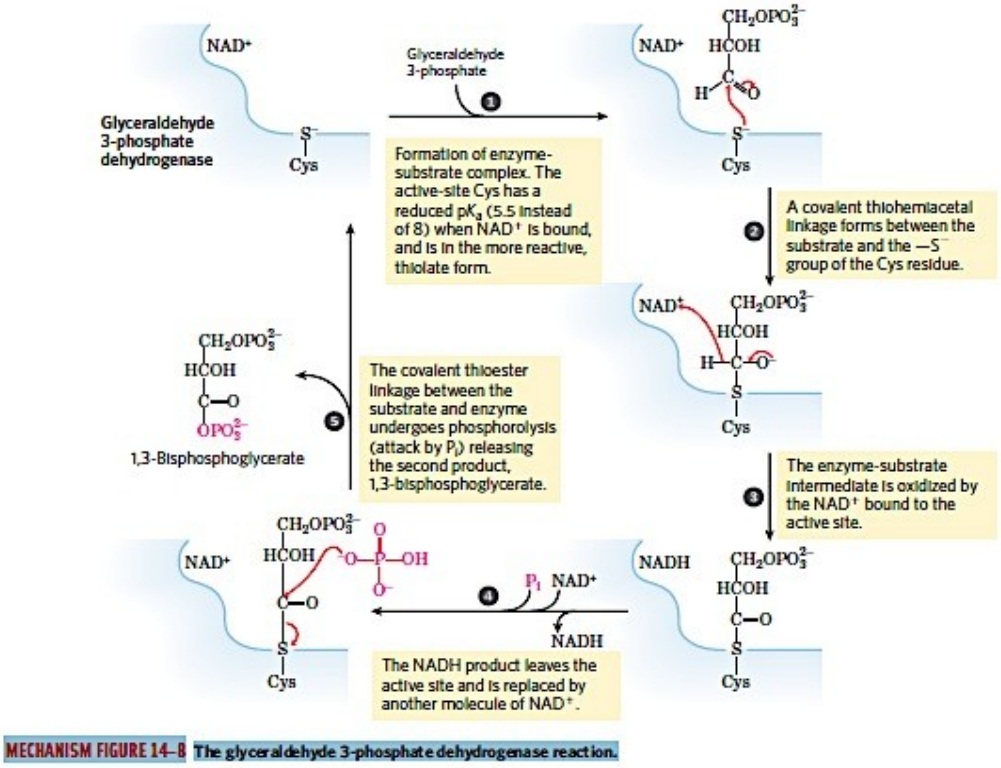
G3P is covalently bound to the dehydrogenase

The aldehyde group of glyceraldehyde 3-phosphate reacts with the —SH group of an essential Cys residue in the active site, in a reaction analogous to the formation of a hemiacetal producing a thiohemiacetal.

Reaction of the essential Cys residue with a heavy metal such as Hg+2 irreversibly inhibits the enzyme.

The amount of NAD+ in a cell is far smaller than the amount of glucose metabolized in a few minutes.

Glycolysis would soon come to a halt if the NADH formed in this step of glycolysis were not continuously reoxidized and recycled.



MECHANISM FIGURE 14–8 The glyceraldehyde 3-phosphate dehydrogenase reaction.

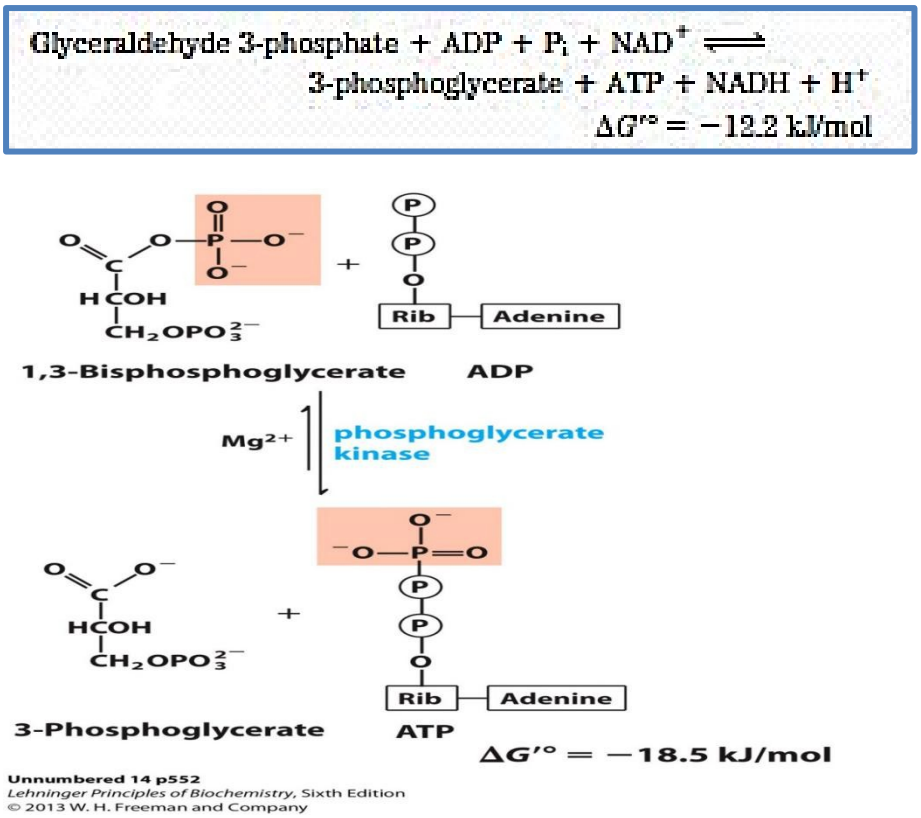
7. Phosphoryl Transfer from 1,3-Bisphosphoglycerate to ADP

The enzyme phosphoglycerate kinase transfers the high-energy phosphoryl group from the carboxyl group of 1,3-bisphosphoglycerate to ADP, forming ATP and 3-phosphoglycerate.

phosphoglycerate kinase is named for the reverse reaction, in which it transfers a phosphoryl enzymes, it catalyzes the reaction in both directions.

In glycolysis, the reaction it catalyzes proceeds in the direction of ATP synthesis.

Steps 6 and 7 of glycolysis together constitute an energy-coupling process in which 1,3-bisphosphoglycerate is the common intermediate; it is formed in the first reaction (which would be endergonic in isolation), and its acyl phosphate group is transferred to ADP in the second reaction (which is strongly exergonic). Overall reaction is exergonic.. The sum of these two reactions is



The outcome of these reactions, both reversible under cellular conditions, is that the energy released on oxidation of an aldehyde to a carboxylate group is conserved by the coupled formation of ATP from ADP and Pi.

The formation of ATP by phosphoryl group transfer from a substrate such as 1,3-bisphosphoglycerate is referred to as a substrate-level phosphorylation, to distinguish this mechanism from respiration-linked phosphorylation.

Substrate-level phosphorylation

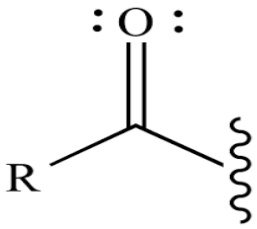
Substrate-level phosphorylation is a metabolic reaction that results in the formation of ATP or GTP by the direct transfer of a phosphoryl (PO3) group to ADP or GDP from another phosphorylated compound.

Substrate-level phosphorylations involve soluble enzymes and chemical intermediates (1,3-bisphosphoglycerate in this case).

Respiration-linked phosphorylations, involve membrane-bound enzymes and transmembrane gradients of protons.

Acyl group

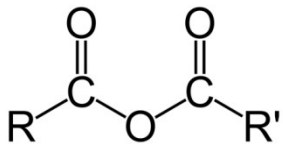
It contains a double bonded oxygen atom and an alkyl group. (R-C=O group is called Acyl Group). In organic chemistry, the acyl group (IUPAC name: alkanoyl) is usually derived from a carboxylic acid.

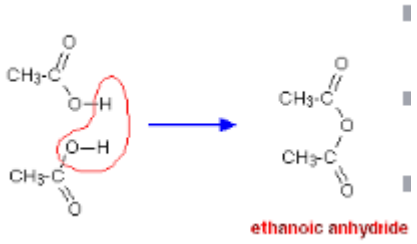


Acid anhydride

An acid anhydride is a compound that has two acyl groups bonded to the same oxygen atom.

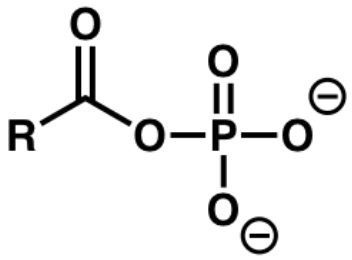
Thus, (CH3CO)2O is called acetic anhydride. Mixed (or unsymmetrical) acid anhydrides, such as acetic formic anhydride.





Acyl phosphate

A phosphate group is connected to the acyl group by a single bond and that the bond occurs between one of the oxygen atoms of the phosphate and the carbonyl carbon of the acyl group.



Alkyl group

:alkane with 1 hydrogen missing

TOPIC#23

Last steps of glycolysis. -8-10

The Payoff Phase of Glycolysis Yields ATP and NADH-2

Step 8-10

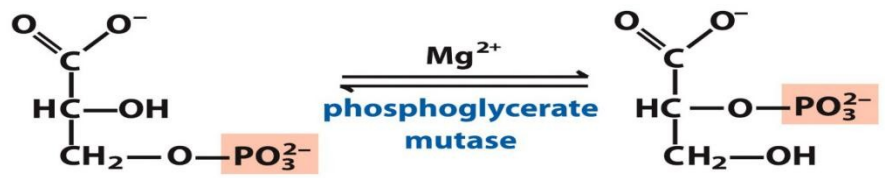
8. Conversion of 3-Phosphoglycerate to 2-Phosphoglycerate

The enzyme phosphoglycerate mutase catalyzes a reversible shift of the phosphoryl group between C-2 and C-3 of glycerate

Mg+2 is essential for this reaction. The reaction occurs in two steps A phosphoryl group initially attached to a His residue of the mutase is transferred to the hydroxyl group at C-2 of 3-phosphoglycerate, forming 2,3-bisphosphoglycerate (2,3-BPG).

The phosphoryl group at C-3 of 2,3-BPG is then transferred to the same His residue, producing 2-phosphoglycerate and regenerating the phosphorylated enzyme.

Phosphoglycerate mutase is initially phosphorylated by phosphoryl transfer from 2,3-BPG, which is required in small quantities to initiate the catalytic cycle and is continuously regenerated by that cycle



Unnumbered 14 p554a
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

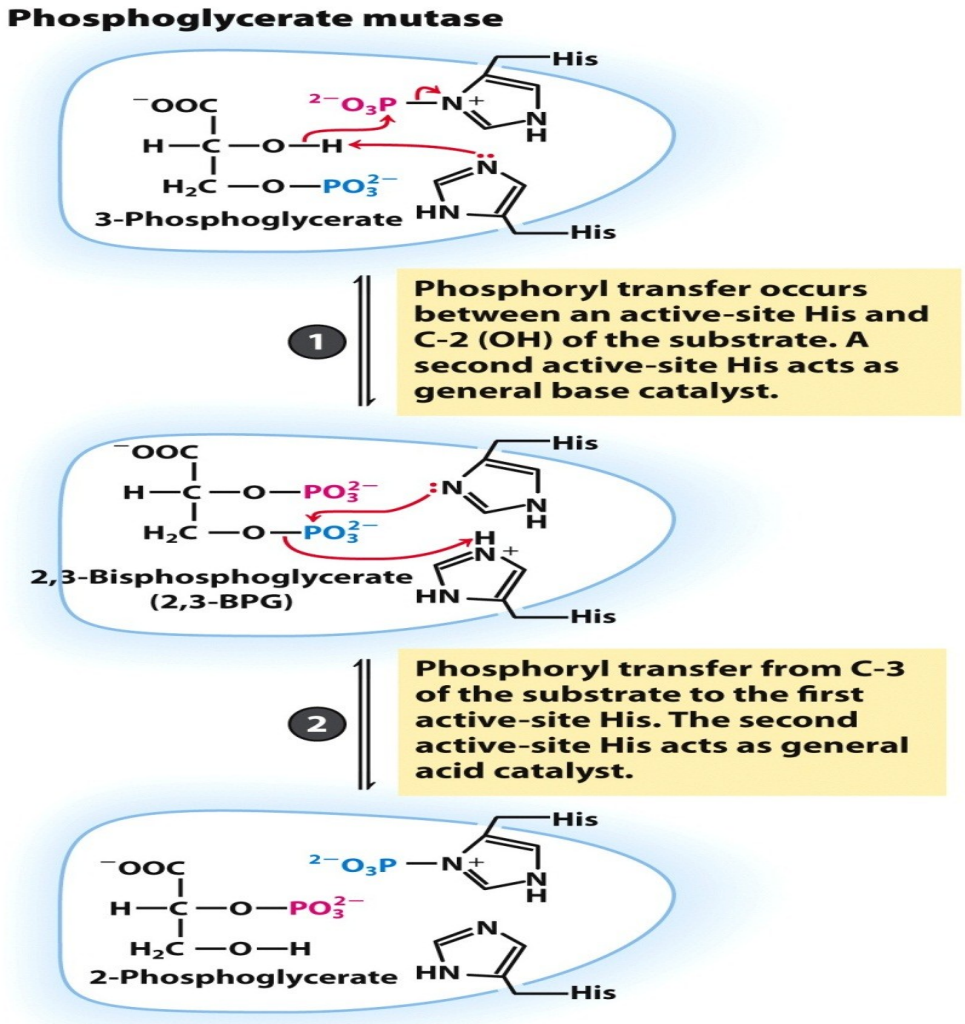


Figure 14-9
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

The phosphoglycerate mutase reaction

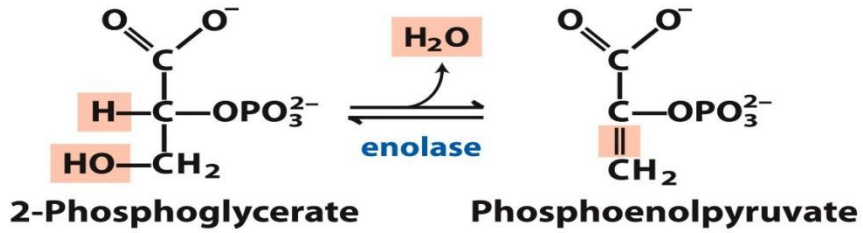
9. <u>Dehydration of 2-Phosphoglycerate to Phosphoenolpyruvate</u>

In the second glycolytic reaction that generates a compound with high phosphoryl group transfer potential

(the first was step 6), enolase promotes reversible removal of a molecule of water from 2-phosphoglycerate

The mechanism of the enolase reaction involves an enolic intermediate stabilized by Mg+2

The reaction converts a compound with a relatively low phosphoryl group transfer potential (✎G' ▼ for hydrolysis of 2-phosphoglycerate is -17.6 kJ/mol) to one with high phosphoryl group transfer potential (✎G' ▼ for PEP hydrolysis is -61.9 kJ/mol)



Unnumbered 14 p554b
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

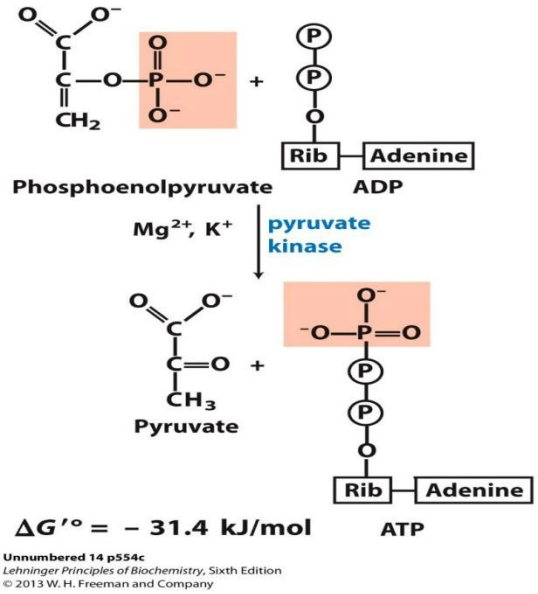
10. Transfer of the Phosphoryl Group from Phosphoenolpyruvate to ADP

The last step in glycolysis is the transfer of the phosphoryl group from phosphoenolpyruvate to ADP, catalyzed by pyruvate kinase, which requires K+ (potassium ions) and either Mg+2 or Mn+2

In this substrate-level phosphorylation, the product pyruvate first appears in its enol form, then tautomerizes rapidly and nonenzymatically to its keto form, which predominates at pH 7.

The overall reaction change, spontaneous conversion of the enol form of pyruvate to the keto form has a large, negative standard free energy

About half of the energy released by PEP hydrolysis (-61.9 kJ/mol) is conserved in the formation of the phosphoanhydride bond of ATP (-30.5 kJ/mol), and the rest (-31.4 kJ/mol) constitutes a large driving force pushing the reaction toward ATP synthesis.



Unnumbered 14 p554c
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

TOPIC#24

Energy generated during Glycolysis- part 1 and 2

GLYCOLYSIS-ATP Formation Coupled to Glycolysis

ATP Formation Coupled to Glycolysis During glycolysis some of the energy of the glucose molecule is conserved in the form of ATP, while much remains in the product, pyruvate.

The overall equation for glycolysis is

$$\text{Glucose} + 2\text{NAD}^+ + 2\text{ADP} + 2\text{Pi} \longrightarrow 2\ \text{pyruvate} + 2\text{NADH} + 2\text{H}^+ + 2\text{ATP} + 2\text{H}_2\text{O} \quad \textbf{(Eqn I)}$$

For each molecule of glucose degraded to pyruvate, two molecules of ATP are generated from ADP and Pi.

We can now resolve the equation of glycolysis into two processes:

1) the conversion of glucose into pyruvate is exergonic:

$$\text{Glucose} + 2\text{NAD}^+ \longrightarrow 2\ \text{pyruvate} + 2\text{NADH} + 2\text{H}^+ \ ,\Delta G'^{\circ}{}_1 = -146\ \text{kJ/mol} \quad \textbf{(Eqn II)}$$

2) and the formation of ATP from ADP and Pi is endergonic:

$$2ADP + 2Pi \longrightarrow 2ATP + 2H_2O\ ,\ \Delta G^{\circ\prime}{}_2 = \mathbf{2(30.5\ kJ/mol) = 61\ kJ/mol}\ \textbf{(Eqn III)}$$

If we now write the sum of Equations (II) and (III), we can also determine the overall standard free-energy change of glycolysis (Eqn I), including ATP formation, as the algebric sum, ΔG°'S, of ΔG°‘1 and ΔG°'2

$$\mathbf{\Delta G^{\circ\prime}{}_s = \Delta G^{\circ\prime}{}_1' + \Delta G^{\circ\prime}{}_2 = -146\ kJ/mol + 61\ kJ/mol = -85\ kJ/mol}$$

Under standard and intracellular conditions, glycolysis is an essentially irreversible process, driven to completion by this large net decrease in free energy.

Energy Remaining in the Pyruvate Produced by Glycolysis

The two molecules of pyruvate formed by glycolysis still contain most of the chemical potential energy of glucose

This energy that can be extracted by oxidative reactions in the citric acid cycle and oxidative phosphorylation

4 electrons..in the form of 2 hydride ions are transferred from 2 G3P to 2NAD+

2 molecules of NADH are reoxidized to NAD+ by transfer of electrons to electron transport chain (ETC) within mitochondria

ETC passes these electrons to oxygen, that provides energy for ATP synthesis during respiration

“Pasteur effect”

The ATP yield from glycolysis under anaerobic conditions (2 ATP per molecule of glucose) is much smaller than that from the complete oxidation of glucose to CO2 under aerobic conditions (30 or 32 ATP per glucose)

About 15 times more glucose would be consumed in anaerobic conditions to produce same amount of energy (ATP)

This was discovered by Louis Pasteur while studying process of fermentation in yeast, and is known as the “Pasteur effect”

"Warburg effect"

The German biochemist Otto Warburg first observed in 1928 that tumors of nearly all types carry out glycolysis at a much higher rate than normal tissue, even when oxygen is available.

This is known as "Warburg effect" and is the basis for several methods of detecting and treating cancer

Isolated 7 enzymes of glycolysis from tumor tissues

Warburg manometer: experimental tool to measure amount oxygen consumed ..so oxidative activity

Nobel prize of Physiology in 1931

Why glycolysis in tumor tissue?

Most tumor cells grow under hypoxic conditions (i.e., with limited oxygen supply) because, at least initially, they lack the capillary network to supply sufficient oxygen.

Cancer cells located more than 100 to 200 um from the nearest capillaries must depend on glycolysis alone (without further oxidation of pyruvate) for much of their ATP production. The energy yield (2 ATP per glucose) is far lower than can be obtained by the complete oxidation of pyruvate to CO2 in mitochondria (about 30 ATP per glucose)

To make the same amount of ATP, tumor cells must take up much more glucose than do normal cells, converting it to pyruvate and then to lactate as they recycle NADH.

In general, the more aggressive the tumor, the greater is its rate of glycolysis. This increase in glycolysis is achieved at least in part by increased synthesis of the glycolytic enzymes
and of the plasma membrane transporters GLUT1 and GLUT3 that carry glucose into cells.

The hypoxia-inducible transcription factor (HIF-1) is a protein stimulates the production of at least eight glycolytic enzymes and the glucose transporters when oxygen supply is limited

Another protein induced by HIF-1 is the peptide hormone VEGF (vascular endothelial growth factor), which stimulates the growth of blood vessels (angiogenesis) toward the tumor

This heavier reliance of tumors than of normal tissue on glycolysis suggests a possibility for anticancer therapy: inhibitors of glycolysis might target and kill tumors by depleting their supply of ATP. Three inhibitors of hexokinase have shown promise as chemotherapeutic agents: 2-

deoxyglucose, lonidamine, and 3-bromopyruvate.

Glucose Uptake Is Deficient in Type 1 Diabetes Mellitus

The metabolism of glucose in mammals is limited by the rate of glucose uptake into cells and its phosphorylation by hexokinase.

Glucose uptake from the blood is mediated by the GLUT family of glucose transporters

The transporters of hepatocytes (GLUT1, GLUT2) and of brain neurons (GLUT3) are always present in plasma membranes.

In contrast, the main glucose transporter in the cells of skeletal muscle, cardiac muscle, and adipose tissue (GLUT4) is sequestered in small intracellular vesicles and moves into the plasma membrane only in response to an insulin signal

Thus in skeletal muscle, heart, and adipose tissue, glucose uptake and metabolism

depend on the normal release of insulin by pancreatic beta cells in response to elevated blood glucose

Individuals with type 1 diabetes mellitus have too few cellsand cannot release sufficient insulin to trigger glucose uptake by the cells of skeletal muscle, heart, or adipose tissue.

Thus, after a meal containing carbohydrates, glucose accumulates to abnormally high levels in the blood, a condition known as hyperglycemia. Unable to take up glucose, muscle and fat tissue use the fatty acids of stored triacylglycerols as their principal fuel.

In the liver, acetyl-CoA derived from this fatty acid breakdown is converted to “ketone bodies”—acetoacetate and -hydroxybutyrate—which are exported and carried to other tissues to be used as fuel

These compounds are especially critical to the brain, which uses ketone bodies as alternative fuel when glucose is unavailable. (Fatty acids cannot pass through the blood-brain barrier and thus are not a fuel for brain neurons.)

In untreated type 1 diabetes, overproduction of acetoacetate and hydroxybutyrate leads to their accumulation in the blood, and the consequent lowering of blood pH produces ketoacidosis, a life-threatening condition.

Insulin injection reverses this sequence of events:

GLUT4 moves into the plasma membranes of hepatocytes and adipocytes, glucose is taken up into the cells and phosphorylated, and the blood glucose level falls, greatly reducing the production of ketone bodies.

Importance of Phosphorylated Intermediates

Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated. The phosphate groups appear to have three functions.

1. The phosphate groups are ionized at pH 7, thus giving each of the intermediates of glycolysis a net negative charge.
2. Because the plasma membrane is impermeable to molecules that are charged, the phosphorylated intermediates cannot diffuse out of the cell.
3. After the initial phosphorylation, the cell does not have to spend further energy in retaining phosphorylated intermediates despite the large difference between the intracellular and extracellular concentrations of these compounds.

Phosphate groups are essential components in the enzymatic conservation of metabolic energy.

Energy released in the breakage of phosphoric acid anhydride bonds (such as those in ATP) is partially conserved in the formation of phosphate esters such as glucose-6 phosphate.

High-energy phosphate compounds formed in glycolysis (1,3-bisphosphoglycerate and phosphoenol pyruvate) donate phosphate groups to ADP to form ATP.

Binding of phosphate groups to the active sites of enzymes provides binding energy that contributes to lowering the activation energy and increasing the specificity of enzyme-catalyzed reactions.

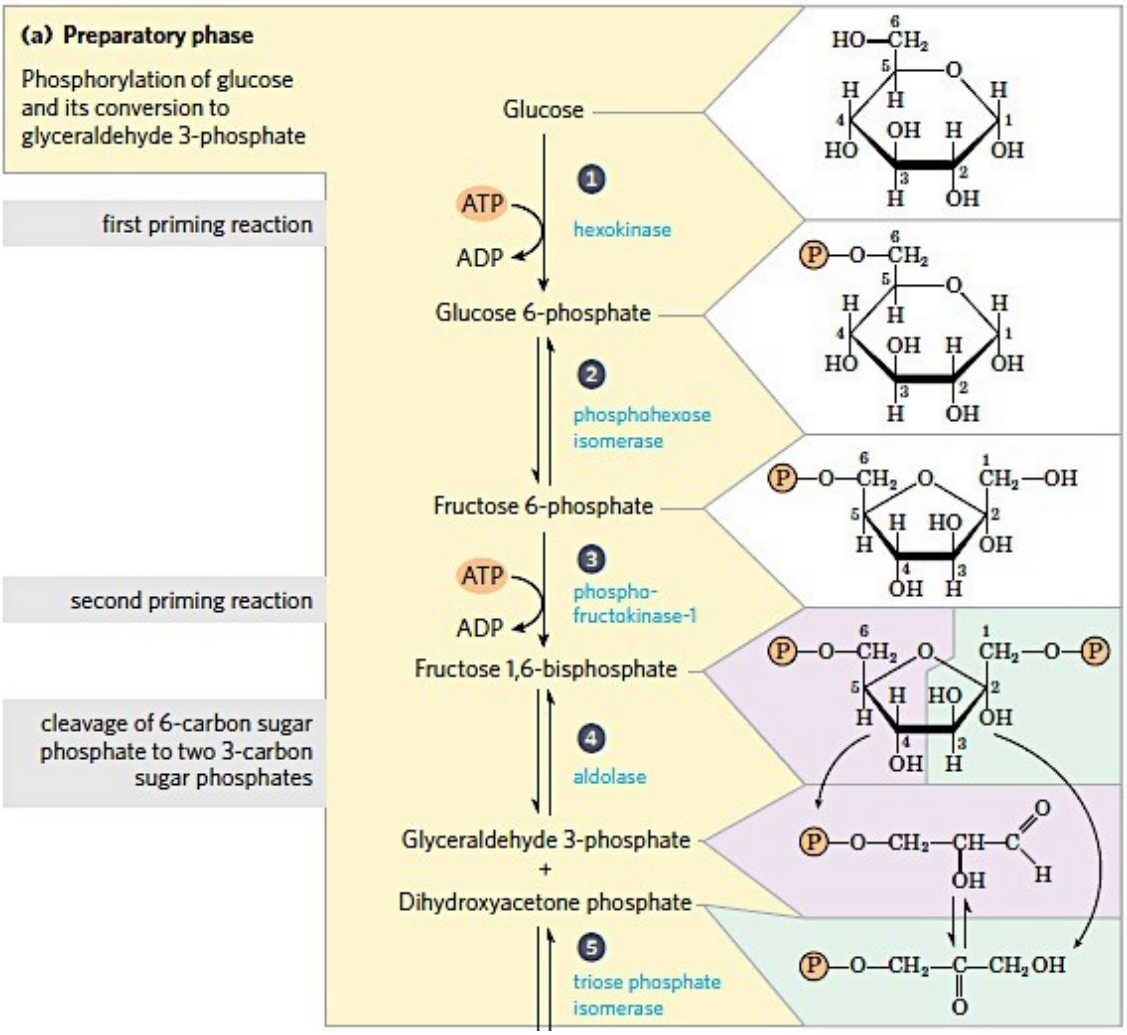
The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with Mg^{2+}, and the substrate binding sites of many of the glycolytic enzymes are specific for these Mg^{2+} complexes.

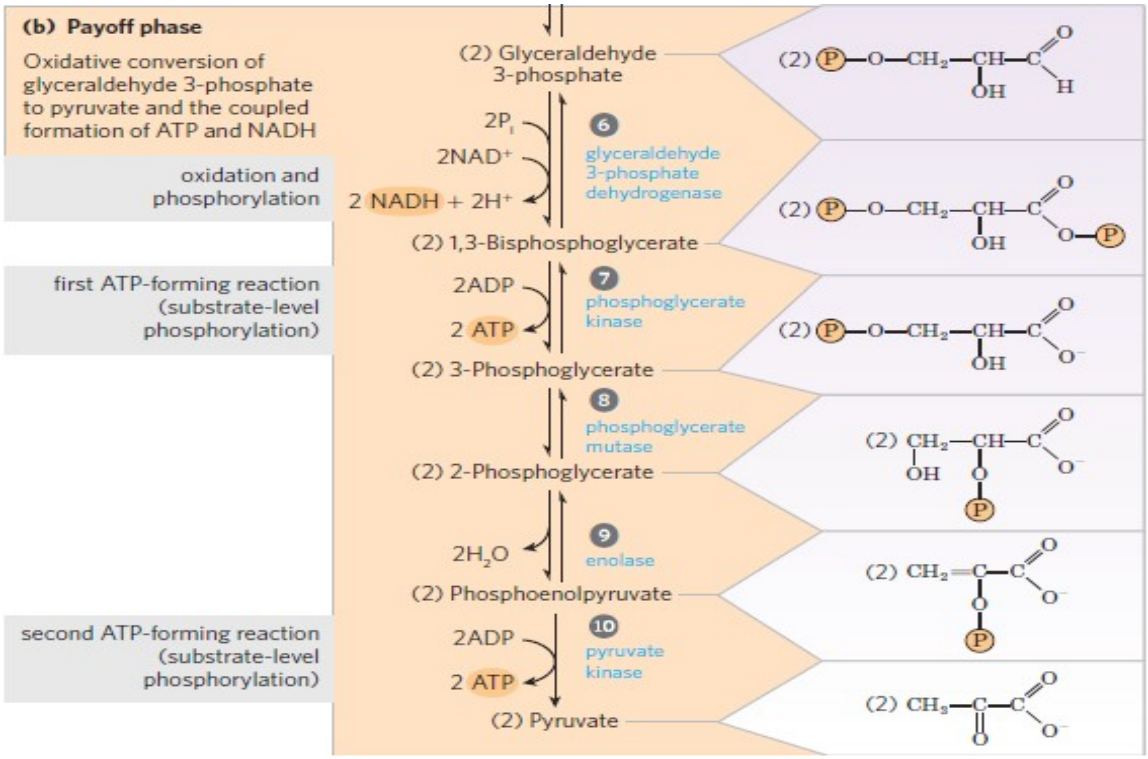
Nearly all the glycolytic enzymes require Mg^{2+} for activity.

Topic#25

Importance of Phosphorylated Intermediates

GLYCOLYSIS:Importance of Phosphorylated Intermediates


(a) Preparatory phase
Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate
first priming reaction
second priming reaction
cleavage of 6-carbon sugar phosphate to two 3-carbon sugar phosphates
Glucose
ATP
ADP
1
hexokinase
Glucose 6-phosphate
2
phosphohexose isomerase
Fructose 6-phosphate
ATP
ADP
3
phospho-fructokinase-1
Fructose 1,6-bisphosphate
4
aldolase
Glyceraldehyde 3-phosphate
+
Dihydroxyacetone phosphate
5
triose phosphate isomerase



Importance of Phosphorylated Intermediates

Each of the nine glycolytic intermediates between glucose and pyruvate is phosphorylated. The phosphate groups appear to have three functions.

- The phosphate groups are ionized at pH 7, thus giving each of the intermediates of glycolysis a net negative charge.

Because the plasma membrane is impermeable to the phosphorylated sugars and charged moieties.. cannot diffuse out of the cell.

After the initial phosphorylation, the cell does not have to spend further energy in retaining phosphorylated intermediates despite the large difference between the intracellular and extracellular concentrations of these compounds.

- Phosphate groups are essential components in the enzymatic conservation of metabolic energy.

Energy released in the breakage of phosphoric acid anhydride bonds (such as those in ATP) is partially conserved in the formation of phosphate esters such as glucose-6 phosphate.

High-energy phosphate compounds formed in glycolysis (1,3-bisphosphoglycerate and phosphoenol pyruvate) donate phosphate groups to ADP to form ATP.

- Binding of phosphate groups to the active sites of enzymes:

provides binding energy that contributes to lowering the activation energy and increasing the specificity of enzyme-catalyzed reactions.

The phosphate groups of ADP, ATP, and the glycolytic intermediates form complexes with Mg^{2+}, and the substrate binding sites of many of the glycolytic enzymes are specific for these Mg^{2+} complexes.

Nearly all the glycolytic enzymes require Mg^{2+} for activity.

Glucose Uptake Is Deficient in Type 1 Diabetes Mellitus

The metabolism of glucose in mammals is limited by the rate of glucose uptake into cells and its phosphorylation by hexokinase.

Glucose uptake from the blood is mediated by the GLUT family of glucose transporters

The transporters of hepatocytes (GLUT1, GLUT2) and of brain neurons (GLUT3) are always present in plasma membranes.

In contrast, the main glucose transporter in the cells of skeletal muscle, cardiac muscle, and adipose tissue (GLUT4) is sequestered in small intracellular vesicles and moves into the plasma membrane only in response to an insulin signal

Thus in skeletal muscle, heart, and adipose tissue glucose uptake and metabolism depend on the normal release of insulin by pancreatic beta cells in response to elevated blood glucose

Patients with type 1 diabetes mellitus have too few cells and cannot release sufficient insulin to trigger glucose uptake by the cells of skeletal muscle, heart, or adipose tissue

Thus, after a meal containing carbohydrates, glucose accumulates to abnormally high levels in the blood, a condition known as hyperglycemia

Unable to take up glucose, muscle and fat tissue use the fatty acids of stored triacylglycerols as their principal fuel

In the liver, acetyl-CoA derived from this fatty acid breakdown is converted to “ketone bodies”—acetoacetate and –hydroxybutyrate

These compounds are especially critical to the brain, which uses ketone bodies as alternative fuel when glucose is unavailable. (Fatty acids cannot pass through the blood-brain barrier and thus are not a fuel for brain neurons)

In untreated type 1 diabetes, overproduction of acetoacetate and hydroxybutyrate leads to their accumulation in the blood, and the consequent lowering of blood pH produces ketoacidosis, a life-threatening condition.

Insulin injection reverses this sequence of events:

GLUT4 moves into the plasma membranes of hepatocytes and adipocytes, glucose is taken up into the cells and phosphorylated, and the blood glucose level falls, greatly reducing the production of ketone bodies.

Topic#26

Fates of Pyruvate

Fates of Pyruvate

The pyruvate formed by glycolysis is further metabolized via one of three catabolic routes.

In aerobic organisms or tissues, under aerobic conditions, glycolysis is only the first stage in the complete degradation of glucose

1O.xidation of Pyruvate:

Pyruvate is oxidized, with loss of its carboxyl group as CO2, to yield the acetyl group of acetyl-coenzyme A; the acetyl group is then oxidized completely to CO2 by the citric acid cycle. The electrons from these oxidations are passed to O2 through a chain of carriers in mitochondria, to form H2O. The energy from the electron-transfer reactions drives the synthesis of ATP in mitochondria

2. Reduction to lactate

The second route for pyruvate is its reduction to lactate via lactic acid fermentation.

When vigorously contracting skeletal muscle must function under low oxygen conditions (hypoxia), NADH cannot be reoxidized to NAD+, but NAD+ is required as an electron acceptor for the further oxidation of pyruvate.

Under these conditions pyruvate is reduced to lactate, accepting electrons from NADH and thereby regenerating the NAD+ necessary for glycolysis to continue. Certain tissues and cell types (retina and erythrocytes, for example) convert glucose to lactate even under aerobic conditions.

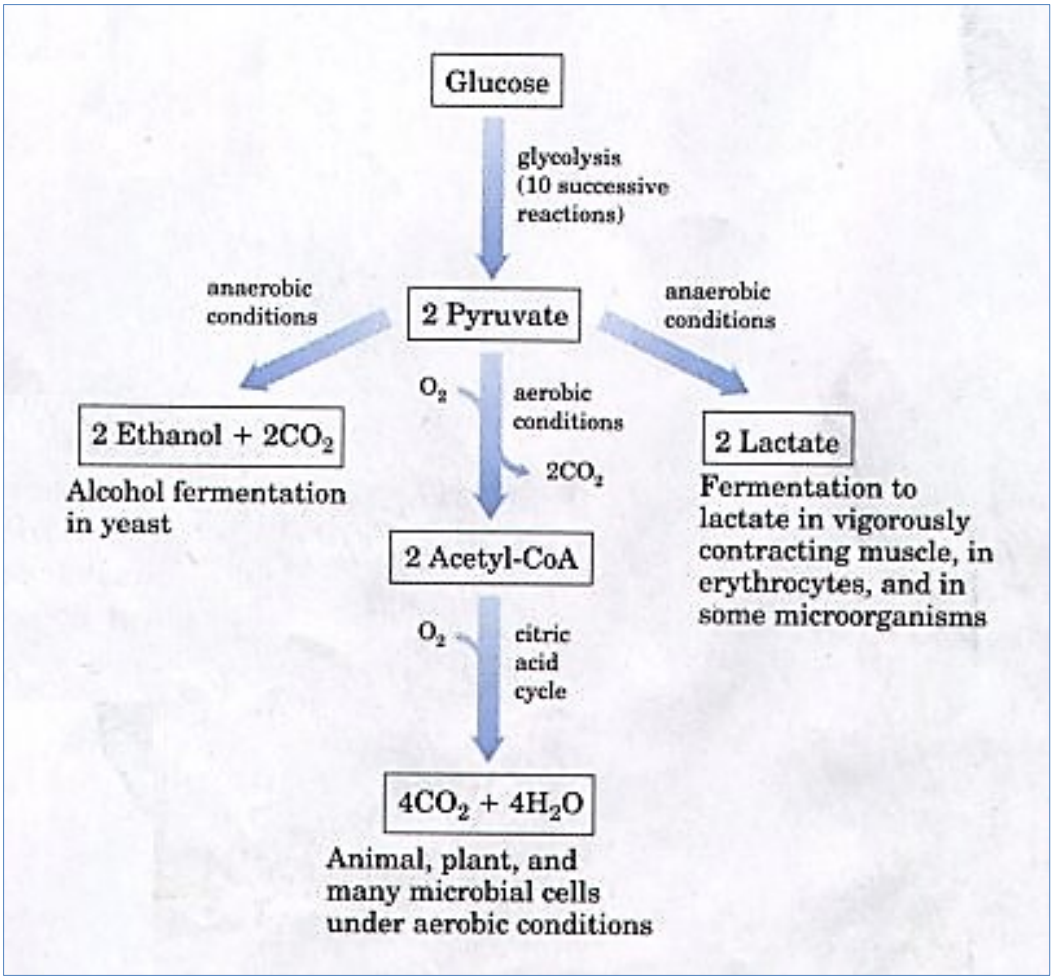
Lactate is also the product of glycolysis under anaerobic conditions in some microorganisms.

3. ethanol (alcohol) fermentation

The third major route of pyruvate catabolism leads to formation of ethanol.

In some plant tissues and in certain invertebrates, protists, and microorganisms such as yeast, pyruvate is converted under hypoxic or anaerobic conditions to ethanol and CO_2, a process called ethanol (alcohol) fermentation

The oxidation of pyruvate is an important catabolic process, but pyruvate has anabolic fates as well. It can, provide the carbon skeleton for the synthesis of the amino acid alanine or for the synthesis of fatty acids.



Three possible catabolic fates of the pyruvate formed in the payoff phase of glycolysis

Topic#27

Fates of Pyruvate -Lactic acid fermentation

Fates of Pyruvate -1

Fates of Pyruvate under Aerobic Conditions

Under Aerobic conditions, the pyruvate formed in the final step of glycolysis is oxidized to acetate (acetyl- CoA), which enters the citric acid cycle and is oxidized to CO_2 and H_2O.

The NADH formed by dehydrogenation of glyceraldehyde 3-phosphate is ultimately reoxidized to NAD+ by passage of its electrons to O_2 in mitochondrial respiration.

Fates of Pyruvate under Anaerobic Conditions

Under hypoxic (low-oxygen) conditions, as in very active skeletal muscle, in submerged plant tissues, in solid tumors, or in lactic acid bacteria—NADH generated by glycolysis cannot be reoxidized by O_2.

Failure to regenerate NAD+ can leave the cell with no electron acceptor for the oxidation of glyceraldehyde 3-phosphate, and the energy-yielding reactions of glycolysis would stop.

There must be another way to regenerated NAD+

modern organisms continually regenerate NAD+ during anaerobic glycolysis by transferring electrons from NADH to form a reduced end product such as lactate or ethanol (by fermentation).

Fermentation

Fermentation actually comes from Latin word “fervere” which means “to boil”

So the word “Fermentation” is referred to physical state of boiling/bubbling

The bubbling appearance in fermentation is due to the production of CO_2 bubbles caused by anaerobic catabolism of sugars

Fermentation is a metabolic process that converts sugars to acids, gases, or alcohol

The science of fermentation is known as ZYMOLOGY

French microbiologist Louis Pasteur is often remembered for his insights into fermentation and its microbial causes.

It occurs in yeast and bacteria, and also in oxygen-starved muscle cells

In terms of BIOCHEMISTRY Fermentation relates to the generation of energy by catabolism of organic compounds

Industrial Fermentation is used to refer to the bulk growth of microorganisms on a growth medium, often with the goal of producing a specific chemical product.

Lactic Acid Fermentation

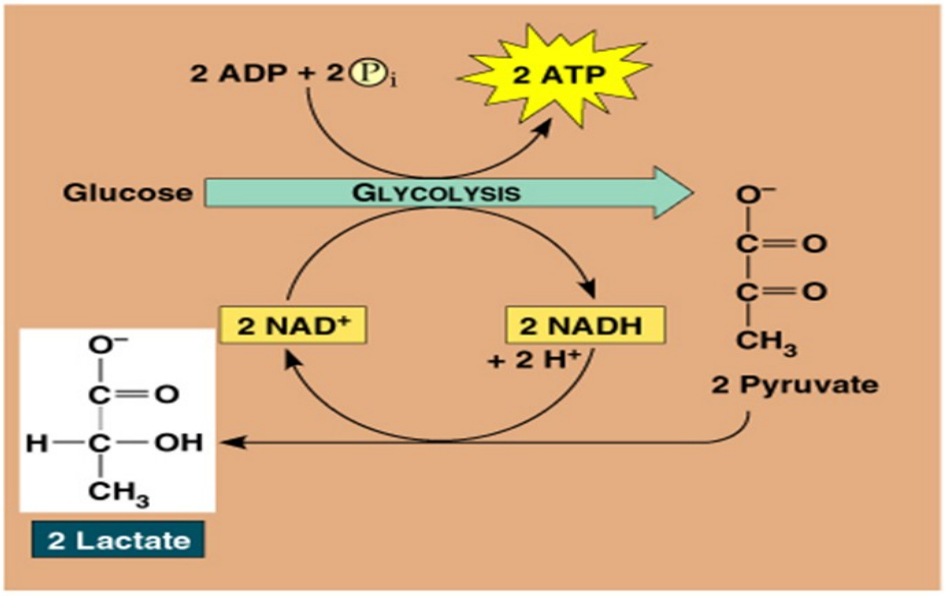
Lactic acid fermentation is done by some fungi, some bacteria like in yogurt, and sometimes by our muscles.

Glucose ⟶ carbon dioxide + lactic acid

In the process of lactic acid fermentation, the 3-carbon pyruvic acid molecules are turned into lactic acid

Allows glycolysis to continue

Normally our muscles do cellular respiration like the rest of our bodies, using O2 supplied by our lungs and blood. However, under greater exertion when the oxygen supplied by the lungs and blood system can't get there fast enough to keep up with the muscles' needs, our muscles can switch over and do lactic acid fermentation.



(b) Lactic acid fermentation

©1999 Addison Wesley Longman, Inc.

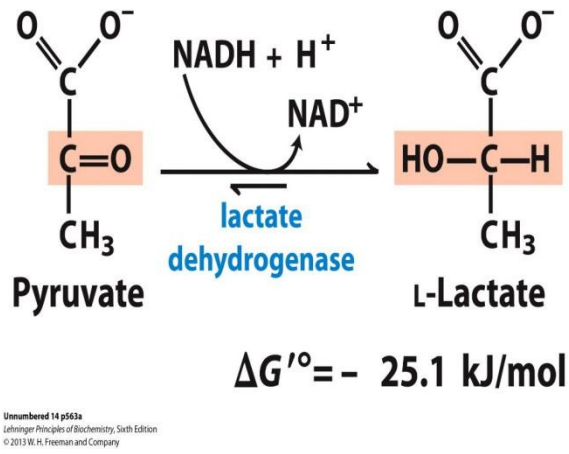
Pyruvate Is the Terminal Electron Acceptor in Lactic Acid Fermentation

NAD+ is regenerated from NADH by the reduction of pyruvate to lactate.

some tissues and cell types (such as erythrocytes, which have no mitochondria and cannot oxidize pyruvate to CO2) produce lactate from glucose even under aerobic conditions.

The reduction of pyruvate in this pathway is catalyzed by lactate dehydrogenase, which forms lactate at pH 7.

- **Under <u>aerobic</u> conditions NADH transfers its electrons eventually to oxygen, regenerating NAD+.**
- **Under <u>anaerobic</u> conditions pyruvate is reduced to lactate to regenerate NAD+ for glycolysis to continue.**



Unnumbered 14 p563a
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

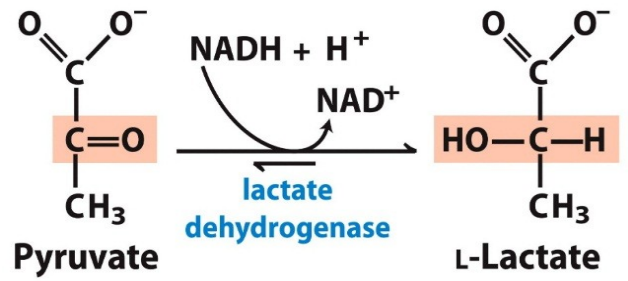
The overall equilibrium of the reaction strongly favors lactate formation, as shown by the large negative standard free-energy change.

In glycolysis, dehydrogenation of the two molecules of glyceraldehyde 3-phosphate derived from each molecule of glucose converts two molecules of NAD+ to two of NADH.

Because the reduction of two molecules of pyruvate to two of lactate regenerates two molecules of NAD+, there is no net change in NAD+ or NADH.

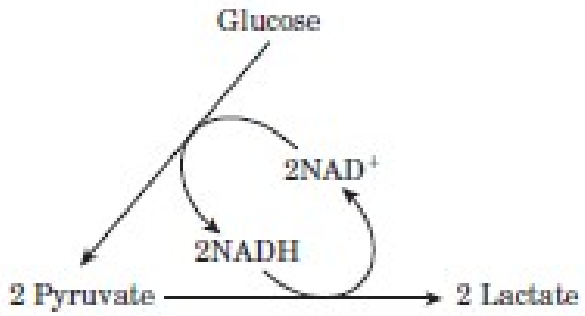
The lactate formed by active skeletal muscles (or by erythrocytes) can be recycled; it is carried in the blood to the liver, where it is converted to glucose during the recovery from strenuous muscular activity.

When lactate is produced in large quantities during vigorous muscle contraction (during a sprint, for example), the acidification that results from ionization of lactic acid in muscle and blood limits the period of vigorous activity.



Unnumbered 14 p563a
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Although conversion of glucose to lactate includes two oxidation-reduction steps, there is no net change in the oxidation state of carbon; in glucose (C6H12O6) and lactic acid (C3H6O3), the H:C ratio is the same.



Lactic Acid Fermentation

Homolactic Fermentation

HOMOLACTIC In homolactic fermentation, one glucose molecule is converted into two molecules of lactic acid

$C_6H_{12}O_6 \rightarrow 2\ CH_3CHOHCOOH$

Heterolactic Fermentation

In Heterolactic Fermentation, one glucose molecule is converted into one molecule of lactic acid, one molecule of ethanol and one molecule of carbondioxide

$C_6H_{12}O_6 \rightarrow CH_3CHOHCOOH + C_2H_5OH + CO_2$

Products of Lactic Acid Fermentation



Yogurt

Topic#28

Ethanol Fermentation

Alcohol fermentation is done by yeast and some kinds of bacteria.

Products: glucose undergoes fermentation to produce ethanol and CO2, instead of lactate.

Substrate: glucose

$$6O_2 + \text{Glucose} \longrightarrow \textit{Pyruvate} \longrightarrow CO_2 + \textit{Ethanol}$$

Humans have long taken advantage of this process in making bread, beer, and wine.

Alcohol fermentation is done by yeast and some kinds of bacteria.

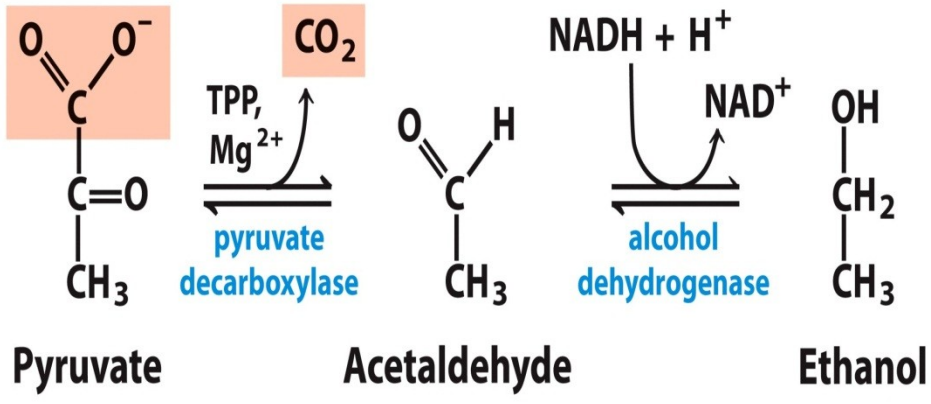
Products: glucose undergoes fermentation to produce ethanol and CO2, instead of lactate.

Substrate: glucose

two-step process:

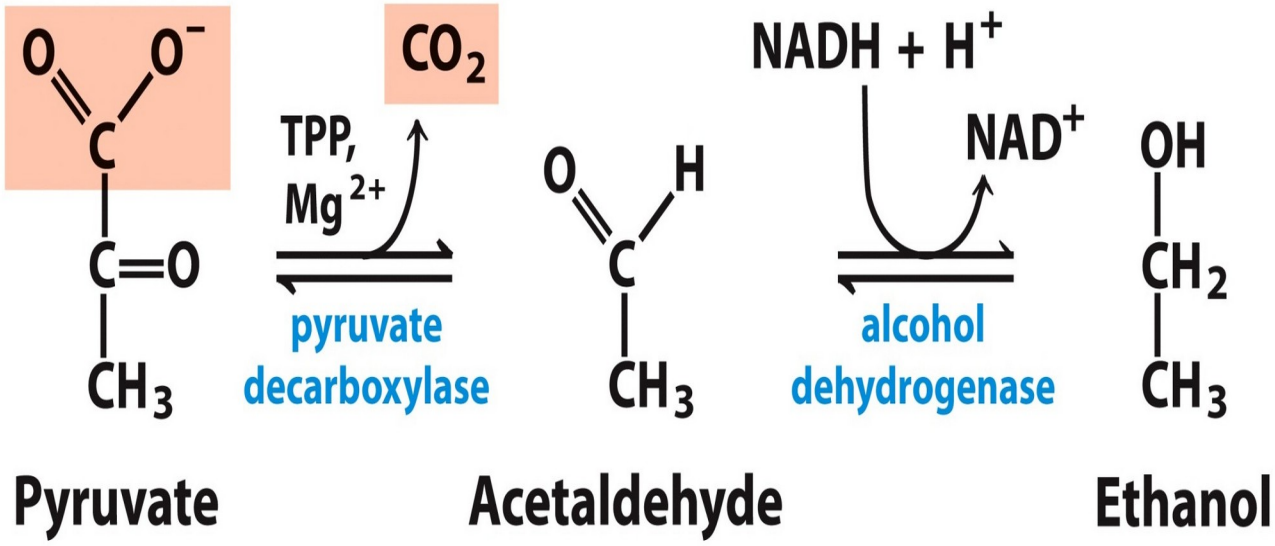
Yeast contain the enzyme pyruvate decarboxylase that decarboxylates pyruvate to form acetaldehyde.

.. Acetaldehyde is reduced to ethanol under anaerobic conditions.



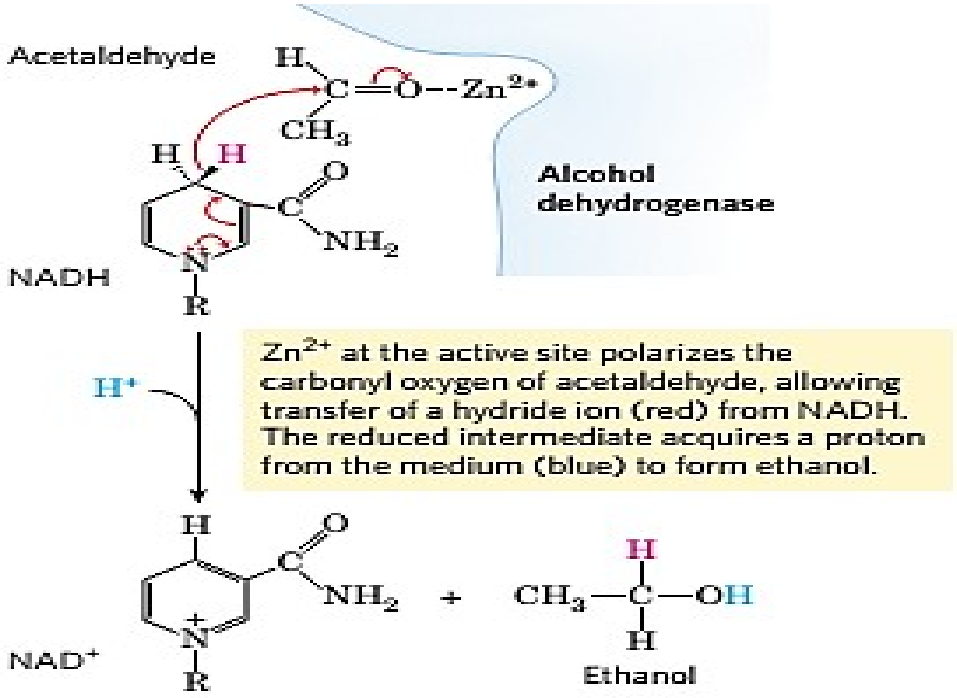
Unnumbered 14 p565
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Ethanol Is the Reduced Product in Ethanol Fermentation



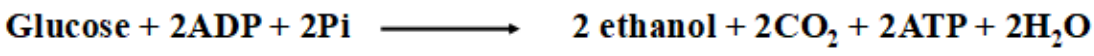
Unnumbered 14 p565
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Ethanol Is the Reduced Product in Ethanol Fermentation



The alcohol dehydrogenase reaction

Final equation:



As in lactic acid fermentation, there is no net change in the ratio of hydrogen to carbon atoms

when glucose (H:C ratio = 12/6 = 2) is fermented to two ethanol and two CO2 (combined H:C ratio = 12/6 = 2).

In all fermentations, the H:C ratio of the reactants and products remains the same.

Alcohol dehydrogenase is present in many organisms that metabolize ethanol, including humans.

In the liver it catalyzes the oxidation of ethanol, either ingested or produced by intestinal microorganisms, with the reduction of NAD+ to NADH.

Pyruvate decarboxylase is present in brewer's andbaker's yeast (Saccharomyces

cerevisiae) and in all other organisms that ferment glucose to ethanol,

including some plants.

The CO2 produced by pyruvate decarboxylation in brewer's yeast is responsible for the characteristic carbonation of champagne (ethanol fermentation).

In baking, CO2 released by pyruvate decarboxylase when yeast is mixed with a fermentable sugar causes dough to rise.

The enzyme is absent in organisms that carry out lactic acid fermentation.

TOPIC#29

Uses of Fermentation in Industry

Industrial Fermentation

Oldest form of microbiology and biotechnology which was used to make wine, beer, bread with use of bacteria and yeasts without knowing scientific basis

Fermentation has different meaning to biochemistry and industrial microbiologists.

Biochemical meaning relates to the generation of energy by the catabolism of organic compounds

Microbiologists have extended the term - fermentation - to describe: “A process that generates a product by the mass culture of microorganism.”

In Industrial Biotechnology, fermentation means a process in which microorganisms that are cultured on a large-scale under aerobic or anaerobic conditions, convert a substrate into a product which is useful to man.

The industrial microorganisms are grown under controlled conditions with an aim of optimizing the growth of the organism for production of a target microbial product.

Use of Fermentation In Industry

In 1910 Chaim Weizmann (later to become the first president of Israel) discovered that the bacterium Clostridium acetobutyricum ferments starch to butanol and acetone.

This discovery opened the field of industrial fermentations, in which some readily available material rich in carbohydrate (corn starch or molasses, for example) is supplied to a pure culture of a specific microorganism, which ferments it into a product of greater commercial value.

Some important fermentation products

Fermentations Are Used to Produce Some Common Foods and Industrial Chemicals

Product	Organism	Use
Ethanol	*Saccharomyces cerevisiae*	Industrial solvents, beverages
Glycerol	*Saccharomyces cerevisiae*	Production of explosives
Lactic acid	*Lactobacillus bulgaricus*	Food and pharmaceutical
Acetone and butanol	*Clostridium acetobutylicum*	Solvents
α-amylase	*Bacillus subtilis*	Starch hydrolysis

Yogurt, is produced when the bacterium Lactobacillus bulgaricus ferments the carbohydrate in milk, producing lactic acid; the resulting drop in pH causes the milk proteins to precipitate, producing the thick texture and sour taste of unsweetened yogurt.

Cheese is produced by using Propionibacterium freudenreichii, which ferments milk to produce propionic acid and CO2; the propionic acid

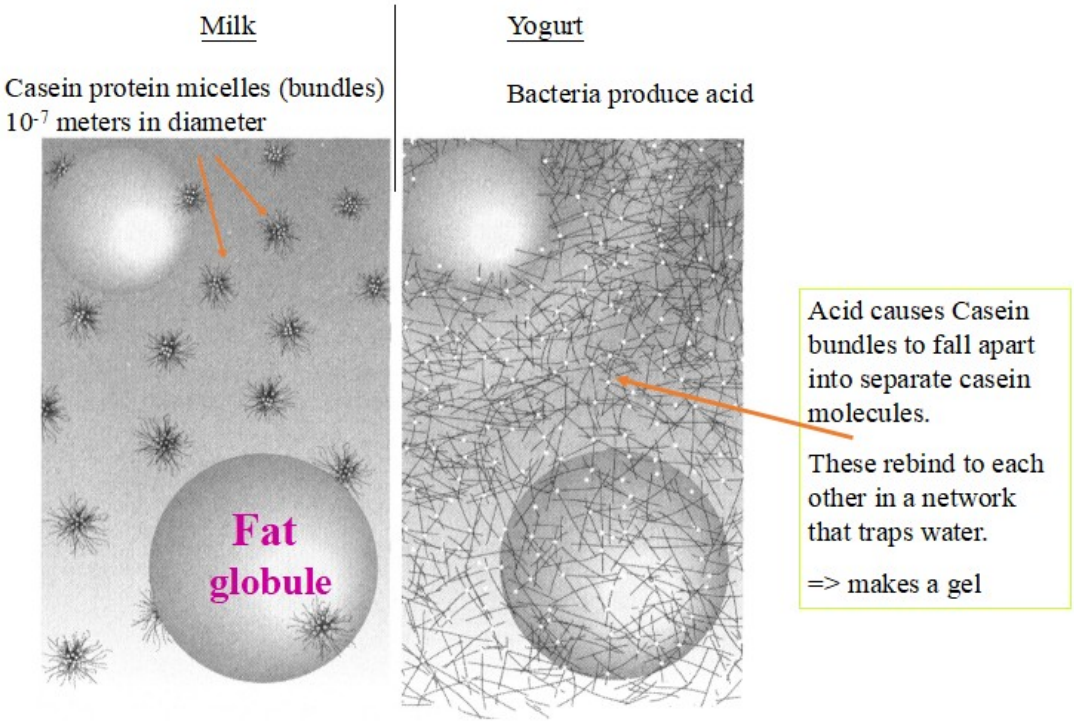
precipitates milk proteins, and bubbles of CO2 cause the holes characteristic of Swiss Cheese.

Commercial Yogurt

Contains 2 species of bacteria specialized to grow well in milk (but can't survive inside the human body):

These bacteria work in symbiosis. Each bacterium stimulates the growth of the other => acidifies the milk more rapidly than either partner on its own.





Use of Fermentation In agriculture

In agriculture, plant byproducts such as corn stalks are preserved for use as animal feed by packing them into a large container (a silo) with limited access to air; microbial fermentation produces acids that lower the pH.

The silage that results from this fermentation process can be kept as animal feed for long time without spoilage.

TOPIC#30

Industrial Fermentation-steps of fermentation.

Fermentation Products

There are five major groups of commercially important fermentations:

1) Those that produce microbial cells (or biomass) as the product.
2) Those that produce microbial metabolites.
3) Those that produce microbial enzymes.
4) Those that produce recombinant products.
5) Those that modify a compound which is added to the fermentation – the transformation process.

Fermentation Products

Biomass Production

- Baker's Yeast
- SCP (Single Cell Protein)
- Probiotics
- Bio fertilizers
- Vaccines
- Bioweapons

1. Upstream processing

Upstream processing includes formulation of the fermentation medium, sterilisation of air, fermentation medium and the fermenter, inoculum preparation and inoculation of the medium. The fermentation medium should contain an energy source, a carbon source, a nitrogen source and micronutrients required for the growth of the microorganism along with water and oxygen, if necessary.
A medium which is used for a large scale fermentation, in order to ensure the sustainability of the operation, should have the following characteristics;
1. It should be cheap and easily available
2. It should maximise the growth of the microorganism, productivity and the rate of formation of the desired product
3. It should minimise the formation of undesired products

Usually, waste products from other industrial processes, such as molasses, cheese whey and corn steep liquor, after modifying with the incorporation of additional nutrients, are used as the substrate for many industrial fermentation

Sterilisation is essential for preventing the contamination with any undesired microorganisms. Air is sterilised by membrane filtration while the medium is usually heat sterilised. Any nutrient component which is heat labile is filter-sterilised and later added to the sterilised medium. The fermenter may be sterilised together with the medium or separately.

Inoculum build up is the preparation of the seed culture in amounts sufficient to be used in the large fermenter vessel. This involves growing the microorganisms obtained from the pure stock culture in several consecutive fermenters. This process cuts down the time required for the growth of microorganisms in the fermenter, thereby increasing the rate of productivity. Then the seed culture obtained through this process is used to inoculate the fermentation medium.

2. The fermentation process

The fermentation process involves the propagation of the microorganism and production of the desired product. The fermentation process can be categorised depending on various parameters.

It can be either aerobic fermentation, carried out in the presence of oxygen or anaerobic fermentation, carried out in the absence of oxygen. Many industrial fermentation are carried out under aerobic conditions where a few processes such as ethanol production by yeast require strictly anaerobic environments.

3. Downstream Processing

Downstream Processing includes the recovery of the products in a pure state and the effluent treatment. Product recovery is carried out through a series of operations including cell separation by settling, centrifugation or filtration; product recovery by disruption of cells (if the product is produced intracellularly); extraction and purification of the product. Finally, the effluents are treated by chemical, physical or biological methods.

TOPIC#31

Industrial Fermentation-cont-5

What is fermentation technique?

Techniques for large-scale production of microbial products. It must both provide an optimum environment for the microbial synthesis of the desired product and be economically feasible on a large scale.

What is fermentation technique?

Fermentation technique can be divided into following on the basis of moisture requirement

- surface (emersion) fermentation
- submersion fermentation technique

The latter may be run in batch, fed batch, continuous reactors

1. Surface techniques:

In the surface techniques, the microorganisms are cultivated on the surface of a liquid or solid substrate.

These techniques are very complicated and used in industry for producing bread, cocoa



2. Submersion processes

In the submersion processes, the microorganisms grow in a liquid medium provided with the substrate

Except in traditional beer and wine fermentation, the medium is held in fermenters and stirred to obtain a homogeneous distribution of cells and medium.

Most processes are aerobic, and for these the medium must be vigorously aerated.

Used for pickling vegetables and producing soy sauce.

All important industrial processes (production of biomass and protein, antibiotics, enzymes are carried out by submersion processes.



Fermenter/Bioreactors

The heart of the fermentation process is the fermenter.

A biorector is a device in which the organisms are cultivated and motivated to form a desired product

Closed vessel or containment designed to give a right environment for optimal growth and metabolic activity of the organism

Fermenter: for microbes/ Bioreactor: for eukaryotic cells

Size variable ranging from 20-250 million litres or Large scale production (..to1000-million L capacity)

Design of a Fermenter

The success of a fermentation process is highly dependent on environmental factors

Factors to consider when designing a fermenter

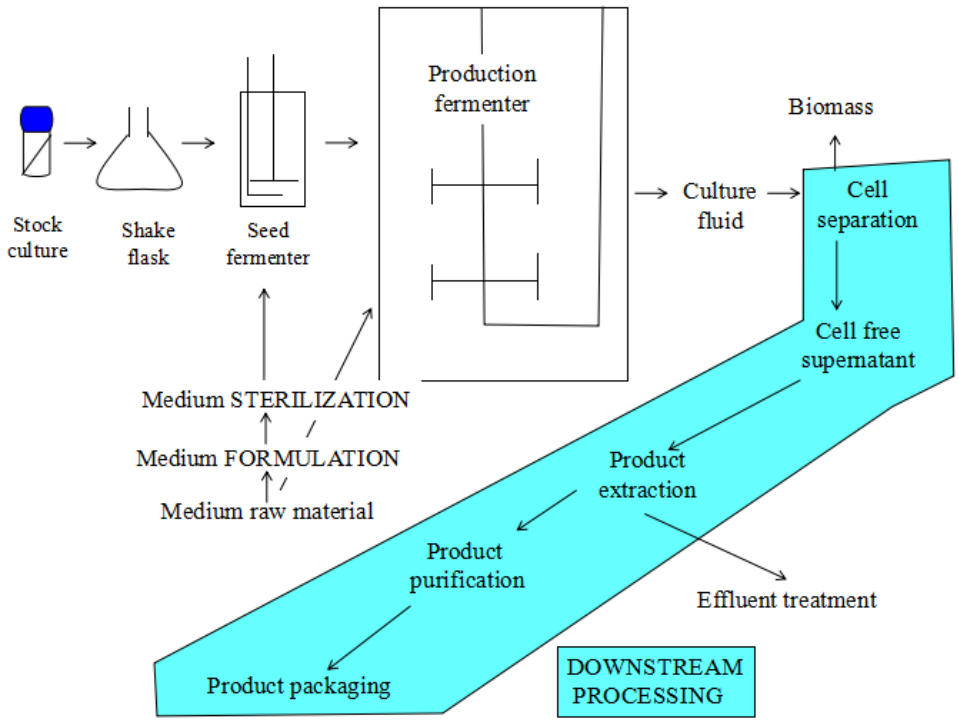
- Aseptic and long-term reliability
- Adequate for aeration and agitation
- Low power consumption
- Temperature and pH controls
- Sampling facilities
- drain or overflow
- control systems
- sensors
- cooling to achieve maximum microbial yield

(14 L fermenter shown is a copyright of New Brunswick Scientific)

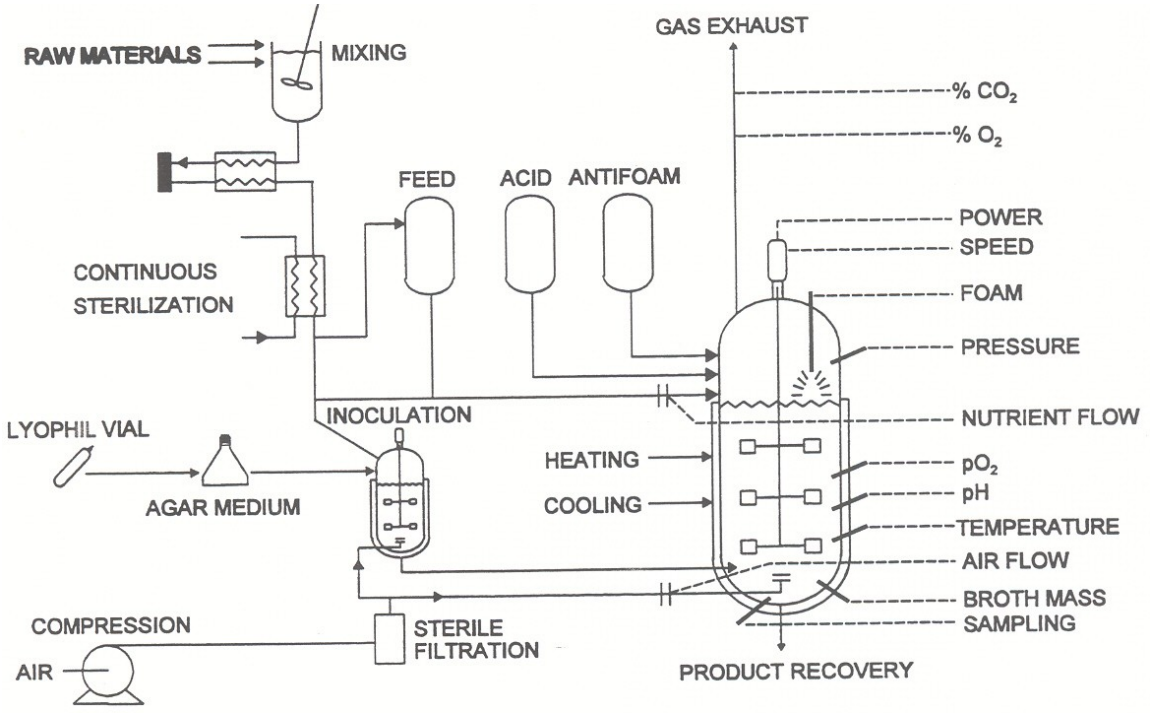


Types of Bioreactors

- Simple fermenters (batch and continuous)
- Fed batch fermenter
- Air-lift or bubble fermenter
- Cyclone column fermenter
- Tower fermenter
- Packed bed bioreactor
- photobioreactor



Flow sheet of a multipurpose fermenter and its auxiliary equipment



Topic#32

Batch fermentation

Basic modes for operating a fermenter

Types of fermentation on the basis of Process

Batch fermentation

Closed system

Batch fermentation refers to “A technique used to grow microorganisms or cells.

A limited supply of Sterile nutrients (substrate) for growth is inoculated (starter culture) when these are used up, or some other factor becomes limiting, the culture declines.

The time for which fermentation occurs is called ‘fermentation time’ or ‘batch time’

The only material added and removed during the course of a batch fermentation is oxygen (in the form of air, if the process is aerobic) and pH control solutions.

The culture is allowed to grow until no more of the product is being made

Cells, or products that the organisms have made, are "harvested" and fermenter is cleaned out for another run.

The principal disadvantage of batch processing is the high proportion of unproductive time (down-time) between batches, comprising the charge and discharge of the fermenter vessel, the cleaning, sterilization and re-start process

We take the example of Aspergillus niger and Lactobacillus are the microbes used to commercially produce citric acid and lactic acid, respectively.

The production takes place in a batch fermenter.

Rules of fermentation are applied on batch fermentation

- Microbial Growth Kinetics

- Media for Industrial Fermentations
- Sterilization
- The Development of Inocula for Industrial Fermentations
- Design of a Fermenter
- Instrumentation and Control
- Aeration and Agitation

Phases of Batch fermentation

- lag phase (adapt to their surroundings)
- exponential growth (grow in numbers)
- stationary phase (stop growing)
- death phase

Microbial Growth Kinetics

Lag Phase

This is the first phase in the fermentation process

The cells have just been injected into a new environment and they need time to adjust accordingly.Cell growth is minimal in this phase.

Exponential Phase

The second phase in the fermentation process

The cells have adjusted to their environment and rapid growth takes place

Cell growth rate is highest in this phase

At some point the cell growth rate will level off and become constant

The most likely cause of this leveling off is substrate limited inhibition

Substrate limited inhibition means that the microbes do not have enough nutrients in the medium to continue multiplying.

Stationary phase

This is the third phase in the fermentation process

The cell growth rate has leveled off and become constant

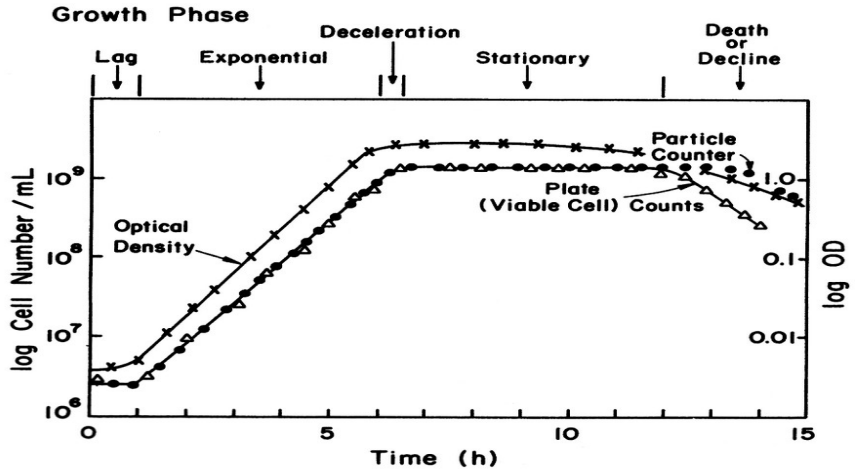
The number of cells multiplying equals the number of cells dying

Death phase

The fourth phase in the fermentation process

The number of cells dying is greater than the number of cells multiplying

The cause of the death phase is usually that the cells have consumed most of the nutrients in the medium and there is not enough left for sustainability



TOPIC#33

Continuous Fermentation-6

2. Continuous Fermentation

A Continuous fermentation is a process in which fresh medium (substrate) is continuously added in the bioreactor , and biomass or products containing left over nutrients and microorganisms are continuously removed at the same rate to keep the culture volume constant.

Under these conditions the cells remain in the logarithmic phase of growth

3. Fed-batch Fermentation

The fed-batch technique was originally devised by yeast producers in the early 1900s to regulate the growth in batch culture of Saccharomyces Yeast producers observed that in the presence of high concentrations of malt, a by-product - ethanol - was produced, while in low concentrations of malt, the yeast growth was restricted. The problem was then solved by a controlled feeding regime, so that yeast growth remained substrate limited.

The concept was then extended to the production of other products, such as some enzymes, antibiotics, growth hormones, microbial cells, vitamins, amino acids and other organic acids.

Fed-batch culture is, in the broad sense, defined as an operational technique in biotechnological processes where one or more nutrients (substrates) are fed (supplied) to the bioreactor during cultivation and in which the product(s) remain in the bioreactor until the end of the run.

An alternative description of the method is that of a culture in which "a base medium supports initial cell culture and a feed medium is added to prevent nutrient depletion".

It is also a type of semi-batch culture. In some cases, all the nutrients are fed into the bioreactor.

Advantage of the fed-batch culture is that one can control concentration of fed-substrate in the culture liquid at arbitrarily desired levels (in many cases, at low levels).

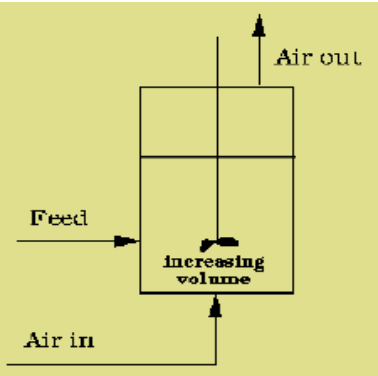
Generally speaking, fed-batch culture is superior to conventional batch culture when controlling concentrations of a nutrient (or nutrients) affect the yield or productivity of the desired metabolite.

Basically, cells are grown under a batch regime for some time, usually until close to the end of the exponential growth phase.

At this point, the reactor is fed with a solution of substrates, without the removal of culture fluid.

This feed should be balanced enough to keep the growth of the microorganisms at a desired specific growth rate and reducing simultaneously the production of by-products (that can be growth or product production inhibitory and make the system not as effective).

By products may lead to cell death



TOPIC#34

Gluconeogenesis

GLUCONEOGENESIS

The process of synthesis of glucose from non-carbohydrate sources is known as Gluconeogenesis (abbreviated GNG) ("new formation of sugar"), OR

Gluconeogenesis is the pathway, which converts pyruvate and related non-carbohydrate carbon compounds to glucose.

Why the body needs Gluconeogenesis?

For the human brain and nervous system, as well as the erythrocytes, testes, renal medulla, and embryonic tissues, glucose from the blood is the sole or major fuel source.

The brain alone requires about 120 g of glucose each day

more than half of all the glucose stored as glycogen in muscle and liver

between meals and longer fasts, after vigorous exercise, glycogen is depleted

For these times, organisms need a method for synthesizing glucose from noncarbohydrate precursors.

Site of Gluconeogenesis?

Organisms doing Gluconeogenesis: all animals, plants, fungi, and microorganisms

In many microorganisms, gluconeogenesis starts from simple organic compounds of two or three carbons, such as acetate, lactate, and propionate, in their growth medium.

Precursors of glucose in Gluconeogenesis: The important precursors of glucose in animals are three-carbon compounds such as lactate, pyruvate, and glycerol, as well as certain amino acids

Organs doing Gluconeogenesis:

mainly in the liver

To a lesser extent in renal cortex

in the epithelial cells that line the inside of the small intestine

In cytosol and mitochondria depending upon the substrate

Where does this glucose go?

The glucose produced passes into the blood to supply other tissues.

Nature: Anabolic: 4 ATP are required for synthesis of 1 glucose from 1 pyruvate or lactate

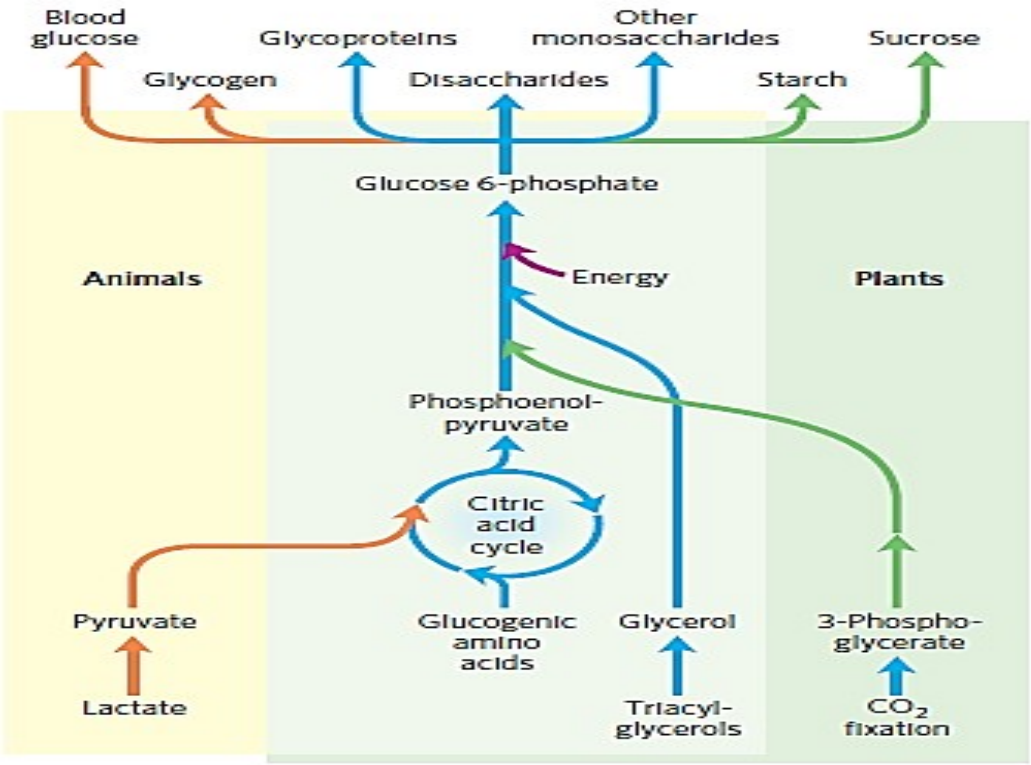
Pathways involved in Gluconeogenesis

- Reverse glycolysis (PEP to G6P)
- Tri carbocyclic acid cycle (TCA) or Kreb's cycle
- Cori cycle
- Glucose-alanine cycle

Precursors of glucose

Gluconeogenesis occurs in all animals, plants, fungi, and microorganisms.

The reactions are essentially the same in all tissues and all species.



TOPIC#35

Gluconeogenesis -cont

Biomedical importance of Gluconeogenesis

- To maintain the blood glucose concentration during prolonged fasting or starvation when sufficient carbohydrate is not available from the diet or glycogen reserves.
- It maintains the level of intermediates of the TCA cycle even when fatty acids are the main source of acetyl coA in the tissues.
- It clears the lactate produced by muscle and erythrocytes and glycerol produced by adipose tissue.

Cori cycle (the Lactic acid cycle)

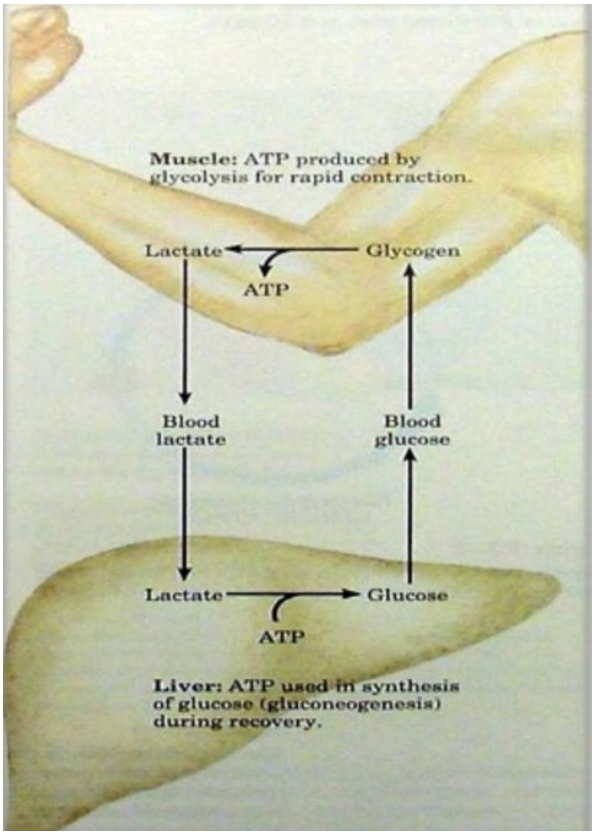
The pathway through which lactate produced by anaerobic glycolysis in skeletal muscle returns to the liver and is converted to glucose

This moves back to muscle and is converted to glycogen is called the Cori cycle

Site: Liver

Substrate: Lactate

"Occurs due to absence of glucose-6-phosphate in liver"



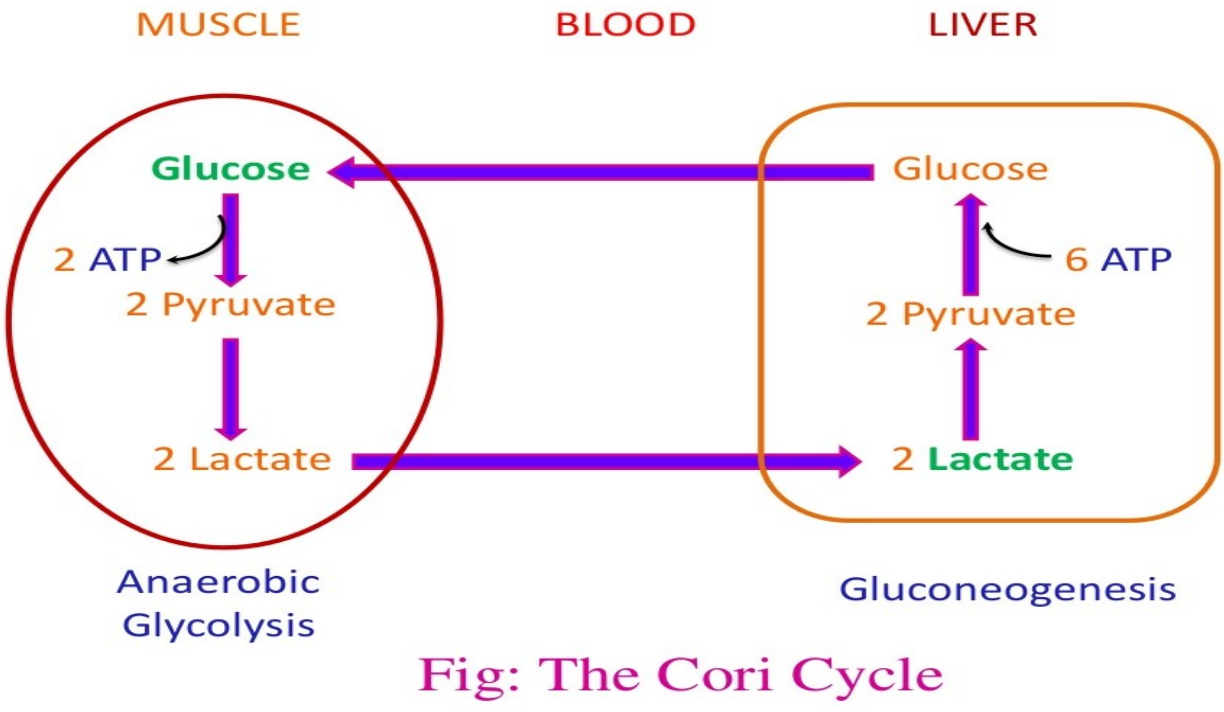


Fig: The Cori Cycle

Significance of Cori cycle

Prevents lactic acidosis in the muscle under anaerobic conditions

The cycle is also important in producing ATP, an energy source, during muscle activity

The Cori cycle is a much more important source of substrate for gluconeogenesis than food.

The drug METFORMIN can cause lactic acidosis in patients because metformin inhibits the hepatic gluconeogenesis of the Cori cycle

Glucose-Alanine cycle

1. When in extrahepatic tissues amino acids are used for energy, pyruvate, derived from GLYCOLYSIS, is used as amino group acceptor, forming alanine
2. Alanine diffuses into the bloodstream and reaches the liver
3. In the liver, the amino group of alanine is transferred to α-ketoglutarate to form pyruvate
4. The amino group enters the urea cycle, and in part acts as a nitrogen donor in many biosynthetic pathways.
5. Pyruvate enters the GLUCONEOGENESIS and is used for glucose synthesis
6. The newly formed glucose diffuses into the bloodstream and reaches the peripheral tissues where, due to GLYCOLYSIS, is converted into pyruvate that can accept amino groups from the free amino acids, thus closing the cycle.
7. consists of a series of steps through which extrahepatic tissues, for example the skeletal muscle, export pyruvate and amino groups as alanine to the liver, and receive glucose from the liver via the bloodstream.
   The main steps of the glucose-alanine cycle are summarized below.

When in extrahepatic tissues amino acids are used for energy, pyruvate, derived from the glycolytic pathway, is used as amino group acceptor, forming alanine, a nonessential amino acid.

Alanine diffuses into the bloodstream and reaches the liver.

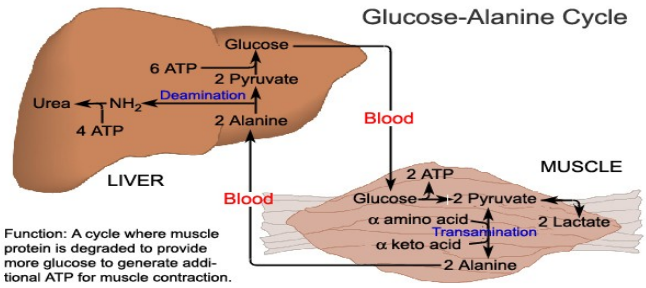
In the liver, the amino group of alanine is transferred to α-ketoglutarate to form pyruvate and glutamate, respectively.

The amino group of glutamate mostly enters the urea cycle, and in part acts as a nitrogen donor in many biosynthetic pathways.
Pyruvate enters the gluconeogenesis pathway and is used for glucose synthesis.

The newly formed glucose diffuses into the bloodstream and reaches the peripheral tissues where, due to glycolysis, is converted into pyruvate that can accept amino groups from the free amino acids, thus closing the cycle.

Therefore, the glucose-alanine cycle provides a link between carbohydrate and amino acid metabolism, as schematically described below.

Glucose → Pyruvate → Alanine → Pyruvate → Glucose



Glucose → Pyruvate → Alanine → Pyruvate → Glucose

<u>Significance of Glucose-Alanine cycle</u>

1. The Alanine cycle is less productive than the Cori cycle, which uses lactate, since a byproduct of energy production from alanine is production of Urea
2. Removal of the urea is energy-dependent, requiring four "high-energy" phosphate bonds
3. This pathway requires the presence of Alanine aminotransferase, restricted to mucle, liver and intestine
4. Therefore, this pathway is used instead of the Cori cycle only when an aminotransferase is present, or a need to transfer ammonia to the liver
5. The alanine cycle also Recycles carbon skeletons between muscle and liver
6. Transports ammonium to the liver and is converted into urea.

<u>Gluconeogenesis and glycolysis</u>

Gluconeogenesis and glycolysis are not identical pathways running in opposite directions, although they do share several steps

7 of the 10 enzymatic reactions of gluconeogenesis are the reverse

of glycolytic reactions

Three reactions of glycolysis are essentially irreversible in vivo and cannot be used in gluconeogenesis:

1) the conversion of glucose to glucose 6-phosphate by hexokinase,
2) the phosphorylation of fructose 6-phosphate to fructose 1,6-bisphosphate by phosphofructokinase-1
3) the conversion of phosphoenolpyruvate to pyruvate by pyruvate kinase

Glycolysis VS Gluconeogenesis

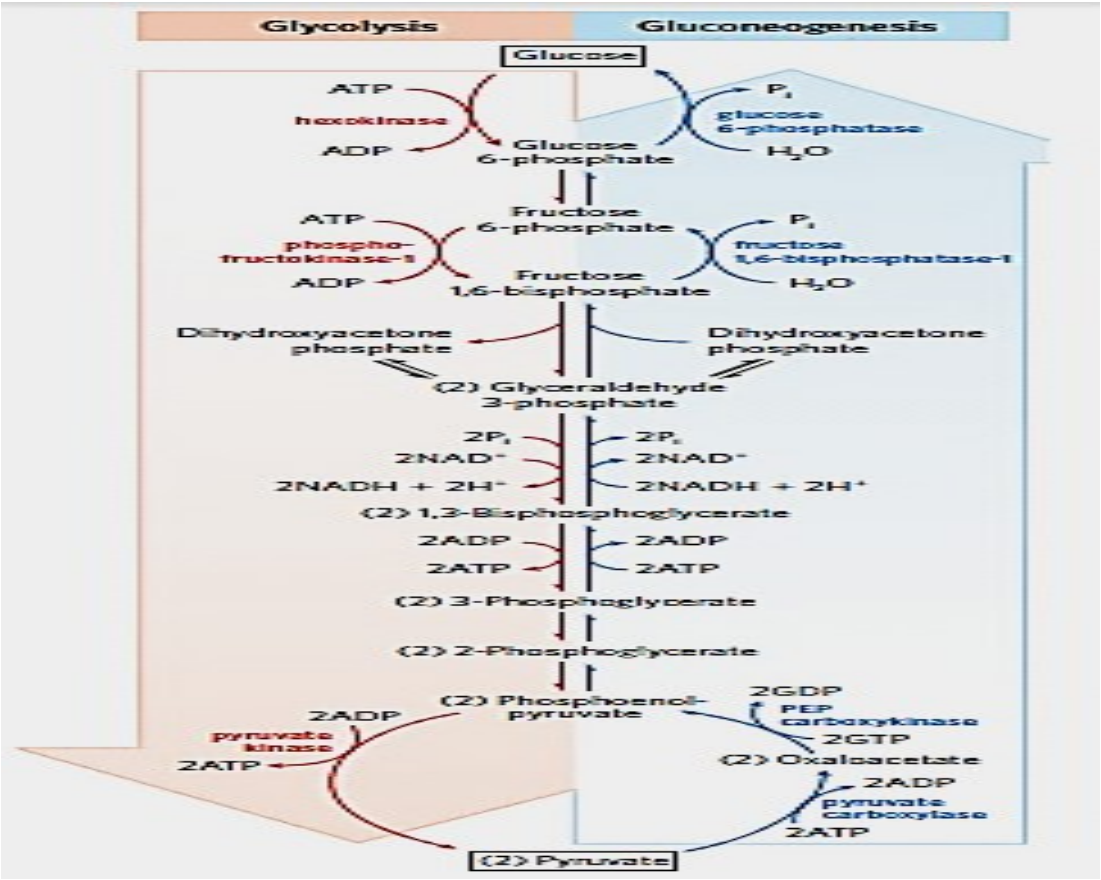


FIGURE 14–17 (Lehninger Ed. 6)

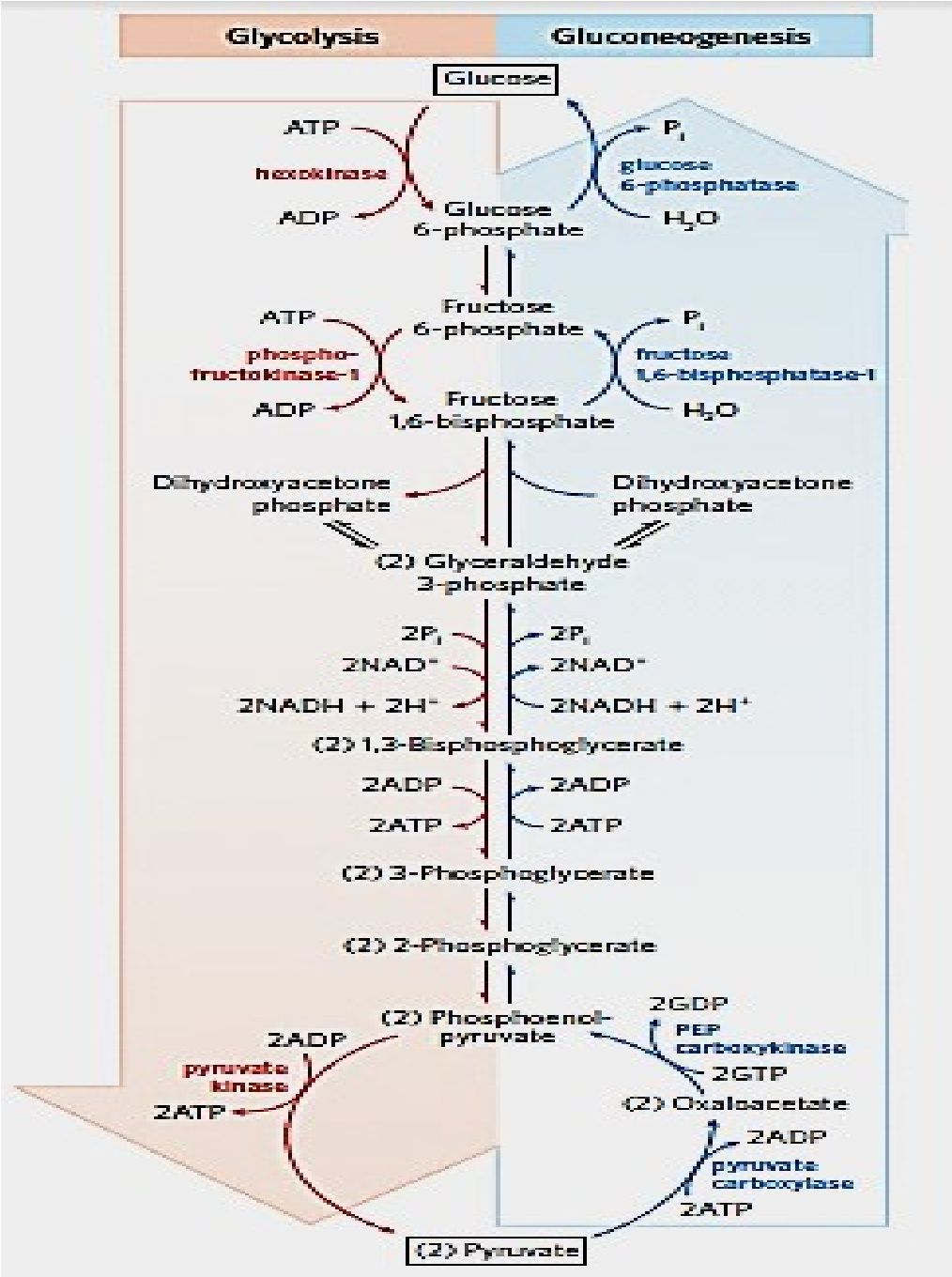
The reactions of glycolysis are on the left side, in red; the opposing pathway of gluconeogenesis is on the right, in blue.

Topic#36

Gluconeogenesis - 1st by pass pathway-complete

Bypass reactions in Gluconeogenesis

Glycolysis VS Gluconeogenesis



The reactions of glycolysis are on the left side, in red; the opposing pathway of gluconeogenesis is on the right, in blue.

Both are energy generating pathways-NOT identical.At one time only one of 2 can take place within a cell

Glycolysis takes place essentially in the cytoplasm.Gluconeogenesis occurs mutually in the cytosol and mitochondria.Requirement of gluconeogenesis-Glycogen

Gluconeogenesis from pyruvate share 7 reversible steps of glycolysis.Seven out of the 10 enzymes used in the glycolytic pathway are used..

The 3 irreversible steps of glycolysis that proceed with a large negative free energy change are bypassed during gluconeogenesis by using different enzymes.

These three bypass reactions (during glycolysis) are catalyzed by:

- Pyruvate kinase
- Phosphofructokinase-1 (PFK-1)
- Hexokinase/glucokinase

TABLE 14–2 Free-Energy Changes of Glycolytic Reactions in Erythrocytes

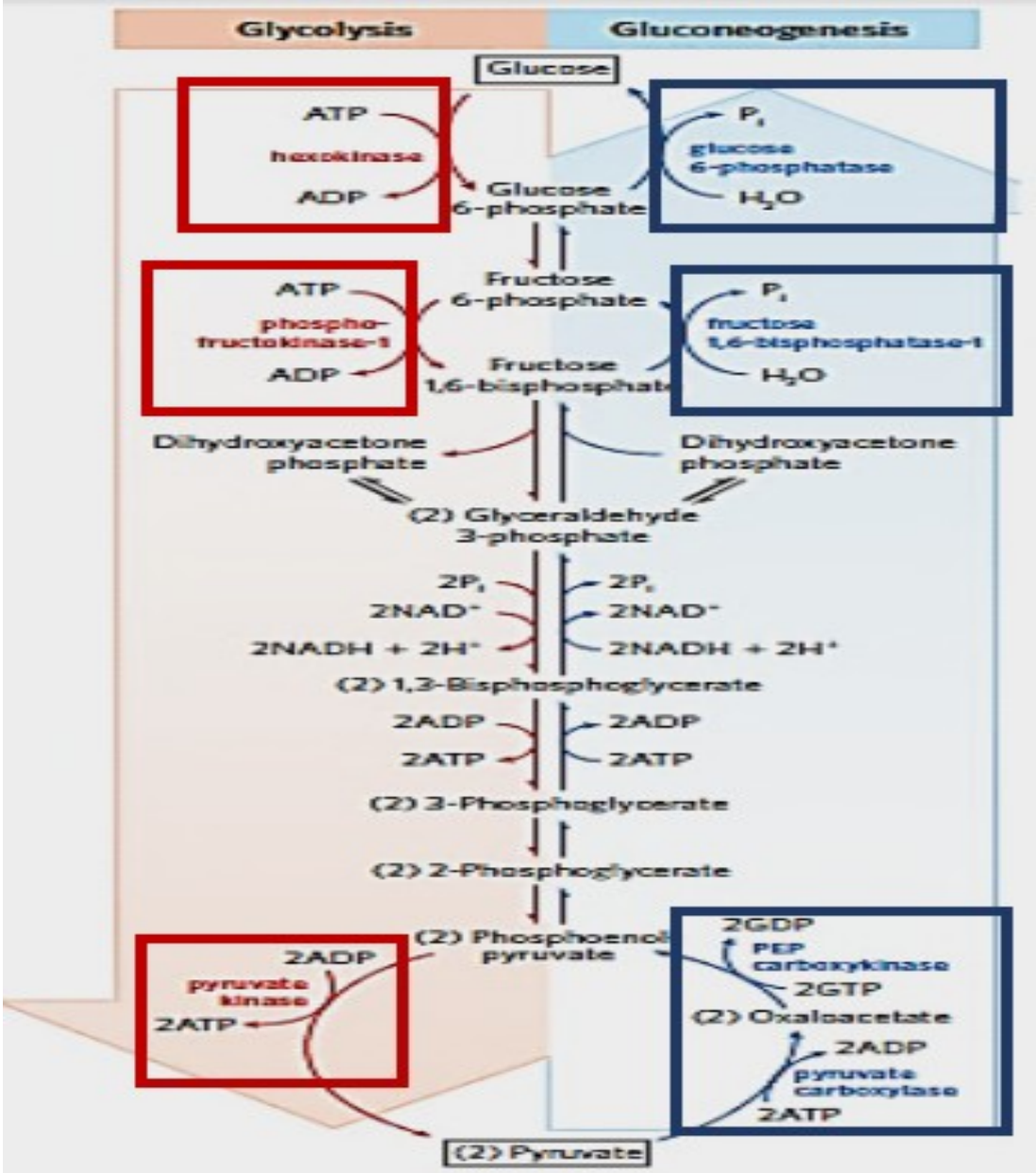
Glycolytic reaction step	$\Delta G'^{\circ}$ (kJ/mol)	ΔG (kJ/mol)
1 Glucose + ATP ⟶ glucose 6-phosphate + ADP	−16.7	−33.4
2 Glucose 6-phosphate ⇌ fructose 6-phosphate	1.7	0 to 25
3 Fructose 6-phosphate + ATP ⟶ fructose 1,6-bisphosphate + ADP	−14.2	−22.2
4 Fructose 1,6-bisphosphate ⇌ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate	23.8	−6 to 0
5 Dihydroxyacetone phosphate ⇌ glyceraldehyde 3-phosphate	7.5	0 to 4
6 Glyceraldehyde 3-phosphate + P_i + NAD^+ ⇌ 1,3-bisphosphoglycerate + NADH + H^+	6.3	−2 to 2
7 1,3-Bisphosphoglycerate + ADP ⇌ 3-phosphoglycerate + ATP	−18.8	0 to 2
8 3-Phosphoglycerate ⇌ 2-phosphoglycerate	4.4	0 to 0.8
9 2-Phosphoglycerate ⇌ phosphoenolpyruvate + H_2O	7.5	0 to 3.3
10 Phosphoenolpyruvate + ADP ⟶ pyruvate + ATP	−31.4	−16.7

Note: $\Delta G'^{\circ}$ is the standard free-energy change, as defined in Chapter 13 (pp. 507–508). ΔG is the free-energy change calculated from the actual concentrations of glycolytic intermediates present under physiological conditions in erythrocytes, at pH 7. The glycolytic reactions bypassed in gluconeogenesis are shown in red. Biochemical equations are not necessarily balanced for H or charge (p. 517).

Table 14-2

Lehninger Principles of Biochemistry, Sixth Edition

© 2013 W. H. Freeman and Company



1. Conversion of Pyruvate to Phosphoenolpyruvate

The first of the bypass reactions in gluconeogenesis is the conversion of pyruvate to phosphoenolpyruvate (PEP).

This reaction cannot occur by simple reversal of the pyruvate kinase reaction

Instead, the phosphorylation of pyruvate is achieved by a sequence of reactions that in eukaryotes requires 2 enzymes in both the cytosol and mitochondria.

Pyruvate is first transported from the cytosol into mitochondria or is generated from alanine within mitochondria by transamination, in which the alpha -amino group is transferred from alanine (leaving pyruvate) to an alpha-keto carboxylic acid

In mitochondria, pyruvate is converted to oxaloacetate in a biotin (coenzyme)-requiring reaction catalyzed by pyruvate carboxylase, a mitochondrial enzyme.

$$\text{Pyruvate} + HCO_3^- + \text{ATP} \longrightarrow \text{oxaloacetate} + \text{ADP} + P_i$$

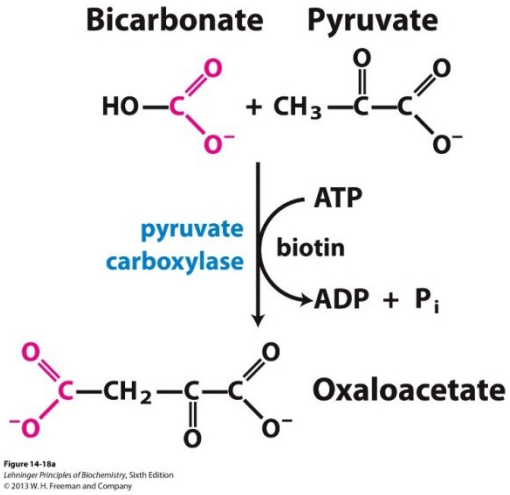


Figure 14-18a
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Pyruvate carboxylase is the first regulatory enzyme in the gluconeogenic pathway.

Pyruvate carboxylase requires acetyl-CoA as a positive effector. (Acetyl-CoA is produced by fatty acid oxidation)

the mitochondrial membrane has no transporter for oxaloacetate, before export to the cytosol the oxaloacetate formed from pyruvate must be reduced to malate by mitochondrial malate dehydrogenase, at the expense of NADH

<u>Alternative paths from pyruvate to phosphoenolpyruvate</u>

Oxalo acetic acid (OAA) produced in mitochondria is impermeable through mitochondrial membrane

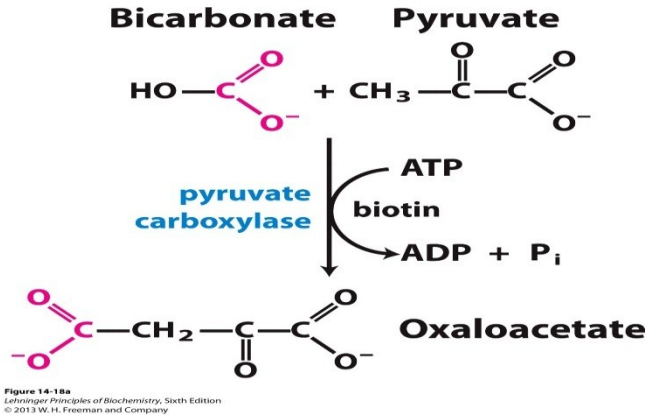
$$\text{Oxaloacetate} + \text{NADH} + H^+ \longleftrightarrow \text{L-malate} + NAD^+$$

Therefore, Oxalo acetic acid is converted into malate, exported from the mitochondrion into the cytoplasm, converted back to oxaloacetate inorder to allow gluconeogenesis

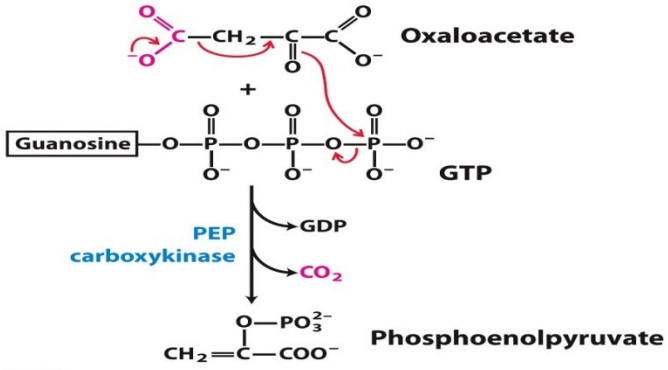
In both mitochondria and cytoplasm, the transporter of malate is malate dehydrogenase

In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP carboxykinase.

The CO2 incorporated in the pyruvate carboxylase reaction is lost here as CO2. The decarboxylation leads to a rearrangement of electrons that facilitates attack of the carbonyl oxygen of the pyruvate moiety on the phosphate of GTP



a)



b)

Synthesis of phosphoenolpyruvate from pyruvate.

(a) In mitochondria, pyruvate is converted to oxaloacetate in a biotin-requiring reaction catalyzed by pyruvate carboxylase.

(b) In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP carboxykinase.

The CO_2 incorporated in the pyruvate carboxylase reaction is lost here as CO_2.

The decarboxylation leads to a rearrangement of electrons that facilitates attack of the carbonyl oxygen of the pyruvate moiety on the phosphate of GTP.

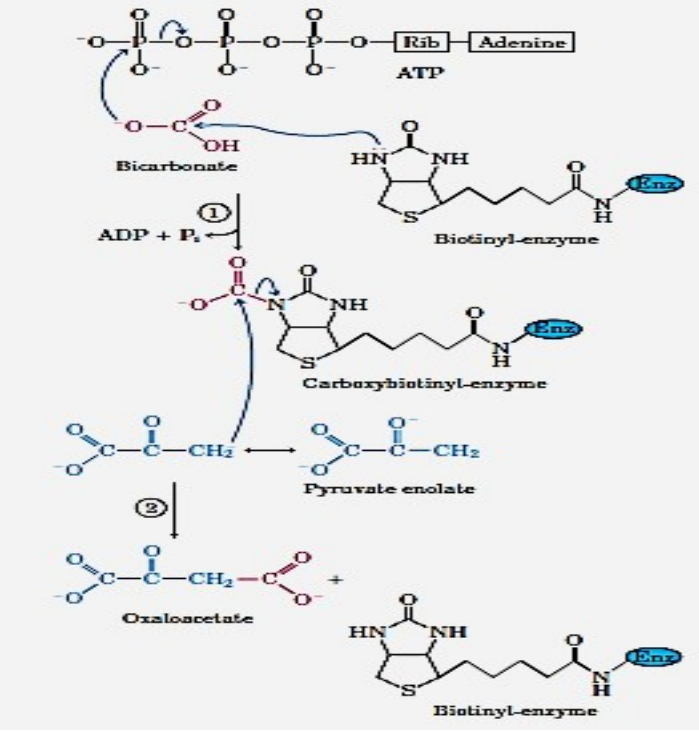
Role of biotin in the pyruvate carboxylase reaction.

The cofactor biotin is covalently attached to the enzyme through an amide linkage to the amino group of a Lys residue, forming a biotinylenzyme.

The reaction occurs in two phases, which occur at two different sites in the enzyme.

At catalytic site 1, bicarbonate ion is converted to CO_2 at the expense of ATP. Then CO_2 reacts with biotin, forming carboxybiotinyl- enzyme.

The long arm composed of biotin and the Lys side chain to which it is attached then carry the CO_2 of carboxybiotinyl enzyme to catalytic site 2 on the enzyme surface, where CO_2 is released and reacts with the pyruvate, forming oxaloacetate and regenerating the biotinyl-enzyme.



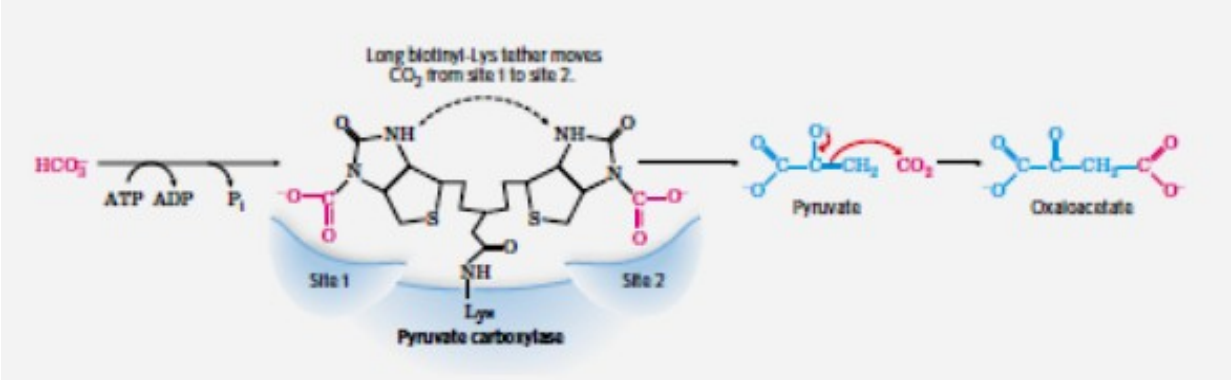
Role of biotin in the pyruvate carboxylase reaction.

The cofactor biotin is covalently attached to the enzyme through an amide linkage to the ´-amino group of a Lys residue, forming a biotinylenzyme.

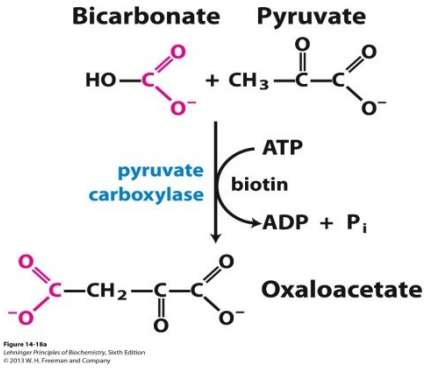
The reaction occurs in two phases, which occur at two different sites in the enzyme.

At catalytic site 1, bicarbonate ion is converted to CO2 at the expense of ATP. Then CO2 reacts with biotin, forming carboxybiotinyl- enzyme.

The long arm composed of biotin and the Lys side chain to which it is attached then carry the CO2 of carboxybiotinyl enzyme to catalytic site 2 on the enzyme surface, where CO2 is released and reacts with the pyruvate, forming oxaloacetate and regenerating the biotinyl-enzyme.



a.



b.

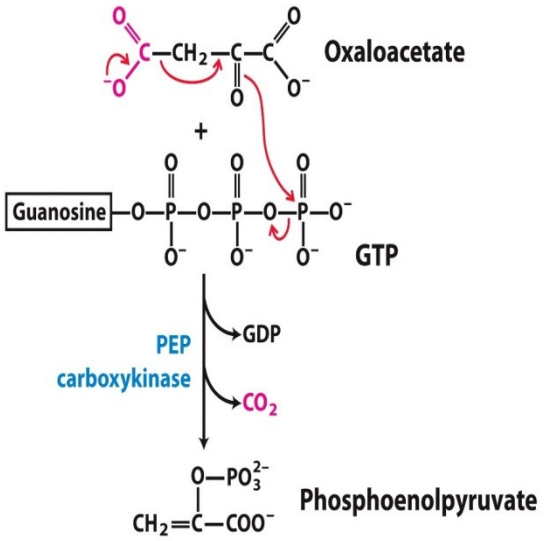
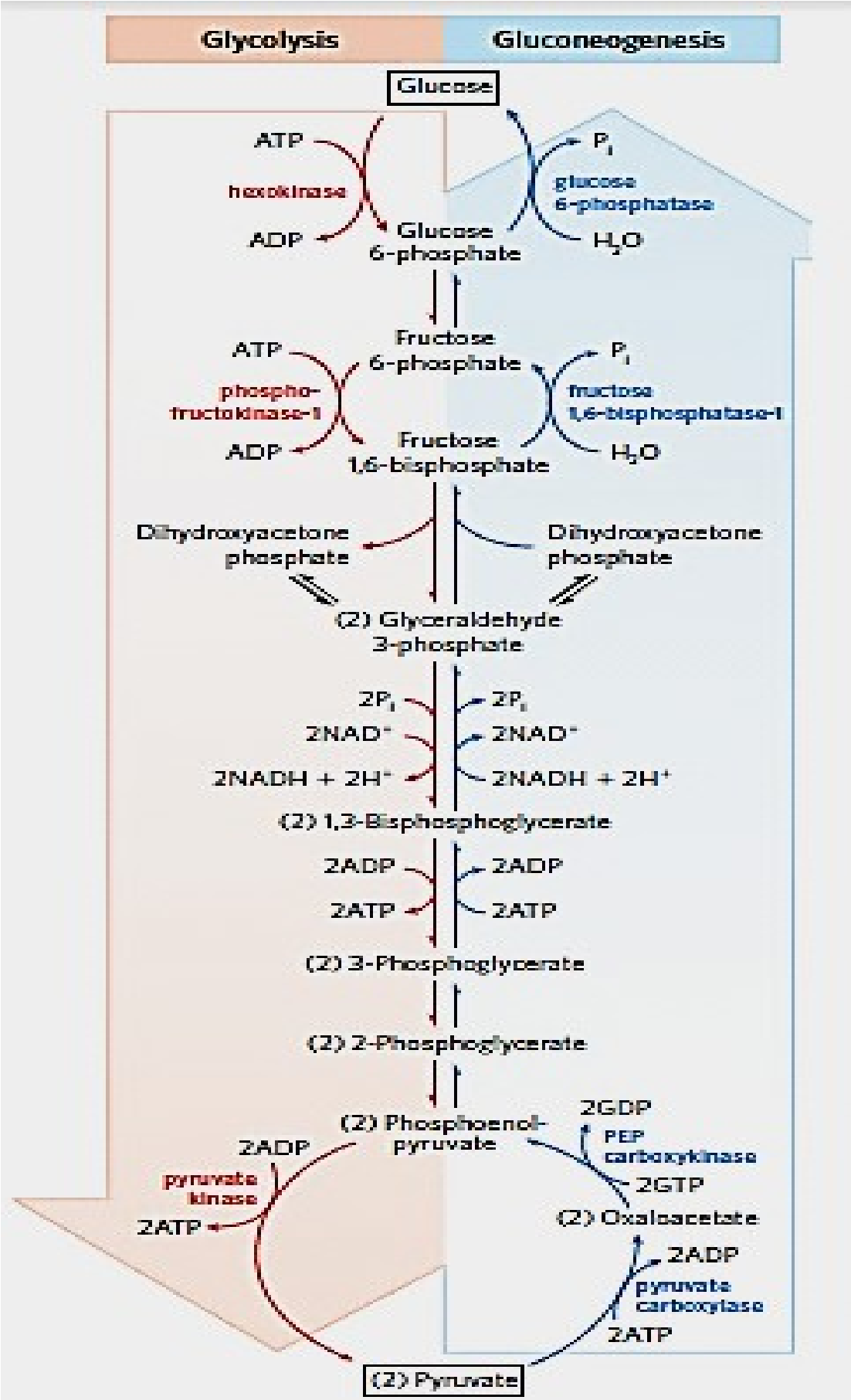


Figure 14-18b
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company


Glycolysis
Gluconeogenesis
Glucose
ATP
hexokinase
ADP
Glucose 6-phosphate
P_i
glucose 6-phosphatase
H_2O
Fructose 6-phosphate
ATP
phospho-fructokinase-1
ADP
Fructose 1,6-bisphosphate
P_i
fructose 1,6-bisphosphatase-1
H_2O
Dihydroxyacetone phosphate
Dihydroxyacetone phosphate
(2) Glyceraldehyde 3-phosphate
$2P_i$
2NAD+
2NADH + 2H+
$2P_i$
2NAD+
2NADH + 2H+
(2) 1,3-Bisphosphoglycerate
2ADP
2ATP
2ADP
2ATP
(2) 3-Phosphoglycerate
(2) 2-Phosphoglycerate
(2) Phosphoenol-pyruvate
2ADP
pyruvate kinase
2ATP
2GDP
PEP carboxykinase
2GTP
(2) Oxaloacetate
2ADP
pyruvate carboxylase
2ATP
(2) Pyruvate

There is a logic to the route of these reactions through the mitochondrion.

The [NADH]/[NAD+] ratio in the cytosol is 8 X 10-4, about 10 times lower than in mitochondria.

Because cytosolic NADH is consumed in gluconeogenesis (in the conversion of 1,3-bisphosphoglycerate to glyceraldehyde 3-phosphate

Glucose biosynthesis cannot proceed unless NADH is available.

The transport of malate from the mitochondrion to the cytosol and its reconversion there to oxaloacetate effectively moves NAD to the cytosol, and converts it into NADH.

This path from pyruvate to PEP therefore provides an important balance between NADH produced and consumed in the cytosol during gluconeogenesis.

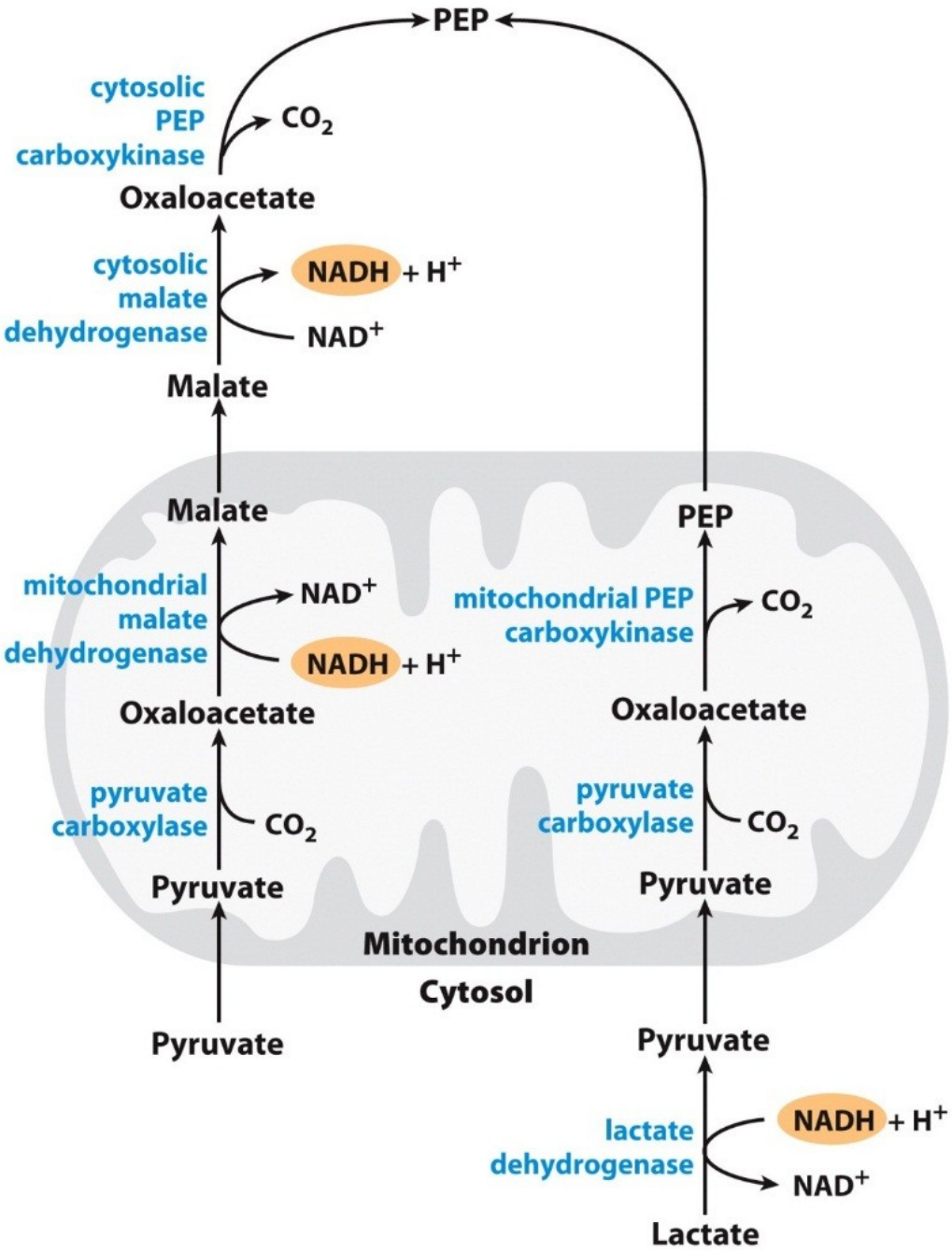


Figure 14-20
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Alternative paths from pyruvate to phosphoenolpyruvate

The relative importance of the two pathways depends on the availability of lactate or pyruvate and the cytosolic requirements for NADH for gluconeogenesis.

The path on the right predominates when lactate is the precursor, because cytosolic NADH is generated in the lactate dehydrogenase reaction and does not have to be shuttled out of the mitochondrion

The requirements of ATP for pyruvate carboxylase and GTP for PEP carboxykinase are omitted for simplicity.

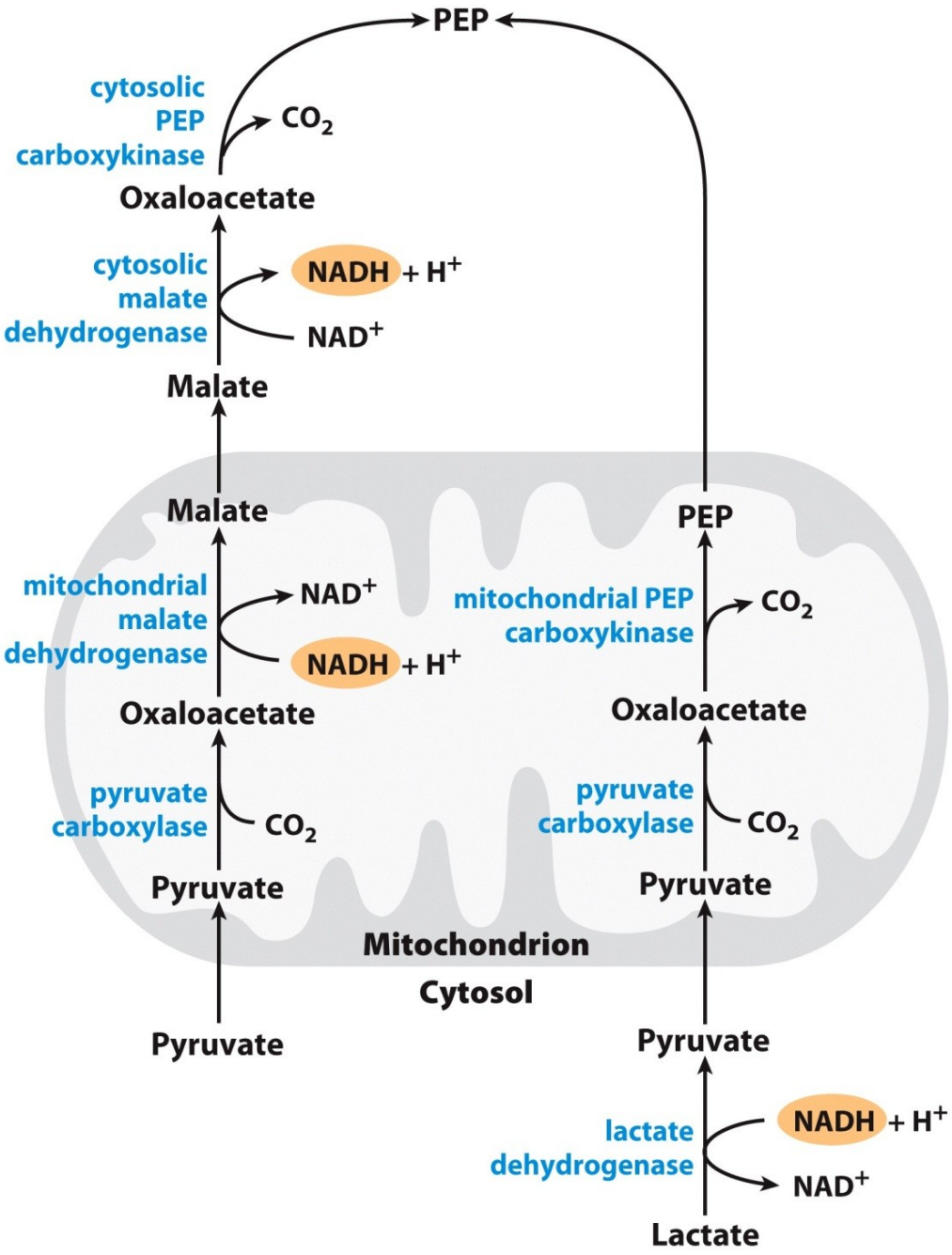


Figure 14-20
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Topic#37

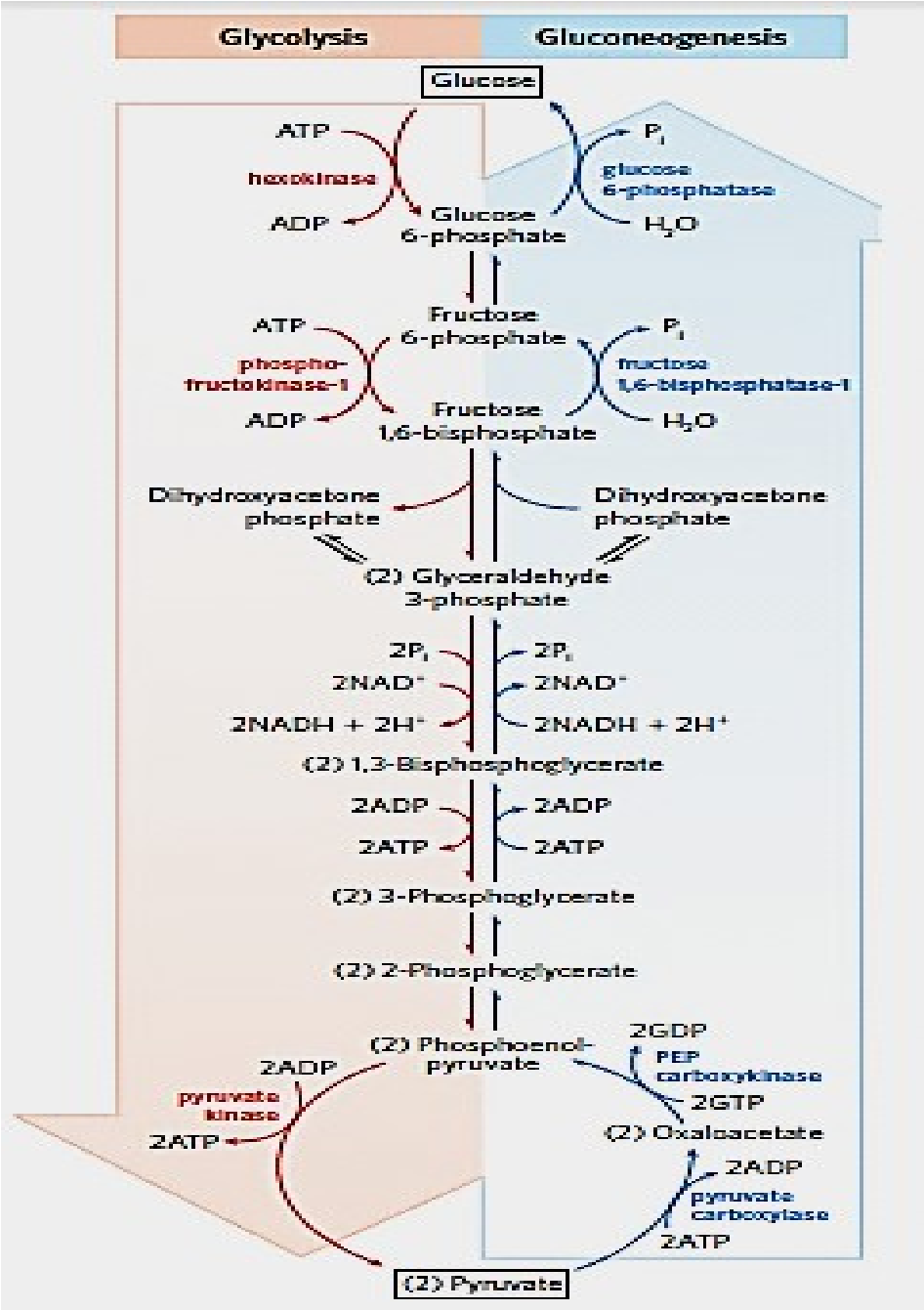
Gluconeogenesis - 2nd-3rd by pass pathway

Bypass reactions in Gluconeogenesis-2

Conversion of Fructose 1,6-Bisphosphate to Fructose 6-Phosphate Is the Second Bypass

The second glycolytic reaction that cannot participate in gluconeogenesis is the phosphorylation of fructose 6-phosphate by PFK-1

Because this reaction is highly exergonic and therefore irreversible, the generation of fructose 6-phosphate from fructose 1,6-bisphosphate is catalyzed by a different enzyme, fructose 1,6-bisphosphatase (FBPase-1)



fructose 1,6-bisphosphatase (FBPase-1) is Mg+2-dependent

FBPase-1 promotes the essentially irreversible hydrolysis of the C-1 phosphate (not phosphoryl group transfer to ADP.

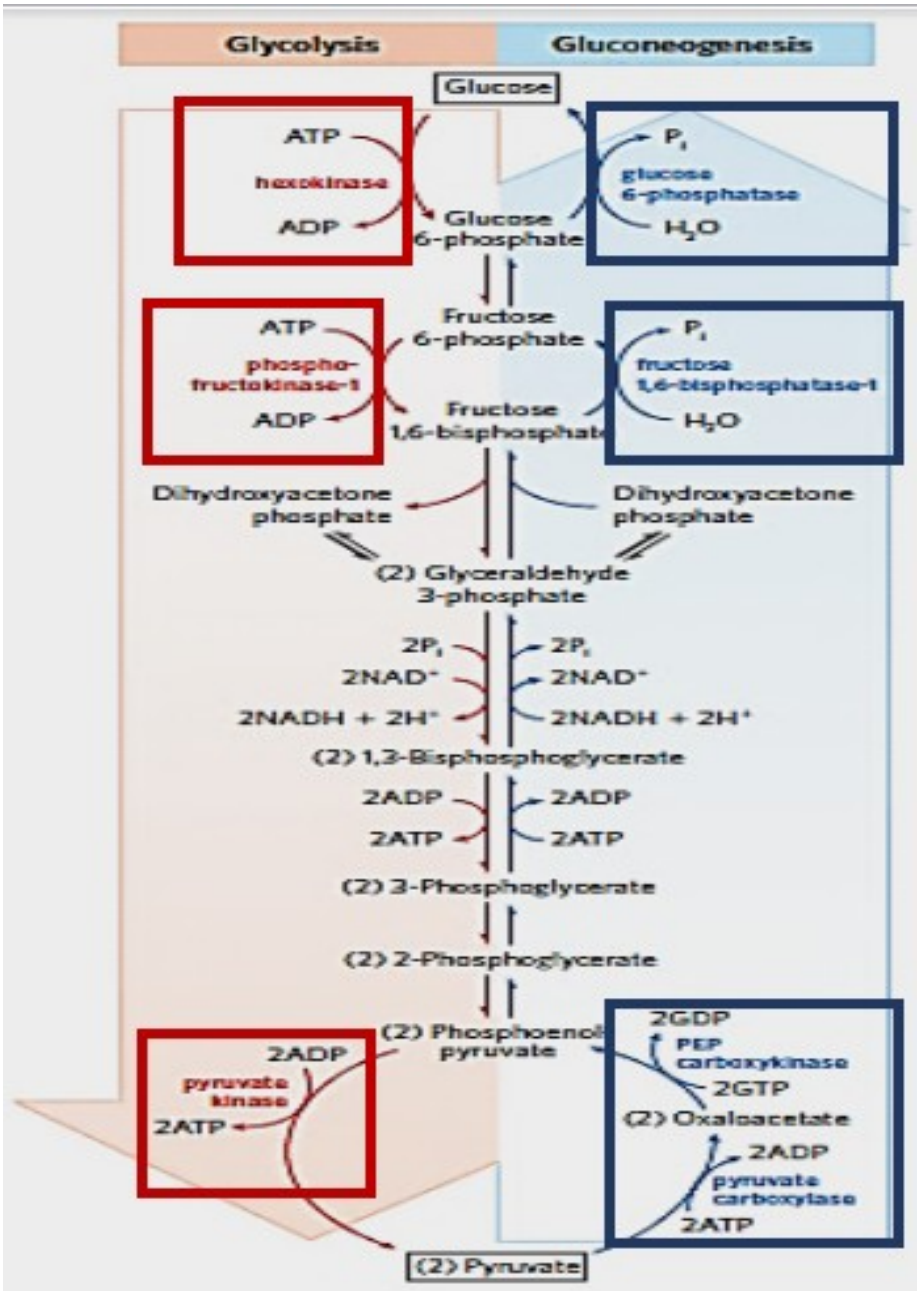
fructose 1,6-bisphosphate + H2O → fructose 6-phosphate + Pi

<u>Conversion of Glucose 6-Phosphate to Glucose Is the Third Bypass</u>

The third bypass is the final reaction of gluconeogenesis, the dephosphorylation of glucose 6-phosphate to yield glucose

Reversal of the hexokinase reaction would require phosphoryl group transfer from glucose 6-phosphate to ADP, forming ATP… An energetically unfavorable reaction

The reaction catalyzed by glucose 6-phosphatase does not require synthesis of ATP; it is a simple hydrolysis of a phosphate ester



Glucose 6-phosphatase + H20 → Glucose + Pi

This Mg+2-activated enzyme (glucose 6-phosphatase) is found on the luminal side of the endoplasmic reticulum of hepatocytes, renal cells, and epithelial cells of the small intestine but not in other tissues, therefore other organs are unable to supply glucose to the blood.

If other tissues had glucose 6-phosphatase, this enzyme's activity would hydrolyze the glucose 6-phosphate needed within those tissues for glycolysis.

Glucose produced by gluconeogenesis in the liver or kidney or ingested in the diet is delivered to these other tissues, including brain and muscle, through the bloodstream.

Topic#38

Gluconeogenesis - cont - 4

Gluconeogenesis Is Energetically Expensive, but Essential

The sum of the biosynthetic reactions leading from pyruvate to free blood glucose is as follows

$$2\ \text{Pyruvate} + 4\text{ATP} + 2\text{GTP} + 2\text{NADH} + 2\text{H}^+ + 4\text{H}_2\text{O} \longrightarrow \text{glucose} + 4\text{ADP} + 2\text{GDP} + 6\text{P}_i + 2\text{NAD}^+ \quad (14\text{–}9)$$

For each molecule of glucose formed from pyruvate, six high-energy phosphate groups are required, four from ATP and two from GTP.

In addition, two molecules of NADH are required for the reduction of two molecules of 1,3-bisphosphoglycerate.

Clearly, Glycolysis is not simply the reverse of the equation for conversion of glucose to pyruvate by glycolysis, which would require only two molecules of ATP

$$\text{Glucose} + 2\text{ADP} + 2\text{P}_i + \text{NAD}^+ \longrightarrow 2\ \text{pyruvate} + 2\text{ATP} + 2\text{NADH} + 2\text{H}^+ + 2\text{H}_2\text{O}$$

The synthesis of glucose from pyruvate is a relatively expensive process

this high energy cost is necessary to ensure the irreversibility of gluconeogenesis

Under intracellular conditions, the overall free-energy change of glycolysis is at least -63 kJ/mol.

Under the same conditions the overall DeltaG of gluconeogenesis is -16 kJ/mol.

Thus both glycolysis and gluconeogenesis are essentially irreversible processes in cells.

A second advantage to investing energy to convert pyruvate to glucose is that if pyruvate were instead excreted, its considerable potential for ATP production by complete, aerobic oxidation would be lost (more than 10 ATP are produced per pyruvate

TABLE 14-3 Sequential Reactions in Gluconeogenesis Starting from Pyruvate

Pyruvate + HCO_3^- + ATP ⟶ oxaloacetate + ADP + P_i	×2
Oxaloacetate + GTP ⇌ phosphoenolpyruvate + CO_2 + GDP	×2
Phosphoenolpyruvate + H_2O ⇌ 2-phosphoglycerate	×2
2-Phosphoglycerate ⇌ 3-phosphoglycerate	×2
3-Phosphoglycerate + ATP ⇌ 1,3-bisphosphoglycerate + ADP	×2
1,3-Bisphosphoglycerate + NADH + H^+ ⇌ glyceraldehyde 3-phosphate + NAD^+ + P_i	×2
Glyceraldehyde 3-phosphate ⇌ dihydroxyacetone phosphate	
Glyceraldehyde 3-phosphate + dihydroxyacetone phosphate ⇌ fructose 1,6-bisphosphate	
Fructose 1,6-bisphosphate ⟶ fructose 6-phosphate + P_i	
Fructose 6-phosphate ⇌ glucose 6-phosphate	
Glucose 6-phosphate + H_2O ⟶ glucose + P_i	
Sum: 2 Pyruvate + 4ATP + 2GTP + 2NADH + $2H^+$ + $4H_2O$ ⟶ glucose + 4ADP + 2GDP + $6P_i$ + $2NAD^+$	

The bypass reactions are in red; all other reactions are reversible steps of glycolysis. The figures at the right indicate that the reaction is to be counted twice, because two three-carbon precursors are required to make a molecule of glucose. The reactions required to replace the cytosolic NADH consumed in the glyceraldehyde 3-phosphate dehydrogenase reaction (the conversion of lactate to pyruvate in the cytosol or the transport of reducing equivalents from mitochondria to the cytosol in the form of malate) are not considered in this summary. Biochemical equations are not necessarily balanced for H and charge.

TOPIC#39

Gluconeogenesis - cont -5

Gluconeogenesis and Glycolysis are Reciprocally regulated

Reciprocal Regulation of Gluconeogenesis and Glycolysis in the Liver

Gluconeogenesis and glycolysis are coordinated so that within a cell one pathway is relatively inactive while the other is highly active.

The rate of glycolysis is determined by the concentration of glucose

The rate of gluconeogenesis by the concentrations of lactate and other precursors of glucose

Pyruvate kinase is inhibited by phosphorylation during starvation.

The level of fructose 2,6-bisphosphate is high in the fed state and low in starvation.

The interconversion of fructose 6-phosphate and fructose 1,6-bisphosphate is stringently controlled

AMP stimulates phosphofructokinase, whereas ATP and citrate inhibit it.

Fructose 1,6-bisphosphatase, is inhibited by AMP and activated by citrate.

A high level of AMP indicates that the energy charge is low and signals the need for ATP generation.

Conversely, high levels of ATP and citrate indicate that the energy charge is high and that biosynthetic intermediates are abundant.

Under these conditions, glycolysis is nearly switched off and gluconeogenesis is promoted.

Phosphofructokinase and fructose 1,6-bisphosphatase are also reciprocally controlled by fructose 2,6-bisphosphate in the liver

The level of F-2,6-BP is low during starvation and high in the fed state, because of the antagonistic effects of glucagon and insulin on the production and degradation of this signal molecule.

Fructose 2,6-bisphosphate strongly stimulates phosphofructokinase and inhibits fructose 1,6-bisphosphatase.

Hence, glycolysis is accelerated and gluconeogenesis is diminished in the fed state.

During starvation, gluconeogenesis predominates because the level of F-2,6-BP is very low. Glucose formed by the liver under these conditions is essential for the viability of brain and muscle.

The interconversion of phosphoenolpyruvate and pyruvate also is precisely regulated.

High levels of ATP and alanine, which signal that the energy charge is high and that building blocks are abundant, inhibit the enzyme in liver.

Conversely, pyruvate carboxylase, which catalyzes the first step in gluconeogenesis from pyruvate, is activated by Acetyl CoA and inhibited by ADP.

Likewise, ADP inhibits phosphoenolpyruvate carboxykinase.

Hence, gluconeogenesis is favored when the cell is rich in biosynthetic precursors and ATP

Regulatory effect of hormones

The regulators in this case are hormones.

Hormones affect gene expression primarily by changing the rate of transcription, as well as by regulating the degradation of mRNA

Insulin, which rises subsequent to eating, stimulates the expression of phosphofructokinase, pyruvate kinase, and the bifunctional enzyme that makes and degrades F-2,6-BP.

Glucagon, which rises during starvation, inhibits the expression of these enzymes and stimulates instead the production of two key gluconeogenic enzymes, phosphoenolpyruvate carboxykinase and fructose 1,6-bisphosphatase.

TOPIC#40

Pentose Phosphate Pathway - PPP -1

Pentose Phosphate Pathway (PPP) of Glucose Oxidation

The oxidative pentose phosphate pathway

- Phosphogluconate pathway
- Hexose monophosphate pathway

The major catabolic fate of glucose 6-phosphate is glycolytic breakdown to pyruvate, much of which is then oxidized via the citric acid cycle, leading to ATP formation .

Glucose 6-phosphate does have other catabolic fates, which lead to specialized products needed by the cell.

E.g. the oxidation of glucose 6-phosphate to pentose phosphates by the pentose phosphate pathway.

Since it results in the synthesis of pentoses and NADPH, it is considered as an anaerobic and anabolic pathway. Occurs in cytosol.

Outputs of PPP

Reducing NADP+ to NADPH and producing pentose phosphates. .

brings about oxidation and decarboxylation of glucose 6-phosphate (G6P) at C-1, to form pentoses.. (bone marrow, skin, intestinal mucosa) to make RNA, DNA, ATP, NADH, FADH2 and coenzyme A

NADPH: to counter damaging effects of oxygen radicals (liver, adipose, erythrocytes, cornea)

Erythrose 4 phosphate (E4P) used in the synthesis of aromatic amino acids

Two phases of PPP

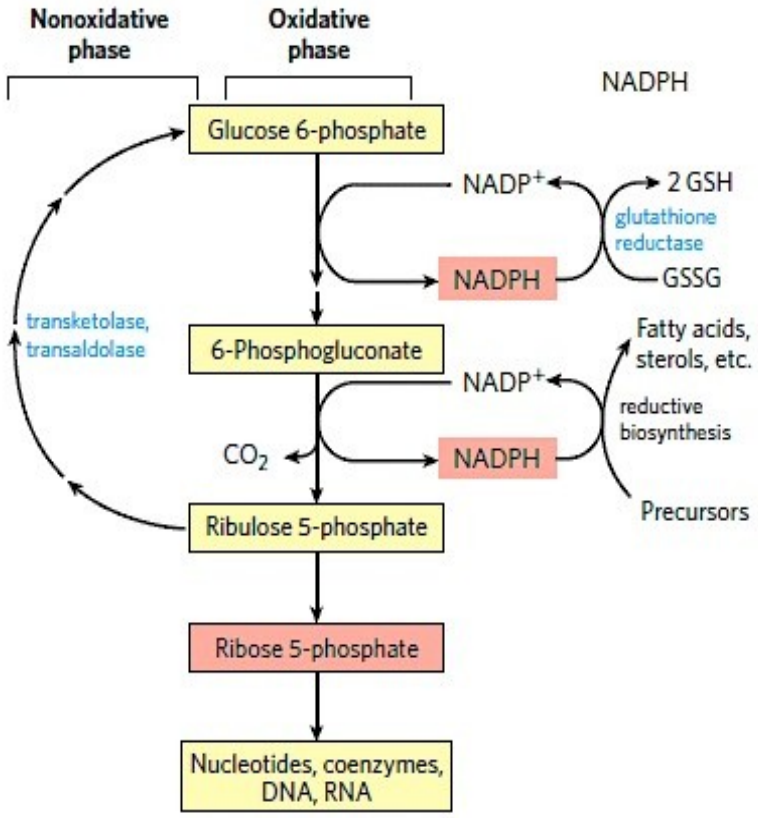
There are two phases of the pentose phosphate pathway:

1. The oxidative phase
2. The non-oxidative phase

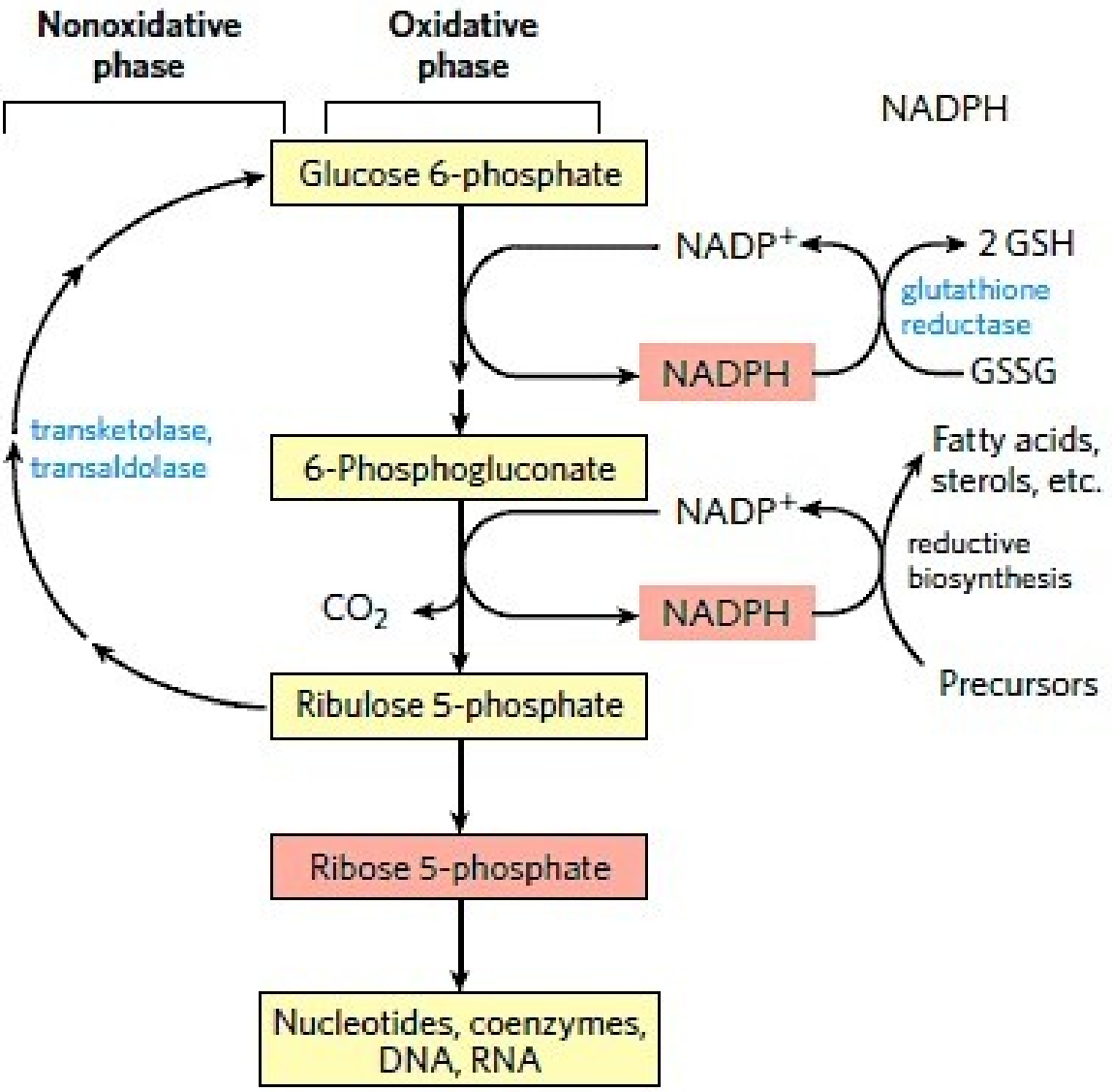
1. The Oxidative Phase

1) The Oxidative Phase Produces Pentose Phosphates and NADPH
2) The first reaction of the pentose phosphate pathway is the oxidation of glucose 6-phosphate by glucose 6-phosphate dehydrogenase (G6PD) to form 6-phosphoglucono-delta-lactone, an intramolecular ester.
3) NADP+ is the electron acceptor, and overall equilibrium is in the direction of NADPH formation.

4) The lactone is hydrolyzed to the free acid 6-phosphogluconate by a specific enzyme lactonase, then 6-phosphogluconate undergoes oxidation and decarboxylation by 6-phosphogluconate dehydrogenase



Pentose Phosphate Pathway NADPH formed in the oxidative phase is used to reduce glutathione to GSSG (Glutathione disulfide) and to support reductive biosynthesis. The other product of the oxidative phase is ribose 5-phosphate, which serves as a precursor for nucleotides, coenzymes, and nucleic acids. In cells that are not using ribose 5-phosphate for biosynthesis, the non-oxidative phase recycles six molecules of the pentose into five molecules of the hexose glucose 6-phosphate, allowing continued production of NADPH and converting glucose 6-phosphate to CO2.



Pentose Phosphate Pathway (PPP) …cont

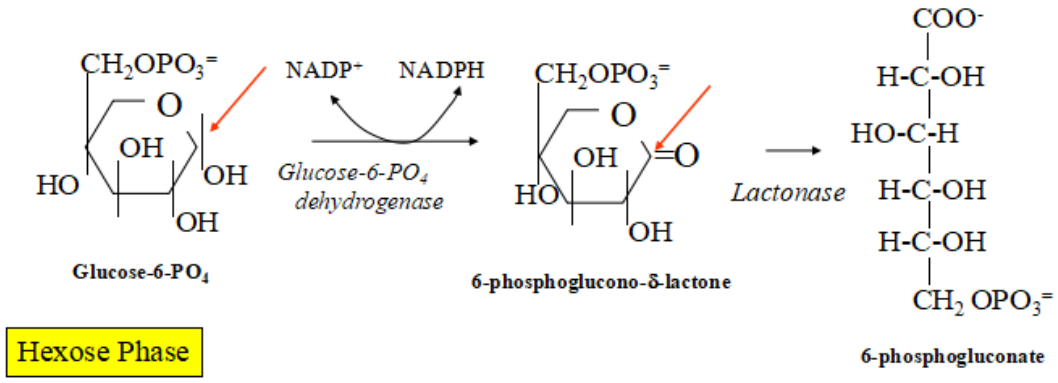
The reaction generates a second molecule of NADPH.

Phosphopentose isomerase converts ribulose 5-phosphate to its aldose isomer, ribose 5-phosphate.

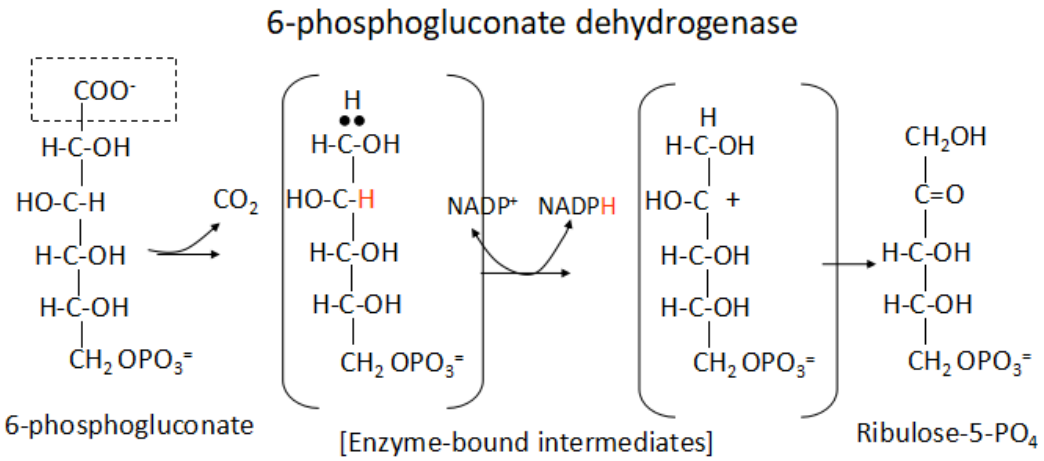
In some tissues, the pentose phosphate pathway ends at this point, and its overall equation for reactions is

G6-P + 2NADP$^+$+ H_2O ribulose 5-phosphate + CO2 + 2NADPH + 2H $^+$

The net result is the production of NADPH, a reducing agent for biosynthetic reactions, and ribose 5-phosphate, a precursor for nucleotide synthesis.



Hexose Phase



Reactants	Products	Enzyme	Description
Glucose 6-phosphate + NADP+	→ 6-phosphoglucono-δ-lactone + **NADPH**	glucose 6-phosphate dehydrogenase	Dehydrogenation. The hydroxyl on carbon 1 of glucose 6-phosphate turns into a carbonyl, generating a lactone, and, in the process, NADPH is generated.
6-phosphoglucono-δ-lactone + H2	→ 6-phosphogluconate+ H+	6-phosphogluconolactonase	Hydrolysis

O			
6-phosphogluconate+ NADP+	→ ribulose 5-phosphate +**NADPH** + CO2	6-phosphogluconate dehydrogenase	Oxidative decarboxylation. NADP+ is the electron acceptor, generating another molecule of NADPH, a CO2, and ribulose 5-phosphate.

TOPIC#61

Lyases

IUB Classification of Enzymes:

3) Hydrolases

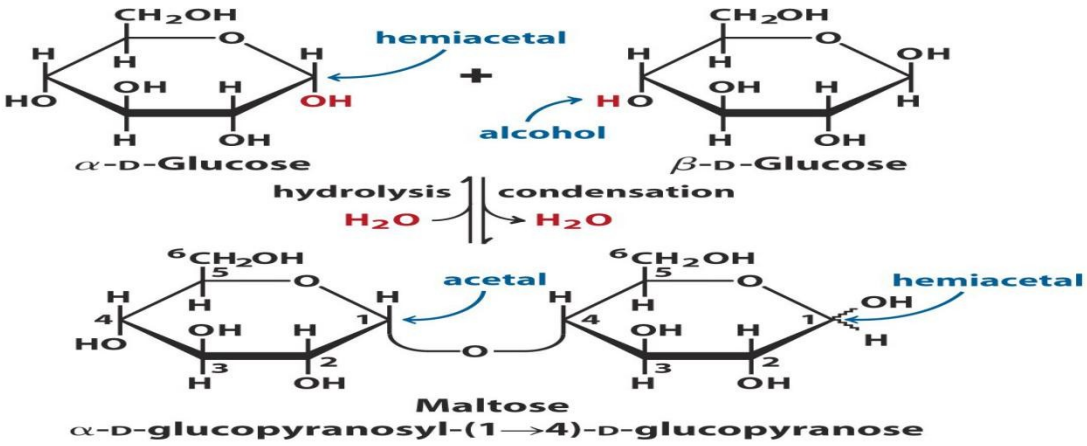
These enzymes catalyze hydrolysis, i.e.

add water molecule to the substrate which is simultaneously decomposed; the

functional group of substrate is transferred to water.

Common example of hydrolases are:

- Protein hydrolyzing Enzymes (peptidases).
- Carbohydrases
- Lipid hydrolyzing enzymes e.g. Lipases and
- Phospholipases.

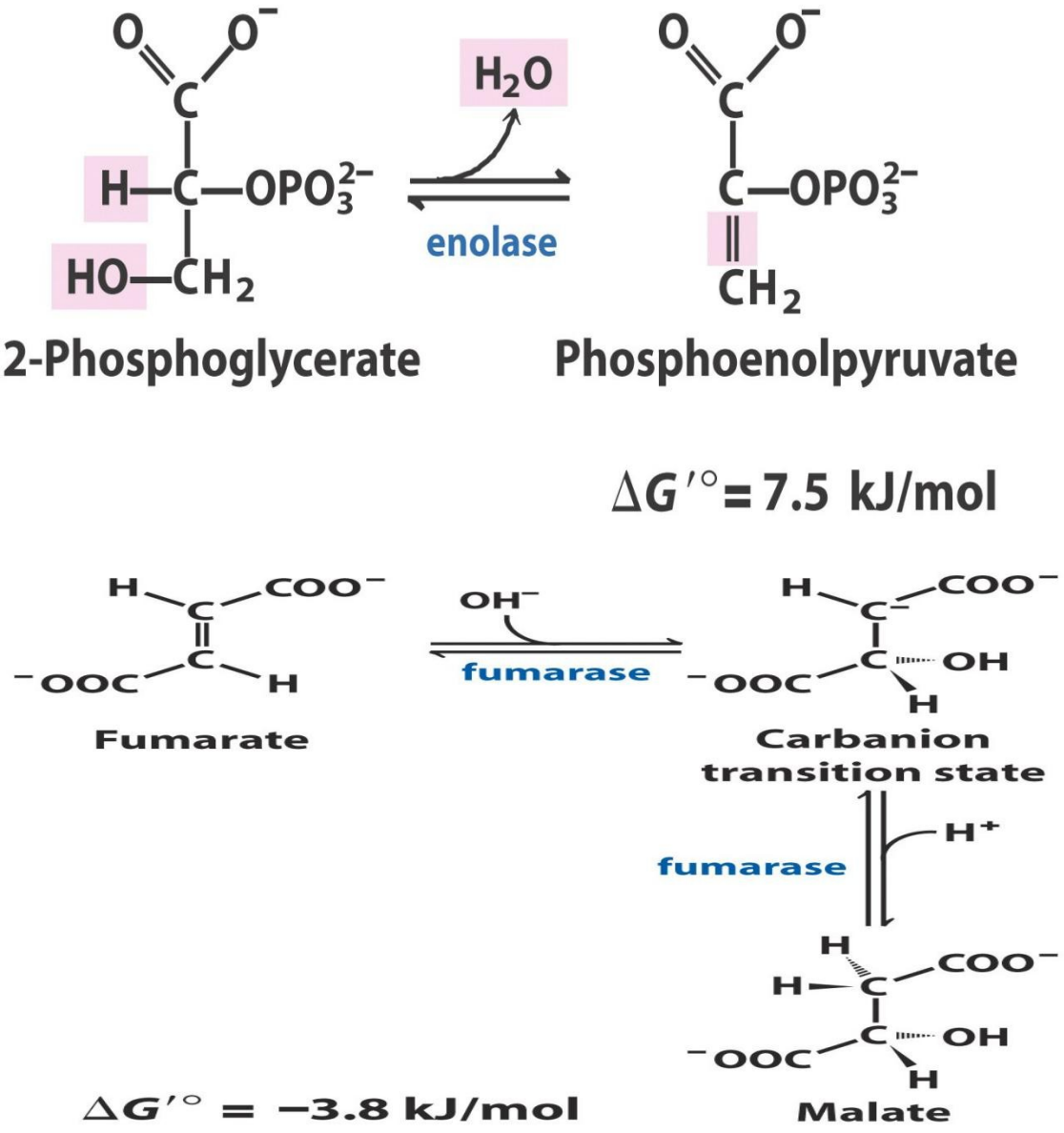


4) Lyases

Theses enzymes catalyze the addition of

- NH_3,
- H_2O or
- CO_2 to double bonds or

- the removal of these groups leaving behind double bonds.



Lyases are included in a separate class because they catalyze these reactions by means other than hydrolysis or oxidation.

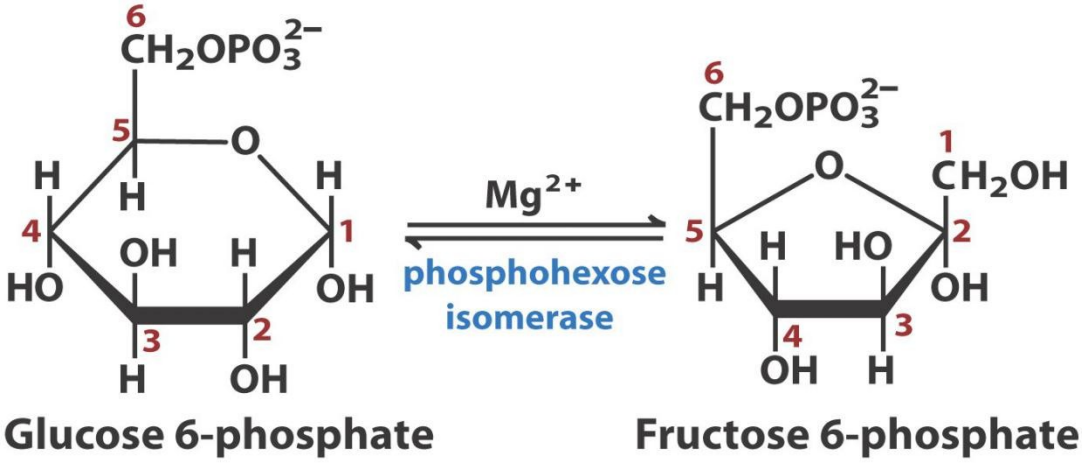
TOPIC#62

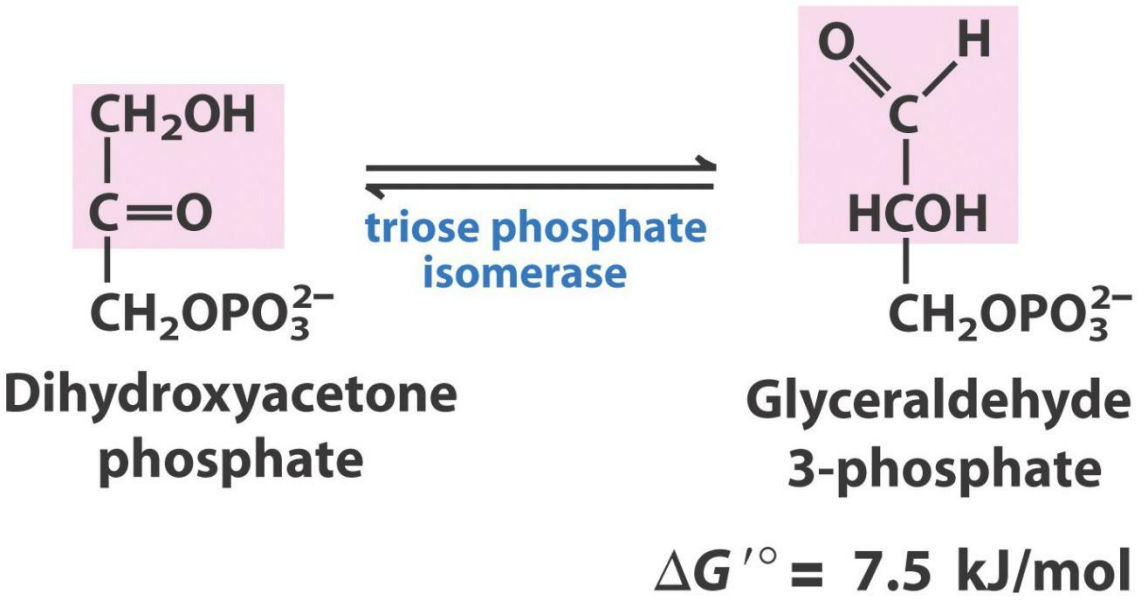
Ligases

IUB Classification of Enzymes:

5) Isomerases:

These enzymes catalyze the structural change within a single molecule by the transfer of groups within it, resulting in the formation of an isomeric form of substrate.

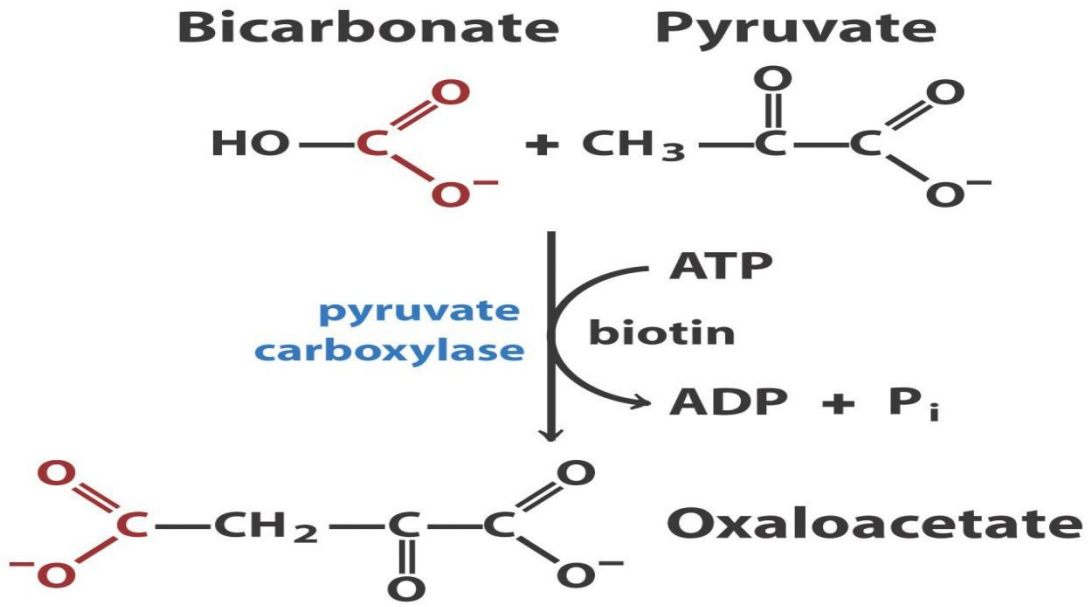




6) Ligases

These enzymes catalyze condensation reactions joining two molecules by forming

- C-O,
- C-S,
- C-N and
- C-C bonds.



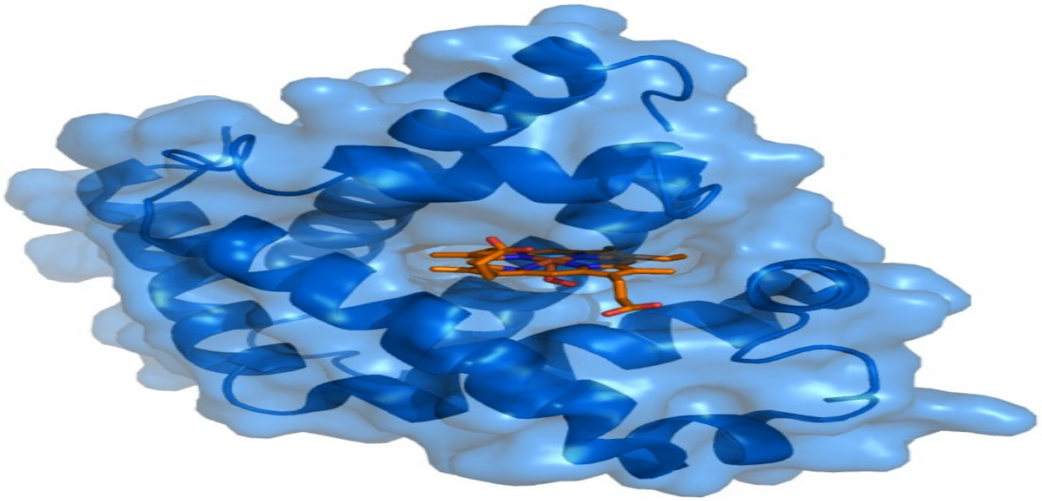
The energy for condensation is provided by cleavage of high energy phosphates, e.g. ATP, GTP etc.

TOPIC#63

Ligand

Ligand:

In biochemistry and pharmacology, a ligand (from the Latin ligandum, binding) is a substance (usually a small molecule),that forms a complex with a biomolecule to serve a biological purpose.



In a narrower sense, it is a signal triggering molecule, binding to a site on a target protein

A molecule bound reversibly by a protein is called a ligand.

Ligands include substrates, inhibitors, activators, and neurotransmitters

A ligand may be any kind of molecule, including another protein.A ligand binds at a site on the protein called the **binding site,**

binding site is complementary to the ligand in

- size,
- shape,
- charge, and
- hydrophobic or
- hydrophilic character.

Furthermore, the interaction is specific: the protein can discriminate among the thousands of different molecules in its environment and selectively bind only one or a few.

The binding of a protein and ligand is often coupled to a conformational change in the protein that makes the binding site more complementary to the ligand, permitting tighter binding called induced fit.

A given protein may have separate binding sites for several different ligands.

In a multi-subunit protein, a conformational change in one subunit often affects the conformation of other subunits.

Interactions between ligands and proteins may be regulated, through interactions with additional ligands.

These other ligands may cause conformational changes in the protein that affect the binding of the first ligand.

TOPIC#64

Mechanism of enzyme action

Mechanism of Enzyme Action:

Enzymes bind and chemically transform other molecules— they catalyze reactions.

The molecules acted upon by enzymes are called reaction substrates rather than ligands.

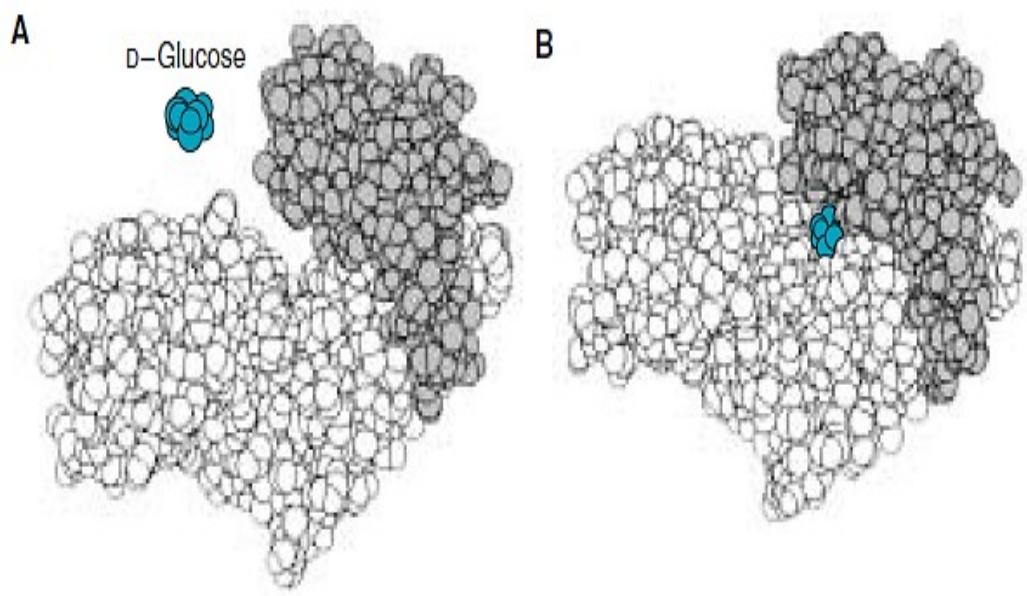
Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17}.

The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the active site.

The surface of the active site is lined with amino acid residues with side chains that complement and bind the substrate and catalyze its chemical transformation.

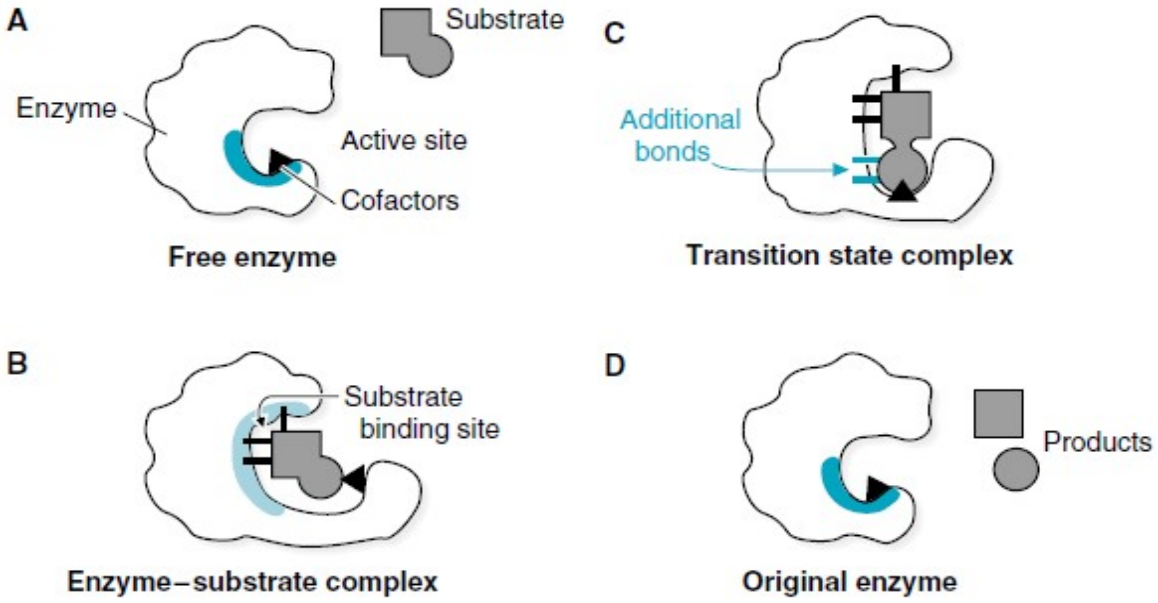
Often, the active site encloses a substrate, sequestering it completely from solution.

The enzyme substrate complex, whose existence is central to the action of enzymes.



Enzyme-catalyzed reactions have three basic steps:

- binding of substrate: E+S↔ES
- conversion of bound substrate to bound product: ES ↔ EP
- release of product : EP ↔ E+P



To understand catalysis, we must first appreciate the important distinction between

reaction equilibria and reaction rates.

TOPIC#65

Mechanism of enzyme action..cont -1

Mechanism of Enzyme Action:

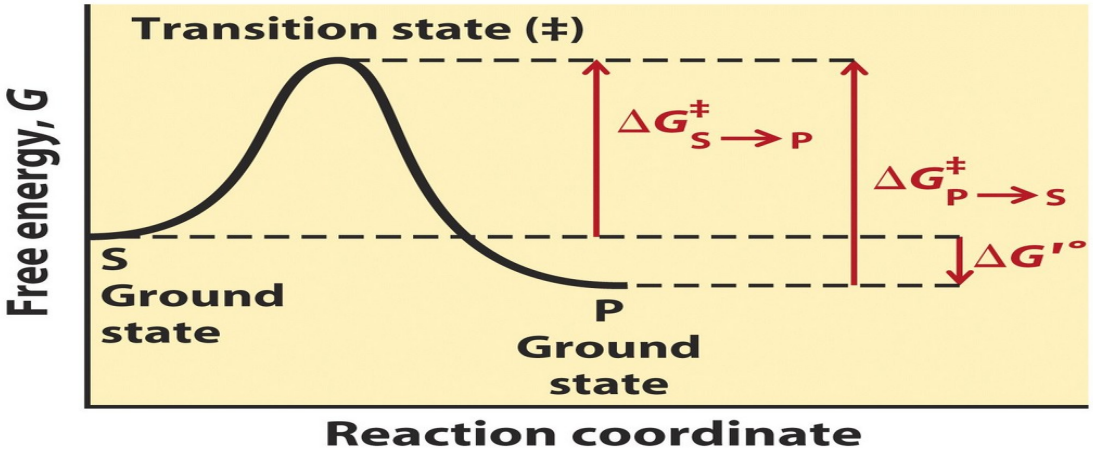
Reaction Equilibria:

The function of a catalyst is to increase the rate of a reaction.

Catalysts do not affect reaction equilibria.Any reaction, such as S ↔ P, can be described by a reaction coordinate diagram.The free energy of the system is plotted against the progress of the reaction S → P.

A diagram of this kind is a description of the energy changes during the reaction.

The horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., bond breakage or formation) as S is converted to P.



Reaction coordinate diagram for a chemical reaction.The free energy of the system is plotted against the progress of the reactionS→P. A diagram of this kind is a description of the energy.Changes during the reaction, and the horizontal axis (reaction coordinate).Reflects the progressive chemical changes (e.g., bond breakage

or formation) as S is converted to P. The activation energies, G‡, for

the S→P and P→S reactions are indicated. G is the overall standard

free-energy change in the direction S→P.

Activation energies, G‡,

The activation energies, G‡, for the

S → P and P → S reactions are indicated.

G′⁰ is the overall standard free-energy change in the direction S → P

Double dagger . Δ G‡.

Free-energy change ΔG:

When a reacting system is not at equilibrium, the tendency to move toward equilibrium represents a driving force

the magnitude of which can be expressed as the free-energy change for the reaction, ΔG.

Standard free-energy change, ΔG^{o}:

Under standard conditions (298 K = 25 C)

when reactants and products are initially present at 1 M concentrations or

for gases, at partial pressures of 101.3 (kPa), or 1 atm

the force driving the system toward equilibrium is defined as the standard free-energy change, ΔG^{o}.However, because the conditions in the body systems are different from standard conditions .Energy in biological systems is described in terms of free energy, G′⁰.

TOPIC#66

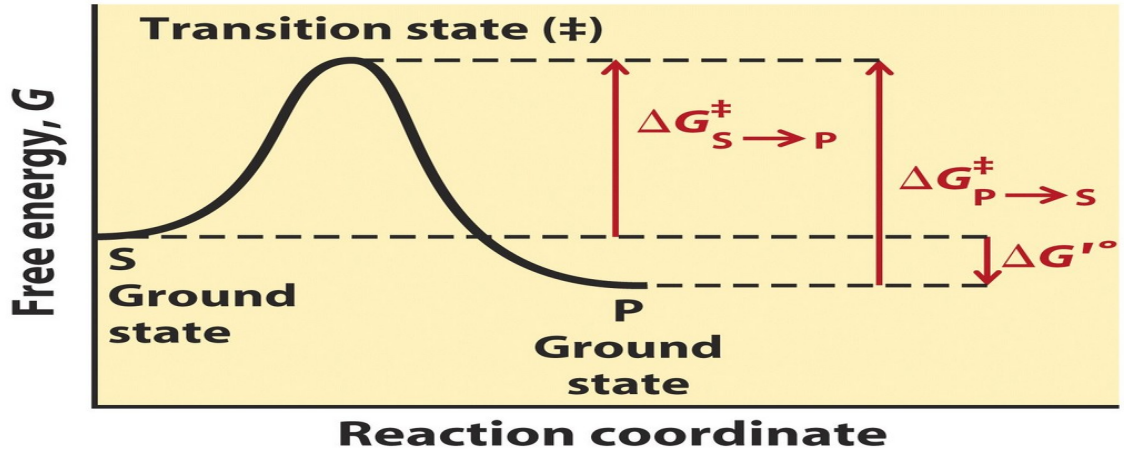
Mechanism of enzyme action..cont -2

Mechanism of Enzyme Action (Contd.)

Ground State & Transition State:

The starting point for either the forward or the reverse reaction is called the ground state.The equilibrium between S and P reflects the difference in the free energies of their ground states.

The free energy of the ground state of P is lower than that of S.So G′⁰ for the reaction is negative and the equilibrium favors P.The position and direction of equilibrium are not affected by any catalyst.



Reaction coordinate diagram for a chemical reaction.The free energy of the system is plotted against the progress of the reaction S→P. A diagram of this kind is a description of the energy.Changes during the reaction, and the horizontal axis (reaction coordinate).reflects the progressive chemical changes (e.g., bond breakage

or formation) as S is converted to P. The activation energies, G‡, for

the S→P and P→S reactions are indicated. G is the overall standard

free-energy change in the direction S→P.

But there is an energy barrier between S and P: The energy required for

- alignment of reacting groups
- formation of transient unstable charges
- bond rearrangements
- and other transformations

This is illustrated by the energy “hill”

To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a higher energy level.This is called the transition state.It is simply a fleeting molecular moment in which events such as

- bond breakage,
- bond formation, and

- charge development

Have proceeded to the precise point at which decay to either substrate or product is equally likely.

The difference between the energy levels of the ground state and the transition state is the activation energy, G‡.

At the top of the energy hill is a point at which decay to the S or P state is equally probable (it is downhill either way).

A substance that modifies the transition state to lower the activation energy is termed a catalyst; a biological catalyst is termed an enzyme.

TOPIC#67

Mechanism of enzyme action..cont -3

<u>Activation energy Δ G‡.:</u>

The difference between the energy levels of the ground state and the transition state is the activation energy Δ G‡.

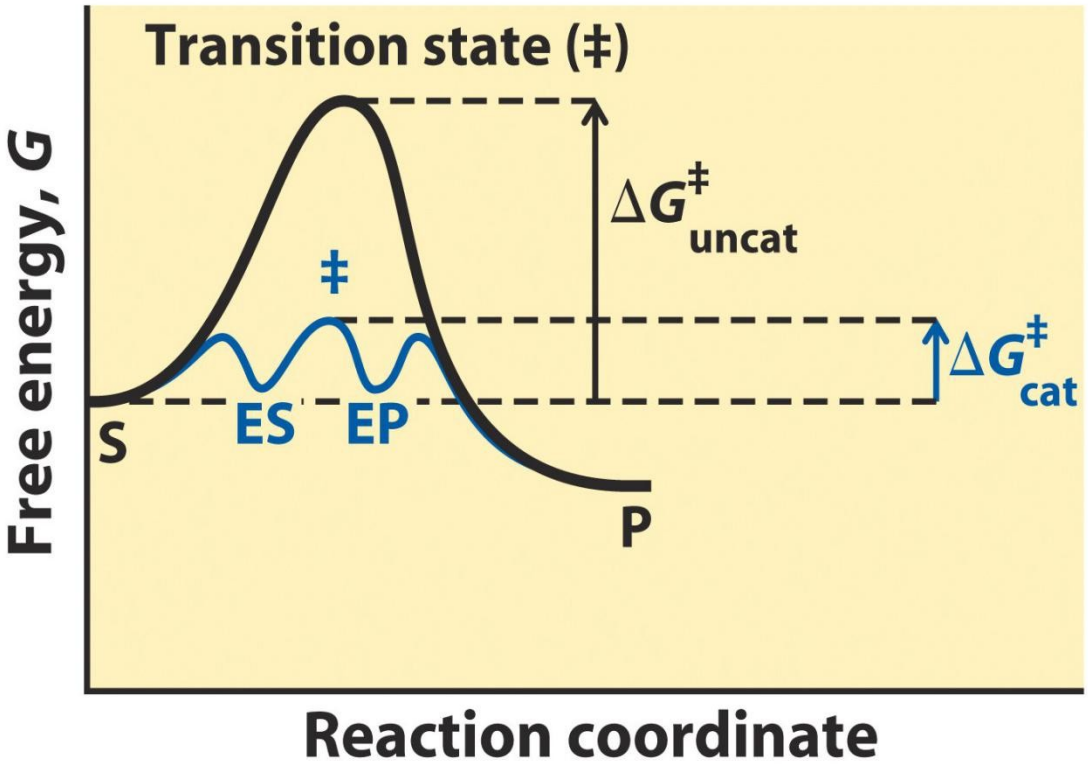
Δ ***G‡. Double*** dagger.The least amount of energy needed for a chemical reaction to take place. Some elements and compounds react together naturally just by being close to each other, and their activation energy is zero. Others will react together only after a certain amount of energy is added to them.

The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction.

Striking a match on the side of a matchbox, for example, provides the activation energy (in the form of heat produced by friction) necessary for the chemicals in the match to ignite

Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier.

Alternatively, the activation energy can be lowered by adding a catalyst.Catalysts enhance reaction rates by lowering activation energies.



Reaction coordinate diagram comparing enzyme catalyzedand uncatalyzed reactions. In the reaction S→P, the ESand EP intermediates occupy minima in the energy progress curve of

the enzyme-catalyzed reaction. The terms G‡ uncat and G‡ cat correspond to the activation energy for the uncatalyzed reaction and the overall activation energy for the catalyzed reaction, respectively.

The activation energy is lower when the enzyme catalyzes the reaction.

The role of enzymes is to accelerate the inter-conversion of S and P.i.e enzymes lower the energy of activation, Δ G‡, of a reaction. The enzyme is not used up in the process, and the equilibrium point is unaffected.

However, the reaction reaches equilibrium much faster when the appropriate enzyme is present, because the rate of the reaction is increased.

TOPIC#68

Cofactor, coenzyme and prosthetic groups

The Induced Fit Hypothesis:

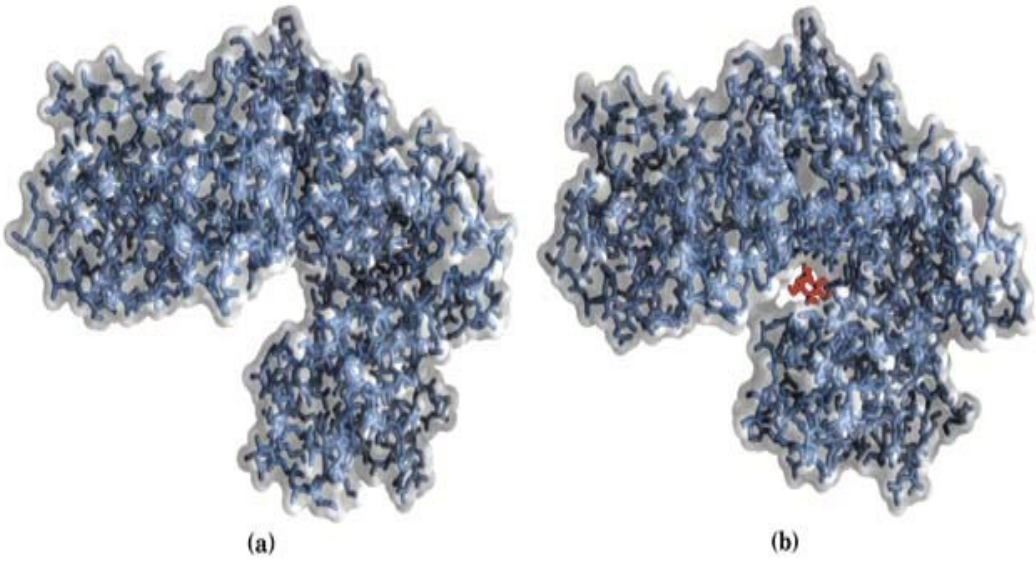
Some proteins can change their shape (conformation)

When a substrate combines with an enzyme, it induces a change in the enzyme's conformation

This change in conformation when the substrate binds is induced by multiple weak interactions with the substrate.

There may also be rearrangements of covalent bonds during an enzyme-catalyzed reaction.This conformational change is referred to as induced fit.

The Induced Fit Hypothesis

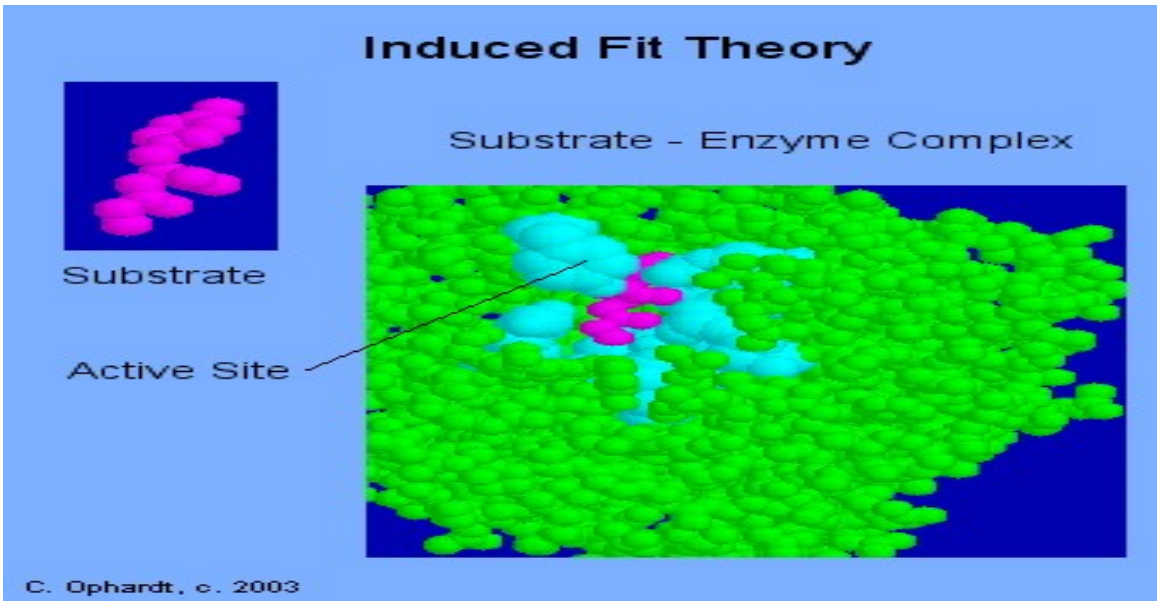


Hexokinase (a) without **(b) with glucose substrate**

Chemical reactions of many types take place between substrates and enzyme's functional groups (specific amino acid side chains, metal ions, and coenzymes.Induced fit serves to bring specific functional groups on the enzyme into the proper position to catalyze the reaction.

Induced Fit Theory:

Enzyme is not rigid, changes shape with substrate.



The active site is also moulded into a precise conformation.Making the chemical environment suitable for the reaction

TOPIC#69

Cofactor, coenzyme and prosthetic groups-1

Cofactors, Coenzymes and Prosthetic groups:

Some enzymes require no chemical groups for activity other than their amino acid residues. Whereas some enzymes require molecules other than proteins for enzymic activity.

If the non-protein moiety is a metal ion such as Zn2+ or Fe2+(Fe+2=Ferrous

), it is called a cofactor. If it is a complex organic molecule or metallo-organic compound it is termed a coenzyme.

Some Coenzymes That Serve as Transient Carriers

Coenzyme	*Examples of chemical groups transferred*
Biocytin	CO_2
Coenzyme A	Acyl groups
5′-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups
Flavin adenine dinucleotide	Electrons
Lipoate	Electrons and acyl groups
Nicotinamide adenine dinucleotide	Hydride ion ($:H^-$)

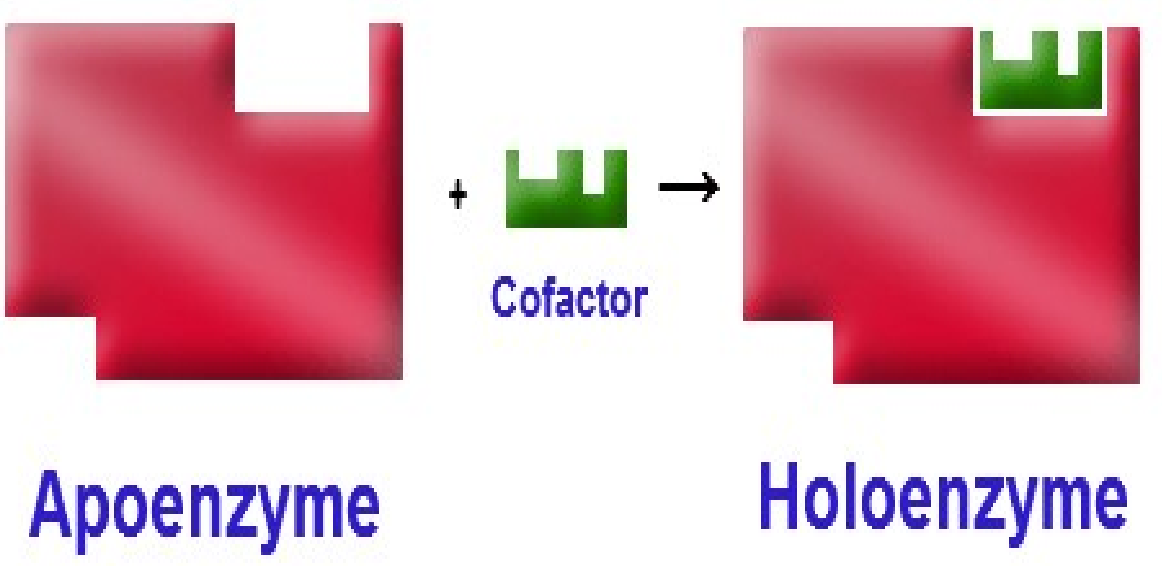
Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase

Cu2+= Cupric. Ferrous=Fe2+. Ferric=Fe3+.

A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called a prosthetic group.

The term holoenzyme refers to the active enzyme with its non-protein component, whereas the enzyme without its non-protein moiety is termed an apoenzyme (apoprotein) and is inactive.



Coenzymes serve as recyclable shuttles that transport many substrates from one point within the cell to another.

The function of these shuttles is twofold.

First, they stabilize species

such as hydrogen atoms (FADH) or hydride ions (NADH)

that are too reactive to persist for any significant time in the presence of the water or organic molecules that permeate cells.

Second, they serve as an adaptor or handle that facilitates the

recognition and binding of small chemical groups, such as acetate (coenzyme A) or glucose (UDP), by their target enzymes

TOPIC#70

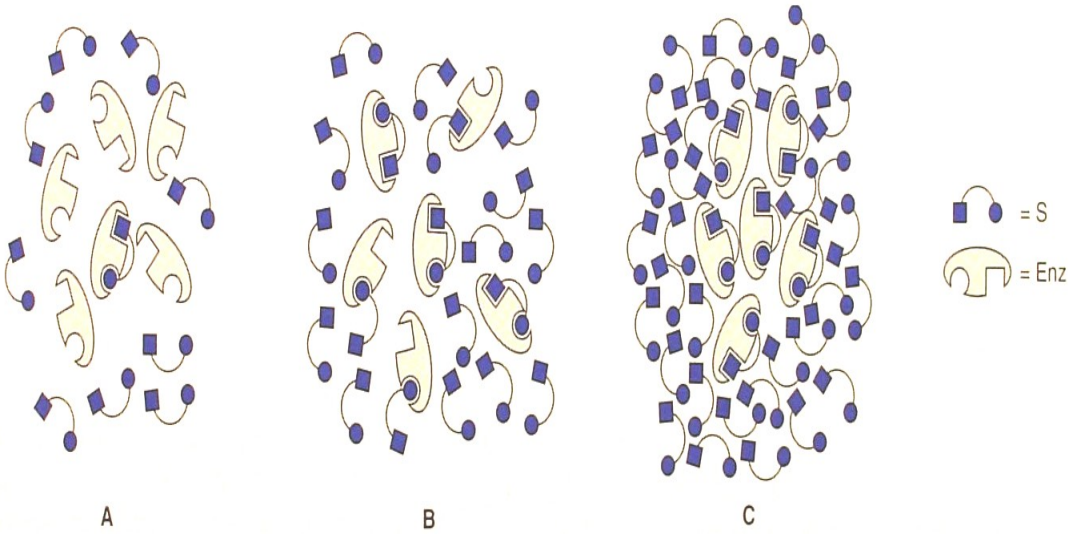
Reaction rate and order of reaction

Reaction Rates and Order of Reactions

Reaction Velocity (v):

The rate or velocity of a reaction (v) is the number of substrate molecules converted to product per unit time; Velocity is usually expressed as μmol of product formed per minute.

The rate of an enzyme-catalyzed reaction increases with substrate concentration until a maximal velocity (Vmax) is reached- reflecting the saturation with substrate of all available binding sites are occupied on the enzyme molecules present.



(Vmax) is reached- reflecting the saturation with substrate of all available binding sites are occupied on the enzyme molecules present.

An equilibrium such as S ↔ P is described by an equilibrium constant, Keq, or simply K.Under the standard conditions used to compare biochemical processes, an equilibrium constant is denoted K´eq (or K´).

The rate of any reaction is determined by the concentration of the reactant (or reactants) and by a rate constant, usually denoted by k.

For the uni molecular reaction S → P, the rate (or velocity) of the reaction, V—representing the amount of S that reacts per unit time—is expressed by a rate equation:

$$V = k[S].$$

In this reaction, the rate depends only on the concentration of S.

This is called a first-order reaction.The factor k is a proportionality constant that reflects the probability of reaction under a given set of conditions (pH, temperature, and so forth).

If a reaction rate depends on the concentration of two different compounds.

or if the reaction is between two molecules of the same compound, the reaction is second order.

The rate equation then becomes

$$V = k[S_1][S_2]$$

TOPIC#71

Factors affecting enzyme activity

Factors Affecting Enzymatic Activity:

Enzymes can be isolated from cells, and their properties studied in a test tube (that is, in vitro).

Factors Affecting Enzymatic Activity:

Different enzymes show different responses to changes in;

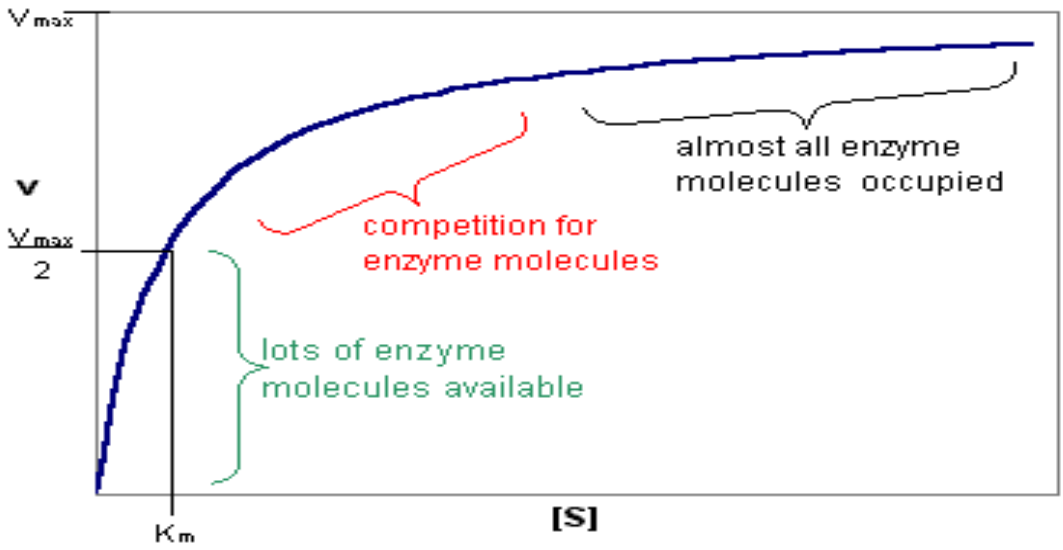
- substrate concentration
- temperature, and
- pH.

Enzymic responses to these factors give us valuable clues as to how enzymes function in living cells (that is, in vivo)

Substrate concentration:

The rate of an enzyme-catalyzed reaction increases with substrate concentration until maximal velocity (Vmax) is reached

The leveling off of the reaction rate at high substrate concentrations reflects the saturation with substrate of all available binding sites on the enzyme molecules present.

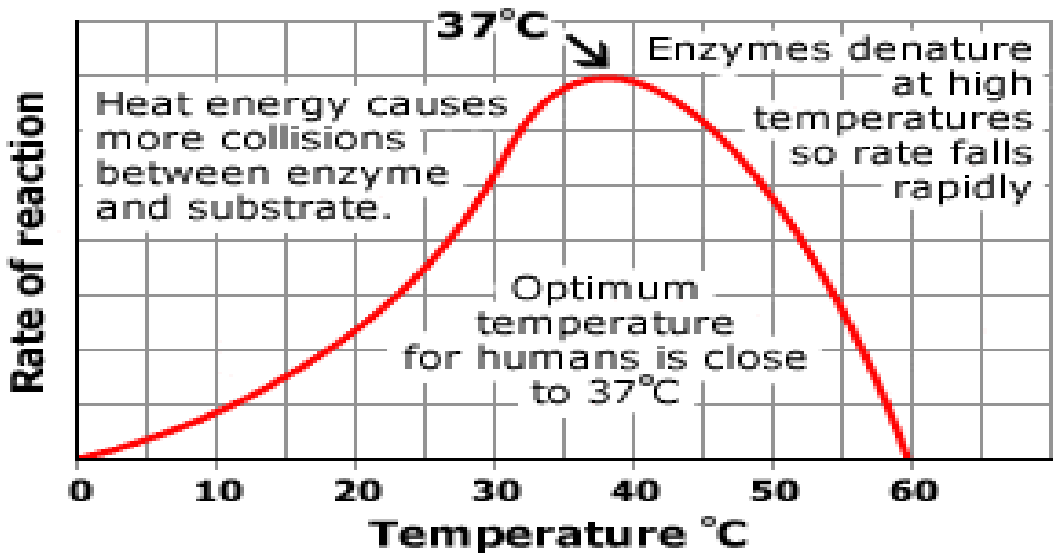


The leveling off of the reaction rate at high substrate concentrations reflects the saturation with substrate of all available binding sites on the enzyme molecules present.

Temperature:

The reaction velocity increases with temperature until a peak velocity is reached

This increase is the result of the increased number of molecules having sufficient energy to pass over the energy barrier and form the products.



Further elevationof the temperature results in a decrease in reaction velocity as a result of temperature-induced denaturation of the enzyme

TOPIC#72

Factors affecting enzyme activity (cont)

Factors Affecting Enzymatic Activity(Contd.)

Effect of pH:

The pH optimum varies for different enzymes:

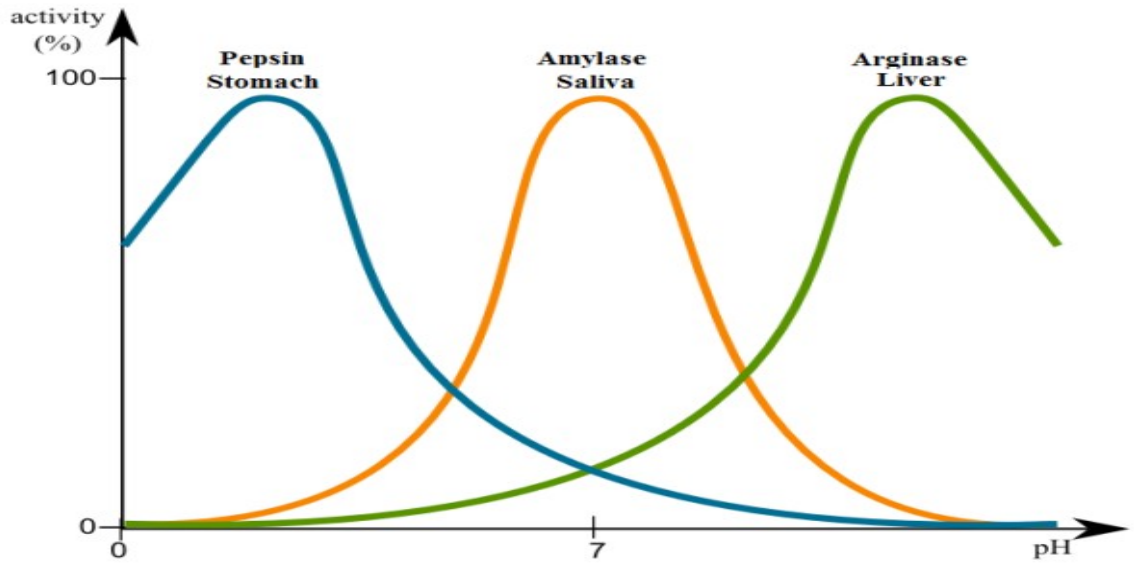
The pH at which maximal enzyme activity is achieved is different for different enzymes, and often reflects the [H+] at which the enzyme functions in the body.

For example, pepsin, a digestive enzyme in the stomach, is maximally active at pH 2.

Whereas other enzymes, designed to work at neutral pH, are denatured by such an acidic environment.

Another examples is that there are two types of phosphatases in the body.

The one that acts in the alkaline pH is called alkaline phosphatase and the other which acts at acidic pH is known as acid phosphatase.



Effect of pH on the ionization of the active site:

The concentration of H+ affects reaction velocity in several ways**.**

First, the catalytic process usually requires that the enzyme and substrate have specific chemical groups in either an ionized or un-ionized state in order to interact.

For example, catalytic activity may require that an amino group of the enzyme be in the protonated form (–NH3 +). .At alkaline pH, this group is deprotonated, and the rate of the reaction, therefore, declines.Extremes of pH can also lead to denaturation of the enzyme.

TOPIC#73

Enzyme kinetics

Enzyme Kinetics:

Substrate Concentration Affects the Rate of Enzyme-Catalyzed Reactions

$E + S \leftrightarrow ES \leftrightarrow EP \leftrightarrow E + P$

A key factor affecting the rate of a reaction catalyzed by an enzyme is the concentration of substrate, [S].

However, studying the effects of substrate concentration is complicated because substrate is converted to product

and because of reversibility of reactions, e.g. conversion of product back to substrate.

One simplifying approach in kinetics experiments is to measure the initial rate (or initial velocity), designated V0, when [S] is much greater than the concentration of enzyme, [E].

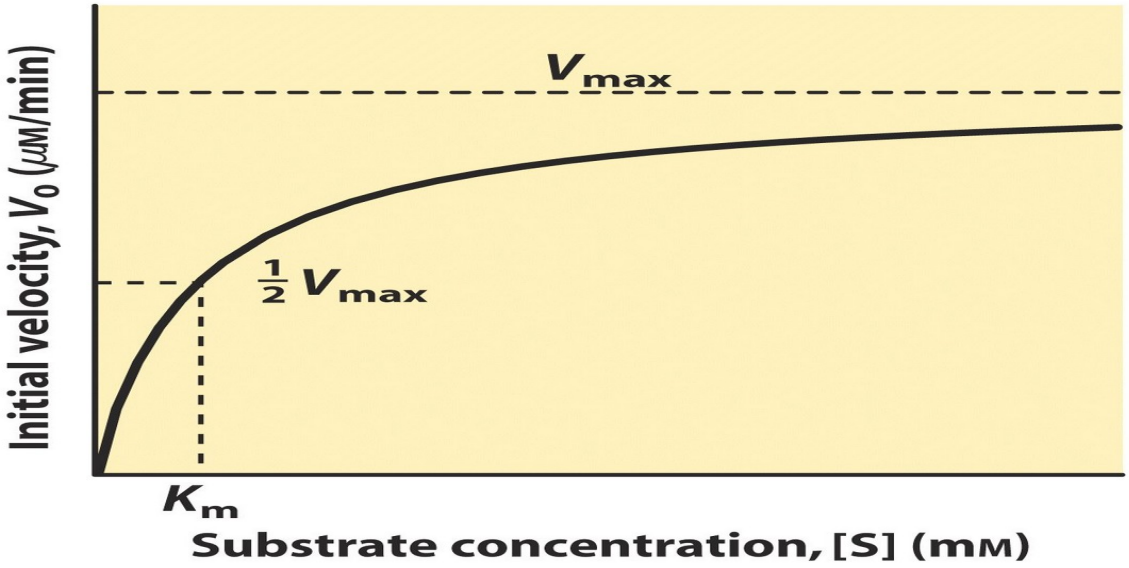
The velocity (v) of a reaction is the rate of product formation

Whereas V0 is the initial velocity and is measured as soon as the reactants and enzymes are mixed.

At the start of a reaction, [S] is in large excess of [P].

Thus the initial velocity of the reaction will be dependent on substrate concentration

At that time, the concentration of product is very small and, therefore, the rate of the back reaction from P to S can be ignored.



Increase in substrate concentration increases V^0 i.e. initial velocity is increased whenever a fixed concentration of enzyme is mixed with an increased concentration of substrate.

TOPIC#74

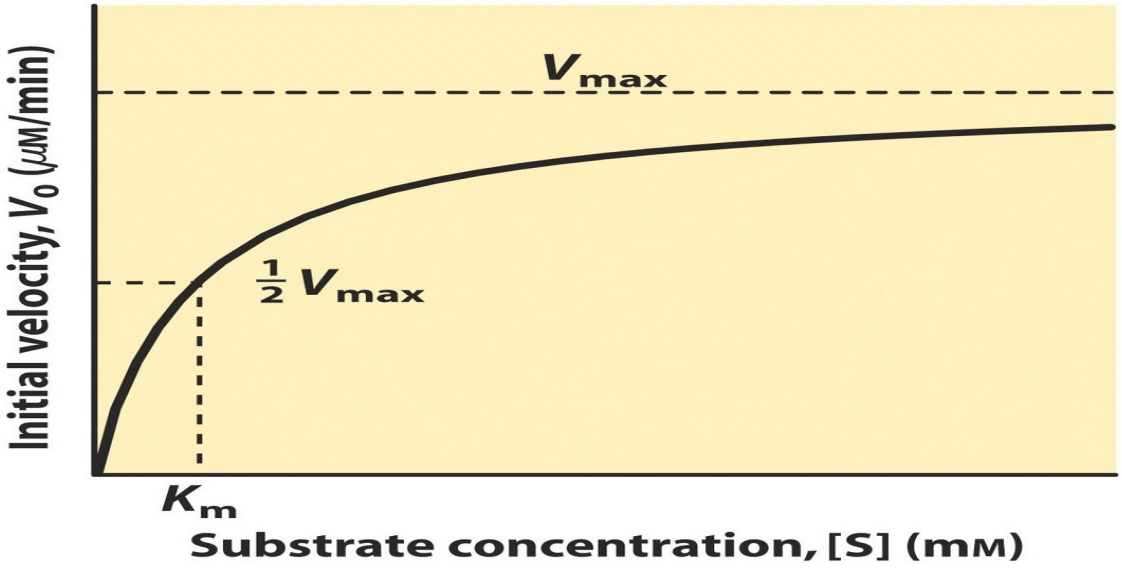
Enzyme kinetics..cont I

Enzyme Kinetics (Contd.)

Maximal Velocity Vmax:

When initial velocity (Vo) is plotted against [S],a hyperbolic curve results,where Vmax represents the maximum reaction velocity.

We have already defined the V and v^o. A ***hyperbola*** is an open curve, meaning that it continues indefinitely to infinity, rather than closing on itself as an ellipse does



The effect of varying [S], on V^O, when the enzyme concentration is held constant.

At relatively low concentrations of substrate, V_o increases almost linearly.

At higher substrate concentrations, V_o increases by smaller and smaller amounts in response to increases in [S].

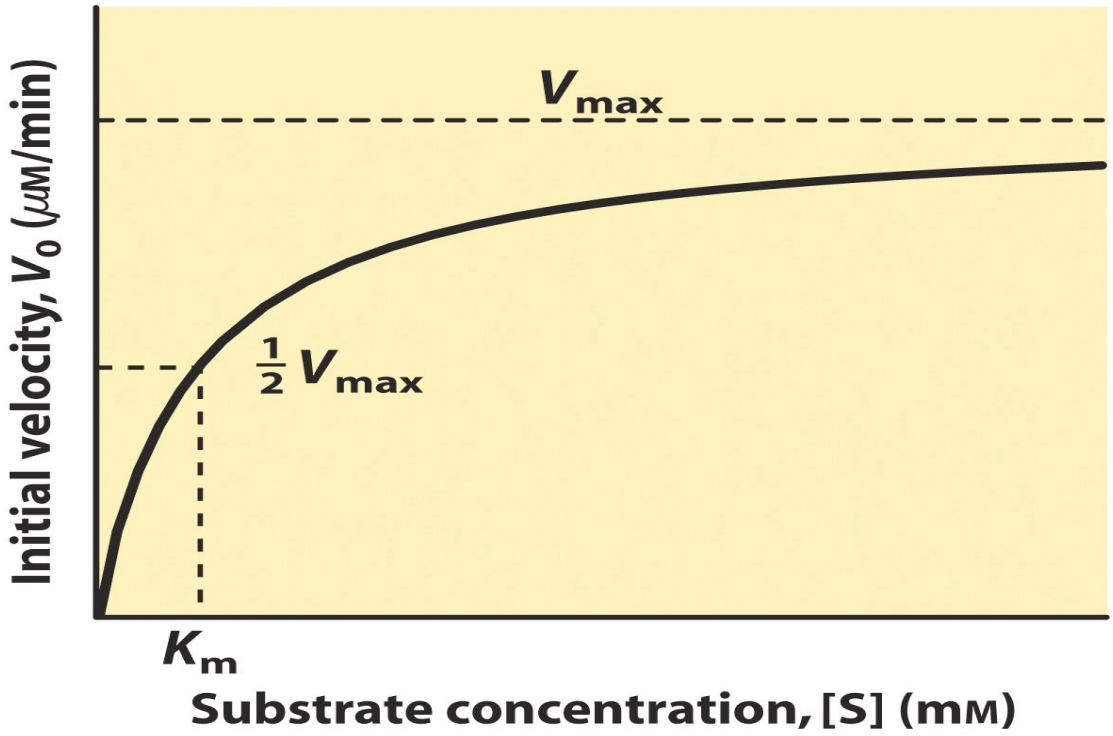
At relatively low concentrations of substrate, V_o increases almost linearly.

At higher substrate concentrations, V_o increases by smaller and smaller amounts in response to increases in [S].

This plateau-like V^o region is close to the maximal velocity, Vmax.

Vmax is extrapolated from the plot, because V_o approaches but never quite reaches Vmax.

Vmax is the theoretical maximal rate of the reaction - but it is NEVER achieved in reality



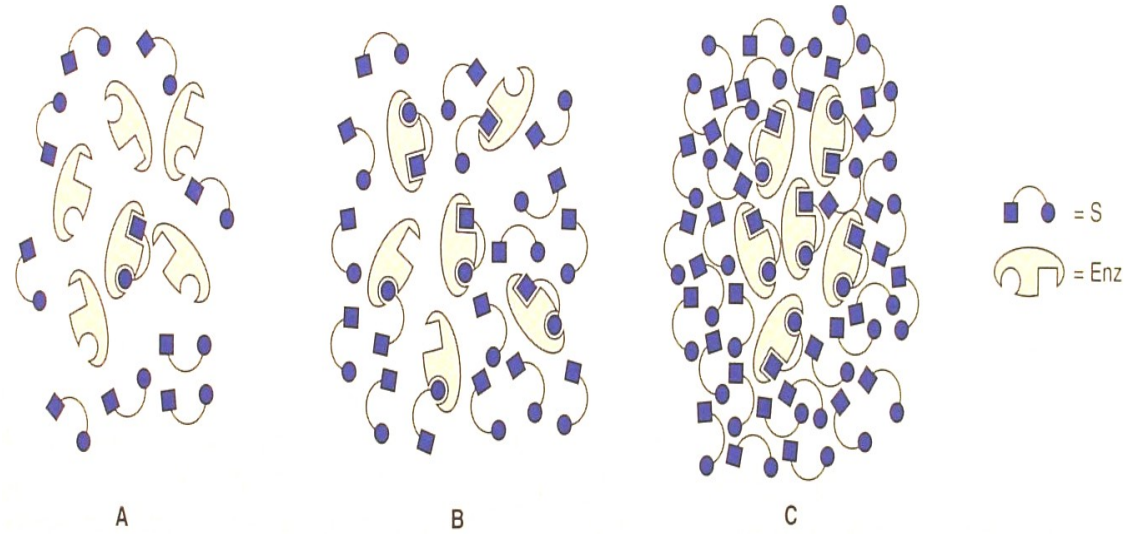
This plateau-like V_0 region is close to the maximal velocity, Vmax.

Vmax is extrapolated from the plot, because V_0 approaches but never quite reaches Vmax.

At this point in the reaction, if [S] >> E, all available enzyme is "saturated" with bound substrate, meaning only the ES complex is present.

This dictates that further increasing the substrate concentration will not result in increased V0

[S] is very very greater than E



A. Low [S] B. 50% [S] or Km C. High, saturating [S]

The leveling off of the reaction rate at high substrate concentrations reflects the saturation with substrate of all available binding sites on the enzyme molecules present.

This finite limit of Vmax is called saturation kinetics.Saturation kinetics is a characteristic property of all rate processes dependent on the binding of a ligand to a protein e.g. membrane transporter proteins.

The process by which a carrier protein transfers a solute molecule across the lipid bilayer resembles an enzyme-substrate reaction, and in many ways carriers behave like enzymes. In contrast to ordinary enzyme-substrate reactions, however, the transported solute is not covalently modified by the carrier protein, but instead is delivered unchanged to the other side of the membrane.

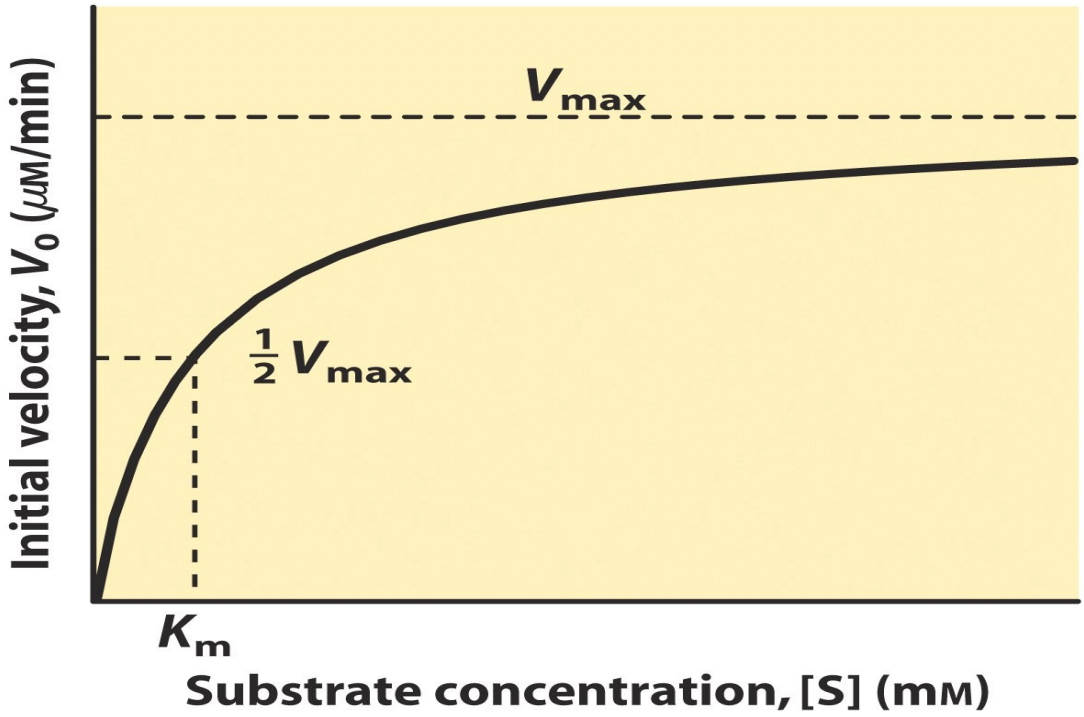
TOPIC#75

Enzyme kinetics..cont II

Enzyme Kinetics (Contd.)

Michaelis constant K_m:

The substrate concentration at which V_o is half maximal is Km, the Michaelis constant.



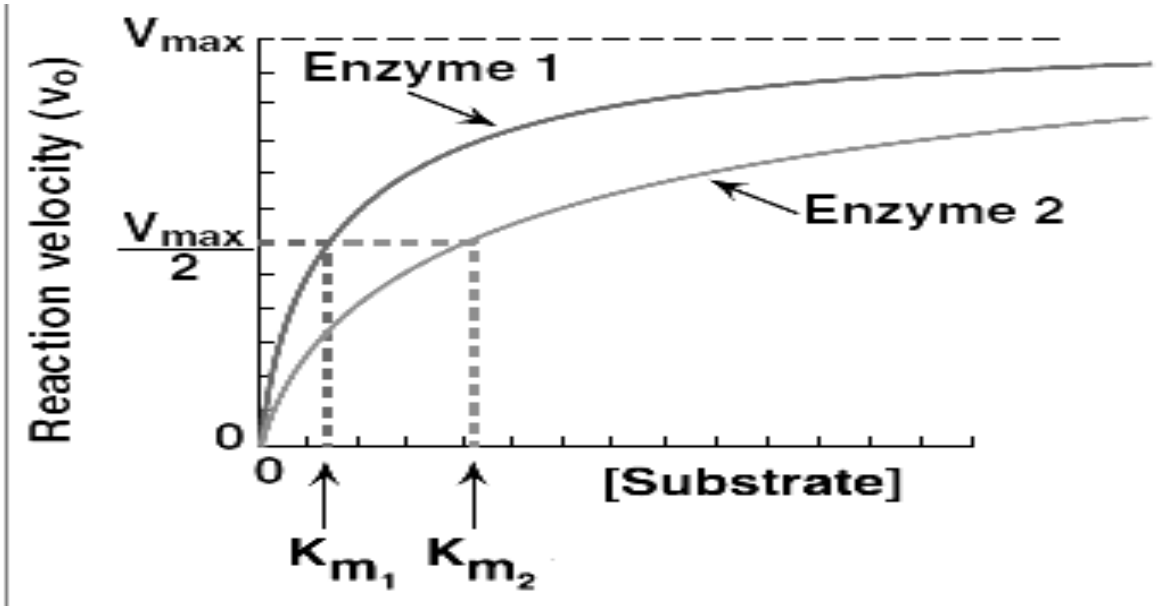
The effect of varying [S], on V_0, when the enzyme concentration is held constant

K_m reflects the affinity of the enzyme for the substrate.

Small K_m reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme i.e. to reach a velocity that is $1/2V_{max}$.

A numerically large (high) Km reflects a low affinity of enzyme for substrate because

a high concentration of substrate is needed to half-saturate the enzyme.



Small K_{m1}: A numerically small (low) Km reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is 1⁄2 Vmax.

Large K_{m2}: A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme

Small K_m: A numerically small (low) Km reflects a high affinity of the enzyme for substrate, because a low concentration of substrate is needed to half-saturate the enzyme—that is, to reach a velocity that is 1⁄2 $V_{max.}$

Large K_m: A numerically large (high) Km reflects a low affinity of enzyme for substrate because a high concentration of substrate is needed to half-saturate the enzyme.

The velocity of an enzyme is most sensitive to changes in substrate concentration over a concentration range below its Km.

At substrate concentrations less than 1⁄10th of the Km, a doubling of substrate concentration nearly doubles the velocity of the reaction

At substrate concentrations 10 times the Km, doubling the substrate concentration has little effect on the velocity.

A comparison between the isozymes of hexokinase illustrates the significance of the Km.

Hexokinase catalyses the first step in glucose metabolism in most cells, the transfer of a phosphate from ATP to glucose to form glucose 6-phosphate.

Hexokinase I, the isozyme in red blood cells has a low Km for glucose of approximately 0.05 mM- helpful in utilizing blood glucose even when the blood glucose concentration is very low.

The isozyme of hexokinase, called glucokinase, which is found in the liver has a much higher Km of approximately 5 to 6 mM- helpful in storing large amounts of "excess" glucose as glycogen or converting it to fat after a carbohydrate meal.

TOPIC#76

Enzymes...Michaelis-Mention Eq.

Michaelis-Menten equation, the rate equation:

Leonor Michaelis and Maud Menten in 1913, proposed a simple model that accounts for most of the features of enzyme-catalyzed reactions.

They postulated that the enzyme first combines reversibly with its substrate to form an enzyme-substrate complex in a relatively fast reversible step:

k1

E + S ↔ ES

K-1

The ES complex then breaks down in a slower second step to yield the free enzyme (E) and the reaction product (P):

K2

ES ↔ E + P

K-2

Early in the reaction, the concentration of the product, [P], is negligible, and we make the simplifying assumption that the reverse reaction, P → S (described by k-2), can be ignored

This assumption is not critical but it simplifies our task.

The overall reaction then reduces to

k1 K2

E + S ↔ ES → E + P

K-I

$$E + S \underset{k_{-1}}{\overset{k_1}{\leftrightarrow}} ES \xrightarrow{k_2} E + P$$

E = Enzyme S = Substrate P = Product

ES = Enzyme-Substrate complex

k_1 rate constant for the forward reaction

k_{-1} = rate constant for the breakdown of the ES to substrate

k_2 = rate constant for the formation of the products

Because the slower second reaction must limit the rate of the overall reaction.

The overall rate must be proportional to the concentration of the species that react in the second step, that is, ES.

The maximum initial rate of the catalyzed reaction (Vmax) is observed when virtually all the enzyme is present as the ES complex and [E] is vanishingly small.

Under these conditions, the enzyme is “saturated” with its substrate, so that further increases in [S] have no effect on rate.

This condition exists when [S] is sufficiently high that essentially all the free enzyme has been converted to the ES form.

After the ES complex breaks down to yield the product P,the enzyme is free to catalyze reaction of another molecule of substrate.

TOPIC#77

Enzymes...Michaelis-Mention Eq.cont-1

Michaelis-Menten equation, the rate equation (Contd.):

$$V_o = \frac{V_{max}[S]}{\mathbf{Km + [S]}}$$

Where;

V_o = initial reaction velocity

Vmax = maximal velocity

Km = Michaelis constant = K1+K2 / K-1

[S] = substrate concentration

This is the Michaelis-Menten equation-statement of the quantitative relationship between the;

- initial velocity V_o,
- the maximum velocity Vmax, and
- the initial substrate concentration [S], all related through the Michaelis constant Km

An important numerical relationship emerges from the Michaelis-Menten equation in the special case;

when V_o is exactly one-half Vmax

$$V_o = \frac{V_{max}[S]}{Km + [S]}$$

$$\frac{V_{max}}{2} = \frac{V_{max}[S]}{Km + [S]}$$

On dividing by Vmax, we obtain

1/2 = [S] / Km + [S]

Solving for Km, we get

Km + [S] = 2[S], or

Km = [S], when V_o = 1/2 Vmax

An important numerical relationship emerges from the Michaelis-Menten equation in the special case when V_o is exactly one-half Vmax.

We have already defined by graphical presentation.

That the substrate concentration at which V_o is half maximal is Km, the Michaelis constant.

This is a very useful, practical definition of Km: Km is equivalent to the substrate concentration at which V_o is one-half Vmax.

The equation describes the kinetic behavior of a great many enzymes, and all enzymes that exhibit a hyperbolic dependence of V0 on [S] are said to follow Michaelis- Menten kinetics.

The practical rule that Km= [S] when V0 =1⁄2Vmax holds for all enzymes that follow Michaelis-Menten kinetics.

The most important exceptions to Michaelis-Menten kinetics are the regulatory enzymes.

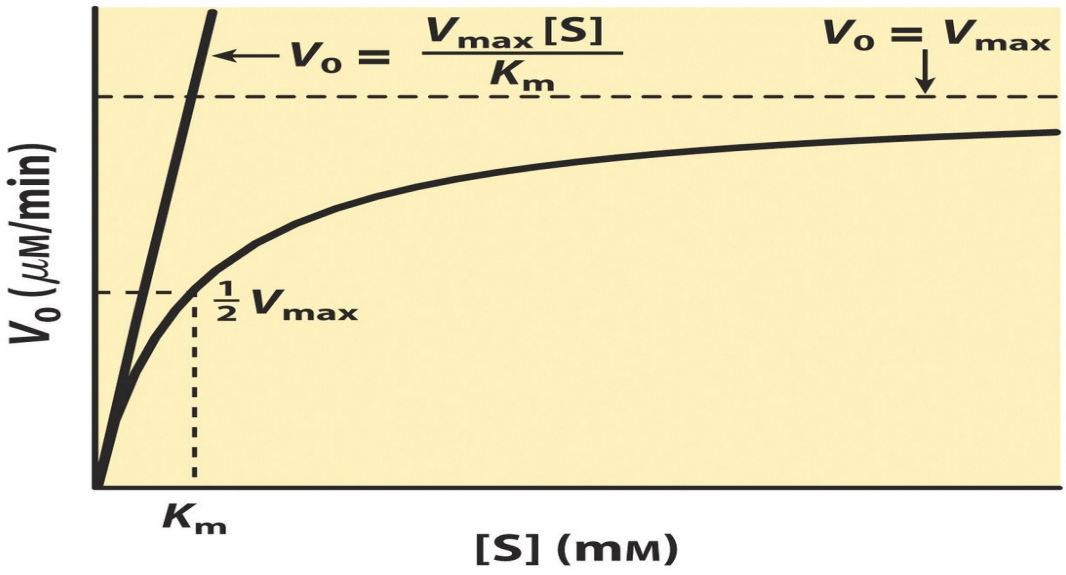
TOPIC#78

Enzymes...Michaelis-Mention Eq.cont-2

The Michaelis-Menton Equation at low [S]

Interpreting Vmax and Km shows a simple graphical method for obtaining an approximate value for Km.

This graph shows the kinetic parameters that define the limits of the curve at high and low [S].



At low [S], Km>>[S] and the [S] term in the denominator of the Michaelis-Menten equation becomes insignificant.

low [S]; Km >> [S]

$V_o = \frac{V_{max}[S]}{Km + [S]}$

$V_o = \frac{V_{max}[S]}{Km}$

Since Vmax and Km are both constants, their ratio is a constant.

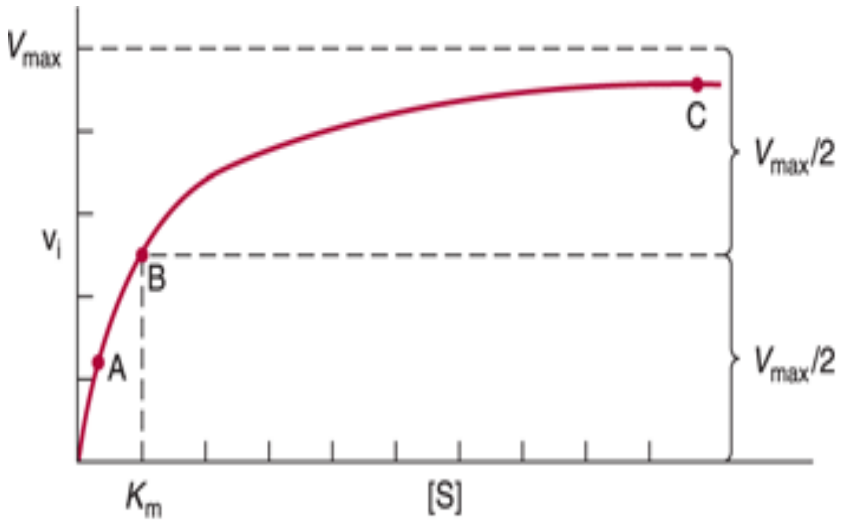
In other words, when [S] is considerably below Km,

V_0 is proportionate to k[S].

The initial reaction velocity, V0 , therefore is directly proportionate to [S].

V_0 exhibits a linear dependence on [S], as observed here.

(First order Reaction)



Source: Murray RK, Bender DA, Botham KM, Kennelly PJ, Rodwell VW, Weil PA: *Harper's Illustrated Biochemistry, 28th Edition:* http://www.accessmedicine.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

When [S] is much less than Km (point A in Figure)

Therefore at concentrations below Km reaction rate is first order i.e. it is directly proportional to the concentration of the substrate.

TOPIC#79

Enzymes...Michaelis-Mention Eq.cont-3

<u>The Michaelis-Menton Equation at high [S]:</u>

At high [S] [S] >> Km

The term Km + [S] is essentially equal to [S].

The Km term in the denominator of the Michaelis- Menten equation becomes insignificant.Replacing Km + [S] with [S] reduces equation.

high [S] [S] >> Km

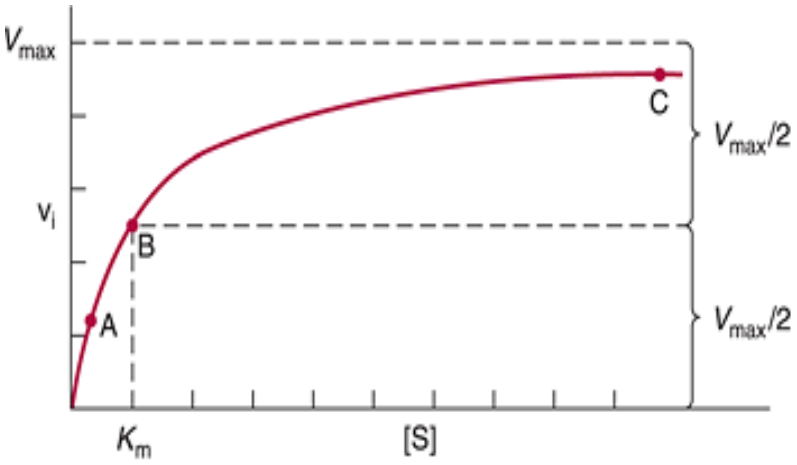
V0 = Vmax [S] Km + [S]

Ignoring Km

V0 = Vmax [S] [S]

V0 = Vmax.

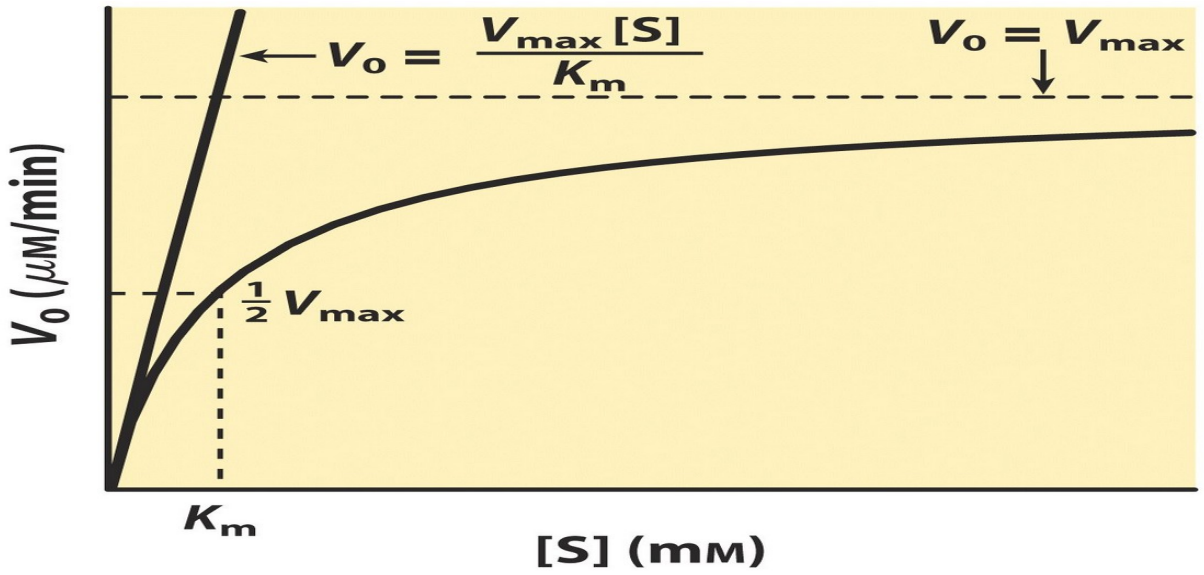
This is consistent with the plateau observed at high [S]. (Zero Order Reaction)



Source: Murray RK, Bender DA, Botham KM, Kennelly PJ, Rodwell VW, Weil PA: *Harper's Illustrated Biochemistry, 28th Edition:* http://www.accessmedicine.com

Copyright © The McGraw-Hill Companies, Inc. All rights reserved.

The rate of reaction is then independent of substrate concentration [S], and is said to be zero order with respect to substrate concentration.



The Michaelis-Menten equation is therefore consistent with the observed dependence of V0 on [S], and the shape of the curve is defined by the terms;

- Vmax/Km at low [S] and
- Vmax at high [S].

TOPIC#80

Enzymes... Order of Reaction

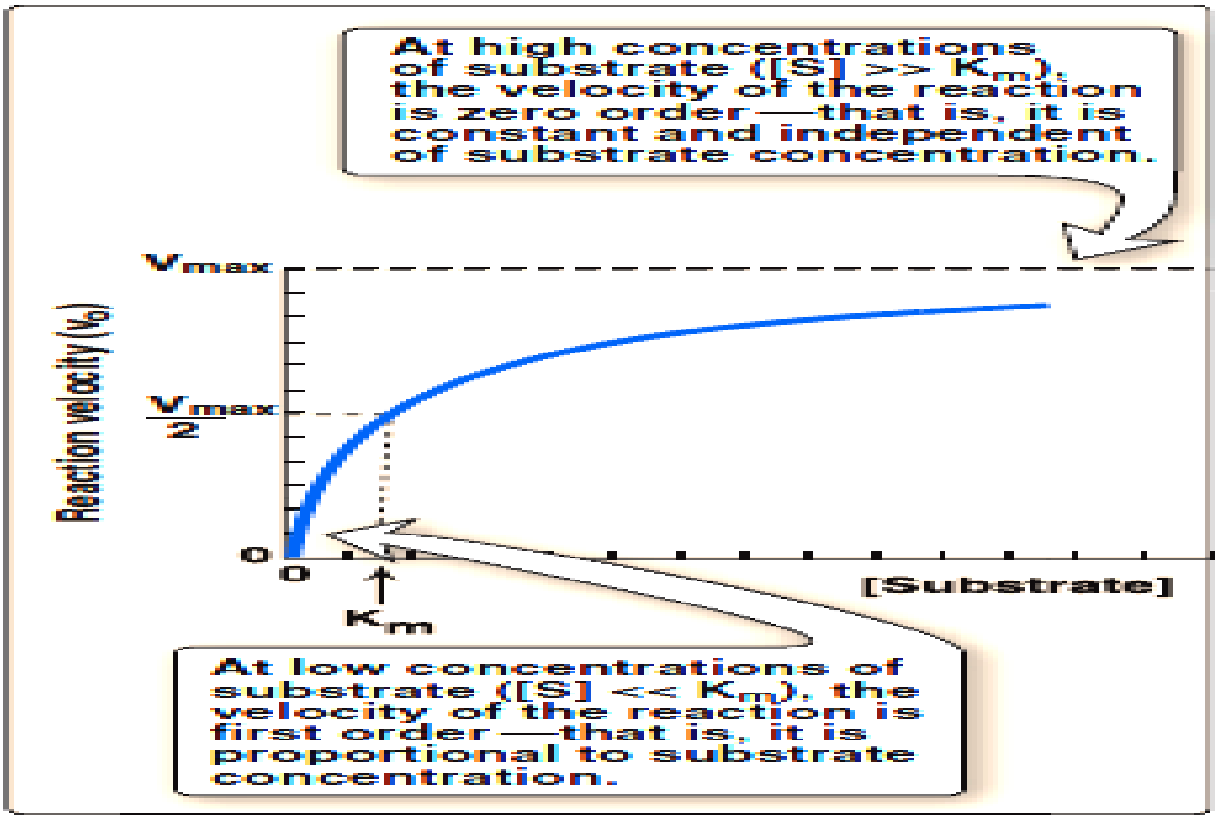
Order of Reaction:

When [S] is much less, then the velocity of the reaction is approximately proportional to the substrate concentration.

The rate of reaction is then said to be first order with respect to substrate.

When [S] is much greater than Km the velocity is constant and equal to Vmax.

The rate of reaction is then independent of substrate concentration, and is said to be zero order with respect to substrate concentration



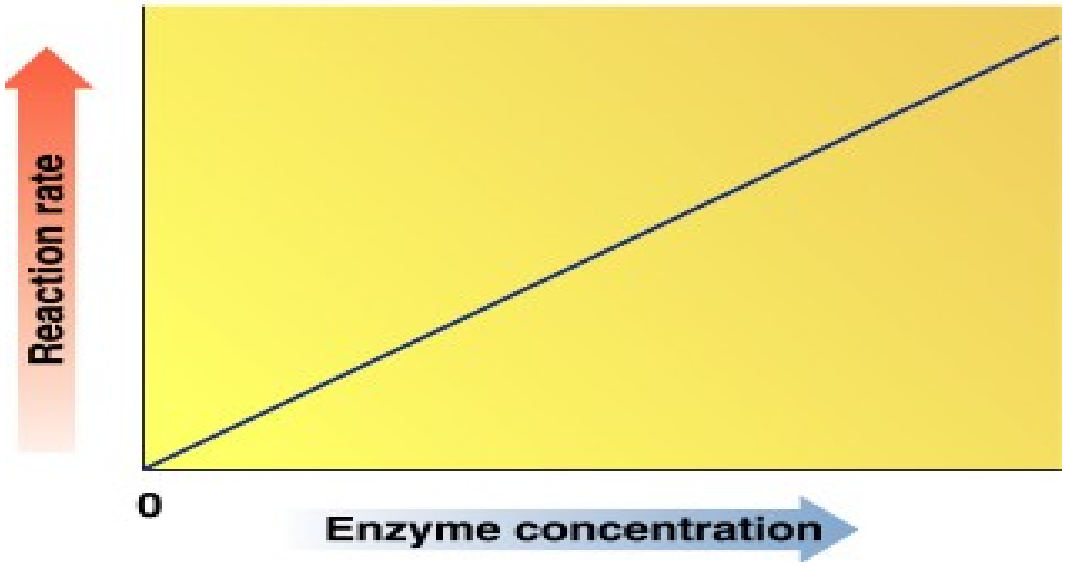
Reaction Orders with Respect to Substrate Concentration:

Order	Rate Equation	Comments
Zero	Rate = k	Rate is independent of substrate concentration
First	First rate = k[S]	Rate is proportional to the first power of substrate concentration
Second	Rate = k[S1][S2]	Rate is proportional to the first power of each of two reactants

Relationship of velocity to enzyme concentration:

The rate of the reaction is directly proportional to the enzyme concentration.

There is a linear relationship between reaction rate and enzyme concentration (at constant substrate concentration)



For example, if the enzyme concentration is halved, the initial rate of the reaction (V_o) as well as that of V_{max} are reduced to one half that of the original.

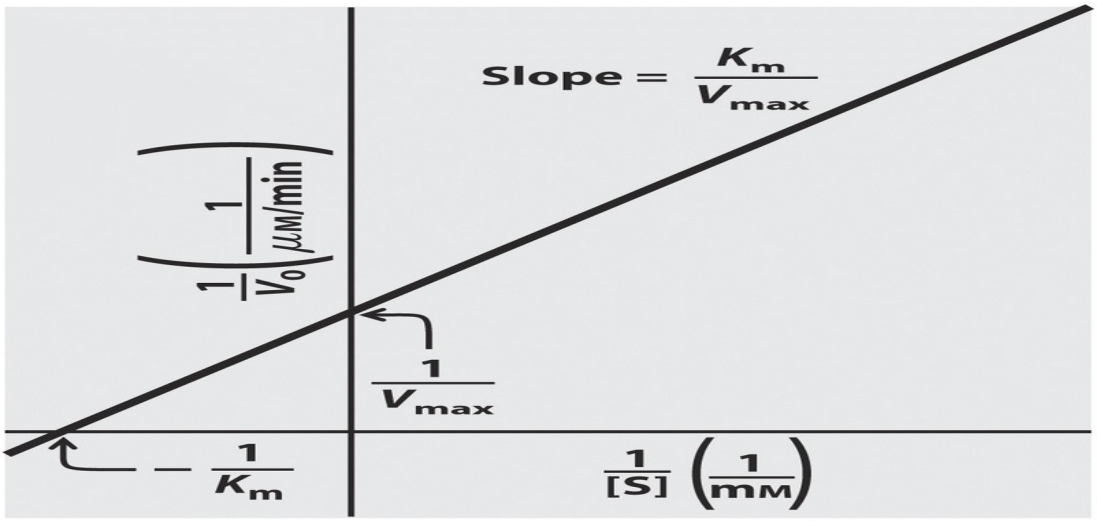
TOPIC#81

Lineweaver-burk plot

Lineweaver-Burke plot:

The Michaelis-Menten equation can be algebraically transformed into Lineweaver-Burke plot, a Double Reciprocal Plot, that is useful in the practical determination of Km and Vmax.

When V_o is plotted against [S], it is not always possible to determine when Vmax has been achieved, because of the gradual upward slope of the hyperbolic curve at high substrate concentrations.



However, if $1/V_o$ is plotted versus 1/[S], a straight line is obtained. This graph, the Lineweaver-Burke plot can be used to calculate Km and Vmax ,as well as to determine the mechanism of action of enzyme inhibitors.

$$V_o = \frac{V_{max}[S]}{K_m + [S]}$$

Lineweaver-Burke transformation is derived simply by taking the reciprocal of both sides of the Michaelis-Menten equation:

$$1 = \frac{K_m + [S]}{}$$

Vo Vmax [S]

1 = Km + [S]

Vo Vmax [S]

1 = Km + [S]

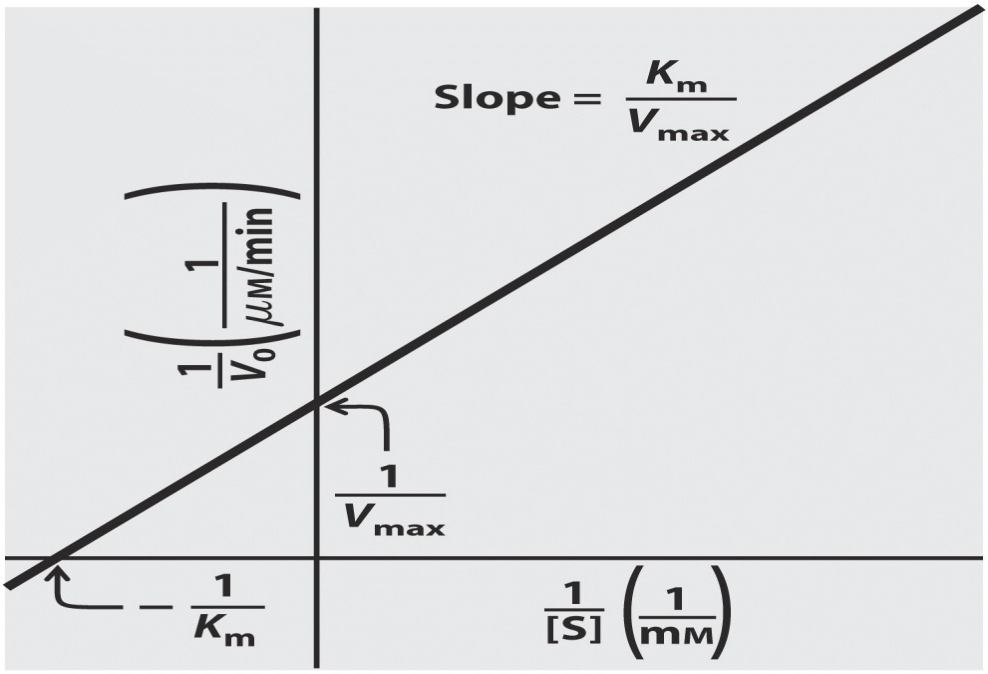
Vo Vmax[S] Vmax[S]

1 = Km + 1_

Vo Vmax[S] Vmax

This form of the Michaelis-Menten equation is called the Lineweaver-Burk equation describing the Lineweaver-Burke plot.

For enzymes obeying the Michaelis-Menten relationship, a plot of 1/Vo versus 1/[S] yields a straight line.



In lineweaver Burke plot y- intercept is 1/Vmax and x-intercept is 1/Km.

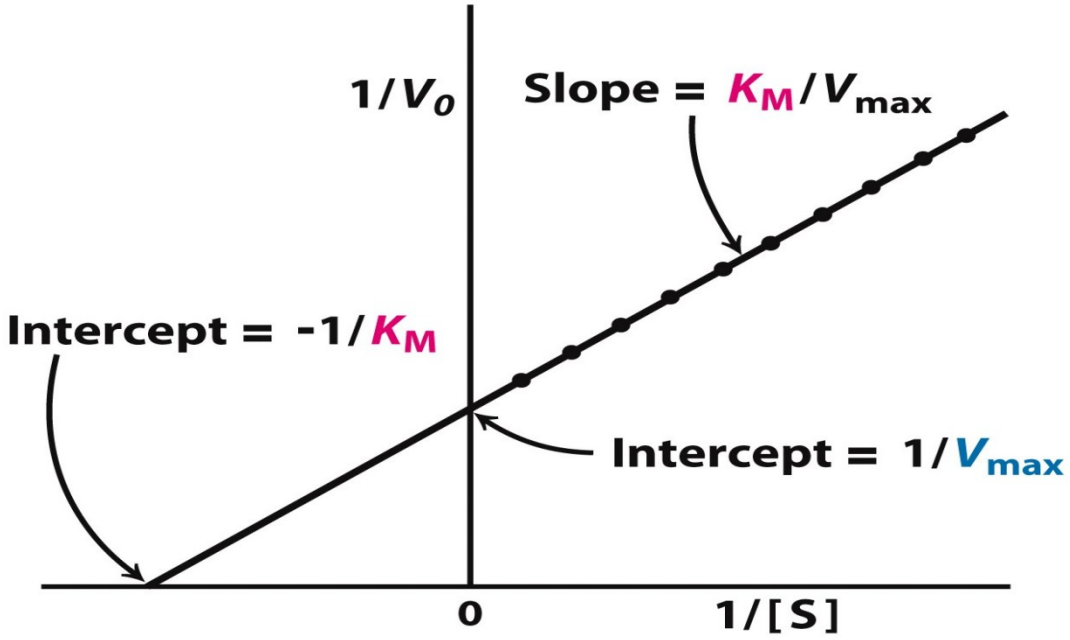


Figure 8.12
***Biochemistry*, Seventh Edition**
© 2012 W. H. Freeman and Company

In lineweaver Burke plot y- Axis intercept is 1/Vmax and x- Axis intercept is -1/Km.

Lineweaver-Burk plot, has the great advantage of allowing a more accurate determination of Vmax,

Which can only be approximated from a simple plot of Vo versus [S].

The double-reciprocal plot of enzyme reaction rates is very useful in distinguishing between certain types of enzymatic reaction mechanisms and in analyzing enzyme inhibition.

TOPIC#82

Inhibition of enzyme activity

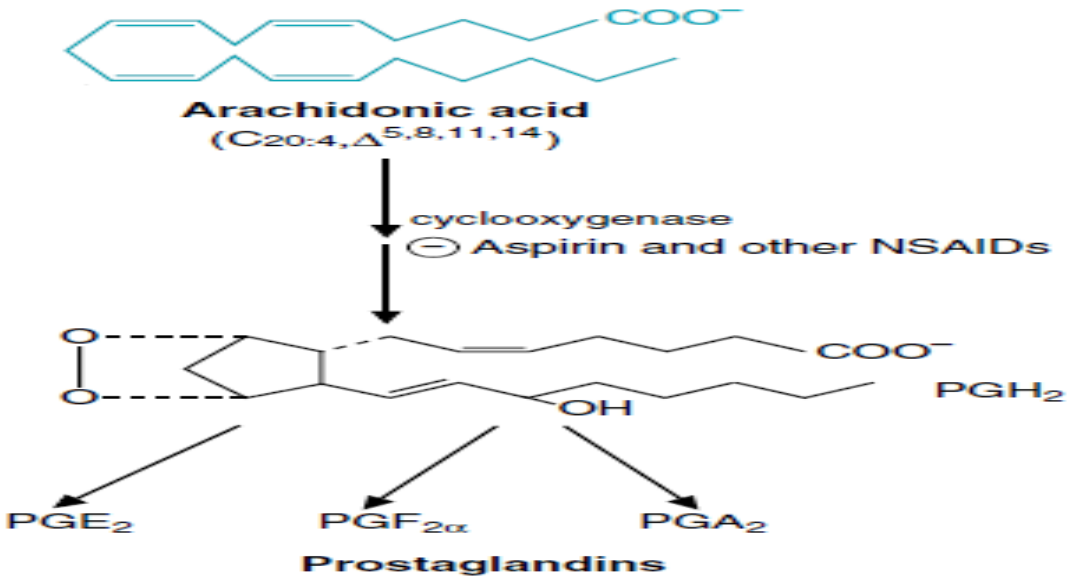
INHIBITION OF ENZYME ACTIVITY:

Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions.

Any substance that can diminish the velocity of an enzyme-catalyzed reaction is called an inhibitor.

Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known.

For example, aspirin inhibits the enzyme (cyclooxygenase) that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain.



For example, aspirin inhibits the enzyme (cyclooxygenase) that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain.

Inhibitors can be classified on the basis of their site of action on the enzyme,on whether they chemically modify the enzyme, or on the kinetic parameters they influence.

Two broad classes of enzyme inhibitors:

1. Reversible

2. Irreversible.

In general, irreversible inhibitors bind to enzymes through covalent bonds.

Reversible inhibitors typically bind to enzymes through noncovalent bonds,

dilution of the enzyme–inhibitor complex results in dissociation of the reversibly

bound inhibitor, and recovery of enzyme activity.

The two most commonly encountered types of reversible inhibition are;

- competitive and
- noncompetitive.

Competitive inhibitors resemble the substrate and compete for binding to the active site of the enzyme.

Noncompetitive inhibitors do not bind at the active site. They bind either free enzyme at a site other than active site or the ES complex.

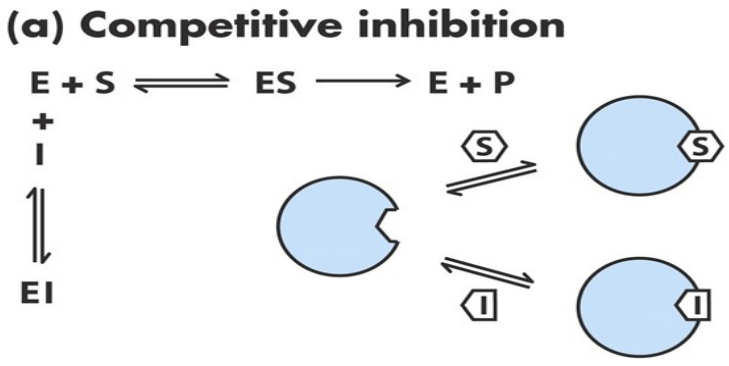
TOPIC#83

Competitive inhibition

Competitive Inhibition:

This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e. active site and, therefore, competes with the substrate for that site.

Competitive inhibitors bind to the enzyme's active site:



While the inhibitor (I) occupies the active site it prevents binding of the substrate to the enzyme. Many competitive inhibitors are compounds that resemble the substrate and combine with the enzyme to form an EI complex, but without leading to catalysis.

Reversible inhibitors bind to enzymes through non covalent bonds.

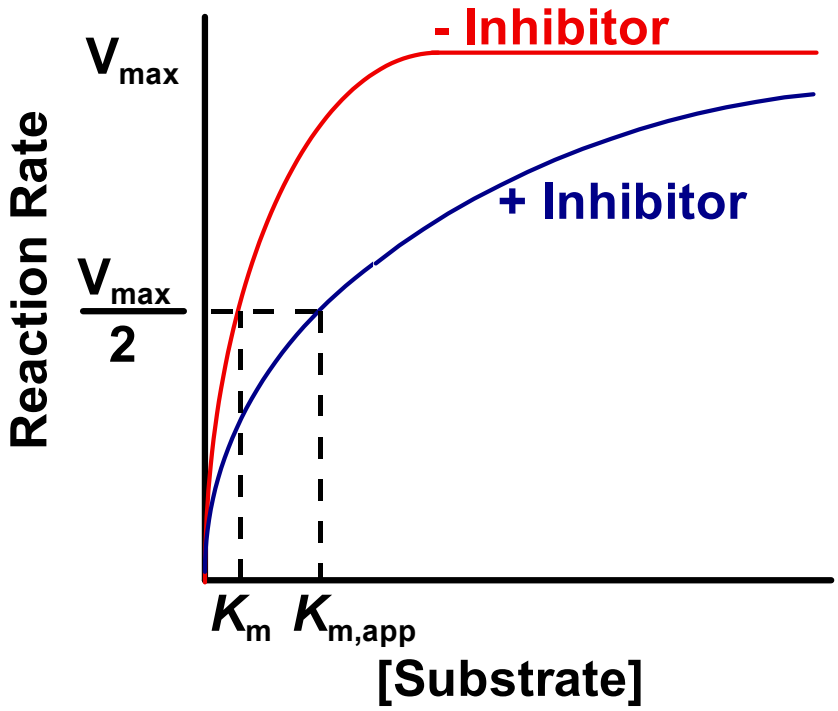
Dilution of the enzyme-inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity.

Since increasing substrate con. results in increasing probability of the formation of ES complexes rather than that of EI complex, increasing [S] decreases the concentration of the EI complex and raises the reaction velocity.

1. Effect on Vmax

The effect of a competitive inhibitor is reversed by increasing [S].

At a sufficiently high substrate concentration, the reaction velocity reaches the Vmax as observed in the absence of inhibitor.



Vmax,app = Vmax Km,app > Km

Even if high conc. of substrate is required Vmax is achieved. Km,app becomes greater than km.

When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal Vmax.Therefore, a competitive inhibitor does not decrease Vmax

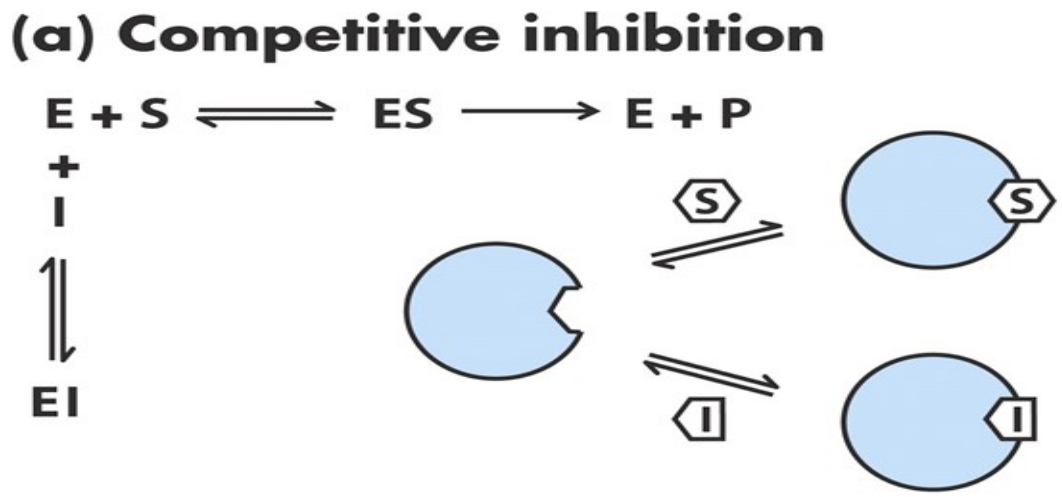
TOPIC#84

Competitive inhibition.. contd

Competitive Inhibition:

This type of inhibition occurs when the inhibitor binds reversibly to the same site that the substrate would normally occupy i.e. active site and, therefore, competes with the substrate for that site.

Competitive inhibitors bind to the enzyme's active site:



While the inhibitor (I) occupies the active site it prevents binding of the substrate to the enzyme. Many competitive inhibitors are compounds that resemble the substrate and combine with the enzyme to form an EI complex, but without leading to catalysis.

Reversible inhibitors bind to enzymes through non covalent bonds.

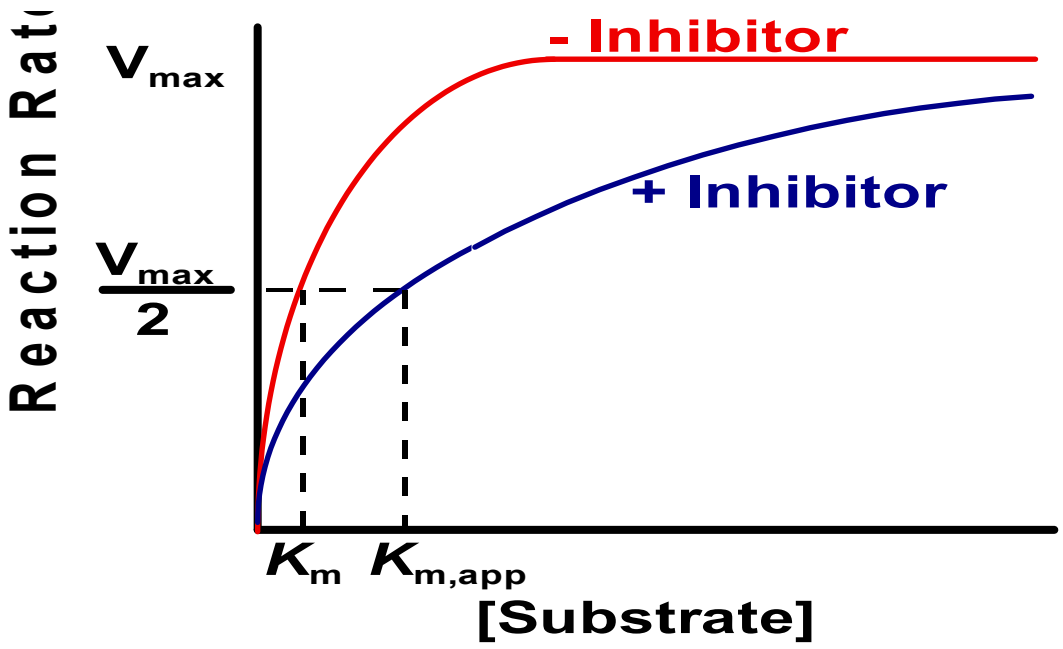
Dilution of the enzyme-inhibitor complex results in dissociation of the reversibly bound inhibitor, and recovery of enzyme activity.

Since increasing substrate con. results in increasing probability of the formation of ES complexes rather than that of EI complex, increasing [S] decreases the concentration of the EI complex and raises the reaction velocity.

1. Effect on Vmax

The effect of a competitive inhibitor is reversed by increasing [S].

At a sufficiently high substrate concentration, the reaction velocity reaches the Vmax as observed in the absence of inhibitor.



Vmax,app = Vmax Km,app > Km

Even if high conc. of substrate is required Vmax is achieved. Km,app becomes greater than km

When [S] far exceeds [I], the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal Vmax.

Therefore, a competitive inhibitor does not decrease Vmax

TOPIC#85

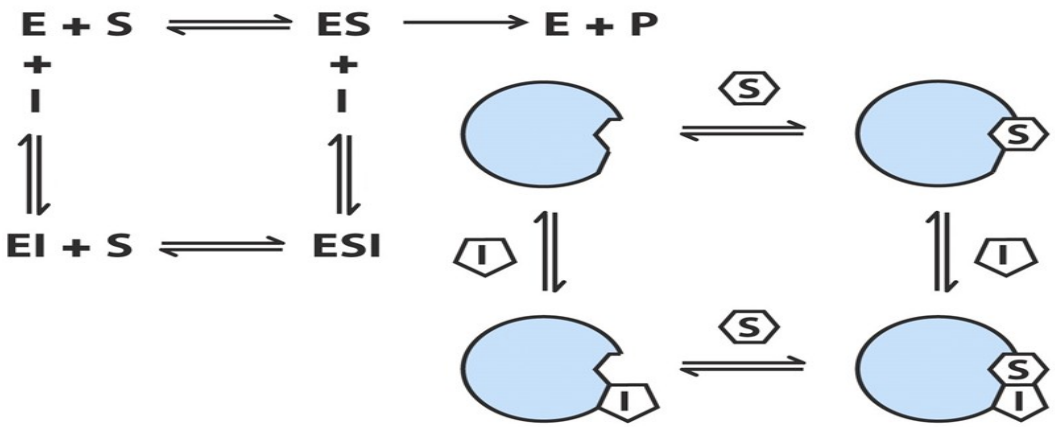
Non-competitive inhibition

Noncompetitive Inhibition:

Inhibitors bind enzymes at sites distinct from the substrate-binding site and

generally bear little or no structural resemblance to the substrate.

Noncompetitive inhibitors bind at a separate site, but may bind to either E or ES.



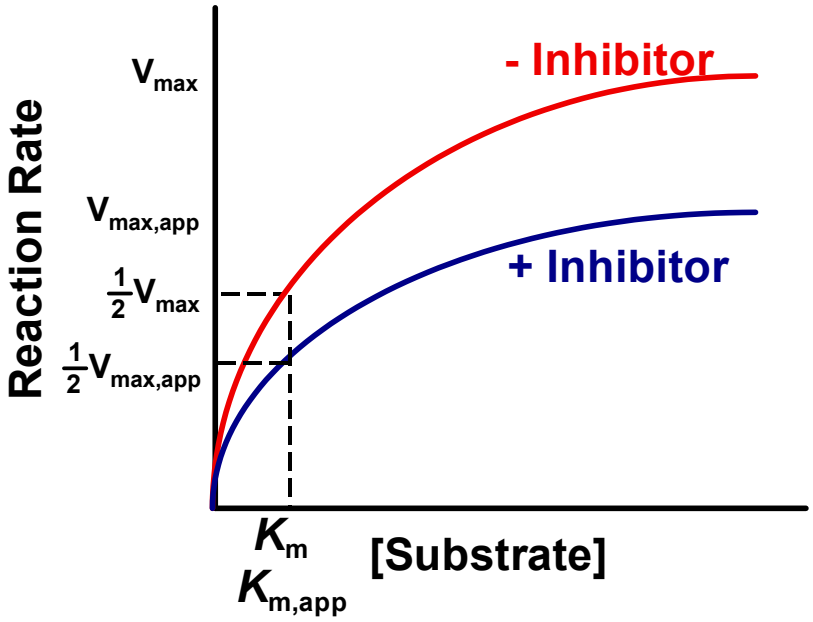
Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.

1. Effect on Vmax

the apparent Vmax changes, because the inhibitor is capable of preventing catalysis regardless of whether the substrate is bound to the enzyme.

Noncompetitive inhibition cannot be overcome by increasing the concentration of substrate.

Thus, noncompetitive inhibitors decrease the Vmax of the reaction.



Vmax decreases in noncompetitive inhibition.

The noncompetitive inhibitor, in effect, lowers the concentration of the active enzyme and therefore decreases the Vmax of the enzyme.

TOPIC#86

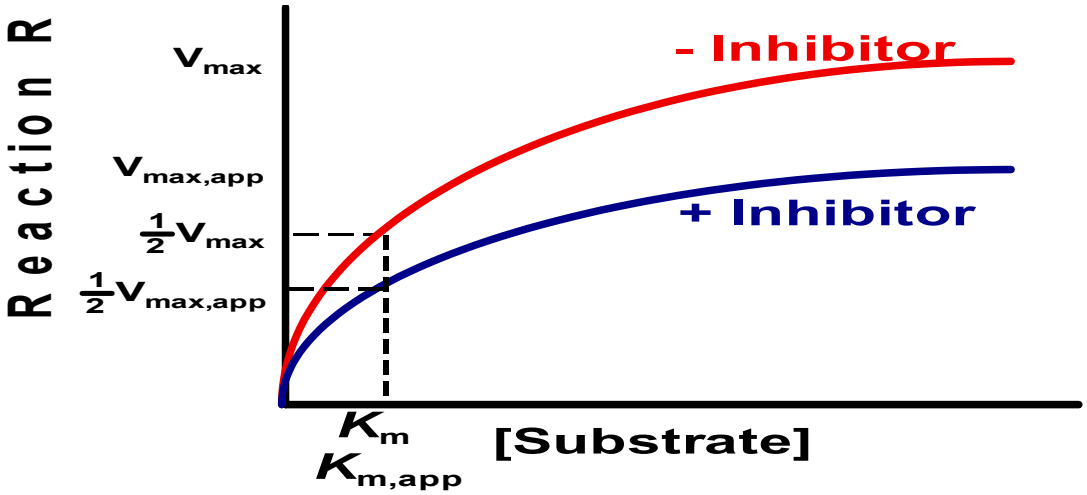
Enzymes.. effect on lineweaver-burk plot

Noncompetitive Inhibition(Contd.)

2. Effect on Km

Noncompetitive inhibitors do not interfere with the binding of substrate to enzyme.

Thus, the enzyme shows the same Km in the presence or absence of the noncompetitive inhibitor.



V max changes but the [S] required to achieve ½ Vmax i.e. Km remains the same

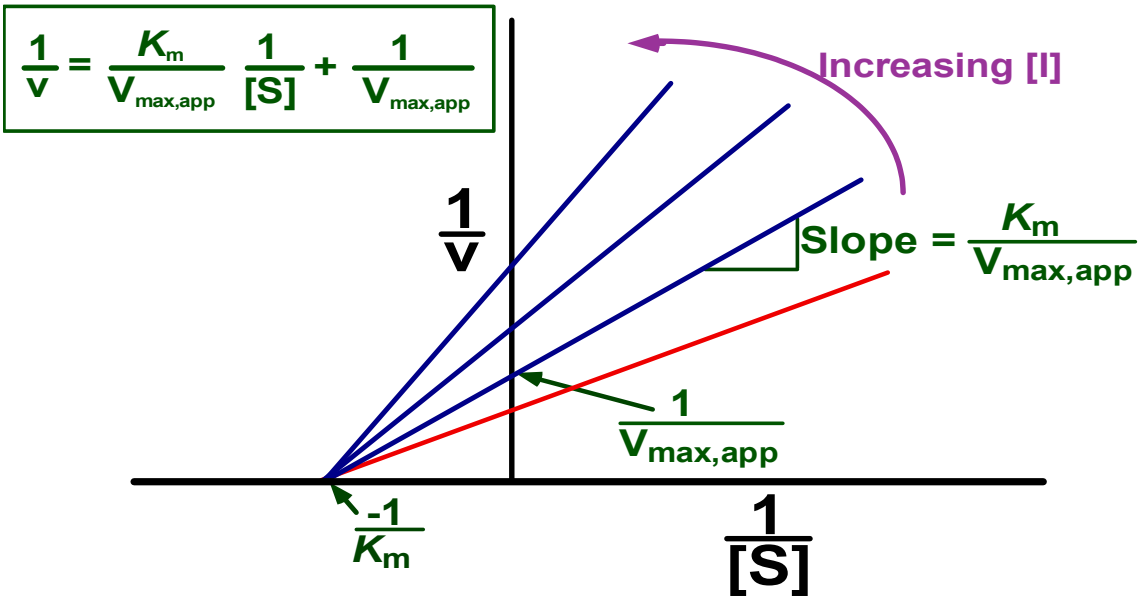
3. Effect on Lineweaver-Burk plot

Noncompetitive inhibition shows a characteristic Lineweaver-Burke plot

the plots of the inhibited and uninhibited reactions intersect at a single point on the x-axis at Km.Km is unchanged

However, the inhibited and uninhibited reactions show; different y axis intercepts

indicating the deccrease in Vmax in the presence of the competitive inhibitor.



While certain inhibitors exhibit characteristics of a mixture of competitive and noncompetitive inhibition, the evaluation of these inhibitors exceeds the scope of our level of study.

TOPIC#87

Water

Water:

Water is the most abundant substance in living systems, making up 70% or more of the weight of most organisms.

The hydrogen bonding between water molecules and the slight tendency of water to ionize are of crucial importance to the structure and function of biomolecules

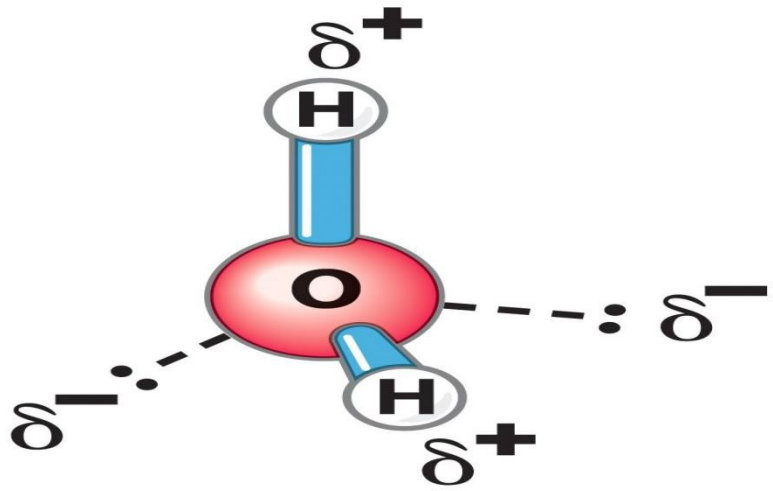
Hydrogen Bonding gives Water its Unusual Properties:

Water has a higher

- melting point,
- boiling point, and
- heat of vaporization than most other common solvents

Each hydrogen atom of a water molecule shares an electron pair with the central oxygen atom

The geometry of the molecule is dictated by the shapes of the outer electron orbitals of the oxygen atom.Valency of Oxygen is 2. atomic no. is 8. octet rule

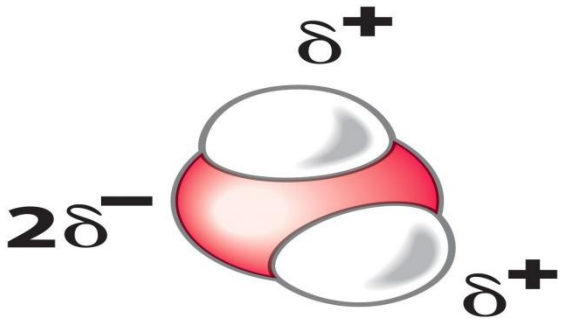


These orbitals describe a rough tetrahedron, with a hydrogen atom at each of two corners and unshared electron pairs at the other two corners.

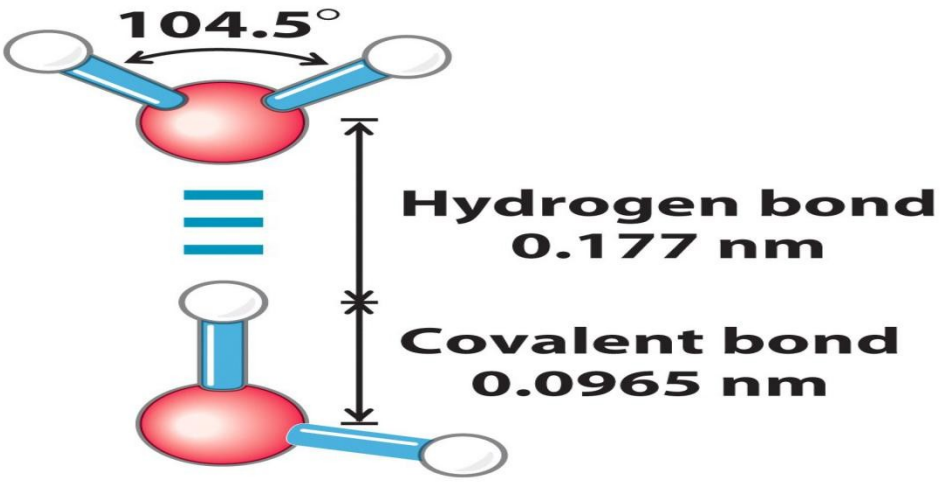
The oxygen nucleus attracts electrons more strongly than does the hydrogen nucleus. That is, oxygen is more electronegative.This means that the shared electrons are more often in the vicinity of the oxygen atom than of the hydrogen

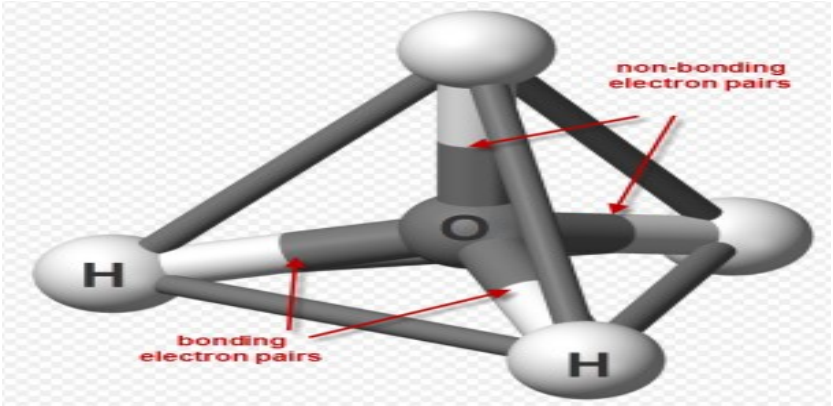
The result of this unequal electron sharing is two electric dipoles in the water molecule.Each hydrogen bears a partial positive charge (δ+), and

the oxygen atom bears a partial negative charge equal to the sum of the two partial positives (2δ-)



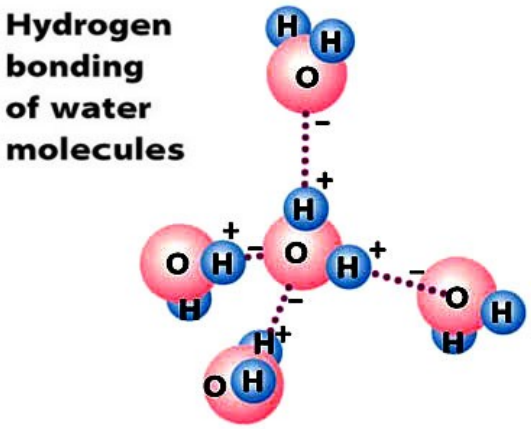
As a result, there is an electrostatic attraction between the oxygen atom of one water molecule and the hydrogen of another, called a hydrogen bond





The nearly tetrahedral arrangement of the orbitals about the oxygen atom allows each water molecule to form hydrogen bonds with as many as four neighboring water molecules

Each water molecule can form two hydrogen bonds involving their hydrogen atoms plus two further hydrogen bonds utilizing the hydrogen atoms attached to neighboring water molecules. These four hydrogen bonds optimally arrange themselves tetrahedrally around each water molecule



Each water molecule can form two hydrogen bonds involving their hydrogen atoms plus two further hydrogen bonds utilizing the hydrogen atoms attached to neighboring water molecules. These four hydrogen bonds optimally arrange themselves tetrahedrally around each water molecule.

TOPIC#88

Ionization of water

Ionization of Water:

Although many of the solvent properties of water can be explained in terms of the uncharged H2O molecule, the small degree of ionization of water into hydrogen ions (H+) and hydroxide ions (OH-) must also be taken into account.Like all reversible reactions, the ionization of water can be described by an equilibrium constant

Pure Water ls Slightly lonized

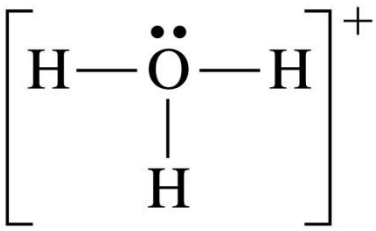
Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium reaction

H2O↔ H+ + OH-

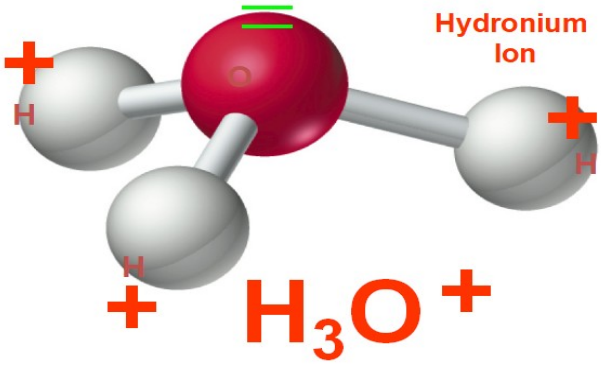
Hydrogen ions formed in water are immediately hydrated to hydronium ions (H3O+)

The ionization of water can be measured by its electrical conductivity

Pure water carries electrical current as H3O+ migrates toward the cathode and OH- toward the anode



Hydronium Ion



The lonization Of Water ls Expressed By an Equilibrium Constant

The degree of ionization of water at equilibrium is small

At 250C only about two out of every 109 molecules in pure water are ionized at any instant

The equilibrium constant for the reversible ionization of water is

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

According to the law of mass action the rate of a chemical reaction is proportional to the product of the masses of the reactants. Necessarily, this implies that for a chemical reaction mixture that is in equilibrium, the ratio between the concentration of reactants and products is constant

$$K_{eq} = \frac{[H^+][OH^-]}{[H_2O]}$$

Since 1 mole (mol) of water weighs 18 g,

1 liter (L) (1000 g) of water contains

1000 ÷ 18 = 55.56 mol

Pure water thus is 55.56 molar(M)

Accordingly, we can substitute 55.5 M in the equilibrium constant expression to yield

Keq = $[H^+][OH^-]/55.5$.Molecular wt. of water is 18.01528 g/mol

Keq = $[H^+][OH^-]/55.5$

On rearranging, this becomes

$(Keq)(55.5)= [H^+][OH^-]= Kw$

Where Kw designates the product (55.5M)(Keq), the ion product of water at 250C

The value for Keq, determined by electrical conductivity measurements of pure water, is 1.8 X 10^{-16} M at 250C

$Kw= (Keq)(55.5)= [H^+][OH^-]$

$Kw = (1.8 \times 10^{-16}M)\ (55.5M)= [H^+][OH^-]$

$Kw = 1.0 \times 10^{-14}\ M^2= [H^+][OH^-]$

$Kw= [H^+][OH^-]= [H^+]^2 = [OH\text{-}]^2$

Solving for$[H^+]$ gives:

$[H^+]= \sqrt{Kw} = \sqrt{}\ 10^{-14}\ M^2$

$[H^+]= 10^{-7}$ M

Thus the product [H+][OH-] in aqueous solutions at 250C always equals 1 x 10-14 M_2

When there are exactly equal concentrations of H+ and OH- , as in pure water, the

solution is said to be at neutral pH.As the ion product of water is constant, whenever [H+] is greater than 1 x 10-7 M, [OH-] must be less than 1 X 10^{-7} M, and vice versa

For example, when [H+] is very high, as in a solution of hydrochloric acid, [OH-] must be very low

TOPIC#89

Buffer System.. working solutions

<u>WORKING EXAMPLES</u>

From the ion product of water we can calculate [H+].If we know [OH-], and vice versa

What is the concentration of H+ in a solution of 0.1 M NaOH?

Solution:

$Kw= [H^+][OH^-]$

With [OH-] = 0.1 M, solving for [H+] gives

$[H^+] = Kw/[OH^-]$

= 1 × 10-14M2/ 0.1M

= 10-14M2/0.1M

= 10-13M

What is the concentration of OH- in a solution with an H+ concentration of 1.3x 10-4M?

Solution:

$Kw= [H^+][OH^-]$

With [H+] = 1.3x 10-4M, solving for [OH-] gives

[OH-] = Kw/[H+]

= 1x10-14M2/ 1.3x 10-4M

= 7.7 x 10-11

The pH Scale:

Designates the H+ and OH- Concentrations.The pH of a solution is defined as the logarithm to the base 10 of the reciprocal of the [H+], i. e, the negative logarithm of the [H+]

pH = log 1/[H+]

= -log[H+]

The symbol p denotes "negative logarithm of"

The pH of water at 25°C, in which H+ and OH– ions are present in equal numbers, is 7.0

pH = -log[1 x 10-7]= 7

For each pH unit less than 7.0, the [H+] is increased tenfold;

for each pH unit above 7.0, it is decreased tenfold

TABLE 2–6 The pH Scale

$[H^+]$ (M)	pH	$[OH^-]$ (M)	pOH*
10^0 (1)	0	10^{-14}	14
10^{-1}	1	10^{-13}	13
10^{-2}	2	10^{-12}	12
10^{-3}	3	10^{-11}	11
10^{-4}	4	10^{-10}	10
10^{-5}	5	10^{-9}	9
10^{-6}	6	10^{-8}	8
10^{-7}	7	10^{-7}	7
10^{-8}	8	10^{-6}	6
10^{-9}	9	10^{-5}	5
10^{-10}	10	10^{-4}	4
10^{-11}	11	10^{-3}	3
10^{-12}	12	10^{-2}	2
10^{-13}	13	10^{-1}	1
10^{-14}	14	10^0 (1)	0

*The expression pOH is sometimes used to describe the basicity, or OH^- concentration, of a solution; pOH is defined by the expression pOH = $-\log [OH^-]$, which is analogous to the expression for pH. Note that in all cases, pH + pOH = 14.

TABLE 2–6 The pH Scale

$[H^+]$ (M)	pH	$[OH^-]$ (M)	pOH*
10^0 (1)	0	10^{-14}	14
10^{-1}	1	10^{-13}	13
10^{-2}	2	10^{-12}	12
10^{-3}	3	10^{-11}	11
10^{-4}	4	10^{-10}	10
10^{-5}	5	10^{-9}	9
10^{-6}	6	10^{-8}	8
10^{-7}	7	10^{-7}	7
10^{-8}	8	10^{-6}	6
10^{-9}	9	10^{-5}	5
10^{-10}	10	10^{-4}	4
10^{-11}	11	10^{-3}	3
10^{-12}	12	10^{-2}	2
10^{-13}	13	10^{-1}	1
10^{-14}	14	10^0 (1)	0

*The expression pOH is sometimes used to describe the basicity, or OH^- concentration, of a solution; pOH is defined by the expression pOH = $-\log [OH^-]$, which is analogous to the expression for pH. Note that in all cases, pH + pOH = 14.

What will be the pH of 0.1 M HCl?

Assuming that being a strong acid HCl is completely dissociated, it's 0.1 M solution will contain 0.1 or 10^{-1} grams H^+ per litre

$pH = - \log [H^+]$

$pH = - \log [10^{-1}]$

$= - [- 1]$

$= 1$

TOPIC#90

Weak acids and bases

Weak Acids and Bases:

Each acid has a characteristic tendency to ionize in an aqueous solution

The stronger the acid, the greater its tendency ionize i.e. to lose its proton

This tendency is measured by an acid dissociation constant

Weak Acids and Bases have Characteristic Acid Dissociation Constants

HCl, H2SO4, and HNO3, commonly called strong acids, are fully ionized in aqueous solutions.The strong bases NaOH and KOH are also completely ionized

Of more interest is the behavior of weak acids and bases-those not completely ionized when dissolved in water

These are ubiquitous in biological systems and play important roles in metabolism and its regulation.

The behavior of aqueous solutions of weak acids and bases is best understood if we first define some terms (in the coming slides)

Acids may be defined as proton donors and bases as proton acceptors

A proton donor and its corresponding proton acceptor make up a conjugate acid-base pair.Acetic acid (CH3COOH), a proton donor, and the acetate anion (CH3COO-), the corresponding proton acceptor, constitute a conjugate acid-base pair, related by the reversible reaction:

CH3COOH ↔ CH3COO- + H+

The tendency of any acid (HA) to lose a proton and form its conjugate base (A-) is defined by the acid dissociation constant (Ka) for the reversible reaction

$$HA \leftrightarrow H+ + A- ;$$

$$Ka = [H+][A-]/[HA]$$

Stronger acids, have larger dissociation constants (Ka) i.e they ionize completely

Weaker acids, have smaller dissociation constants (Ka) i.e. they ionize only partially.

pKa

analogous to pH, pKa is defined by the equation

pKa = log 1/Ka

= -logKa

The stronger the tendency to dissociate a proton, the stronger is the acid and the lower its pKa

TOPIC#91

Water, pH and Buffer Solutions

Working with pKa:

Titration is used to determine the amount of an acid in a given solution.A measured volume of the acid is titrated with a solution of a strong base, usually sodium hydroxide (NaOH), of known concentration.

The NaOH is added in small increments until the acid is consumed (neutralized), as determined with an indicator dye or a pH meter

The concentration of the acid in the original solution can be calculated from the volume and concentration of NaOH added

Consider the titration of a 0.1M solution of acetic acid with 0.1M NaOH at 250C

Two reversible equilibria are involved in the process (here, for simplicity, acetic acid is denoted HAc)

$$H_2O \rightleftharpoons H^+ + OH$$

$$HAc \rightleftharpoons H^+ + Ac^-$$

The equilibria must simultaneously conform to their characteristic equilibrium constants, which are, respectively,

$K_w = [H^+][OH^-]$

$= 1 \times 10^{-14}\ M^2$

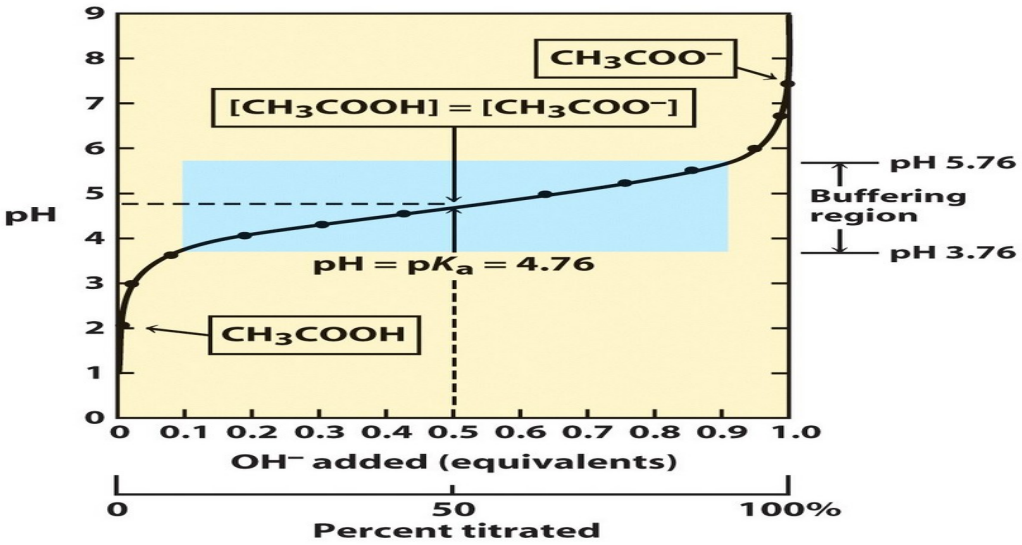
$K_a = [H^+][Ac^-]/[HAc]$

$= 1.74 \times 10^{-5}\ M$

The Dissociation Constant of Acetic Acid is determined by a Conductimetric Method

At the beginning of the titration, the acetic acid is only slightly ionized

As NaOH is gradually added, OH- combines with the free H+ in the solution to form H2O, .As free H+ is removed, HAc dissociates further to satisfy its equilibrium constant



The net result as the titration proceeds is that more and more HAc ionizes, forming Ac-, as the NaOH is added.At the midpoint of the titration, at which exactly 0.5 equivalent of NaOH has been added, one-half of the original acetic acid has undergone dissociation, so that the concentration of the proton donor, [HAc], now equals that of the proton acceptor, [Ac-]

At the midpoint a very important relationship holds:

The pH of the equimolar solution of acetic acid and acetate is exactly equal to the pKa of acetic acid (pKa = 4.76)

At the midpoint of the titration, at which exactly 0.5 equivalent of NaOH has been added, one-half

of the original acetic acid has undergone dissociation, so that the concentration of the proton donor,

[HAc], now equals that of the proton acceptor, [Ac].

At pKa of a weak acid, half of the acid is in dissociated form

whereas other half is un-dissociated.

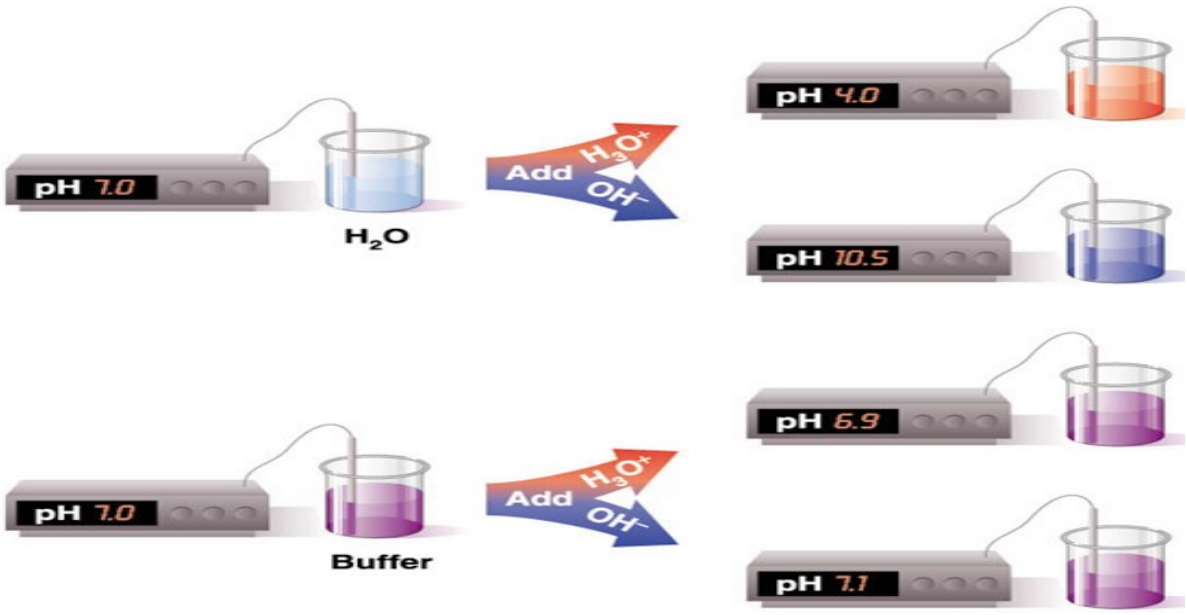
Increasing the pH will result in an increased dissociation and vice versa.

TOPIC#92

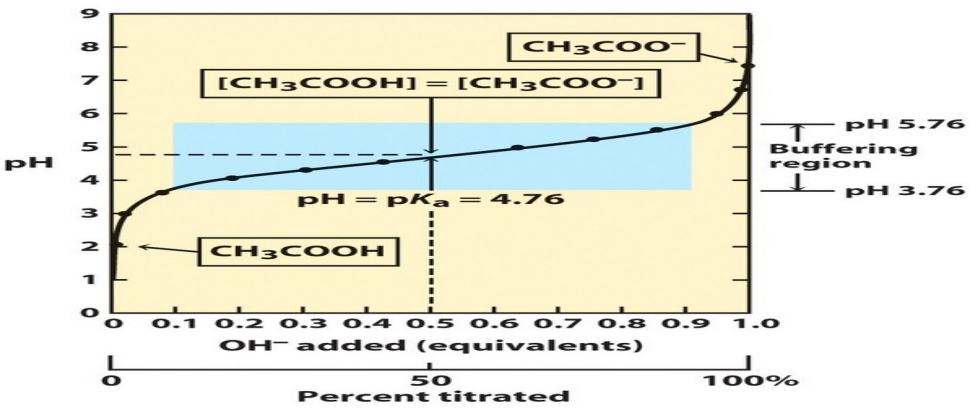
Buffer Solutions

Buffer Solutions:

Buffers are aqueous systems that tend to resist changes in pH when small amounts of acid (H+) or base (OH-) are added.A buffer system consists of a weak acid (the proton donor) and its conjugate base (the proton acceptor),



As an example, a mixture of ;acetic acid andacetate ion, is a buffer system,



If more H+ is added to this solution, it simply shifts the equilibrium to the left, absorbing H+, so the [H+] remains unchanged.

If H+ is removed (e.g. by adding OH-) then the equilibrium shifts to the right, releasing H+ to keep the pH constant

Notice that the titration curve of acetic acid has a relatively flat zone extending about 1 pH unit on either side of its midpoint pH of 4.76.

In this zone, a given amount of H+ or OH- added to the system has much less effect on pH than the same amount added outside the zone

This relatively flat zone is the buffering region of the acetic acid-acetate buffer pair

At the midpoint of the buffering region, where the concentration of the proton donor (acetic acid) exactly equals that of the proton acceptor (acetate), the buffering power is maximal;

The pH at this point in the titration curve of acetic acid is equal to its pKa.The pH of the acetate buffer system does change slightly when a small amount of H+ or OH- is added, but this pH change is very small

compared with the pH change that would result if the same amount of H+ or OH- were added to pure water.

Each conjugate acid-base pair has a characteristic pH zone in which it is an effective buffer.For example, the H2PO4-/HPO42- pair has a pKa of 6.86 and thus can serve as an effective buffer system between approximately pH 5.9 and pH 7.9.

Topic#93

Citric acid cycle-1

Citric Acid Cycle:

Also called Tricarboxylic Acid Cycle (TCA) or Krebs Cycle (after its discoverer, Hans Krebs) . Three names for the same thing.

A series of chemical reactions used by all aerobic organisms to generate energy through the OXIDATION of acetyl CoA … derived from carbohydrates, fatty acids and proteins into carbondioxide, water and chemical energy in the form of ATP

Conversion of pyruvate to activated acetate .Cellular respiration and intermediates for biosynthesis..Conversion of acetate to carbohydrate precursors in the glyoxylate cycle

Cellular Respiration:

For most eukaryotic cells and many bacteria, which live under aerobic conditions and oxidize their organic fuels to CO2 and H2O, glycolysis is the first stage in the complete oxidation of glucose.

Rather than being reduced to lactate, ethanol, or some other fermentation product, ***the pyruvate produced by glycolysis is further oxidized to H2O and CO2. This aerobic phase of catabolism is called respiration.***

Biochemists and cell biologists, however, use the term in a narrower sense to refer to the molecular processes by which cells consume O2 and produce CO2—processes more precisely termed cellular respiration.

Cellular respiration occurs in three major stages.

In the ***first stage***, organic fuel molecules—glucose, fatty acids, and some amino acids are oxidized to yield two-carbon fragments in the form of the acetyl group of acetyl-coenzyme A (acetyl-CoA).

In the ***second stage***, the acetyl groups are fed into the citric acid cycle, which enzymatically oxidizes them to CO2; the energy released is conserved in the reduced electron carriers NADH and FADH2.

In the ***third stage*** of respiration, these reduced coenzymes are themselves oxidized, giving up protons (H+) and electrons. The electrons are transferred to O2—the final electron acceptor—via a chain of electron-carrying molecules known as the respiratory chain.

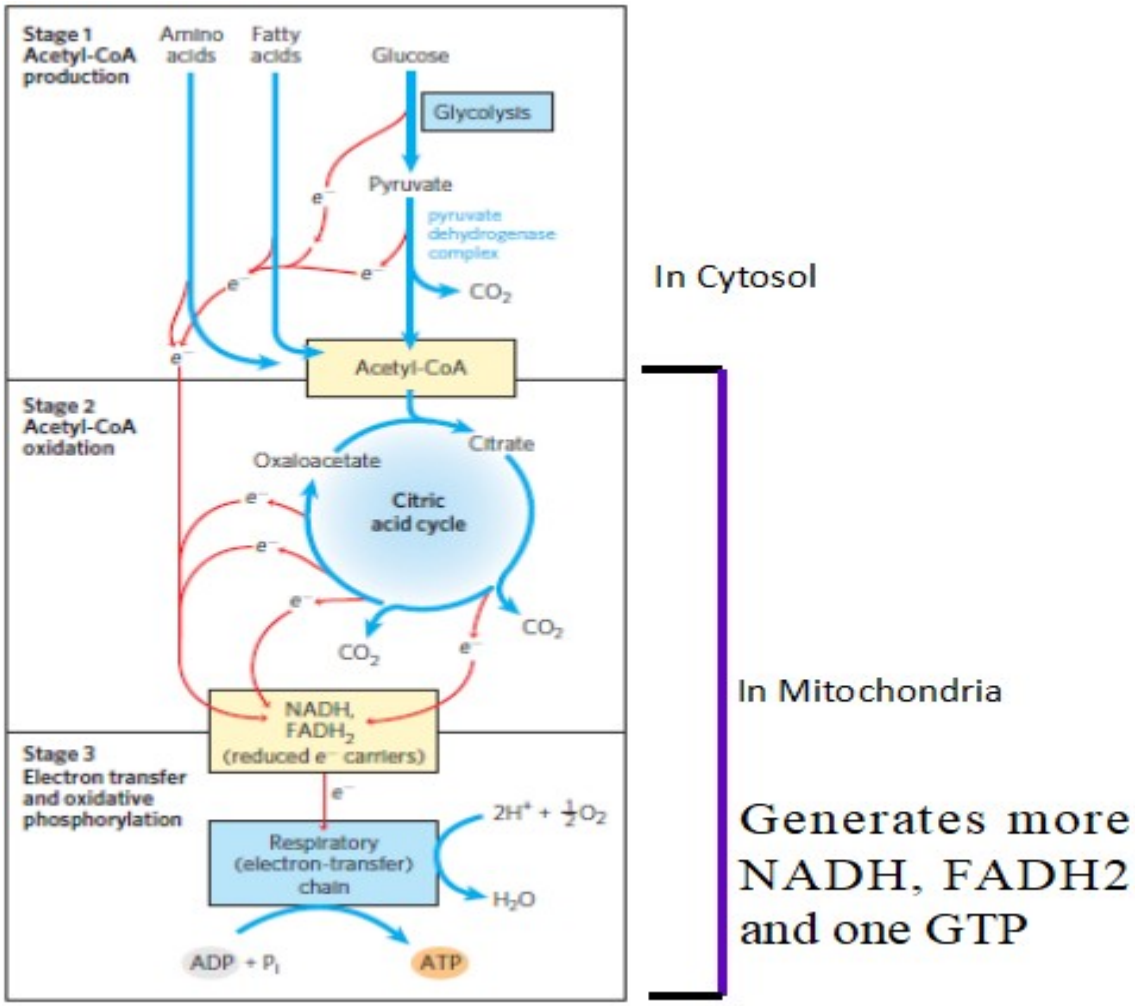


FIGURE 16–1 Catabolism of proteins, fats, and carbohydrates in the three stages of cellular respiration.

Stage 1: oxidation of fatty acids, glucose, and some amino acids yields acetyl-CoA.

Stage 2: oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted.

Stage 3: electrons carried by NADH and FADH2 are funneled into a chain of mitochondrial (or, in bacteria, plasma membrane–bound) electron carriers—the respiratory chain—ultimately reducing O2 to H2O. This electron flow drives the production of ATP.

TOPIC#94

Citric acid cycle-2

Production of Acetyl-CoA:

Pyruvate, the product of glycolysis, is converted to acetyl-CoA.acetyl-CoA is the starting material for the citric acid cycle

Enzyme: Pyruvate dehydrogenase complex (PDH complex)

The PDH complex is composed of multiple copies of three enzymes:

Pyruvate dehydrogenase,

E1, E2 and E3

Mitochondria of eukaryotes and cytosol of bacteria

Pyruvate Is Oxidized to Acetyl-CoA and CO2:

The overall reaction catalyzed by the pyruvate dehydrogenase complex is an ***oxidative decarboxylation***

Irreversible oxidation process in which following things take place

- the carboxyl group is removed from pyruvate as a molecule of CO2 and the two remaining carbons become the acetyl group of acetyl-CoA
- The NADH formed in this reaction gives up a hydride ion (:H2) to the respiratory chain
- NADH carries the two electrons to oxygen or, in anaerobic microorganisms, t an alternative electron acceptor such as nitrate or sulfate.
- The transfer of electrons from NADH to oxygen ultimately generates 2.5 molecules of ATP per pair of electrons.
    - The irreversibility of the PDH complex reaction has been demonstrated by isotopic labeling experiments: the complex cannot reattach radioactively labeled CO2 to acetyl-CoA to yield carboxyl-labeled pyruvate.

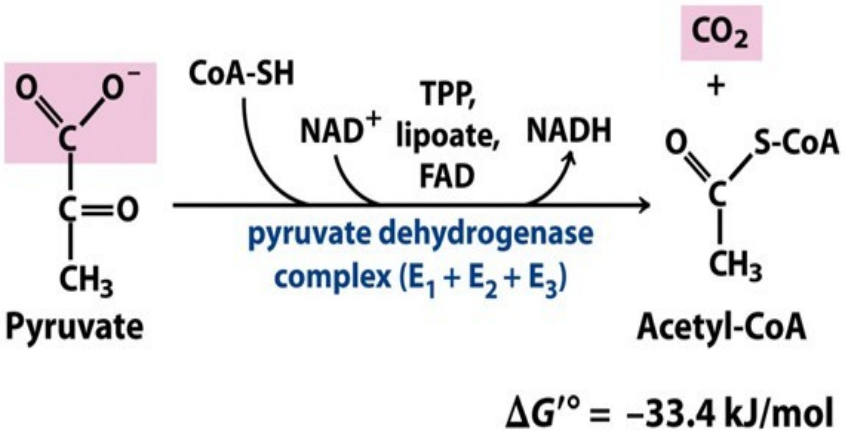


Figure 16-2
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

TOPIC#95

Citric acid cycle-3

The Pyruvate Dehydrogenase Complex (PDH) Requires Five Coenzymes:

The process of conversion of Pyruvate to the acetyl group of acetyl-CoA requires the sequential action of ..

- three different enzymes
- five different coenzymes or prosthetic groups—
- Thiamine pyrophosphate (TPP)
- flavin adenine dinucleotide (FAD)
- Coenzyme A (CoA, sometimes denoted CoA-SH, to emphasize the role of the OSH group)
- Nicotinamide adenine dinucleotide (NAD)
- Lipoate

Four different vitamins required in human nutrition are vital components of this system:

1. thiamine (in TPP),
2. riboflavin (in FAD),

3. niacin (in NAD), and

4. pantothenate (in CoA).

FAD and NAD act as electron carriers ..TPP as the coenzyme of pyruvate decarboxylase.

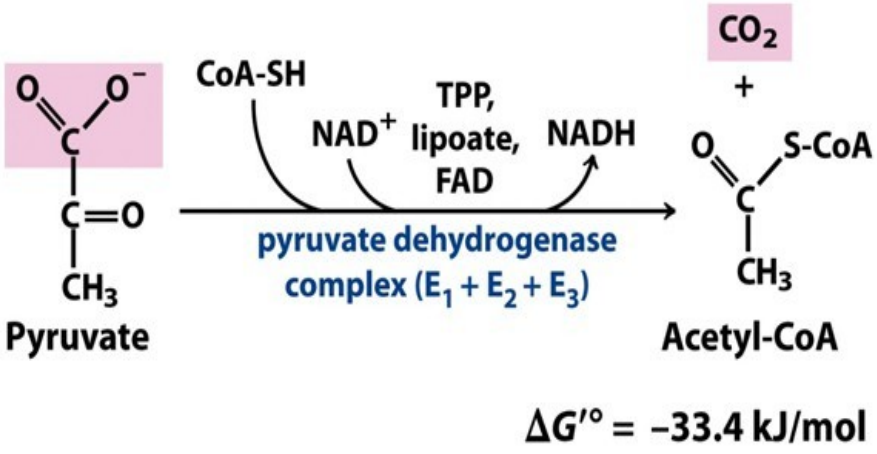


Figure 16-2
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

- **Five coenzymes for the PDH complex**
- **TPP, FAD, NAD, CoA, lipoate**

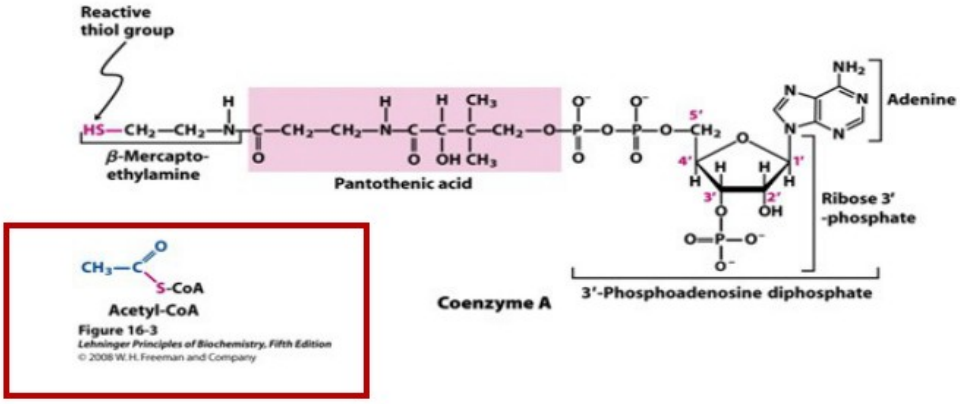
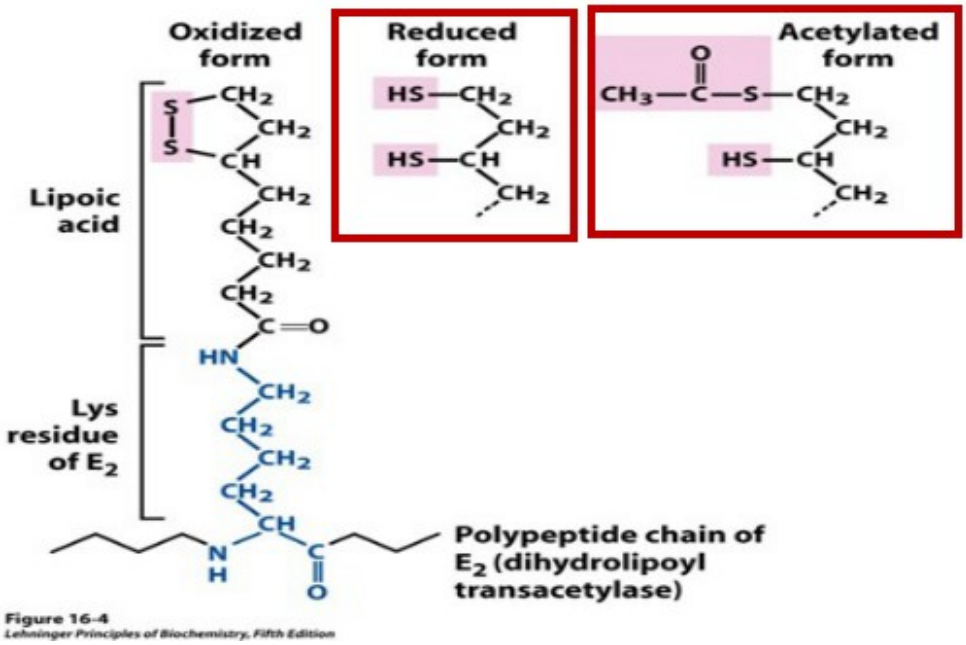


Figure 16-3
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Coenzyme A (CoA). A hydroxyl group of pantothenic acidis joined to a modified ADP moiety by a phosphate ester bond, and its carboxyl group is attached to b-mercaptoethylamine in amide linkage.
The hydroxyl group at the 3' position of the ADP moiety has a phosphoryl group not present in free ADP.

The —SH group forms a thioester with acetate in acetyl-coenzyme A (acetyl-CoA)



Lipoic acid (lipoate) in amide linkage with a Lys residue.

The lipoyllysyl moiety is the prosthetic group of dihydrolipoyl transacetylase (E2 of the PDH complex).The lipoyl group occurs in oxidized (disulfide) and reduced (dithiol) forms and acts as a carrier of both hydrogen and an acetyl (or other acyl) group.

Sulfhydryl (-SH group)

A sulfhydryl is a functional group consisting of a sulfur bonded to a hydrogen atom.
The sulfhydryl group, also called a thiol, is indicated in chemistry nomenclature by "-thiol" as a suffix and "mercapto-" or "sulfanyl" as a prefix.

Thiols have great affinity for soft metals.

Acyl Group

It contains a double bonded oxygen atom and an alkyl group. (R-C=O group is called Acyl Group)

Acetyl Group

In organic chemistry, acetyl is a moiety, the acyl with chemical formula CH_3CO.

(The acetyl group contains a methyl group single-bonded to a carbonyl).

TOPIC#96

PDH complex-Citric acid cycle - 4

Citric Acid Cycle

The Pyruvate Dehydrogenase (PDH) Complex

The PDH Complex Consists of 3 Enzymes

1. Pyruvate dehydrogenase (E1)
2. dihydrolipoyl transacetylase (E2)
3. dihydrolipoyl dehydrogenase (E3)

The PDH complex isolated from mammals is about 50 nm in diameter—more than five times the size of an entire ribosome and big enough to be visualized with the electron microscope

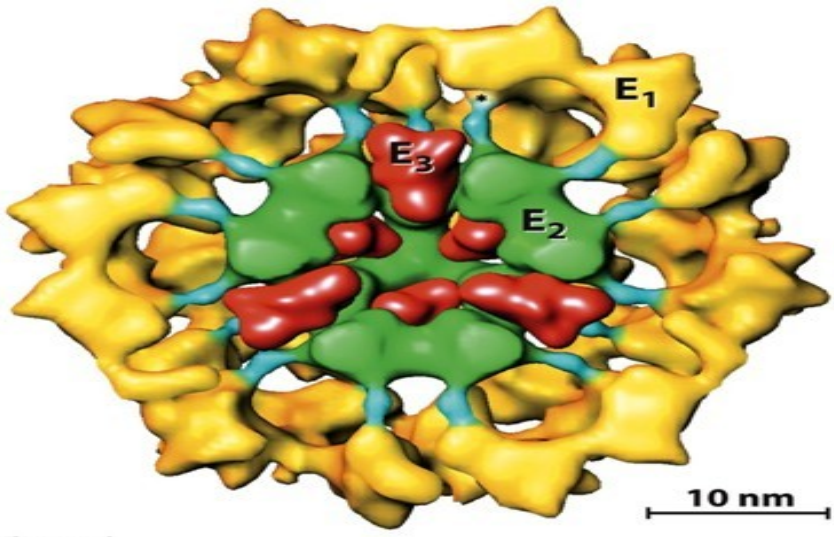


Figure 16-5b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

In the bovine enzyme, 60 identical copies of E2 form a pentagonal dodecahedron (the core) with a diameter of about 25 nm.The core of the Escherichia coli enzyme contains 24 copies of E2.

E2 is the point of connection for the prosthetic group lipoate (blue), attached through an amide bond to the amino group of a Lys residue …reaches outward to touch the active sites of E1 molecules (yellow) arranged on the E2 core.

The domains of E2 are separated by linkers, sequences of 20 to 30 amino acid residues, rich in Ala and Pro and interspersed with charged residues; these linkers tend to assume their extended forms, holding the three domains apart.

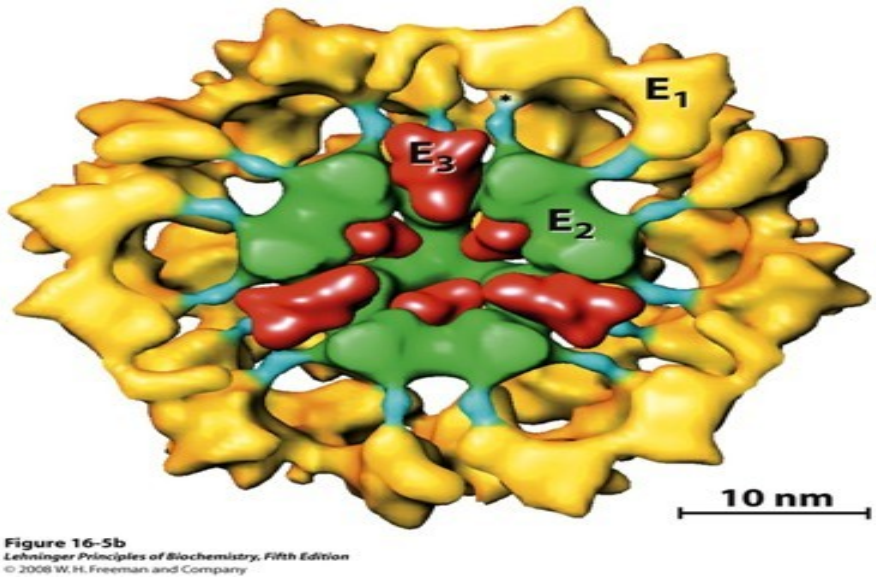
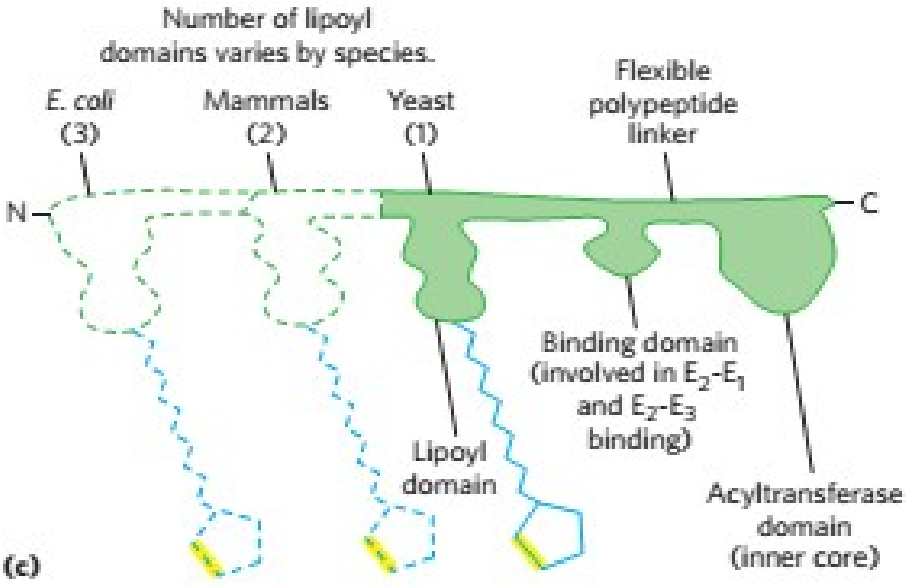


Figure 16-5b
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Several E3 subunits (red) are also bound to the core, where the E2 can reach their active sites. An asterisk marks the site where a lipoyl group is attached to the lipoyl domain of E2. To make the structure clearer, about half of the complex has been cut away from the front. This model was prepared by Z. H. Zhou and colleagues (2001)

E2 has three functionally distinct domains

- the amino-terminal lipoyl domain, containing the lipoyl-Lys residue(s)
- the central E1- and E3-binding domain
- the innercore acyltransferase domain, which contains the acyltransferase active site.



The yeast PDH complex has a single lipoyl domain with a lipoate attached but the mammalian complex has two, and E. coli has three.

The active site of E1 has bound TPP, and that of E3 has bound FAD.

Also part of the complex are two regulatory proteins, a protein kinase and a phosphoprotein phosphatase,

This basic E1–E2–E3 structure has been conserved during evolution and used in a number of similar metabolic reactions.The attachment of lipoate to the end of a Lys side chain in E2 produces a long, flexible arm that can move from the active site of E1 to the active sites of E2 and E3, a distance of 5 nm

The Coenzymes and prosthetic groups of pyruvate dehydrogenase

Cofactor	Location	Function
Thiamine pyrophosphate	Bound to E1	Decarboxylates pyruvate
Lipoic acid	Covalently linked to a Lys on E2 (lipoamide)	Accepts hydroxyethyl carbanion from TPP
CoenzymeA	Substrate for E2	Accepts acetyl group from lipoamide
FAD (flavin)	Bound to E3	reduced by lipoamide
NADH	Substrate for E3	Reduced by FADH2

TOPIC#97 IS NOT AVAIABLE

THIS PAGE IS LEFT BLANK INTENTIONALLY

TOPIC#98

Steps of TCA-8 enzymes-

Citric Acid Cycle-8 enzymes

Acetyl CoA enters TCA and undergoes oxidation.

acetyl-CoA donates its acetyl group to the four-carbon compound oxaloacetate to form the six-carbon citrate.

Citrate is then transformed into isocitrate, also a six-carbon molecule

Isocitrate is dehydrogenated with loss of CO_2 to yield the five-carbon compound a-ketoglutarate (also called oxoglutarate).

a-Ketoglutarate undergoes loss of a second molecule of CO_2 and yields the four-carbon compound succinate..Succinate is then enzymatically converted in three steps into the four-carbon oxaloacetate

oxaloacetate is ready to react with another molecule of acetyl-CoA.

In each turn of the cycle, one acetyl group (two carbons) enters as acetyl-CoA and two molecules of CO_2 leave

one molecule of oxaloacetate (OAA) is used to form citrate and one molecule of oxaloacetate is regenerated. No net removal of oxaloacetate occurs

Four of the eight steps in this process are oxidations, in which the energy of oxidation is very efficiently conserved in the form of the reduced coenzymes NADH and $FADH_2$.

In eukaryotes, cycle takes place in mitochondria - the site of most energy- yielding oxidative reactions and of the coupled synthesis of ATP

In prokaryotes cycle takes place in the cytosol, plasma membrane plays a role analogues to that of inner mitrochondrial membrane in ATP synthesis

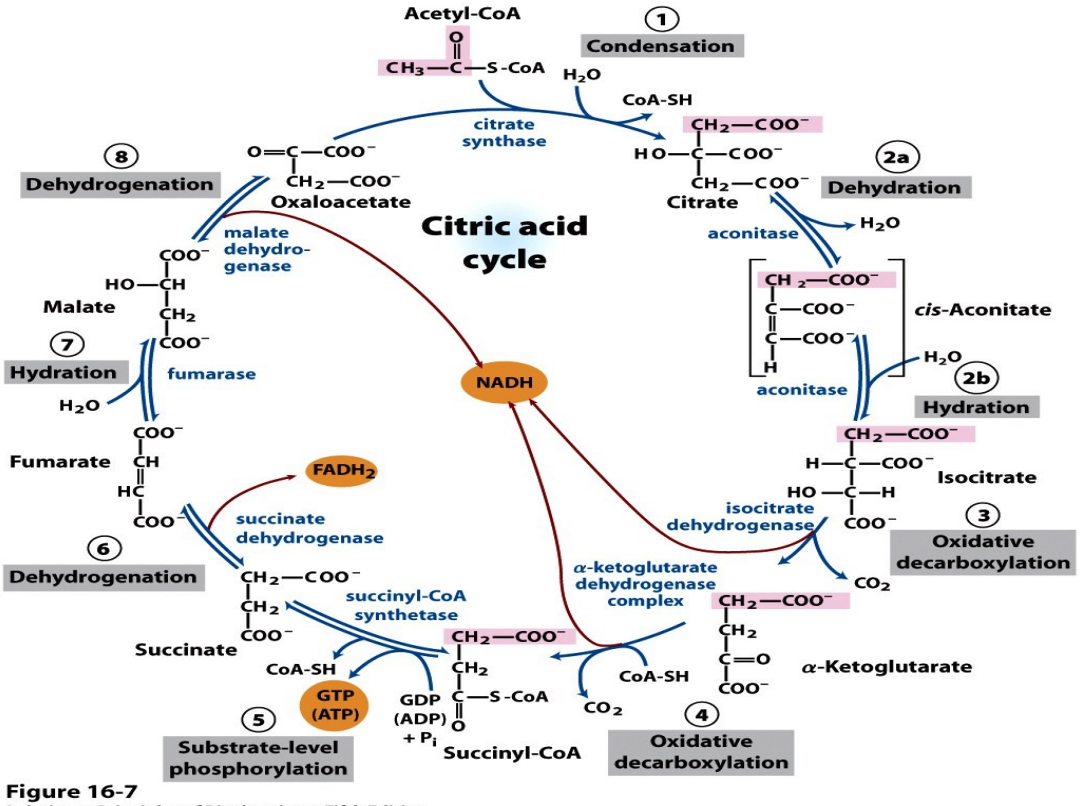


Figure 16-7
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

FIGURE 16-7 Reactions of the citric acid cycle.

The carbon atoms shaded in pink are those derived from the acetate of acetyl-CoA in the first turn of the cycle; these are not the carbons released as CO2 in the first turn. Note that in succinate and fumarate, the two-carbon group derived from acetate can no longer be specifically denoted; because succinate and fumarate are symmetric molecules, C-1 and C-2 are indistinguishable from C-4 and C-3. The red arrows show where energy is conserved by electron transfer to FAD or NAD+, forming FADH2 or NADH + H+. Steps 1, 3, and 4 are essentially irreversible in the cell; all other steps are reversible. The product of step 5 may be either ATP or GTP, depending on which succinyl-CoA synthetase isozyme is the catalyst.

Products

Products of the first turn of the cycle are one GTP (or ATP), three NADH, 1FADH2 and two CO2.

Because two acetyl-CoA are produced from each GLUCOSE molecule, two cycles are required per glucose molecule.

Therefore, at the end of two cycles, the products are: two GTP, six NADH, two FADH2, and four CO2.

	Substrates	Products	Enzyme	Reaction type	Comment
0 / 10	Oxaloacetate + Acetyl CoA + H2O	Citrate + CoA-SH	Citrate synthase	Aldol condensation	irreversible, extends the 4C oxaloacetate to a 6C molecule
1	Citrate	*cis*-Aconitate + H2O	Aconitase	Dehydration	reversible isomerisation
2	*cis*-Aconitate + H2O	Isocitrate		Hydration	
3	Isocitrate + NAD+	Oxalosuccinate + NADH + H+	Isocitrate dehydrogenase	Oxidation	generates NADH (equivalent of 2.5 ATP)
4	Oxalosuccinate	α-Ketoglutarate + CO2		Decarboxylation	rate-limiting, irreversible stage, generates a 5C molecule
5	α-Ketoglutarate	Succinyl-CoA +	α-Ketogluta	Oxidative decarboxylat	irreversible stage, generates NADH (equi

	+ NAD+ + CoA-SH	NADH + H+ + CO2	rate dehydrogenase	ion	valent of 2.5 ATP), regenerates the 4C chain (CoA excluded)

6	Succinyl-CoA + GDP + Pi	Succinate + CoA-SH + GTP	Succinyl-CoA synthetase	substrate-level phosphorylation	or ADP→ATP instead of GDP→GTP,[12] generates 1 ATP or equivalent Condensation reaction of GDP + Pi and hydrolysis of Succinyl-CoA involve the H2O needed for balanced equation.
7	Succinate + ubiquinone (Q)	Fumarate + ubiquinol (QH2)	Succinate dehydrogenase	Oxidation	uses FAD as a prosthetic group (FAD→FADH2in the first step of the reaction) in the enzyme,[12] generates the equivalent of 1.5 ATP
8	Fumarate + H2O	*L*-Malate	Fumarase	Hydration	Hydration of C-C double bond
9	*L*-Malate + NAD+	Oxaloacetate + NADH + H+	Malate dehydrogenase	Oxidation	reversible (in fact, equilibrium favors malate), generates NADH (equivalent of 2.5 ATP)
10	Oxaloacetate + Acetyl CoA + H2O	Citrate + CoA-SH	Citrate synthase	Aldol condensation	This is the same as step 0 and restarts the cycle. The reaction is irreversible and extends the 4C oxaloacetate to a 6C molecule

TOPIC#99

TCA-step 1

Steps of Citric Acid Cycle-1

1. Formation of Citrate (The Citrate Synthase Reaction)

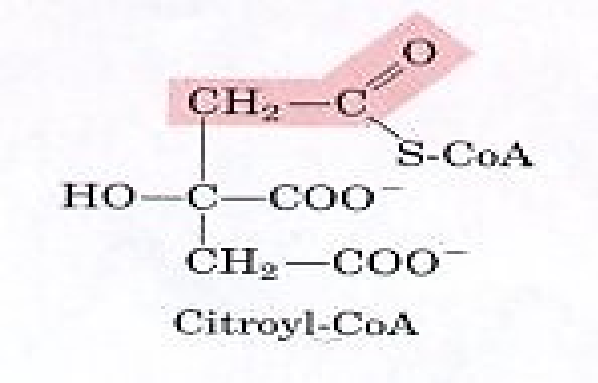
The first reaction of the cycle is the condensation of acetyl-CoA with oxaloacetate to form citrate, catalyzed by citrate synthase

The methyl carbon of the acetyl group is joined to the carbonyl group (C-2) of oxaloacetate.

Citroyl-CoA is a transient intermediate formed on the active site of the enzyme …It rapidly undergoes hydrolysis to release CoA and citrate

The hydrolysis of this high-energy thioester intermediate makes the forward reaction highly exergonic.

The CoA liberated in this reaction is recycled to participate in the oxidative decarboxylation of another molecule of pyruvate by the PDH complex.

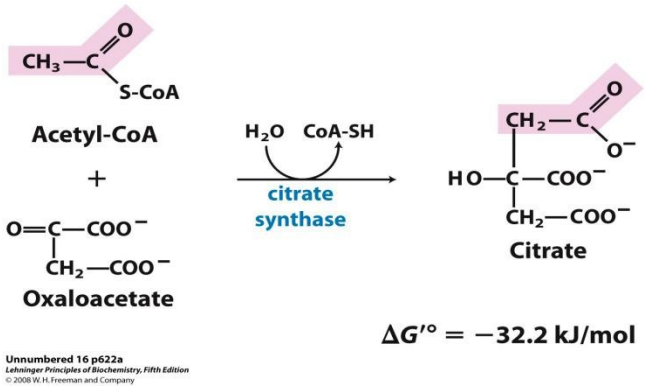


The Citrate Synthase Reaction

Formation of citrate: the condensation of acetyl-CoA with oxaloacetate. -Methyl carbon of acetyl group is joined to the carbonyl group (C-2) of OAA

The hydrolysis of high-energy thioester makes the forward reaction

Citrate sythase: homodimeric enzyme; induce conformational change by OAA binding, creating binding site for acetyl-CoA; when citroyl- CoA has formed, another conformational change brings about thioester hydorlysis; ordered bisubstrate mechanism



The only cycle reaction with C-C bond formation

Essentially irreversible process

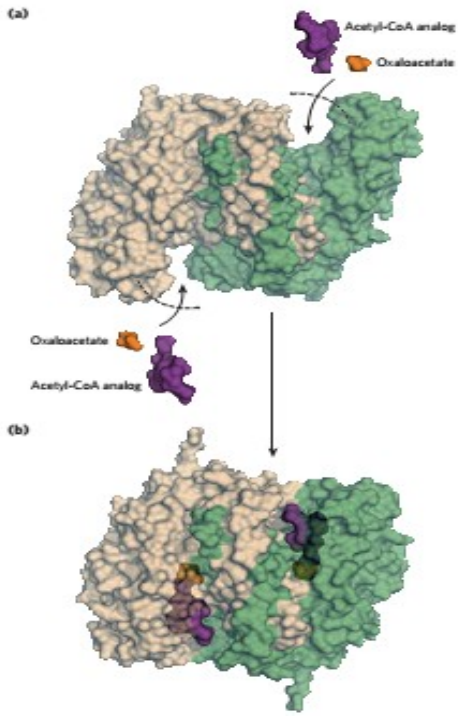
Citrate synthase : the homodimeric enzyme

In these representations one subunit is colored tan and one green.

Each subunit is a single polypeptide with two domains, with the active site between them.

- large and rigid,
- smaller and more flexible
  - Open form of the enzyme alone
  - closed form with bound oxaloacetate and a stable analog of acetyl-CoA

The flexible domain of each subunit undergoes a large conformational change on binding oxaloacetate, creating a binding site for acetyl-CoA.

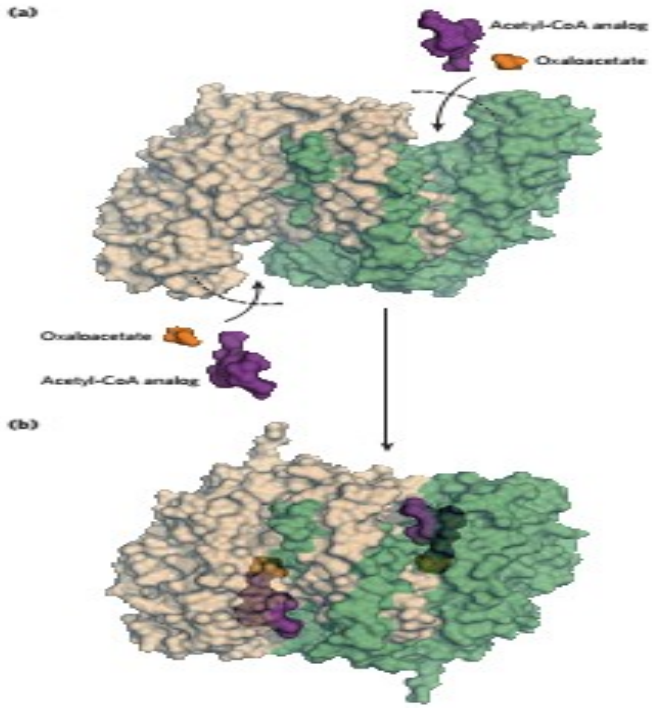


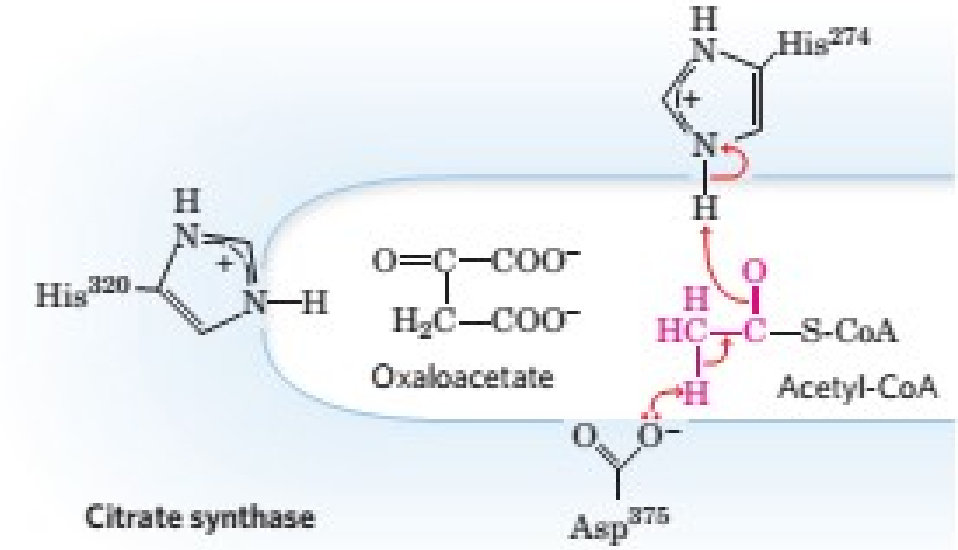
Structure of Citrate Synthase

Ordered bisubstrate mechanism of Citrate synthase.

In the mammalian citrate synthase reaction, oxaloacetate binds first, in a strictly ordered reaction sequence.

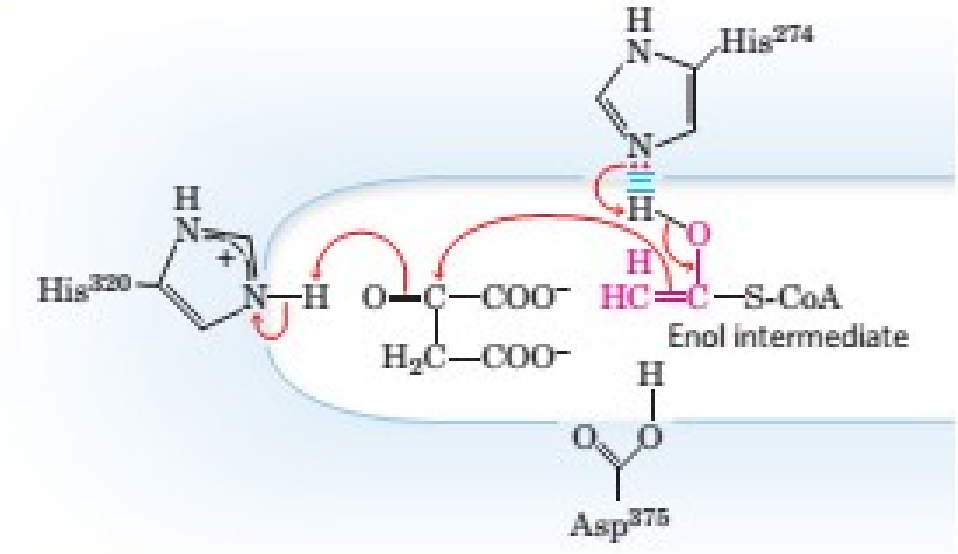
This binding triggers a conformation change that opens up
the binding site for acetyl-CoA. Oxaloacetetate is specifically oriented in the active site of citrate synthase by interaction of its two carboxylates with two positively charged Arg residues


(a)
Acetyl-CoA analog
Oxaloacetate
Oxaloacetate
Acetyl-CoA analog
(b)



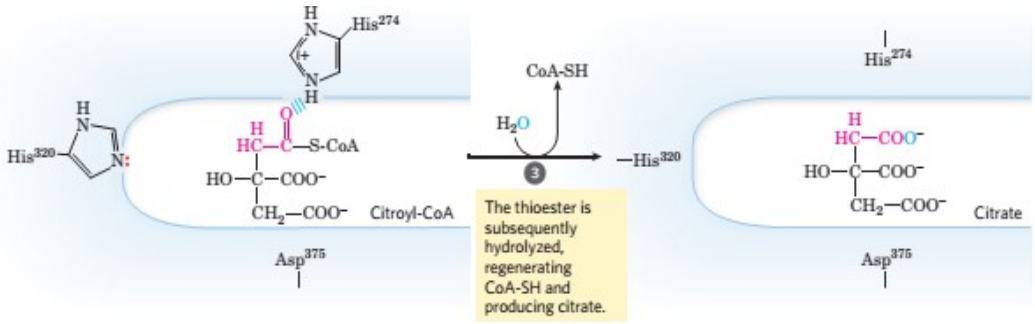
The thioester linkage in acetyl-CoA activates the methyl hydrogens. Asp^{375} abstracts a proton from the methyl group, forming an enolate intermediate. The intermediate is stabilized by hydrogen bonding to and/or protonation by His^{274} (full protonation is shown).

1



The enol(ate) rearranges to attack the carbonyl carbon of oxaloacetate, with His^{274} positioned to abstract the proton it had previously donated. His^{320} acts as a general

2



Ordered bisubstrate mechanism of Citrate synthase.

Oxaloacetate, the first substrate to bind to the enzyme, induces a large conformational change in the flexible domain, creating a binding site for the second substrate, acetyl-CoA.

When citroyl-CoA has formed in the enzyme active site..Another conformational change brings about thioester hydrolysis, releasing CoA-SH.
This induced fit of the enzyme first to its substrate and then to its reaction intermediate decreases the likelihood of premature and unproductive cleavage of the thioester bond of acetyl-CoA.

TOPIC#100

TCA-step2

Steps of Citric Acid Cycle-2

2. Isomerization of Citrate by Aconitase

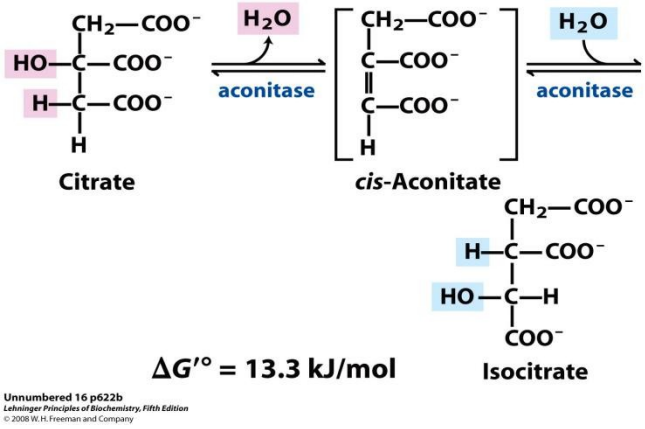
The enzyme aconitase (aconitate hydratase) catalyzes the reversible transformation of citrate to isocitrate

Reaction Intermediate of the tricarboxylic acid is formed..known as cis-aconitate

cis-aconitate does not dissociate from the active site

Aconitase can promote the reversible addition of H2O to the double bond of enzyme-bound cis-aconitate in two different ways,
one leading to citrate and the other to isocitrate

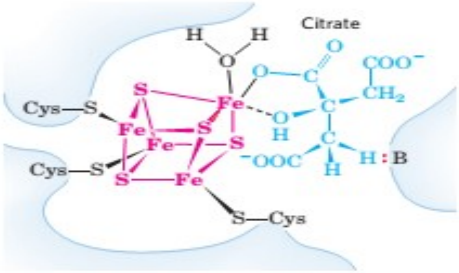
Under physiological conditions in the cell.. Conc. Of isocitrate is very little , the reaction is pulled to the right because isocitrate is rapidly consumed in the next step of the cycle.



Aconitase contains an iron sulfur center which acts both in the binding of the substrate at the active site and in the catalytic addition or removal of H2O.

In iron-depleted cells, aconitase loses its iron-sulfur center and acquires a new role in the regulation of iron homeostasis.

Aconitase is one of many enzymes known to "moonlight" in a second role (protein with 2 jobs)



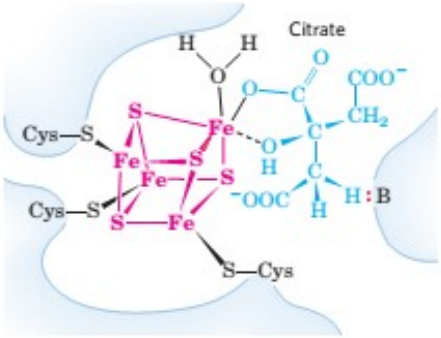
Iron-sulfur center in aconitase.

The iron-sulfur center is in red, the citrate molecule in blue. Three Cys residues of the enzyme bind three iron atoms

the fourth iron is bound to one of the carboxyl groups of citrate and also interacts noncovalently with a hydroxyl group of citrate (dashed bond).

A basic residue (:B) in the enzyme helps to position the citrate in the active site.

The iron-sulfur center acts in both substrate binding and catalytic site.



The Isocitrate Dehydrogenase Reaction

Oxidation of Isocitrate to a-Ketoglutarate and CO2 In the next step

isocitrate dehydrogenase catalyzes oxidative decarboxylation of isocitrate to form a-ketoglutarate

Mn2+ in the active site interacts with the carbonyl group of the intermediate oxalosuccinate, which is formed transiently but does not leave the binding site until decarboxylation converts it to a-ketoglutarate.Mn2+ also stabilizes the enol formed
transiently by decarboxylation.

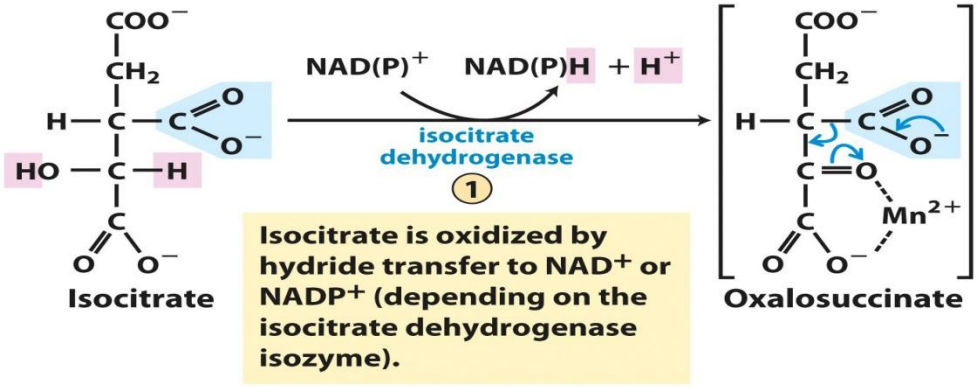
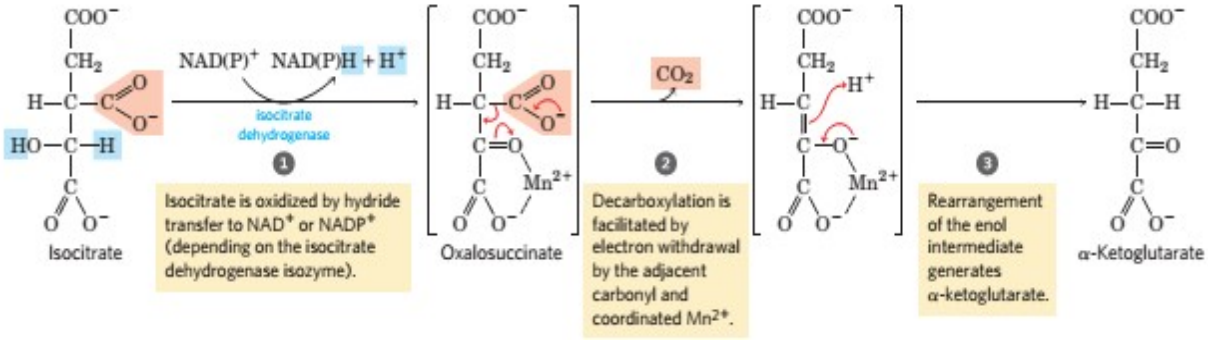


Figure 16-11 part 1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company



Isocitrate dehydrogenase reaction.

In this reaction, the substrate, isocitrate, loses one carbon by oxidative

decarboxylation

<u>Two different forms of isocitrate dehydrogenase (Isozymes)</u>

There are two different forms of isocitrate dehydrogenase in all cells, one requiring NAD+ as electron acceptor and the other requiring NADP+.

The overall reactions are otherwise identical

In eukaryotic cells, the NAD-dependent enzyme occurs in the mitochondrial matrix and serves in the citric acid cycle.

The main function of the NADP-dependent enzyme, found in both the mitochondrial matrix and the cytosol, may be the generation of NADPH, which is essential for reductive anabolic reactions.

TOPIC#101

TCA-STEP 4

Oxidation of →-ketoglutarate

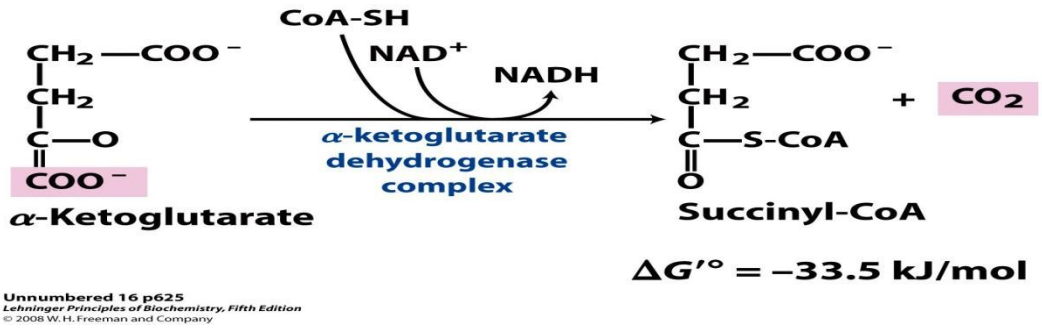
The next step is another oxidative decarboxylation

Oxidation of a-Ketoglutarate to Succinyl-CoA and CO2 .a-ketoglutarate is converted to succinyl-CoA and CO2 by the action of the a-ketoglutarate dehydrogenase complex

NAD+ serves as electron acceptor and CoA as the carrier of the succinyl group.

The energy of oxidation of a-ketoglutarate is conserved in the formation of the thioester bond of succinyl-CoA

This reaction is somehow identical to the pyruvate dehydrogenase reaction and to the reaction sequence responsible for the breakdown of branched chain amino acids



Unnumbered 16 p625
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W.H. Freeman and Company

a-ketoglutarate dehydrogenase complex resembles PDH complex

The a-ketoglutarate dehydrogenase complex closely resembles the PDH complex in both structure and function.

It includes three enzymes, homologous to E1, E2, and E3 of the PDH complex, as well as enzyme-bound TPP, bound lipoate, FAD, NAD, and coenzyme A.

Both complexes are certainly derived from a common evolutionary ancestor.

Although the E1 components of the two complexes are structurally similar, their amino acid sequences differ and, they have different binding specificities

E1 of the PDH complex binds pyruvate, and E1 of the a-ketoglutarate dehydrogenase complex binds a-ketoglutarate. The E2 components of the two complexes are also very similar, both having covalently bound lipoyl moieties.

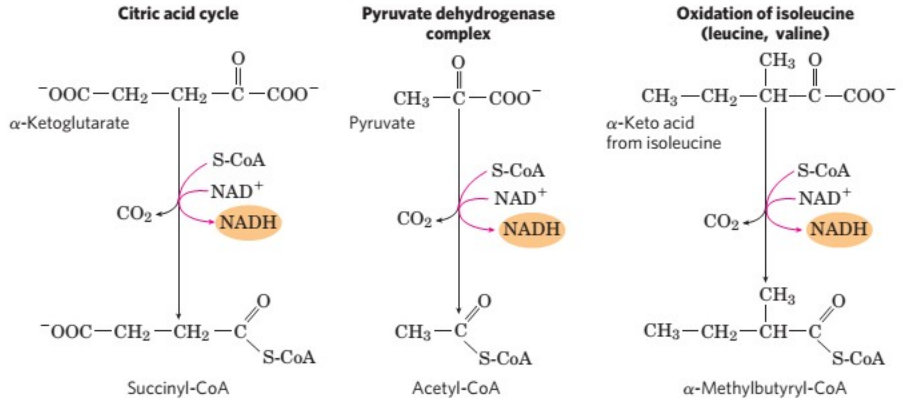
The subunits of E3 are identical in the two enzyme complexes.

The enzymatic complex that degrades branched-chain a-keto acids catalyzes the same reaction sequence using the same five cofactors.

This is a clear case of divergent evolution, in which the genes for an enzyme with one substrate specificity give rise, during evolution, to closely related enzymes with different substrate specificities but the same enzymatic mechanism.

A conserved mechanism for oxidative decarboxylation

The pathways shown use the same five cofactors (thiamine pyrophosphate, coenzyme A, lipoate, FAD, and NAD+) closely similar multienzyme complexes, and the same enzymatic mechanism to carry out oxidative decarboxylations of pyruvate (by the pyruvate dehydrogenase complex) a-ketoglutarate (in the citric acid cycle) and the carbon skeletons of the three branched-chain amino acids, isoleucine leucine, and valine



TOPIC#102

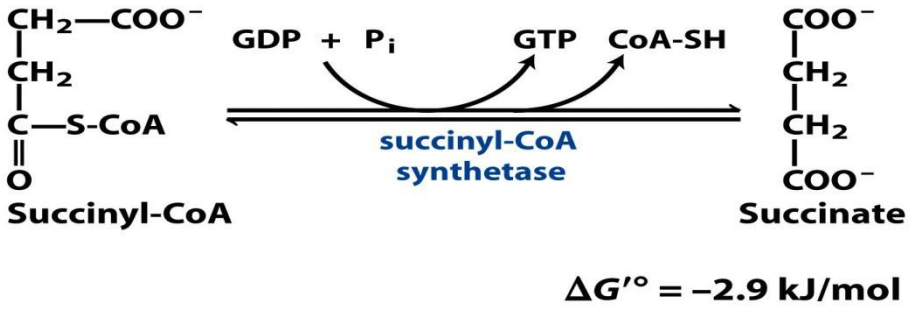
TCA-step 5

Conversion of Succinyl-CoA to Succinate

Succinyl-CoA like acetyl-CoA, has a thioester bond with a strongly negative standard free energy of hydrolysis (G= -36 kJ/mol).

Energy released in the breakage of this bond is used to drive the synthesis of a phosphoanhydride bond in either GTP or ATP, with a net G= - 2.9 kJ/mol.

The enzyme that catalyzes this reversible reaction is called succinyl-CoA synthetase or succinic thiokinase, names indicate the participation of a nucleoside triphosphate in the reaction



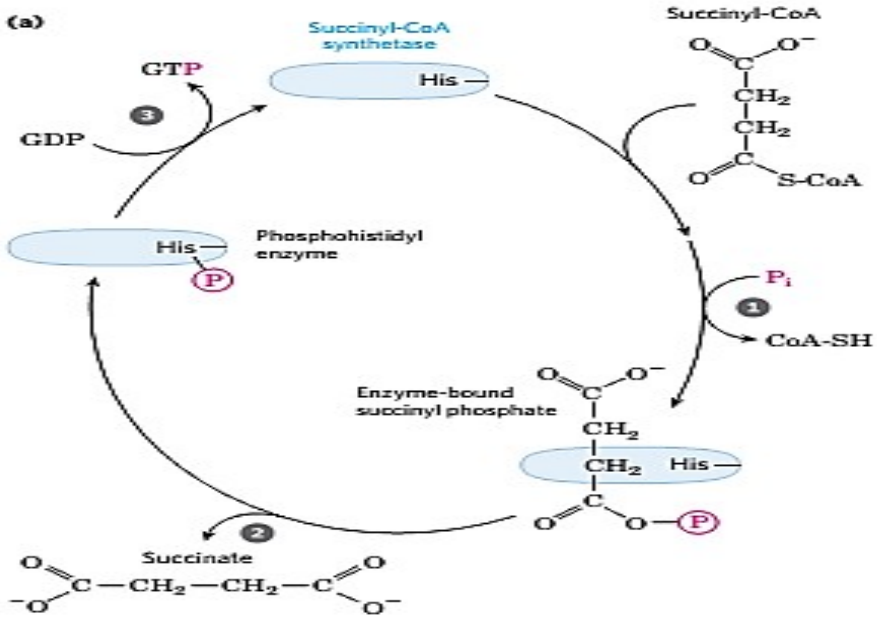
Unnumbered 16 p626
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

The succinyl-CoA synthetase reaction.

In step 1 a phosphoryl group replaces the CoA of succinyl-CoA bound to the enzyme, forming a high-energy acyl phosphate.

In step 2 the succinyl phosphate donates its phosphoryl group to a His residue of the enzyme, forming a high-energy phosphohistidyl enzyme.

In step 3 the phosphoryl group is transferred from the His residue to the terminal phosphate of GDP (or ADP), forming GTP (or ATP).



Nucleoside diphosphate kinase reaction:

The GTP formed by succinyl-CoA synthetase can donate its terminal phosphoryl group to ADP to form ATP, reversible reaction catalyzed by Nucleoside diphosphate kinase reaction

- The enzyme has two subunits,
  a subunit: which has the P –His residue (His246) and the binding site for CoA
- B subunit: confers specificity for either ADP or GDP.

The active site is at the interface between subunits. The crystal structure of succinyl-CoA synthetase reveals two "power helices" (one from each subunit), oriented so that their electric dipoles situate partial positive charges close to the negatively charged P –His (Fig. 16–13b), stabilizing the phosphoenzyme intermediate.

TOPIC#103

TCA-ending

Oxidation of Succinate to Fumarate

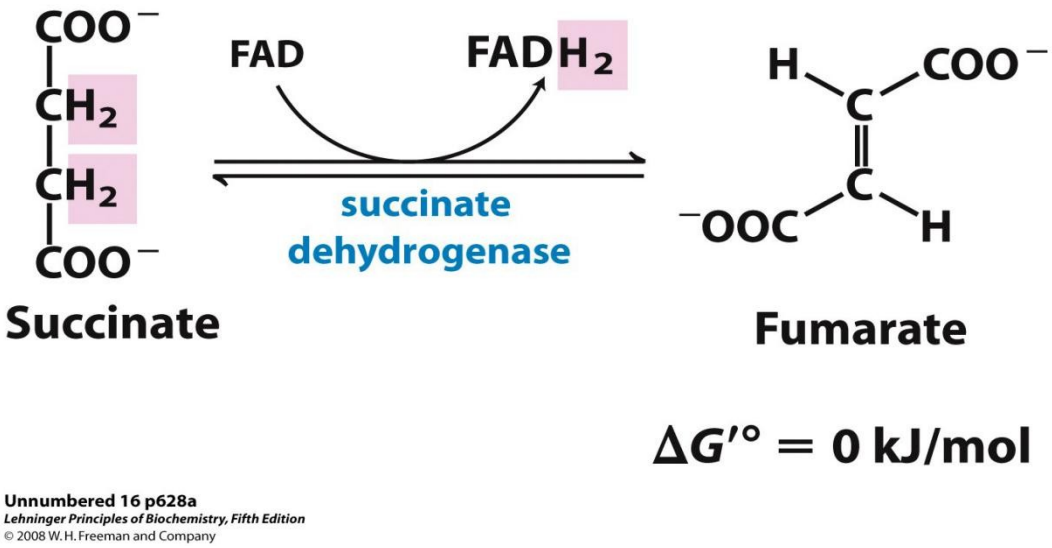
Succinate Dehydrogenase reaction

The succinate formed from succinyl-CoA is oxidized to fumarate by the flavoprotein succinate dehydrogenase

In eukaryotes, succinate dehydrogenase is tightly bound to the mitochondrial inner membrane; in bacteria, to the plasma membrane.

The enzyme contains three different iron-sulfur clusters and one molecule of covalently bound FAD

Electron flow from succinate through these carriers to the final electron acceptor, O2, is coupled to the synthesis of about 1.5 ATP molecules per pair of electrons (respiration-linked phosphorylation).



Unnumbered 16 p628a
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

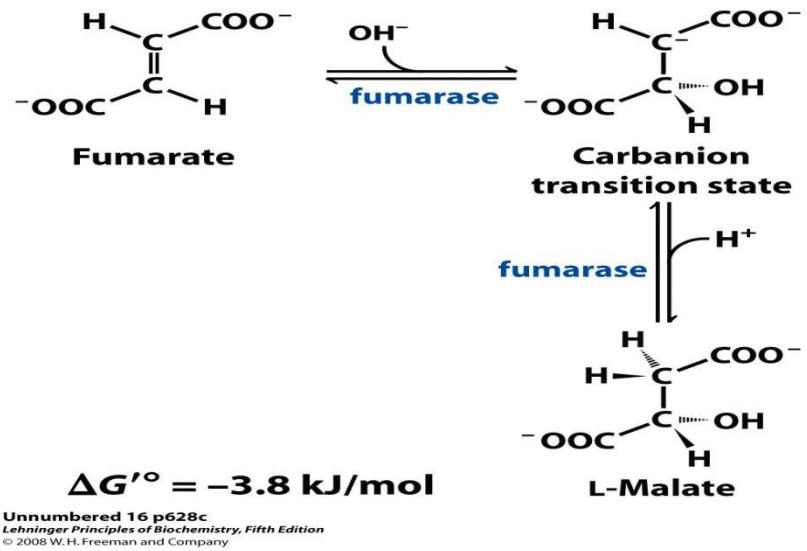
7. Hydration of Fumarate to Malate

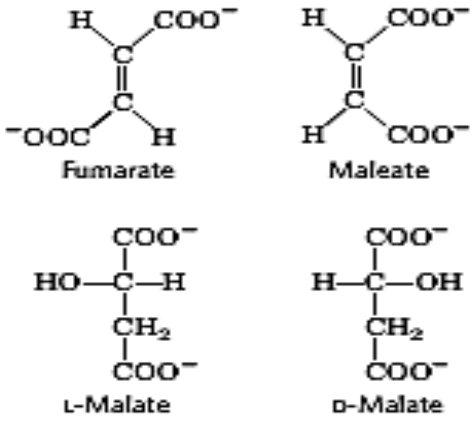
The reversible hydration of fumarate to L-malate is catalyzed by fumarase (fumarate hydratase).

The transition state in this reaction is a carbanion

This enzyme is highly stereospecific; it catalyzes hydration of the trans double bond of fumarate but not the cis double bond of maleate (the cis isomer of fumarate).

In the reverse direction (from L-malate to fumarate),fumarase is equally stereospecific: D-malate is not a substrate.





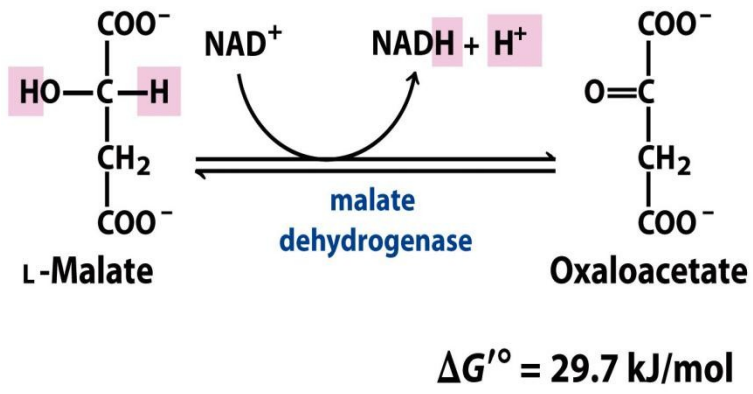
Oxidation of Malate to Oxaloacetate

Oxidation of Malate to Oxaloacetate Is the last reaction of the citric acid cycle

NAD-linked L-malate dehydrogenase catalyzes the oxidation of L-malate to oxaloacetate

In intact cells oxaloacetate is continually removed by the highly exergonic citrate synthase reaction

This keeps the concentration of oxaloacetate in the cell remains extremely low pulling the malate dehydrogenase reaction toward the formation of oxaloacetate.



Unnumbered 16 p628e
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Interesting facts

The individual reactions of the citric acid cycle were initially worked out in vitro, using minced muscle tissue

Later, the pathway and its regulation were studied in vivo.

By using radioactively labeled precursors such as [14C] pyruvate and [14C]acetate

Carbon-14, 14C, or radiocarbon, is a radioactive isotope of carbon with an atomic nucleus containing 6 protons and 8 neutrons.

Researchers have traced the fate of individual carbon atoms through the citric acid cycle.

Direct and Indirect ATP Yield

TABLE 16–1 Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

Reaction	Number of ATP or reduced coenzyme directly formed	Number of ATP ultimately formed*
Glucose ⟶ glucose 6-phosphate	−1 ATP	−1
Fructose 6-phosphate ⟶ fructose 1,6-bisphosphate	−1 ATP	−1
2 Glyceraldehyde 3-phosphate ⟶ 2 1,3-bisphosphoglycerate	2 NADH	3 or 5†
2 1,3-Bisphosphoglycerate ⟶ 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate ⟶ 2 pyruvate	2 ATP	2
2 Pyruvate ⟶ 2 acetyl-CoA	2 NADH	5
2 Isocitrate ⟶ 2 α-ketoglutarate	2 NADH	5
2 α-Ketoglutarate ⟶ 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA ⟶ 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate ⟶ 2 fumarate	2 $FADH_2$	3
2 Malate ⟶ 2 oxaloacetate	2 NADH	5
Total		30–32

*This is calculated as 2.5 ATP per NADH and 1.5 ATP per $FADH_2$. A negative value indicates consumption.

†This number is either 3 or 5, depending on the mechanism used to shuttle NADH equivalents from the cytosol to the mitochondrial matrix; see Figures 19–30 and 19–31.

Table 16-1
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

TABLE 16-1 Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

TOPIC#104

fatty acid metabolism-1

Fatty Acid Catabolism

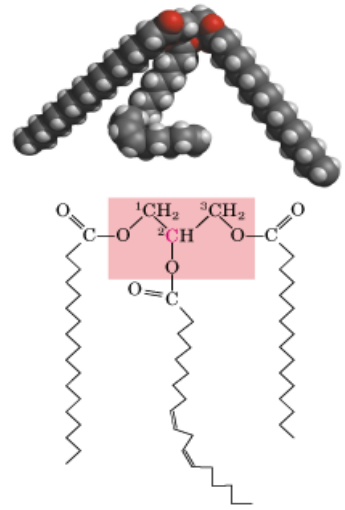
Triacylglycerols

(fats or triglycerides)

Three fatty acidsester-linked to aglycerol molecule

Used as energy storage molecules in eukaryotes

Stored in adipocytes



1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol, a mixed triacylglycerol

Advantages of using triacylglycerols for energy storage

1. Fats are highly reduced hydrocarbons with a large energy of oxidation.
2. Fats are insoluble molecules that aggregate into droplets.They are unsolvated and no storage mass is water.
3. Fats are chemically inert. They can be stored without fear of unfavorable reactions.

Disadvantages of triacylglycerols as energy storage

1. Fats must be emulsified before enzymes can digest them.
2. Fats are insoluble in the blood and must be carried in the blood as protein complexes.

Extraction of energy from fatty acids

Step 1. Oxidation of fatty acids to acetyl-CoA. This generates NADH and FADH2.

Step 2. Oxidation of acetyl-CoA to CO2 in the citric acid cycle. This generates NADH, FADH2 and

GTP (ATP).

Step 3. Transfer of electrons from NADH and FADH2 to O2.

This results in the synthesis of ATP.

TOPIC#105

Anaplerotic reactions of TCA-

Net Effect of the Citric Acid Cycle

Acetyl-CoA + 3NAD+ + FAD + GDP + Pi + 2 H2O $\longrightarrow$

2CO2 +3NADH + FADH2 + GTP + CoA + 3H+

Electrons from this process reduce NAD+ and FAD

One GTP is formed per cycle, this can be converted to ATP

Intermediates in the cycle are not depleted

Anaplerotic reactions

This pathway is utilized for the both catabolic reactions to generate energy as well as for anabolic reactions to generate metabolic intermediates for biosynthesis.

As intermediates of the citric acid cycle are removed to serve as biosynthetic precursors, they are replenished by anaplerotic reactions

Anaplerotic reactions :

These are chemical reactions that form intermediates of a metabolic pathway.

Most common anaplerotic reactions convert either pyruvate or phosphoenolpyruvate to oxaloacetate or malate.

The most important anaplerotic reaction in mammalian liver and kidney is the reversible carboxylation of pyruvate by CO2 to form oxaloacetate, catalyzed by pyruvate carboxylase.

When the citric acid cycle is deficient in oxaloacetate or any other intermediates, pyruvate is carboxylated to produce more oxaloacetate.

The enzymatic addition of a carboxyl group to pyruvate requires energy, supplied by ATP—required to attach a carboxyl group to pyruvate is about equal to the free energy available from ATP.

TABLE 16–2 Anaplerotic Reactions

Reaction	Tissue(s)/organism(s)
$Pyruvate + HCO_3^- + ATP \xrightleftharpoons{\text{pyruvate carboxylase}} oxaloacetate + ADP + P_i$	Liver, kidney
$Phosphoenolpyruvate + CO_2 + GDP \xrightleftharpoons{\text{PEP carboxykinase}} oxaloacetate + GTP$	Heart, skeletal muscle
$Phosphoenolpyruvate + HCO_3^- \xrightleftharpoons{\text{PEP carboxylase}} oxaloacetate + P_i$	Higher plants, yeast, bacteria
$Pyruvate + HCO_3^- + NAD(P)H \xrightleftharpoons{\text{malic enzyme}} malate + NAD(P)^+$	Widely distributed in eukaryotes and bacteria

Table 16-2
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

Pyruvate carboxylase is inactive in the absence of acetyl-CoA.

Whenever acetyl-CoA, the fuel for the citric acid cycle, is present in excess, it stimulates the pyruvate carboxylase reaction to produce more oxaloacetate, enabling the cycle to use more acetylCoA in the citrate synthase reaction

Phosphoenolpyruvate (PEP) carboxylase activated by the glycolytic intermediate fructose 1,6-bisphosphate, which accumulates when the citric acid cycle operates too slowly to process the pyruvate generated by glycolysis.

Role of the Citric Acid Cycle in Anabolism

Intermediates of the citric acid cycle are drawn off as precursors in many biosynthetic pathways.

Pathways that replenish depleted cycle intermediates are shown in red (four anaplerotic reactions)

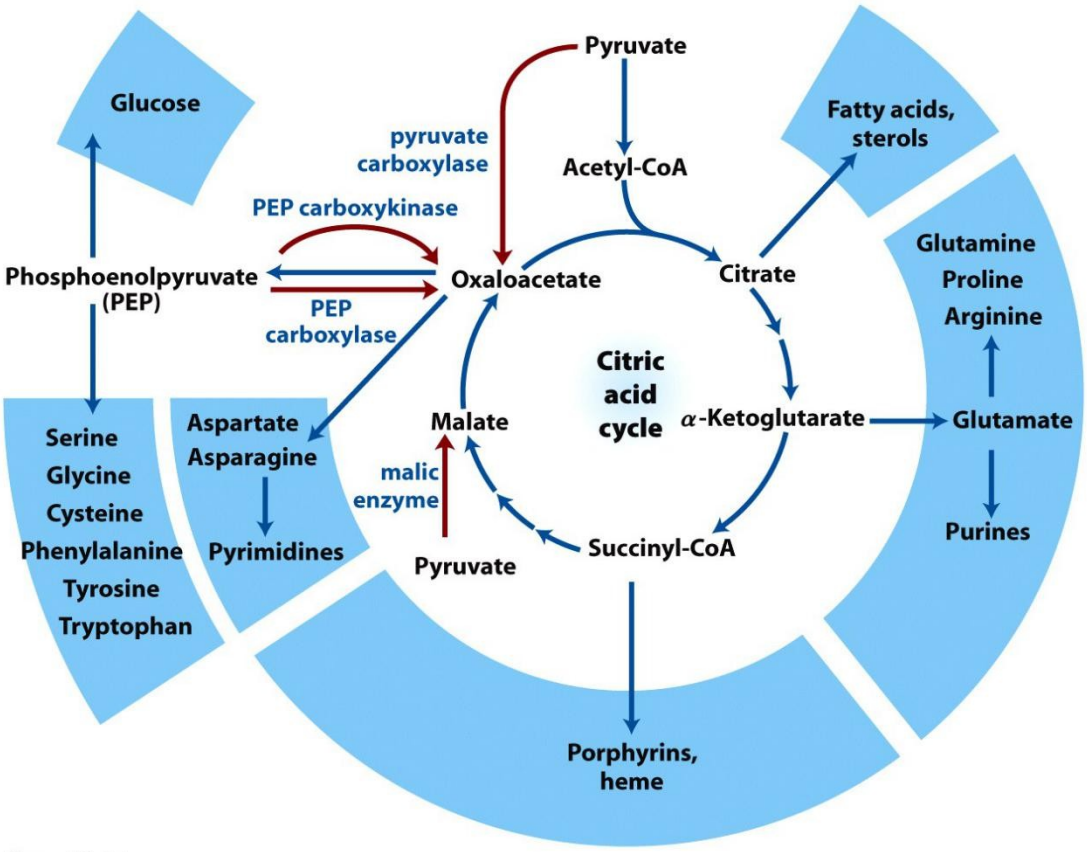


Figure 16-15
Lehninger Principles of Biochemistry, Fifth Edition
© 2008 W. H. Freeman and Company

FIGURE 16-15 Role of the citric acid cycle in anabolism. Intermediates of the citric acid cycle are drawn off as precursors in many biosynthetic pathways. Shown in red are four anaplerotic reactions that replenish depleted cycle intermediates (see Table 16-2)

.

TOPIC#106

Glyoxalate cycle -1

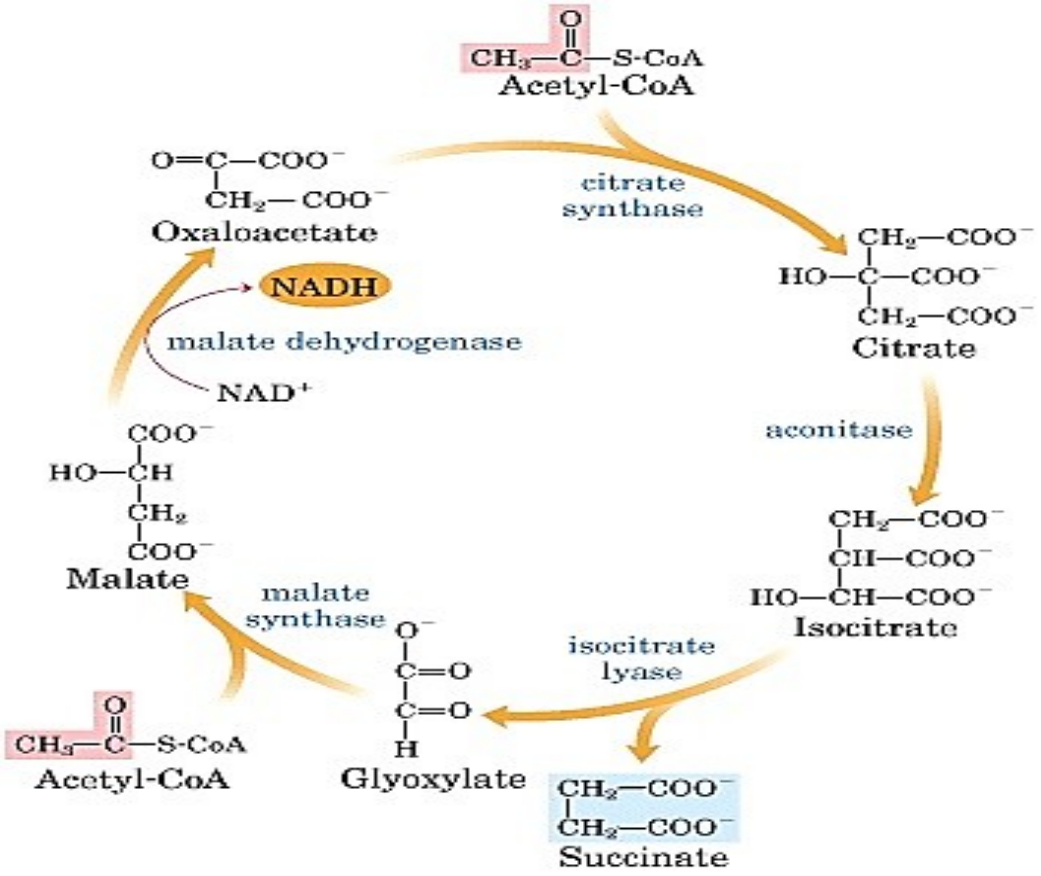
Glyoxylate cycle

The glyoxylate cycle, a variation of the tricarboxylic acid cycle, is an anabolic pathway occurring in plants, bacteria, protists, and fungi. The glyoxylate cycle relies on the conversion of acetyl-CoA to succinate for the synthesis of carbohydrates.

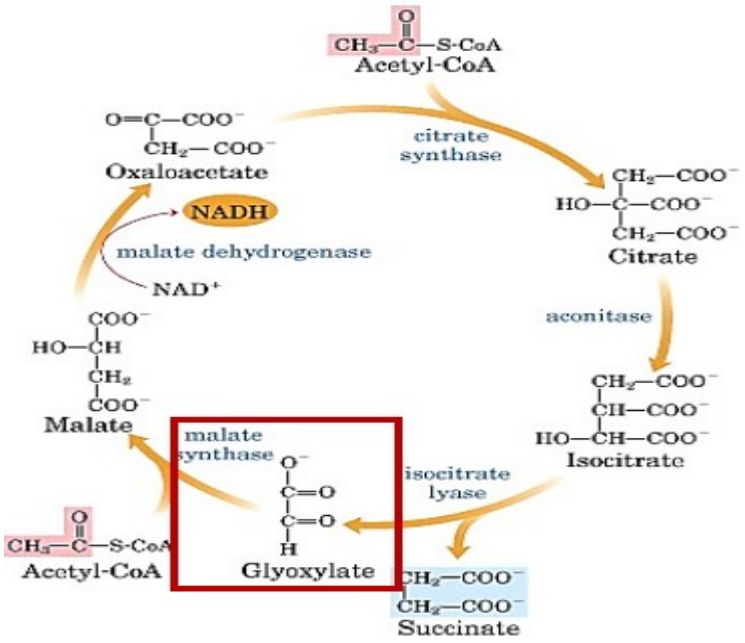
In microorganisms, the glyoxylate cycle allows cells to utilize simple carbon compounds as a carbon source when complex sources such as glucose are not available

Discovered by H. Kornberg and Neil Madsen in the lab. Of Hans Krebs.

Gets name from product “glyoxylate”



The two cycles differ in that in the glyoxylate cycle, isocitrate is converted into glyoxylate and succinate by ICL instead of into α-ketoglutarate.



Similarities with TCA cycle

The glyoxylate cycle utilizes five of the eight enzymes associated with the citric acid cycle:

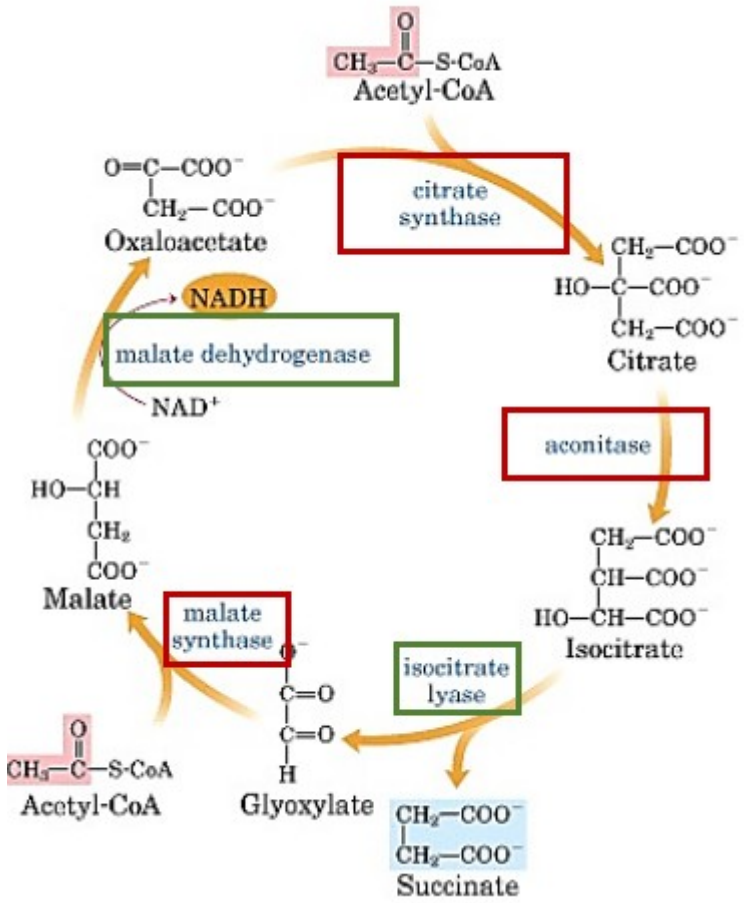
- citrate synthase
- Aconitase
- Succinate dehydrogenase
- Fumarase
- malate dehydrogenase.

Glyoxylate cycle

The citrate synthase, aconitase, and malate dehydrogenase of the glyoxylate cycle are isozymes of the citric acid cycle enzymes

Isocitrate lyase and malate synthase are unique to the glyoxylate cycle.

Notice that two acetyl groups (light red) enter the cycle and four carbons leave as succinate (blue).



The cycle is generally assumed to be absent in animals, with the exception of nematodes at the early stages of embryogenesis.

In recent years, the detection of malate synthase (MS) and isocitrate lyase (ICL), key enzymes involved in the glyoxylate cycle, in some animal tissue has raised questions regarding the evolutionary relationship of enzymes in bacteria and animals and suggests that animals encode alternative enzymes of the cycle that differ in function from known MS and ICL in non-metazoan species.

This bypasses the decarboxylation steps that take place in the TCA cycle, allowing simple carbon compounds to be used in the later synthesis of macromolecules, including glucose.

Glyoxylate is subsequently combined with acetyl-CoA to produce malate, catalyzed by malate synthase

Malate is also formed in parallel from succinate by the action of succinate dehydrogenase and fumarase.

TOPIC#107

Glyoxalate cycle -2

Glyoxylate cycle-2

Steps of Glyoxylate cycle

Acetyl-CoA condenses with oxaloacetate to form citrate .citrate is converted to isocitrate, exactly as in the citric acid cycle.

Then the cleavage of isocitrate by isocitrate lyase, forming succinate and glyoxylate (not the breakdown of isocitrate by isocitrate dehydrogenase)

The glyoxylate then condenses with a second molecule of acetyl-CoA to yield malate, in a reaction catalyzed by malate synthase

The malate is subsequently oxidized to oxaloacetate, which can condense with another molecule of acetyl-CoA to start another turn of the cycle

Each turn of the glyoxylate cycle consumes two molecules of acetyl-CoA and produces one molecule of succinate, which is then available for biosynthetic purposes.

The net result of the glyoxylate cycle is therefore the production of glucose from fatty acids. Succinate generated in the first step can enter into the citric acid cycle to eventually form oxaloacetate.

Role in gluconeogenesis

Fatty acids from lipids are commonly used as an energy source by vertebrates as fatty acids are degraded through beta oxidation into acetate molecules (used in acetyl coA).

This acetate, bound to the active thiol group of coenzyme A, enters the Citric acid cycle (TCA or CCA cycle) where it is fully oxidized to carbon dioxide

To utilize acetate from fat for biosynthesis of carbohydrates, the glyoxylate cycle, whose initial reactions are identical to the TCA cycle, is used.

This pathway thus allows cells to obtain energy from fat.

The oxaloacetate can be converted into Phosphoenolpyruvate, which is the product of phosphoenol pyruvate carboxykinase, the first enzyme in gluconeogenesis…

Function in Plants

In plants the glyoxylate cycle occurs in specialized peroxisomes which are called glyoxysomes. This cycle allows seeds to use lipids as a source of energy to form the shoot during germination.

In plants, certain invertebrates, and some microorganisms (E. coli and yeast) acetate can serve 2 purposes:

- as an energy-rich fuel
- as a source of phosphoenolpyruvate for carbohydrate synthesis

In these organisms, enzymes of the glyoxylate cycle catalyze conversion of acetate to succinate or other four-carbon intermediates of the citric acid cycle

Those enzymes common to the citric acid and glyoxylate cycles have two isozymes, one specific to mitochondria, the other to glyoxysomes.

Glyoxysomes are not present in all plant tissues at all times.

They develop in lipid-rich seeds during germination, before the developing plant acquires the ability to make glucose by photosynthesis.

In addition to glyoxylate cycle enzymes, glyoxysomes contain all the enzymes needed for the degradation of the fatty acids stored in seed oils

Germinating seeds can therefore convert the carbon of stored lipids into glucose

The glyoxylate cycle can also provide plants with another aspect of metabolic diversity.

Acetyl-CoA formed from lipid breakdown is converted to succinate via the glyoxylate cycle, and the succinate is exported to mitochondria, where citric acid cycle enzymes transform it to malate.

A cytosolic isozyme of malate dehydrogenase oxidizes malate to oxaloacetate, a precursor for gluconeogenesis.

Germinating seeds can therefore convert the carbon of stored lipids into glucose.

The four carbon succinate molecule can be transformed into a variety of carbohydrates through combinations of other metabolic processes; the plant can synthesize molecules using acetate as a source for carbon.

The Acetyl CoA can also react with glyoxylate to produce some NADPH from NADP+, which is used to drive energy synthesis in the form of ATP later in the Electron Transport Chain

TOPIC#108

Glyoxalate cycle -3

Glyoxylate cycle-3

Relationship between the glyoxylate and citric acid cycles

The two initial steps of the glyoxylate cycle are identical to those in the citric acid cycle: acetate → citrate → isocitrate.

The glyoxylate cycle bypasses the steps in the citric acid cycle where carbon is lost in the form of CO2.

In the glyoxylate cycle, the bypassing of the two decarboxylation steps of the citric acid cycle makes possible the net formation of succinate, oxaloacetate, and other cycle intermediates from acetyl-CoA.

Oxaloacetate thus formed can be used to synthesize glucose via gluconeogenesis.

In germinating seeds, the enzymatic transformations of dicarboxylic and tricarboxylic acids occur in three intracellular compartments

and there is a continuous interchange of metabolites among these compartments

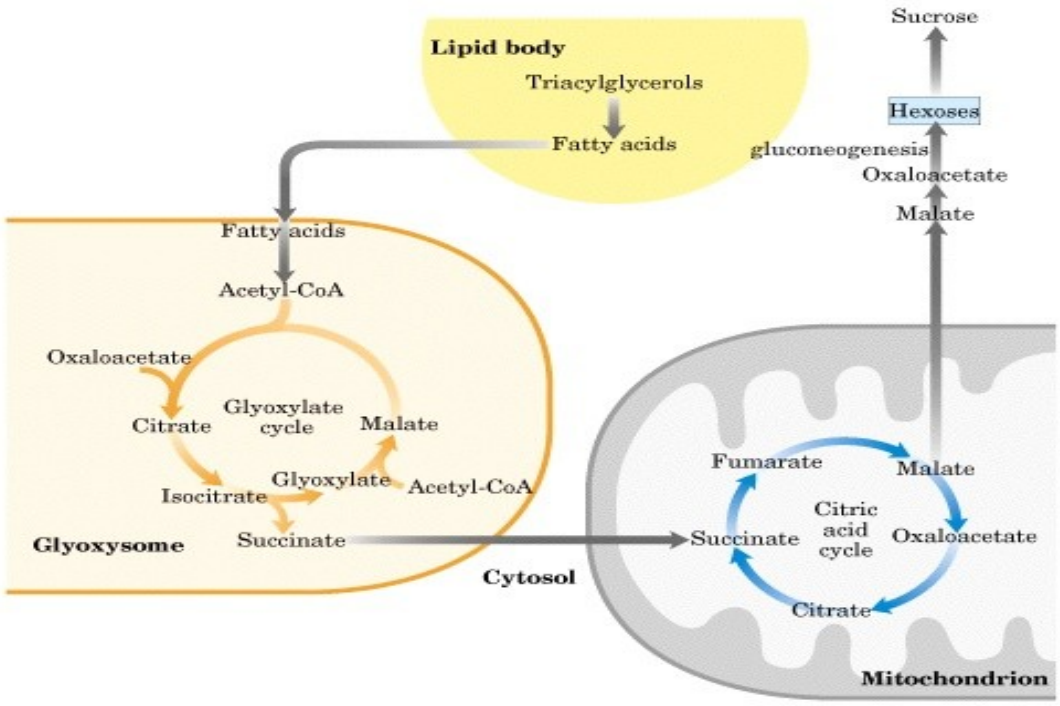
- Mitochondria
- glyoxysomes
- cytosol

Linkage to Gluconeogenesis in Plants

Relationship between the glyoxylate and citric acid cycles.

The reactions of the glyoxylate cycle (in glyoxysomes) proceed simultaneously with, those of the citric acid cycle (in mitochondria) as intermediates pass between these compartments.

The conversion of succinate to oxaloacetate is catalyzed by citric acid cycle enzymes.



The carbon skeleton of oxaloacetate from the citric acid cycle (in the mitochondrion) is carried to the glyoxysome in the form of aspartate.

Aspartate is converted to oxaloacetate, which condenses with acetyl-CoA derived from fatty acid breakdown.

The citrate thus formed is converted to isocitrate by aconitase, then split into glyoxylate

and succinate by isocitrate lyase.

The succinate returns to the mitochondrion, where it reenters the citric acid cycle and is transformed into malate, which enters the cytosol and is oxidized (by cytosolic malate dehydrogenase) to oxaloacetate.

Oxaloacetate is converted via gluconeogenesis into hexoses and sucrose, which can be

transported to the growing roots and shoot.

Four distinct pathways participate in these conversions:

- fatty acid breakdown to acetyl-CoA (in glyoxysomes)
- the glyoxylate cycle (in glyoxysomes)
- the citric acid cycle (in mitochondria)
- gluconeogenesis (in the cytosol)

Coordinated regulation of glyoxylate and citric acid cycles

The central molecules between 2 cycles is isocitrate

The translocation of isocitrate between the citric acid cycle and the glyoxylate cycle is controlled by the activity of isocitrate dehydrogenase, which is regulated by reversible phosphorylation.

A specific protein kinase phosphorylates and inactivates the dehydrogenase. This inactivation shunts isocitrate to the glyoxylate cycle, where it begins the synthetic route toward glucose.

A phosphoprotein phosphatase removes the phosphoryl group from isocitrate dehydrogenase, reactivating the enzyme

Reactivation of enzyme sends more isocitrate through the energy-yielding citric acid cycle.The regulatory protein kinase and phosphoprotein phosphatase are separate enzymatic activities of a single polypeptide.

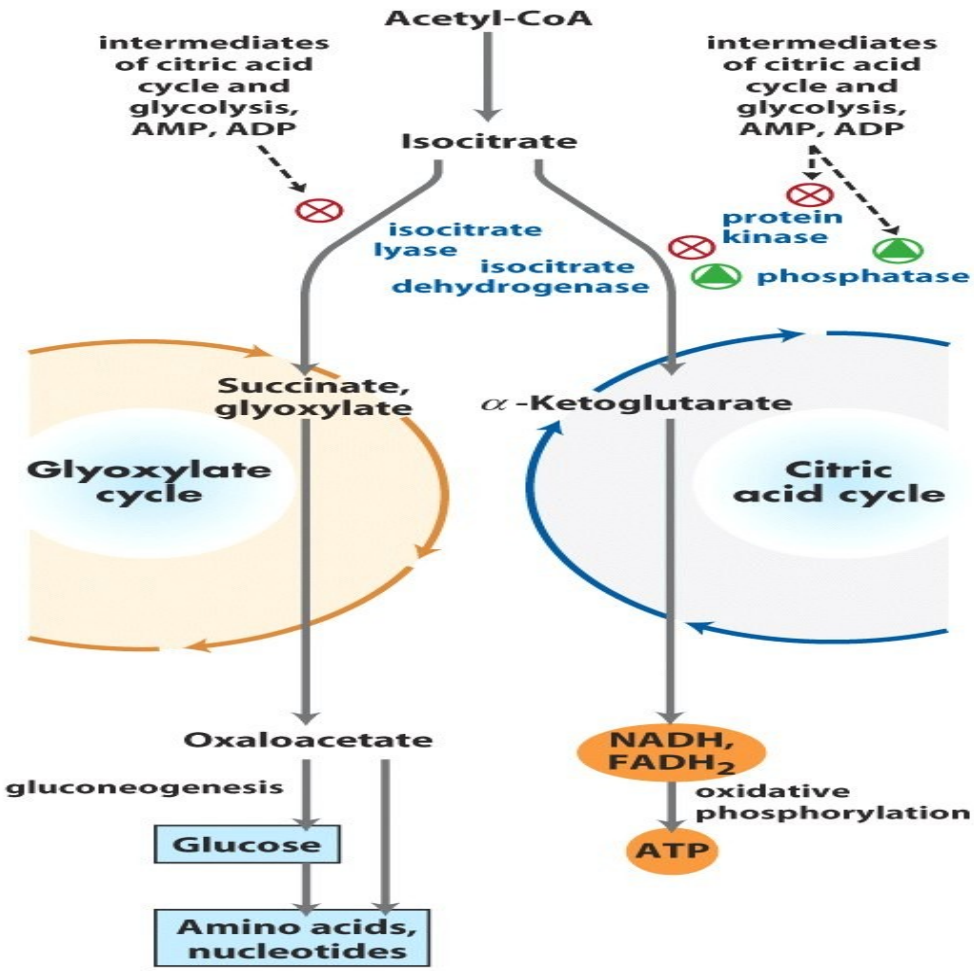
Regulation Linkage

Coordinated regulation of glyoxylate and citric acid cycles.

Regulation of isocitrate dehydrogenase activity determines the partitioning of isocitrate between the glyoxylate and citric acid cycles.

When the enzyme is inactivated by phosphorylation (by a specific protein kinase), isocitrate is directed into biosynthetic reactions via the glyoxylate cycle.

When the enzyme is activated by dephosphorylation (by a specific phosphatase), isocitrate enters the citric acid cycle and ATP is produced.



TOPIC#109

Fatty acid-1

Fatty Acid Catabolism

Extraction of energy from fatty acids

Fatty acids yield the maximum amount of ATP on an energy per gram basis, when they are completely oxidized to CO2 and water by β-oxidation and the citric acid cycle.

Fatty acids (mainly in the form of triglycerides) are the foremost storage form of fuel in most animals, and to a lesser extent in plants.

Beta oxidation

The repetitive four-step process, by which fatty acids are converted into acetyl-CoA called Beta oxidation.

To overcome the relative stability of the C—C bonds in a fatty acid, the carboxyl group at C-1 is activated by attachment to coenzyme A, which allows stepwise oxidation of the fatty acyl group at the C-3, or beta, position-hence the name beta oxidation.

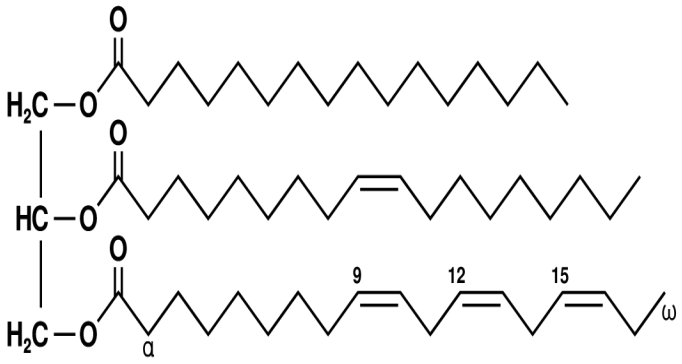
Triacylglycerols

triesters of fatty acids and glycerol with 1 molecule of glycerol (3 –OH).

Three fatty acids ester-linked to a glycerol molecule

Each –OH bonded to –COO group of 1 fatty acid (forming ester)

Triglycerides are the main constituents of body fat in humans and other animals, as well as vegetable fat.Used as energy storage molecules in eukaryotes.Stored in adipocytes



Example of an unsaturated fat triglyceride (C55H98O6).

Left part: glycerol

Right part, from top to bottom: palmitic acid, oleic acid, alpha-linolenic acid.

Advantages of using triacylglycerols for energy storage

- Fats are highly reduced hydrocarbons with a large energy of complete oxidation (38 kJ/g) more than twice that for the same weight of carbohydrate or protein.
- Fats are insoluble molecules that aggregate into droplets.

- They are not dissolved due to which do not raise the osmolarity of the cytosol. (In storage polysaccharides, by contrast, water of solvation makes two-thirds of the overall weight of the stored molecules.)
- Fats are chemically inert (triglycerides or neutral fats)
  that make them especially suitable as storage fuels so they can be stored without fear of unfavorable reactions.

Disadvantages of triacylglycerols as energy storage

- The properties that make triacylglycerols good
  storage compounds present problems in their role as fuels.
- Fats must be emulsified before enzymes can digest them.
- Fats are insoluble in the blood and must be carried in the blood as protein complexes.

Emulsification: The breakdown of fat globules in the duodenum into tiny droplets, which provides a larger surface area on which the enzyme pancreatic lipase can act to digest the fats into fatty acids and glycerol. Emulsification is assisted by the action of the bile salts.

Extraction of energy from fatty acids

Complete oxidation of fatty acids to CO2 and H2O takes place in three stages:

Step 1. Oxidation of fatty acids to acetyl-CoA. This generates NADH and FADH2.

Step 2. Oxidation of acetyl-CoA to CO2 in the citric acid cycle. This generates NADH, FADH2 and GTP (ATP).

Step 3. Transfer of electrons from NADH and FADH2 to O2. This results in the synthesis of ATP.

TOPIC#110

fatty acid metabolism-2

Fatty Acid Catabolism-2

Digestion, Mobilization, and Transport of Fats

Cells can obtain fatty acid fuels from three sources:

- Fats consumed in the diet
- fats stored in cells as lipid droplets
- fats synthesized in one organ for export to another

Some species use all three sources under various circumstances, others use one or two. Vertebrates, obtain fats in the diet, mobilize fats stored in specialized tissue (adipose tissue, consisting of cells called adipocytes), and, in the liver, convert excess dietary carbohydrates to fats for export to other tissues.

40% or more of the daily energy requirement of humans in highly industrialized countries is supplied by dietary triacylglycerols

Triacylglycerols provide more than half the energy requirements of some organs, particularly the liver, heart, and resting skeletal muscle.

In hibernating animals and migrating birds, stored triacylglycerols are sole source of energy

Protists obtain fats by consuming organisms lower in the food chain, and some also store fats as cytosolic lipid droplets.

Vascular plants mobilize fats stored in seeds during germination, but do not otherwise depend on fats for energy.

Dietary Fats Are Absorbed in the Small Intestine

step 1 In vertebrates, before ingested triacylglycerols can be absorbed through the intestinal wall they must be converted from insoluble macroscopic fat particles to microscopic micelles (soluble) by bile salts, such as taurocholic acid
step 2 Lipase action converts triacylglycerols to monoacylglycerols (monoglycerides) and diacylglycerols (diglycerides), free fatty acids, and glycerol .

step 3 These products of lipase action diffuse into the epithelial cells lining the intestinal surface (the intestinal mucosa)

step 4 In the intestinal mucosa they are reconverted to triacylglycerols and packed with dietary cholesterol and specific proteins into lipoprotein aggregates called chylomicrons

Molecular Structure of a Chylomicron

The surface is a layer of phospholipids, with head groups facing the aqueous phase. Triacylglycerols sequestered in the interior (yellow) make up more than 80% of the mass. Several apolipoproteins that protrude from the surface (B-48, C-III, C-II) act as signals in the uptake and metabolism of chylomicron contents.The diameter of chylomicrons ranges from about 100 to 500 nm.

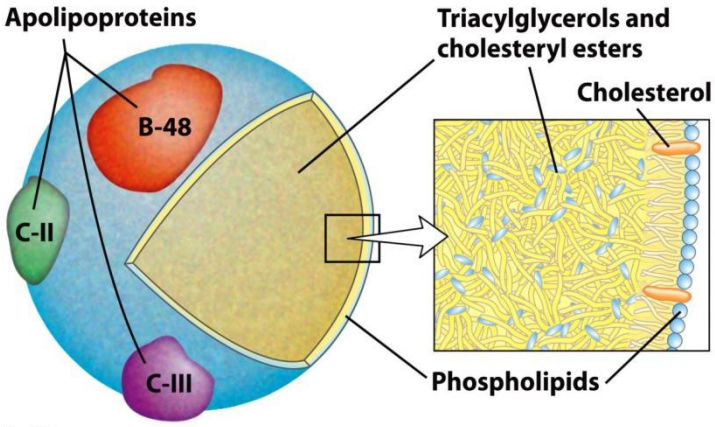


Figure 17-2
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

step 5 The protein moieties of lipoproteins are recognized by receptors on cell surfaces. In lipid uptake from the intestine, chylomicrons, which contain apolipoprotein C-II (apoC-II), move from the intestinal mucosa into the lymphatic system, and then enter the blood, which carries them to muscle and adipose tissue

step 6 In the capillaries of these tissues, the extracellular enzyme lipoprotein lipase, activated by apoC-II, hydrolyzes triacylglycerols to fatty acids and glycerol

step 7 fatty acids and glycerol are taken up by cells in the target tissues .

step 8 In muscle, the fatty acids are oxidized for energy; in adipose tissue, they are reesterified for storage as triacylglycerols

Bile salts are synthesized from cholesterol in the liver, stored in the gallbladder, and released into the small intestine after ingestion of a fatty meal.

Bile salts act as biological detergents, converting dietary fats into mixed micelles of bile salts and triacylglycerols

Micelle formation makes lipid molecules accessible to the action of water-soluble lipases in the intestine

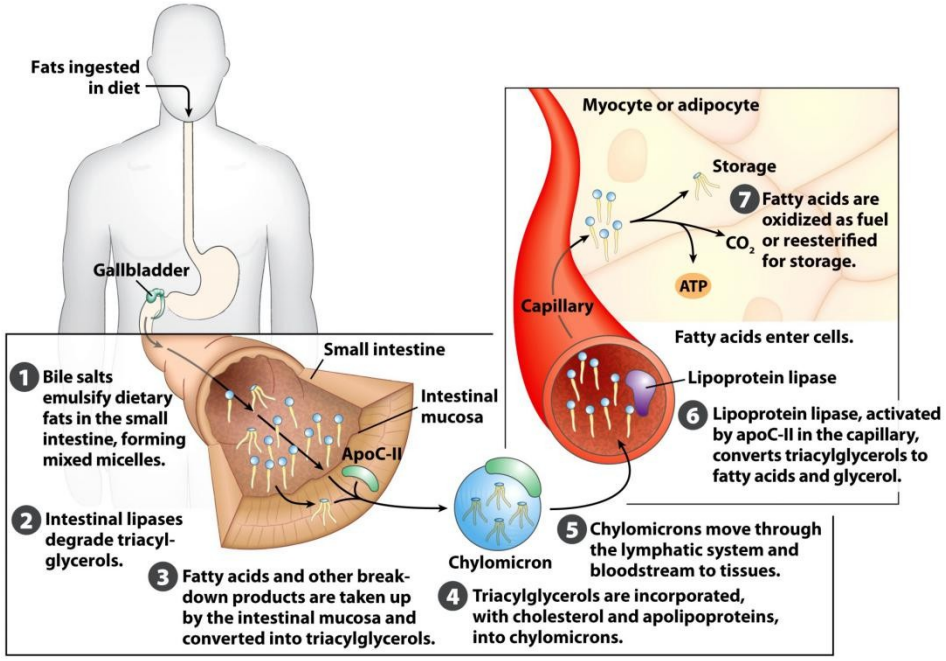


Figure 17-1
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

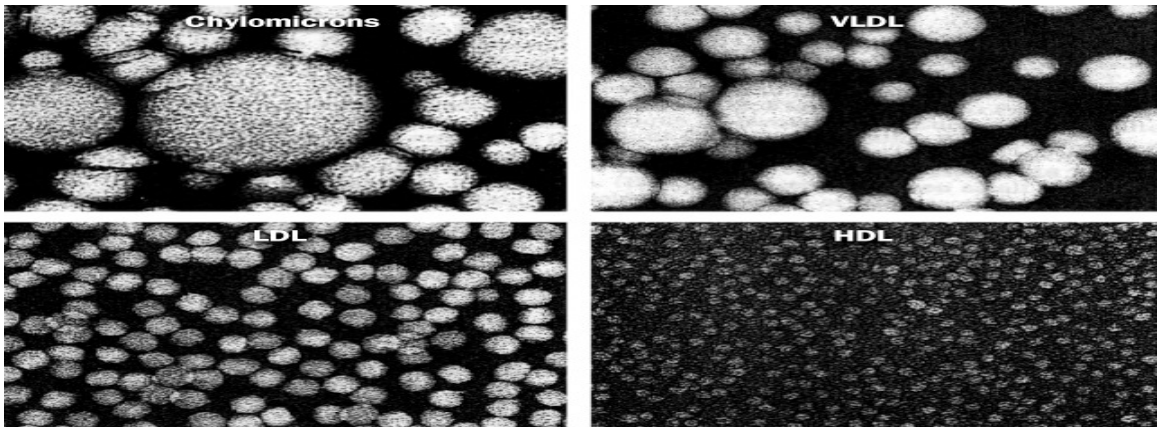
Types of Dietary Fats

Apolipoproteins are lipid-binding proteins in the blood, responsible for the transport of triacylglycerols, phospholipids, cholesterol, and cholesteryl esters between organs. Apolipoproteins ("apo" means "detached" or "separate," designating the protein in its lipid-free form) combine with lipids to form several classes of lipoprotein particles, spherical aggregates with hydrophobic lipids at the core and hydrophilic protein side chains and lipid head groups at the surface.
Various combinations of lipid and protein produce particles of different densities, ranging from chylomicrons

- Lipoproteins are large complexes of lipids and proteins designed to transport lipids in the blood. The lipoproteins are classified by particle density. Very-low-density lipoproteins (VLDL)
- Very-high-density lipoproteins (VHDL)

The protein moieties of lipoproteins are recognized by receptors on cell surfaces.

In lipid uptake from the intestine, chylomicrons, which contain apolipoprotein C-II (apoC-II)



Dietry fats are absorbed in small intestine

1. Chylomicrons (transport dietary cholesterol to tissues)

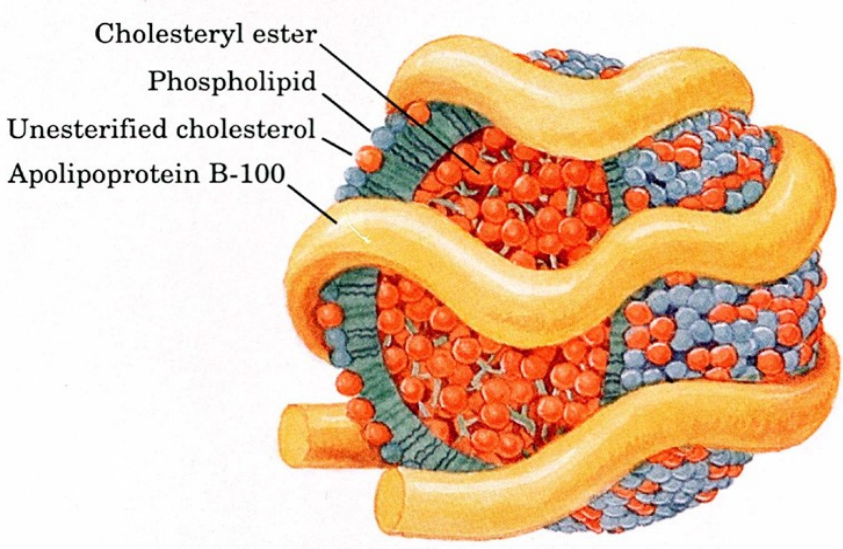
2. Very low density lipoprotein (VLDL) transport cholesterol

3. Intermediate density lipoprotein (IDL) produced by the liver

4. Low density lipoprotein (LDL)

1st four transport cholesterol produced by the liver to tissues

5. High density lipoprotein (HDL) (transport cholesterol from tissue to liver)



TOPIC#111

Fatty acid metabolism-effect of hormones-

Hormones trigger metabolization of Triacylglycerols

Neutral lipids are stored in adipocytes (and in steroid synthesizing cells of the adrenal cortex, ovary, and testis) in the form of lipid droplets, with a core of sterol esters and triacylglycerols surrounded by a monolayer of phospholipids.

The surface of these droplets is coated with perilipins

Perilipins are a family of proteins that restrict access to lipid droplets, preventing untimely lipid mobilization.

When hormones signal the need for metabolic energy, triacylglycerols stored in adipose tissue are mobilized (brought out of storage) and transported to tissues (skeletal muscle, heart, and renal cortex) in which fatty acids can be oxidized for energy production.

Two hormones are secreted in response to low blood glucose levels

- Epinephrine and
- glucagon

Epinephrine and glucagon stimulate the enzyme adenylyl cyclase in the adipocyte plasma membrane (which produces the intracellular second messenger cyclic AMP)

Cyclic AMP–dependent protein kinase (PKA) triggers changes that open the lipid droplet up to the action of three lipases, which act on tri-, di-, and monoacylglycerols, releasing fatty acids and glycerol.

These free fatty acids, (FFA) pass from the adipocyte into the blood, bind to the blood protein serum albumin noncovalently, carried to tissues skeletal muscle, heart, and renal cortex

serum albumin makes up about half of the total serum protein, binds as many as 10 fatty acids per protein monomer

About 95% of the biologically available energy of triacylglycerols resides in their three long-chain fatty acids; only 5% is contributed by the glycerol moiety
The glycerol released by lipase action is phosphorylated by glycerol kinase and glycerol 3-phosphate is formed.Glycerol 3-phosphate is oxidized to dihydroxyacetone phosphate. The glycolytic enzyme triose phosphate isomerase converts this compound to glyceraldehyde 3-phosphate, which is oxidized via glycolysis.

Mobilization of triacylglycerols stored in adipose tissue

When low levels of glucose in the blood trigger the release of glucagon, the hormone binds its receptor in the adipocyte membrane and thus stimulates adenylyl cyclase, via a G protein, to produce cAMP.

This activates PKA, which phosphorylates the hormone-sensitive lipase (HSL) And perilipin molecules on the surface of the lipid droplet.

Phosphorylation of perilipin causes dissociation of the protein CGI from perilipin.

CGI then associates with the enzyme adipose triacylglycerol lipase (ATGL), activating it.

Active ATGL converts triacylglycerols to diacylglycerols.

The phosphorylated perilipin associates with phosphorylated hormone-sensitive lipase HSL, allowing it access to the surface of the lipid droplet, where it converts diacylglycerols to monoacylglycerols.

A third lipase, monoacylglycerol lipase (MGL) hydrolyzes monoacylglycerols.

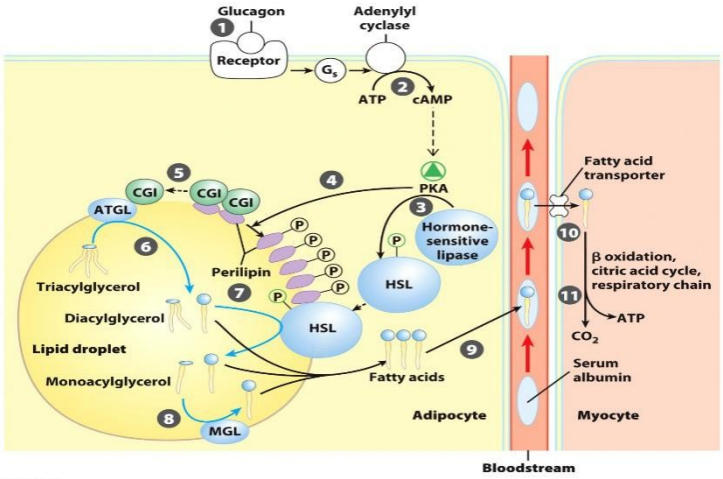


Figure 17-3
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Utilization of stored triacylglycerols

Fatty acids leave the adipocyte, bind serum albumin in the blood, and are carried in the blood; they are released from the albumin and enter a myocyte via a specific fatty acid transporter.In the myocyte, fatty acids are oxidized to CO2.the energy of oxidation is conserved in ATP, which fuels muscle contraction and other.energy-requiring metabolism in the myocyte.

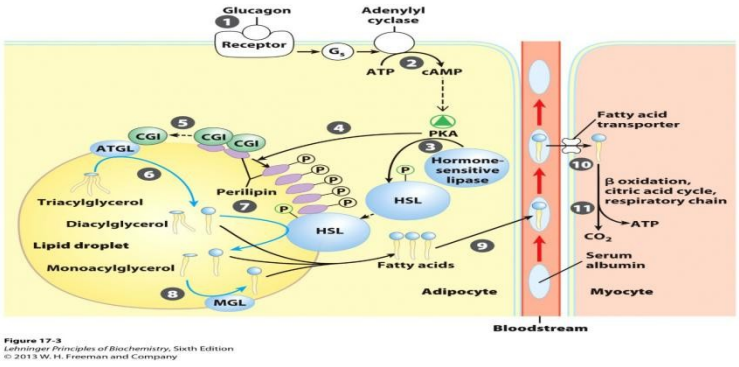


Figure 17-3
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

TOPIC#112
carnitine shuttle

Fatty Acids Are Activated and Transported into Mitochondria

The enzymes of fatty acid oxidation in animal cells are located in the mitochondrial matrix, as demonstrated in 1948 by Eugene P. Kennedy and Albert Lehninger.

The fatty acids with chain lengths of 12 or fewer carbons enter mitochondria without the help of membrane transporters.

The fatty acids with chain lengths of 14 or more carbons, which constitute the majority of the FFA obtained in the diet or released from adipose tissue, cannot pass directly through the mitochondrial membranes

The fatty acids with chain lengths of 14 or more carbons must first undergo the three enzymatic reactions of the carnitine shuttle.

- Esterification to CoA
- Transesterification to carnitine followed by transport
- Transesterification back to CoA

Three enzymatic reactions of the carnitine shuttle

The first reaction

This is catalyzed by a family of isozymes (different isozymes specific for fatty acids having short, intermediate, or long carbon chains) present in the outer mitochondrial membrane, the acyl- CoA synthetases

Acyl–CoA synthetases catalyze the formation of a thioester linkage between the fatty acid carboxyl group and the thiol group of coenzyme A to yield a fatty acyl– CoA, coupled to the cleavage of ATP to AMP and PPi.
(free energy released by cleavage of phosphoanhydride bonds in ATP can be coupled to the formation of a high-energy compound.)

The reaction occurs in two steps and involves a fatty acyl-adenylate intermediate (Fatty acyl–CoAs)

Fatty acyl–CoAs like acetyl-CoA, are high-energy compounds; their hydrolysis to FFA and CoA has a large, negative standard free-energy change (G= -31 kJ/mol).

The formation of a fatty acyl–CoA is made more favorable by the hydrolysis of two high-energy bonds in ATP; the pyrophosphate formed in the activation reaction is immediately hydrolyzed by inorganic pyrophosphatase which pulls the reaction in the direction of fatty acyl–CoA formation. The overall reaction is

Fatty Acyl–CoA Synthetase Mechanism
Conversion of a fatty acid to a fatty acyl–CoA.
The conversion is catalyzed by fatty acyl–CoA synthetase and inorganic pyrophosphatase.
Fatty acid activation by formation of the fatty acyl–CoA derivative occurs in two

steps. The overall reaction is highly exergonic.

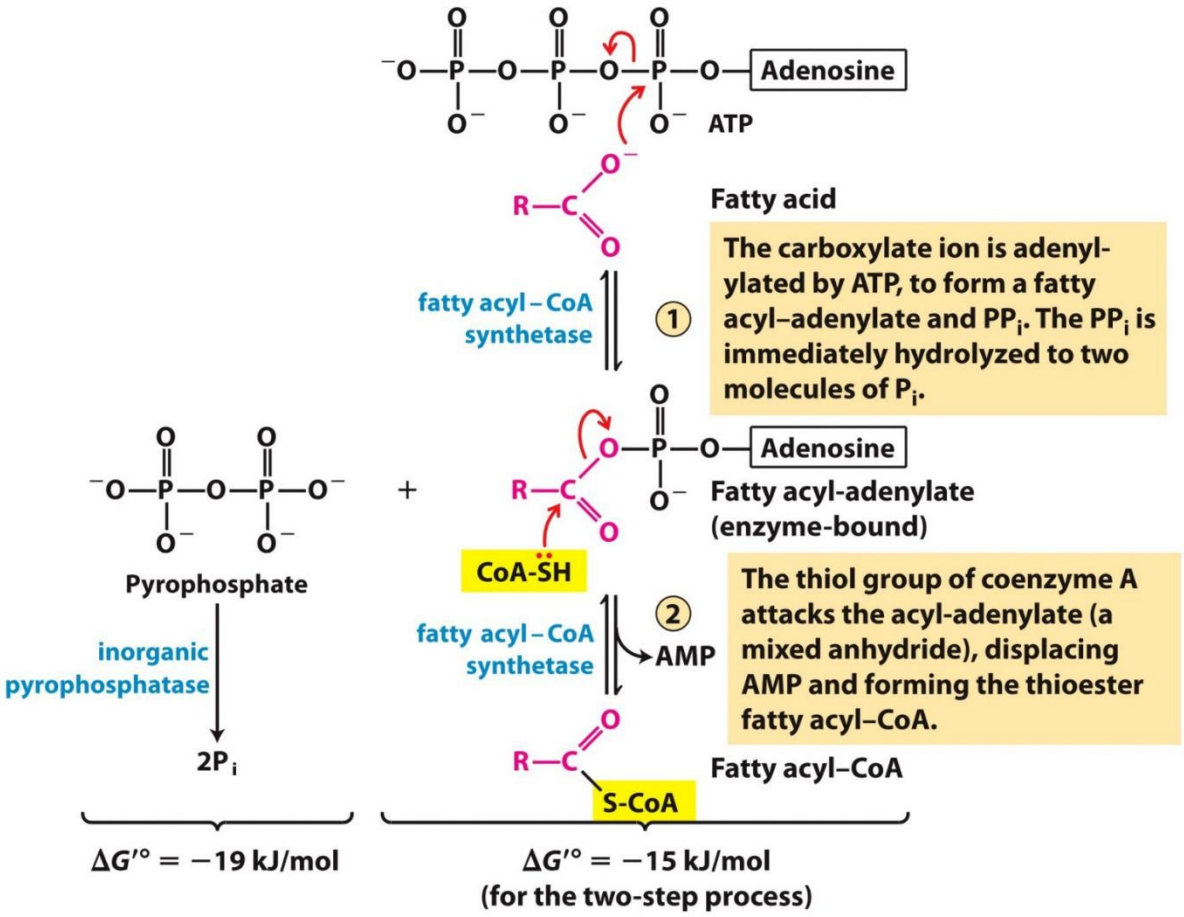


Figure 17-5
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

The second reaction

Fatty acyl–CoA esters formed at the cytosolic side of the outer mitochondrial membrane can be oxidized to produce ATP in the mitochondrion and or they can be used in the cytosol to synthesize membrane lipids.

Fatty acids destined for mitochondrial oxidation are transiently attached to the hydroxyl group of carnitine to form fatty acyl–carnitine— the second reaction of the shuttle.

This transesterification is catalyzed by carnitine acyltransferase I, in the outer membrane.

So the carnitine ester is formed then moved across the outer membrane to the intermembrane space

This passage into the intermembrane space (the space between the outer and inner membranes) occurs through large pores (formed by the protein porin) in the outer membrane.

The fatty acyl–carnitine ester then enters the matrix by facilitated diffusion through the acyl-carnitine/carnitine transporter of the inner mitochondrial membrane

Third (final step) of the carnitine shuttle

The fatty acyl group is enzymatically transferred from carnitine to intramitochondrial coenzyme A by carnitine acyltransferase II.

This isozyme, located on the inner face of the inner mitochondrial membrane, regenerates fatty acyl–CoA and releases it, along with free carnitine, into the matrix .

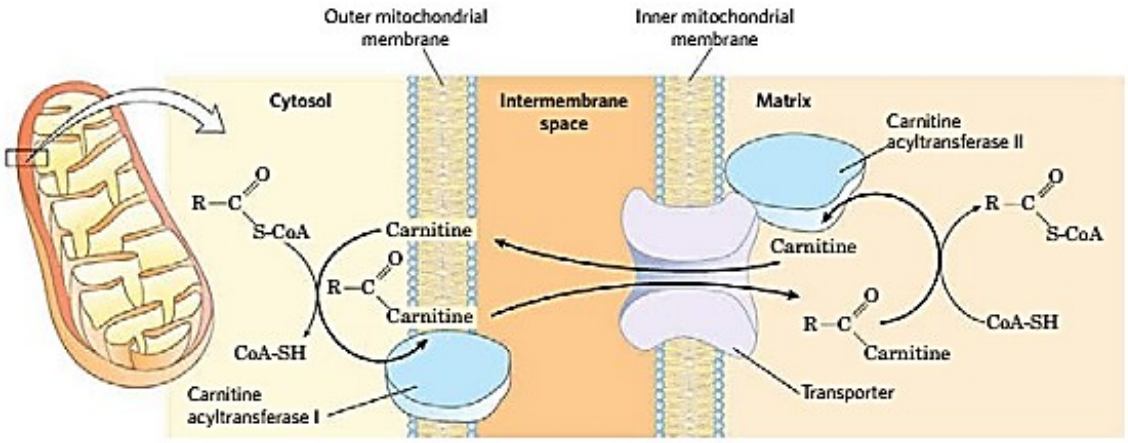
Carnitine reenters the intermembrane space via the acylcarnitine/carnitine transporter.

Fatty acid entry into mitochondria via the acyl-carnitine/ carnitine transporter.

After fatty acyl–carnitine is formed at the outer membrane or in the intermembrane space, it moves into the matrix by facilitated diffusion through the transporter in the inner membrane.

In the matrix, the acyl group is transferred to mitochondrial coenzyme A,

freeing carnitine to return to the intermembrane space through the same transporter.



This three-step process for transferring fatty acids into the mitochondrion—links two separate pools of coenzyme A and of fatty acyl–CoA, one in the cytosol, the other in mitochondria.

These pools have different functions.

Coenzyme A in the mitochondrial matrix is used in oxidative degradation of pyruvate, fatty acids, and some amino acids

Coenzyme A in cytosol.. Is conversely, is used in the biosynthesis of fatty acids

Fatty acyl–CoA in the cytosolic pool can be used for membrane lipid synthesis or can be moved into the mitochondrial matrix for oxidation and ATP production.

Conversion to the carnitine ester commits the fatty acyl moiety to the oxidative fate.

Rate limiting step for oxidation of fatty acids in mitochondria

The carnitine-mediated entry process is the rate limiting step for oxidation of fatty acids in mitochondria.And is a regulation point.Once inside the mitochondrion, the fatty acyl–CoA is acted upon by a set of enzymes in the matrix.

TOPIC#113

Fatty acid metabolism-fatty acid oxidation-3 stages

Stages for Oxidation of fatty acids

Oxidation of fatty acids

The mitochondrial oxidation of fatty acids takes place in three stages

In First Sage fatty acids undergo oxidative removal of successive two-carbon units in the form of acetyl-CoA, starting from the carboxyl end of the fatty acyl chain.

The 16-carbon palmitic acid undergoes seven passes through the oxidative sequence

In each pass two carbons are lost as acetyl-CoA.

At the end of seven cycles the last two carbons of palmitate (originally C-15 and C-16) remain as acetyl-CoA.

The overall result is the conversion of the 16-carbon chain of palmitate to eight two-carbon acetyl groups of acetyl-CoA molecules.

Formation of each acetyl-CoA requires removal of four hydrogen atoms (two pairs of electrons and four H+) from the fatty acyl moiety by dehydrogenases.

Stages of fatty acid oxidation.

Stage 1:

A long-chain fatty acid is oxidized to yield acetyl residues in the form of acetyl-CoA.

This process is called oxidation.

Stage 2:

The acetyl groups are oxidized to CO2 via the citric acid cycle.

Stage 3:

Electrons derived from the oxidations of stages 1 and 2 pass to O2 via the mitochondrial respiratory chain,

providing the energy for ATP synthesis by oxidative phosphorylation.

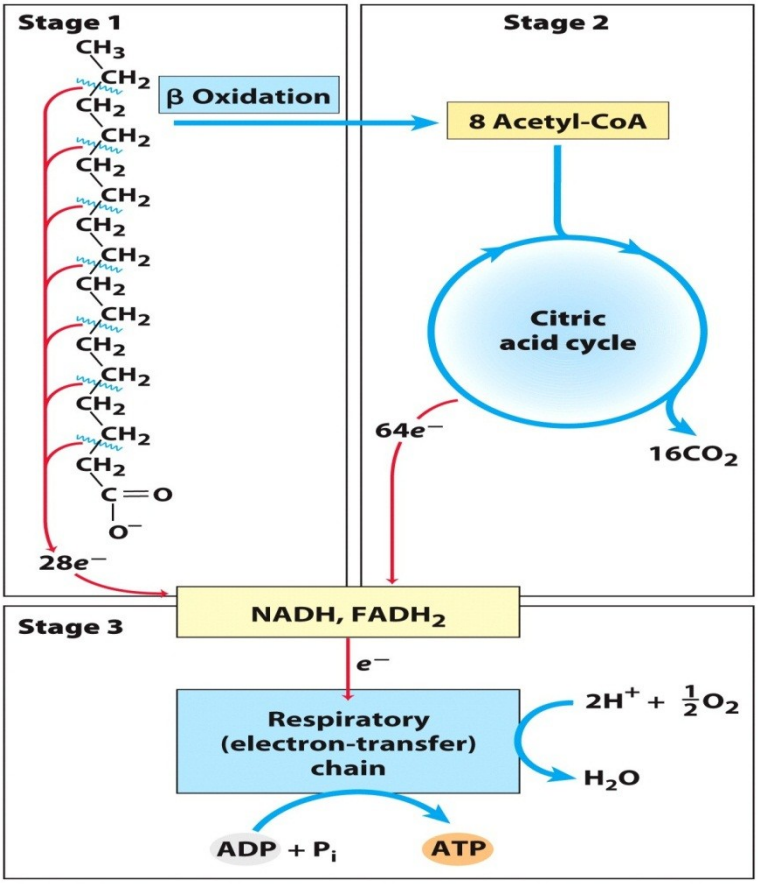


Figure 17-7
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

In the ***second stage*** of fatty acid oxidation, the acetyl groups of acetyl-CoA are oxidized to CO2 in the citric acid cycle, which also takes place in the mitochondrial matrix.

Acetyl-CoA derived from fatty acids thus enters a final common pathway of oxidation with the acetyl-CoA derived from glucose via glycolysis and pyruvate oxidation

Third stage Electrons derived from the oxidations of stages 1 and 2 pass to O2 via the mitochondrial respiratory chain, providing the energy for ATP synthesis by oxidative phosphorylation.

This is due to production of reduced electron carriers NADH and FADH2 in first two stages of fatty acid oxidation

Stages of fatty acid oxidation.

Stage 1:

A long-chain fatty acid is oxidized to yield acetyl residues in the form of acetyl-CoA.

This process is called oxidation.

Stage 2:

The acetyl groups are oxidized to CO2 via the citric acid cycle.

Stage 3:

Electrons derived from the oxidations of stages 1 and 2 pass to O2 via the mitochondrial respiratory chain,

providing the energy for ATP synthesis by oxidative phosphorylation.

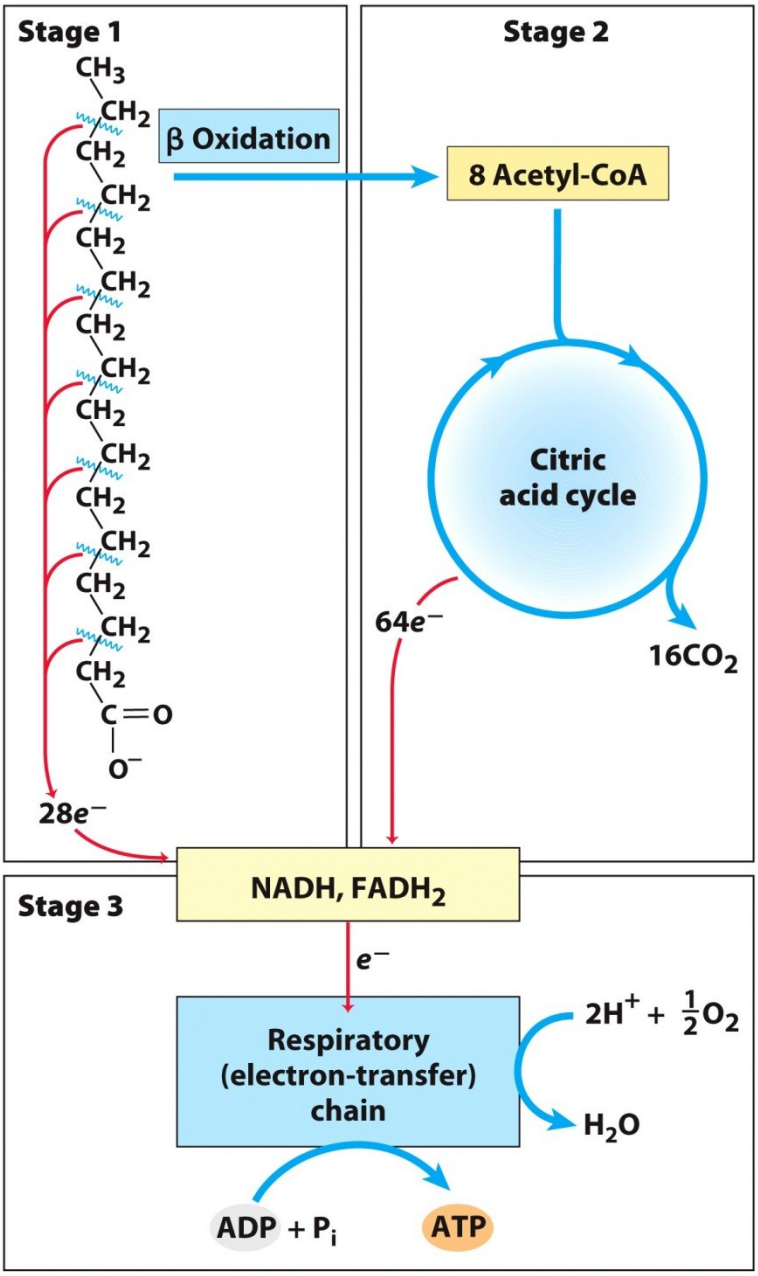


Figure 17-7
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

TOPIC#114

beta oxidation-complete

Steps of Beta Oxidation

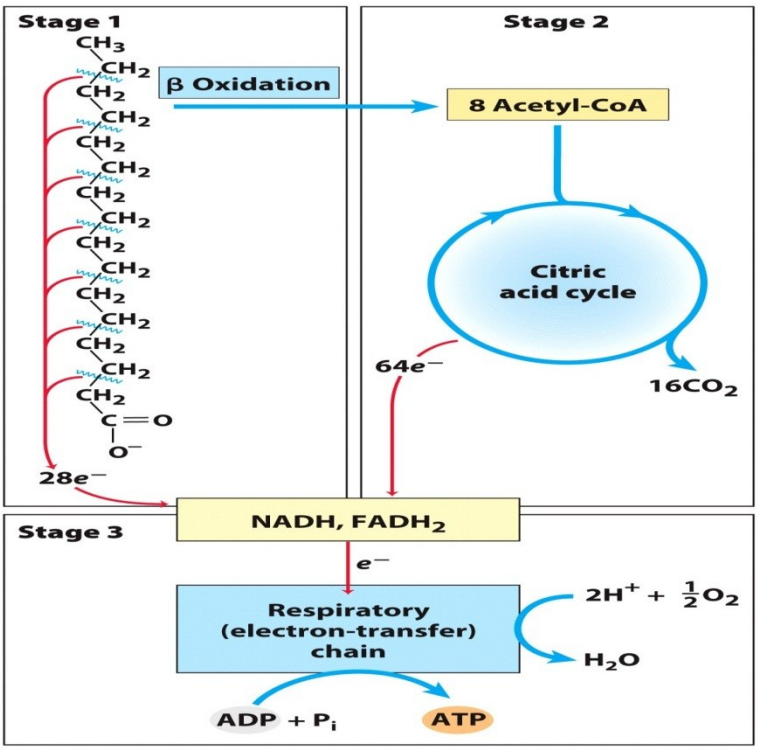


Figure 17-7
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Beta - Oxidation of saturated fatty acids

First, dehydrogenation of fatty acyl–CoA produces a double bond between the alpha and beta carbon atoms (C-2 and C-3), yielding a trans-Δ2-enoyl-CoA (the symbol Δ2 designates the position of the double bond).

This first step is catalyzed by acyl- CoA dehydrogenase

Acyl- CoA dehydrogenase:

Three isozymes of each specific for a range of fatty acyl chain lengths. All three isozymes are flavoproteins with FAD as a prosthetic group.

- Very-long-chain acyl-CoA dehydrogenase (VLCAD), acting on fatty acids of 12 to 18 carbons
- Medium-chain (MCAD), acting on fatty acids of 4 to 14 carbons
- Short-chain (SCAD), acting on fatty acids of 4 to 8 carbons.

The electrons removed from the fatty acyl–CoA are transferred to FAD, the reduced form of the dehydrogenase immediately donates its electrons to an electron carrier of the mitochondrial respiratory chain, the electron-transferring flavoprotein (ETF)

The oxidation catalyzed by an acyl-CoA dehydrogenase is analogous to succinate dehydrogenation in the citric acid cycle in both reactions the enzyme is bound to the inner membrane

A double bond is introduced into a carboxylic acid between alpha and beta carbons

FAD is the electron acceptor, and electrons from the reaction ultimately enter the respiratory chain and pass to O2, with the synthesis of about 1.5 ATP molecules per electron pair.

2nd and 3rd Steps: β-Oxidation of saturated fatty acids

Second Step: In the second step of the beta-oxidation cycle water is added to the double bond of the trans- D2-enoyl-CoA to form the L-stereoisomer of beta-hydroxyacyl- CoA (3-hydroxyacyl-CoA).

Enzyme: Enoyl-CoA hydratase, is analogous to the fumarase reaction in the citric acid cycle, in which H2O adds across an alpha and beta double bond

Third step: In the third step, L-beta- hydroxyacyl-CoA is dehydrogenated to form beta-ketoacyl-CoA, by the action of beta-hydroxyacyl-CoA dehydrogenase

NAD+ is the electron acceptor.

The NADH formed in the reaction donates its electrons to NADH dehydrogenase, an electron carrier of the respiratory chain, and ATP is formed from ADP as the electrons pass to O2.

Enzyme: The reaction catalyzed by beta-hydroxyacyl- CoA dehydrogenase is closely analogous to the malate dehydrogenase reaction of the citric acid cycle

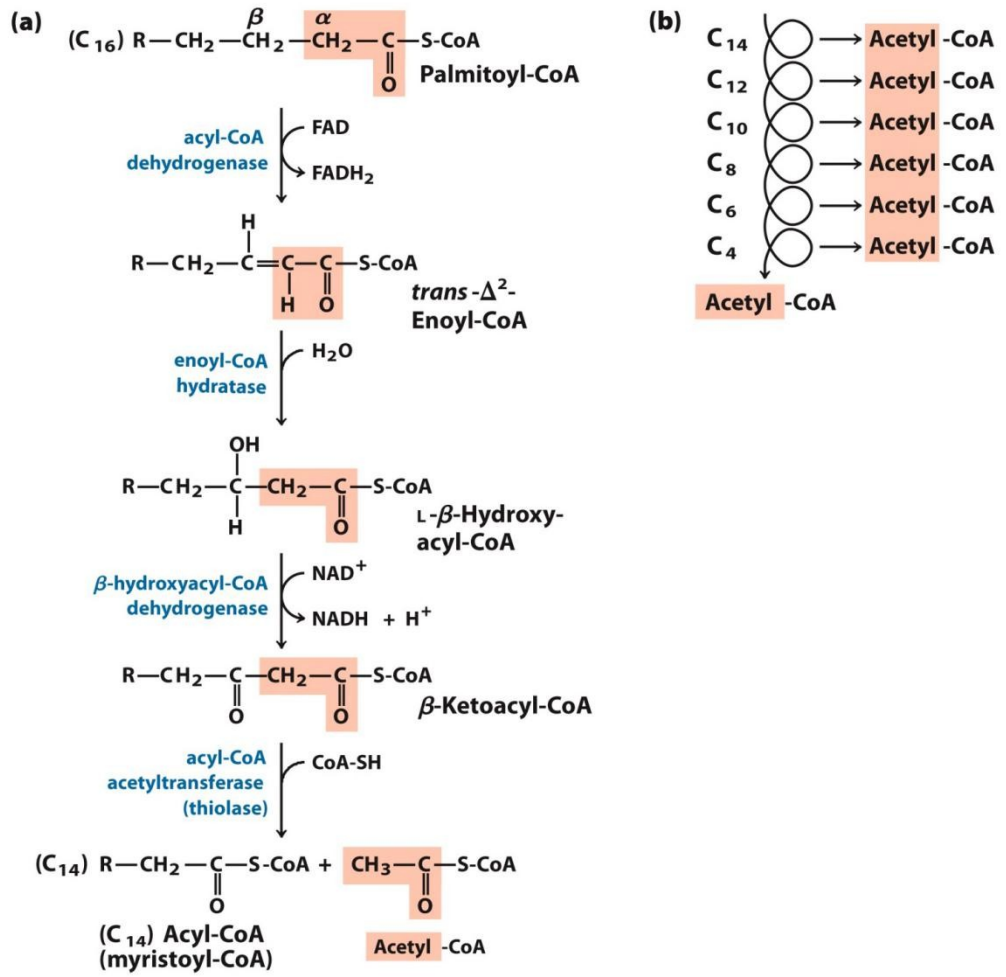


Figure 17-8
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

The beta -oxidation pathway.

(a) In each pass through this four-step sequence, one acetyl residue (shaded in pink) is removed in the form of acetyl-CoA from the carboxyl end of the fatty acyl chain—in this example palmitate (C16) is shown, which enters as palmitoyl-CoA.

(b) Six more passes through the pathway yield seven more molecules of acetyl-CoA, the seventh arising from the last two carbon atoms of the 16-carbon chain. Eight molecules of acetyl-CoA are formed in all.

4th Step: β-Oxidation of saturated fatty acids

Last step is catalyzed by acyl-CoA acetyltransferase, called thiolase, which promotes reaction of beta-ketoacyl-CoA with a molecule of free coenzyme A to split off the carboxyl-terminal two-carbon fragment of the original fatty acid as acetyl-CoA.

Another product is the coenzyme A; thioester of the fatty acid, now shortened by two carbon atoms

This reaction is called thiolysis, by analogy with the process of hydrolysis, because the β -ketoacyl-CoA is cleaved by reaction with the thiol group of coenzyme A.

The thiolase reaction is a reverse Claisen condensation

β Oxidation of saturated fatty acids

The last three steps of this four-step sequence are catalyzed by either of two sets of enzymes, with the enzymes employed depending on the length of the fatty acyl chain.

For fatty acyl chains of 12 or more carbons, the reactions are catalyzed by a multienzyme complex associated with the inner mitochondrial membrane, the trifunctional protein (TFP). TFP is a heterooctamer of α4β4 subunits.

Each subunit contains two activities, the enoyl-CoA hydratase and the -hydroxyacyl-CoA dehydrogenase; the β subunits contain the thiolase activity.

When TFP has shortened the fatty acyl chain to 12 or fewer carbons, further oxidations are catalyzed by a set of four soluble enzymes in the matrix.

… breakage of bonds in saturated fatty acids

The single bond between methylene (—CH2—) groups in fatty acids is relatively stable.

During first three reactions of β -oxidation less stable C—C bonds are created, in which the carbon (C-2) is bonded to two carbonyl carbons (the β -ketoacyl-CoA intermediate).

The ketone function on the carbon (C-3) makes it a good target for nucleophilic attack by the —SH of coenzyme A, catalyzed by thiolase.

SThe acidity of the α- hydrogen and the resonance stabilization of the carbanion generated by the departure of this hydrogen make the terminal —CH2—CO—S-CoA a good leaving group, facilitating breakage of the α-β bond.

TOPIC#115

The four steps are repeated to yield acetyl CoA and ATP

In one pass through the B-oxidation sequence, one molecule of acetyl-CoA, two pairs of electrons, and four protons (H+) are removed from the long-chain fatty acyl–CoA,

shortening it by two carbon atoms. The equation for one pass, beginning with the coenzyme A ester of our example, palmitate, is oxidized as following equation:

$$\text{Palmitoyl-CoA} + \text{CoA} + \text{FAD} + \text{NAD}^+ + \text{H}_2\text{O} \longrightarrow \text{myristoyl-CoA} + \text{acetyl-CoA} + \text{FADH}_2 + \text{NADH} + \text{H}^+$$

Following removal of one acetyl-CoA unit from palmitoyl- CoA, the coenzyme A thioester of the shortened fatty acid (now the 14-carbon myristate) remains

The myristoyl- CoA can now go through the four steps β–oxidation, exactly analogous to the first, to yield a second molecule of acetyl-CoA and lauroyl-CoA, the coenzyme A thioester of the 12-carbon laurate.

Altogether, seven passes through the β-oxidation sequence are required to oxidize one molecule of palmitoyl-CoA to eight molecules of acetyl-CoA

The overall equation is:

$$\text{Palmitoyl-CoA} + 7\text{CoA} + 7\text{FAD} + 7\text{NAD}^+ + 7\text{H}_2\text{O} \longrightarrow 8\ \text{acetyl-CoA} + 7\text{FADH}_2 + 7\text{NADH} + 7\text{H}^+$$

However, when the free-energy changes are calculated from actual concentrations of reactants and products under intracellular conditions the free-energy recovery is more than 60%; the energy conservation is remarkably efficient

Acetyl CoA produced by β-oxidation can enter Citric acid cycle

Acetyl CoA produced by β-oxidation can be oxidized to CO2 and H2O by the citric acid cycle

$$8\ \text{Acetyl-CoA} + 16\text{O}_2 + 80\text{P}_i + 80\text{ADP} \longrightarrow 8\text{CoA} + 80\text{ATP} + 16\text{CO}_2 + 16\text{H}_2\text{O} \qquad (17\text{–}5)$$

$$\text{Palmitoyl-CoA} + 23\text{O}_2 + 108\text{P}_i + 108\text{ADP} \longrightarrow \text{CoA} + 108\text{ATP} + 16\text{CO}_2 + 23\text{H}_2\text{O}$$

TABLE 17–1 Yield of ATP during Oxidation of One Molecule of Palmitoyl-CoA to CO_2 and H_2O

Enzyme catalyzing the oxidation step	Number of NADH or $FADH_2$ formed	Number of ATP ultimately formed*
Acyl-CoA dehydrogenase	7 $FADH_2$	10.5
β-Hydroxyacyl-CoA dehydrogenase	7 NADH	17.5
Isocitrate dehydrogenase	8 NADH	20
α-Ketoglutarate dehydrogenase	8 NADH	20
Succinyl-CoA synthetase		8†
Succinate dehydrogenase	8 $FADH_2$	12
Malate dehydrogenase	8 NADH	20
Total		108

*These calculations assume that mitochondrial oxidative phosphorylation produces 1.5 ATP per $FADH_2$ oxidized and 2.5 ATP per NADH oxidized.

†GTP produced directly in this step yields ATP in the reaction catalyzed by nucleoside diphosphate kinase (p. 526).

Table 17-1
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Acetyl CoA produced by β-oxidation can enter Citric acid cycle:

Acetyl CoA produced by β-oxidation can be oxidized to CO2 and H2O by the citric acid cycle

The energetic cost of activating a fatty acid is equivalent to two ATP, and the net gain per molecule of palmitate is 106 ATP.

Many animals depend on fat stores for energy during hibernation, during migratory periods, and in other situations involving radical metabolic adjustments.

One of the most pronounced adjustments of fat metabolism occurs in hibernating grizzly bears.

These animals remain in a continuous state of dormancy for periods as long as seven months.

Unlike most hibernating species, the bear maintains a body temperature of between 32 and 35C, close to the normal (nonhibernating) level.



Although expending about 25,000 kJ/day (6,000 kcal/day), the bear does not eat, drink, urinate, or defecate for months at a time.

Experimental studies have shown that hibernating grizzly bears use body fat as their sole fuel.

Fat oxidation yields sufficient energy for maintenance of body temperature, active synthesis of amino acids and proteins, and other energy-requiring activities, such as

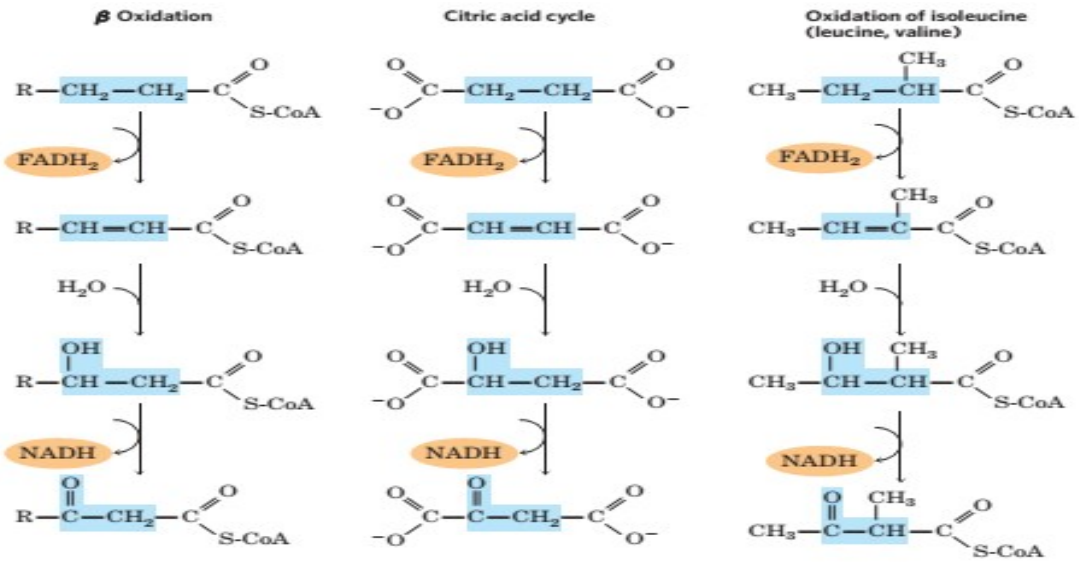
membrane transport. Fat oxidation also releases large amounts of water, as described in the text, which replenishes water lost in breathing.

The glycerol released by degradation of triacylglycerols is converted into blood glucose by gluconeogenesis.

Urea formed during breakdown of amino acids is reabsorbed in the kidneys and recycled, the amino groups reused to make new amino acids for maintaining body proteins.Bears store an enormous amount of body fat in preparation for their long sleep.

An adult grizzly consumes about 38,000 kJ/day during the late spring and summer, but as winter approaches it feeds 20 hours a day, consuming up to 84,000 kJ daily.

This change in feeding is a response to a seasonal change in hormone secretion. Large amounts of triacylglycerols are formed from the huge intake of carbohydrates during the fattening-up period. Other hibernating species, including the tiny dormouse, also accumulate large amounts of body fat.



A conserved reaction sequence to introduce a carbonyl function on the carbon β- to a carboxyl group. The β-oxidation pathway for fatty acyl–CoAs, the pathway from succinate to oxaloacetate in the citric acid cycle, and the pathway by which the

deaminated carbon skeletons from isoleucine, leucine, and valine are oxidized as fuels, use the same reaction sequence.

TOPIC#116

"Oxidation of unsaturated fatty acids requires 2 additional steps "

Oxidation of unsaturated fatty acids requires 2 additional steps

The fatty acid oxidation sequence just described is typical when the incoming fatty acid is saturated (that is, has only single bonds in its carbon chain).

These bonds are in the cis configuration and cannot be acted upon by enoyl-CoA hydratase, the enzyme catalyzing the addition of H2O to the trans double bond of the D2-enoyl-CoA generated during oxidation.

Two auxiliary enzymes are needed for oxidation of the common unsaturated fatty acids:

- an isomerase
- a reductase

However, most of the fatty acids in the triacylglycerolsand phospholipids of animals and plants are unsaturated,having one or more double bonds. These bonds are in the cis configuration and cannot be acted upon.by enoyl-CoA hydratase, the enzyme catalyzing the addition of H2O to the trans double bond of the

We illustrate these auxiliary reactions with two examples.

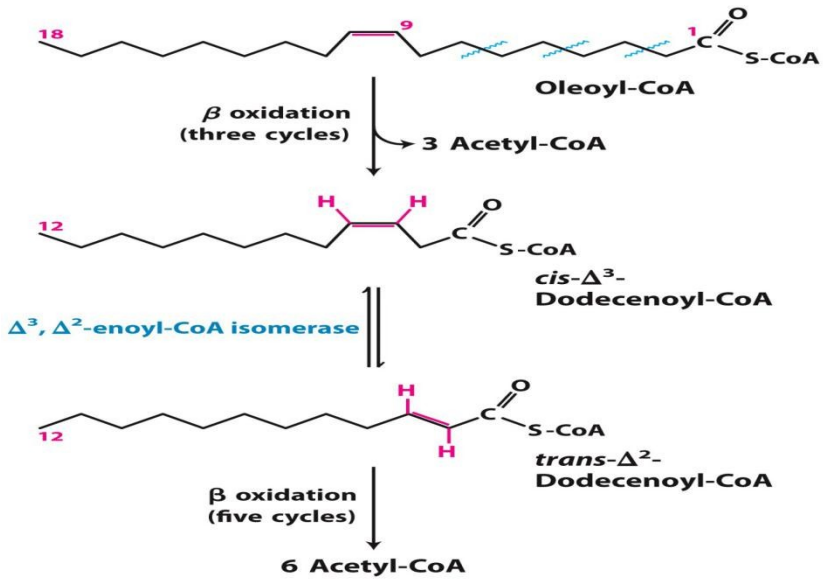


Figure 17-10
Lehninger Principles of Biochemistry, Sixth Edition
© 2013 W. H. Freeman and Company

Beta oxidation of oleate 18:1Δ9

Oxidation of Oleate (example of unsaturated fatty acids)

Oleate is an abundant 18-carbon monounsaturated fatty acid with a cis double bond between C-9 and C-10 (denoted Δ9).

In the first step of oxidation, oleate is converted to oleoyl-CoA and, like the saturated fatty acids, enters the mitochondrial matrix via the carnitine shuttle

Oleoyl-CoA then undergoes three passes through the fatty acid oxidation cycle to yield three molecules of acetyl-CoA and the coenzyme A ester of a D3, 12-carbon unsaturated fatty acid, cis- Δ 3-dodecenoyl-CoA

This product cannot serve as a substrate for enoyl-CoA hydratase, which acts only on trans double bonds.

The auxiliary enzyme D3,D2-enoyl-CoA isomerase isomerizes the cis-D3-enoyl-CoA to the trans-D2-enoyl- CoA, which is converted by enoyl-CoA hydratase intothe corresponding L--hydroxyacyl-CoA (trans-D2-dodecenoyl-CoA). This intermediate is now acted upon by the remaining enzymes of oxidation to yield acetyl-CoA and the coenzyme A ester of a 10-carbon saturated fatty acid, decanoyl-CoA. The latter undergoes four more passes through the -oxidation pathway to yield five more molecules of acetyl-CoA. Altogether, nine acetyl-CoAs are produced from one molecule of the 18-carbon oleate.

The other auxiliary enzyme (a reductase) is required for oxidation of polyunsaturated fatty acids — for example, the 18-carbon linoleate, which has a cis-D9,cis-D12 configuration (Fig. 17–11).

Linoleoyl- CoA undergoes three passes through the –oxidation sequence to yield three molecules of acetyl-CoA and the coenzyme A ester of a 12-carbon unsaturated fatty acid with a cis-D3,cis-D6 configuration.

This intermediate cannot be used by the enzymes of the β–oxidation pathway; its double bonds are in the wrong position

And have the wrong configuration (cis, not trans).

However, the combined action of enoyl-CoA isomerase and 2,4-dienoyl-CoA reductase, as shown in Figure 17–11, allows reentry of this intermediate into the –oxidation pathway and its degradation to six acetyl-CoAs.

The overall result is conversion of linoleate to nine molecules of acetyl-CoA.

Auxiliary enzymes

We investigated the enzyme activity of the blank in the spectrophotometric determination of the aminotransferase activities and aspartate aminotransferase activity. 6 lactate dehydrogenase and 3 malate dehydrogenase preparations from different manufactures and from different organs showed additional and contaminating activity. The additional activity depends upon the 2-oxoglutarate concentration. The contaminating activity is caused by alanine aminotransferase and aspartate aminotransferase in the auxiliary enzymes. We propose that exact definitions must be given for the auxiliary enzymes in the recommendations of standard determinations for enzyme activities.

TOPIC#117

Complete Oxidation of odd-numbered fatty acids requires 3 additional steps

Oxidation of Odd-numbered fatty acids:

Fatty acids with an odd number of carbons are found in the lipids of many plants and marine organisms.

Cattle and other ruminant animals form large amounts of the three-carbon propionate (CH3—CH2—COO2) during fermentation of carbohydrates in the rumen.

The propionate is absorbed into the blood and oxidized by the liver and other tissues

Small quantities of propionate are added as a mold inhibitor to some breads and cereals, thus entering the human diet.

Long-chain odd-number fatty acids are oxidized in the same pathway as the even-number acids, … at the carboxyl end of the chain.

the substrate for the last pass through the β-oxidation sequence is a fatty acyl–CoA with a five-carbon fatty acid.

Oxidation of propionyl-CoA

The oxidation and cleaving of this gives acetyl-CoA and propionyl-CoA as the products

The acetyl-CoA can be oxidized in the citric acid cycle, but propionyl-CoA enters a different pathway having three enzymes.

Propionyl-CoA is first carboxylated to form the D stereoisomer of methylmalonyl-CoA by propionyl-CoA carboxylase, which contains the cofactor biotin. In this enzymatic reaction, as in the pyruvate carboxylase reaction (see Fig. 16–17), CO2 (or its hydrated ion, HCO) is activated by attachment to biotin before its transfer to the substrate, in this case the propionate moiety.Formation of the carboxybiotin intermediate requires energy, which is provided by ATP.

The D-methylmalonyl-CoA thus formed is enzymatically epimerized to its L stereoisomer by methylmalonyl- CoA epimerase (Fig. 17–12).

The L-methylmalonyl-CoA then undergoes an intramolecular rearrangement to form succinyl-CoA, which can enter the citric acid cycle.

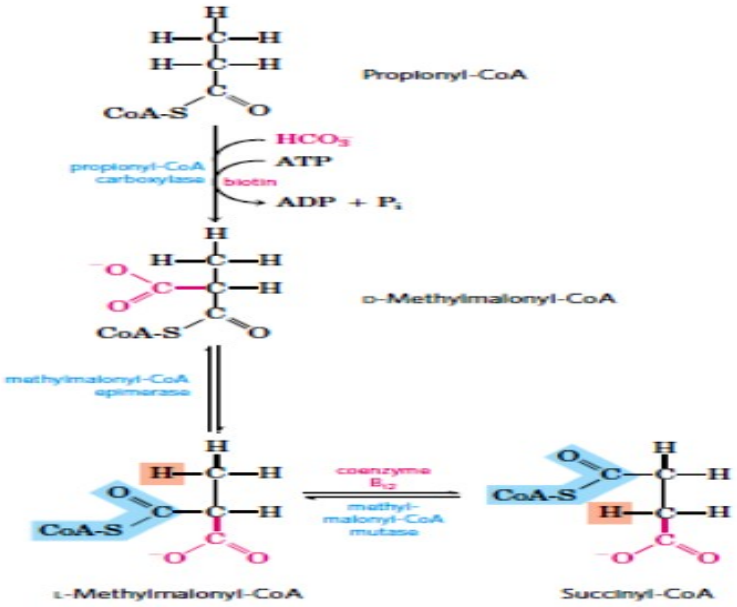
This rearrangement is catalyzed by methylmalonyl- CoA mutase, which requires as its coenzyme 5’-deoxyadenosylcobalamin, or coenzyme B12, which is derived from vitamin B12 (cobalamin).

Oxidation of propionyl-CoA produced by oxidation ofodd-number fatty acids.

The sequence involves the carboxylation of

propionyl-CoA to D-methylmalonyl-CoA and conversion of the latter to succinyl-CoA. This conversion requires epimerization of D- to L-methylmalonyl- CoA,

followed by a reaction in which substituents on adjacent carbon atoms exchange positions



TOPIC#118 IS NOT AVAILABLE

THIS PAGE IS LEFT BLANK INTENTIOANLLY

TOPIC#119

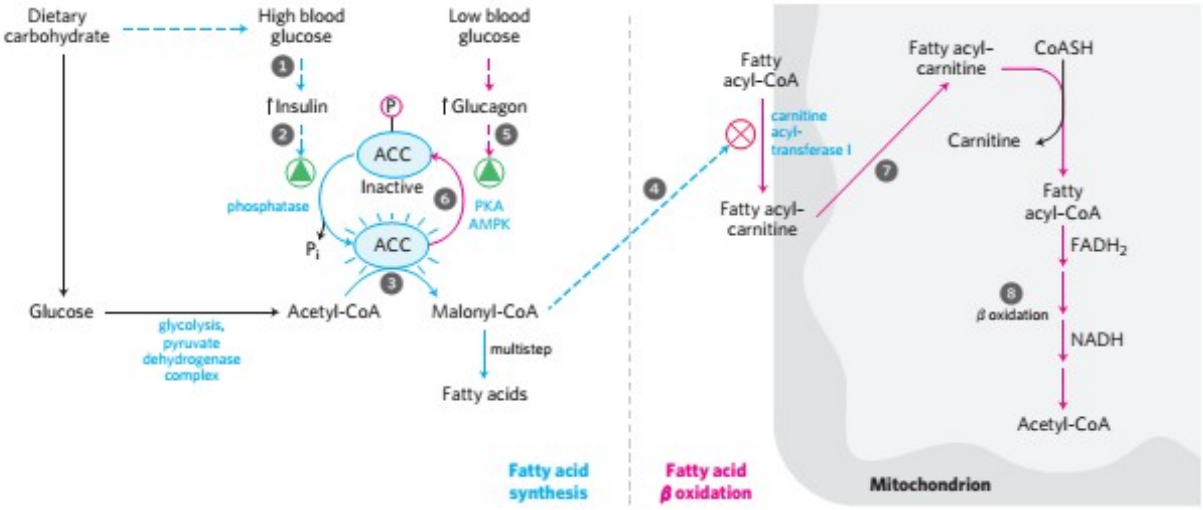
Transcription Factors Turn on the Synthesis of Proteins for Lipid Catabolism

Transcription Factors Turn on the Synthesis of Proteins for Lipid Catabolism

In addition to the various short-term regulatory mechanisms that modulate the activity of existing enzymes, transcriptional regulation can change the number of molecules of the enzymes of fatty acid oxidation on a longer time scale, minutes to hours.

The PPAR family of nuclear receptors are transcription factors that affect many metabolic processes in response to a variety of fatty acid–like ligands. (They were originally recognized as peroxisome proliferator-activated receptors, then were found to function more broadly.)

PPAR acts in muscle, adipose tissue, and liver to turn on a set of genes essential for fatty acid oxidation, including the fatty acid transporter, carnitine acyltransferases I and II, fatty acyl–CoA dehydrogenases for short, medium, long, and very long acyl chains, and related enzymes.



When the diet provides a ready source of carbohydrate as fuel, oxidation of fatty acids is unnecessary and is therefore downregulated. Two enzymes are key to the coordination of fatty acid metabolism: acetyl-CoA carboxylase (ACC), the first enzyme in the synthesis of fatty acids, and carnitine acyltransferase I, which limits the transport of fatty acids into the mitochondrial matrix for oxidation
Ingestion of a high-carbohydrate meal raises the blood glucose level and thus

1. triggers the release of insulin.
2. Insulin dependent protein phosphatase dephosphorylates ACC, activating it.
3. ACC catalyzes the formation of malonyl-CoA (the first intermediate of fatty acid synthesis), and
4. malonyl-CoA inhibits carnitine acyltransferase I, thereby preventing fatty acid entry into the mitochondrial matrix. When blood glucose levels drop between meals,
5. Glucagon release activates cAMP-dependent protein kinase (PKA), which
6. phosphorylates and inactivates ACC.
7. The concentration of malonylCoA falls, the inhibition of fatty acid entry into mitochondria is relieved,
   and fatty acids enter the mitochondrial matrix
8. become the major fuel. Because glucagon also triggers the mobilization of fatty acids in adipose tissue, a supply of fatty acids begins arriving in the blood.

This response is triggered when a cell or organism has an increased demand for energy from fat catabolism, such as during a fast between meals or under conditions of longer-term starvation.

Glucagon, released in response to low blood glucose, can act through cAMP and the transcription factor CREB to turn on certain genes for lipid catabolism.
Another situation that is accompanied by major changes in the expression of the enzymes of fatty acid oxidation is the transition from fetal to neonatal metabolism in the heart.

In the fetus the principal fuels are glucose and lactate, but in the neonatal heart, fatty acids are the main fuel.

At the time of this transition, PPAR is activated and in turn activates the genes essential for fatty acid metabolism.

The major site of fatty acid oxidation, at rest and during exercise, is skeletal muscle.

Endurance training increases PPAR expression in muscle, leading to increased levels of fatty acid–oxidizing enzymes and increased oxidative capacity of the muscle.

Genetic Defects in Fatty Acyl–CoA Dehydrogenases Cause Serious Disease

Stored triacylglycerols are typically the chief source of energy for muscle contraction, and an
inability to oxidize fatty acids from triacylglycerols has serious consequences for health. The most common genetic defect in fatty acid catabolism in U.S. and northern European populations is due to a mutation in the gene encoding the medium-chain acyl-CoA dehydrogenase (MCAD).

Among northern Europeans, the frequency of carriers (individuals with this recessive mutation on one of the two homologous chromosomes) is about 1 in 40, and about 1 individual in 10,000 has the disease—that is, has two copies of the mutant MCAD allele and is unable to oxidize fatty acids of 6 to 12 carbons.

The disease is characterized by recurring episodes of a syndrome that includes fat accumulation in the liver, high blood levels of octanoic acid (8:0), low blood glucose (hypoglycemia), sleepiness, vomiting, and coma.

The pattern of organic acids in the urine helps in the diagnosis of this disease: the urine commonly contains high levels of 6-carbon to 10-carbon dicarboxylic acids (produced by oxidation) and low levels of urinary ketone bodies (we discuss oxidation below and ketone bodies in Section 17.3).

Although individuals may have no symptoms between episodes, the episodes are very serious; mortality from this disease is 25% to 60% in early childhood.

If the genetic defect is detected shortly after birth, the infant can be started on a low-fat, high-carbohydrate diet.

With early detection and careful management of the diet—including avoiding long intervals between meals, to prevent the body from turning to its fat reserves for

energy—the prognosis for these individuals is good.
More than 20 other human genetic defects in fatty acid transport or oxidation have been documented, most much less common than the defect in MCAD.

One of the most severe disorders results from loss of the long-chain Beta-hydroxyacyl-CoA dehydrogenase activity of the trifunctional protein, TFP.

Other disorders include defects in the alpha or Beta subunits that affect all three activities of TFP and cause serious heart disease and abnormal skeletal muscle.

TOPIC#120

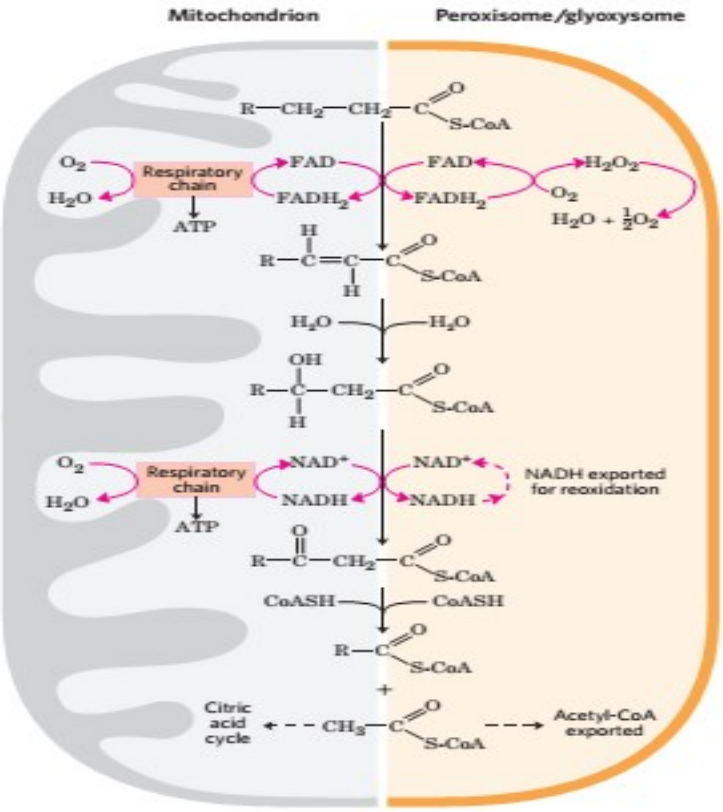
Oxidation in peroxisomes -

Peroxisomes Also Carry Out beta Oxidation

The mitochondrial matrix is the major site of fatty acid oxidation in animal cells, but in certain cells other compartments also contain enzymes capable of oxidizing fatty acids to acetyl-CoA, by a pathway not identical to that in mitochondria.

In plant cells, the major site of beta oxidation is not mitochondria but peroxisomes. In peroxisomes, membrane-enclosed organelles of animal and plant cells, the intermediates for oxidation of fatty acids are coenzyme A derivatives, and the process consists of four steps, as in mitochondrial beta oxidation

- Dehydrogenation
- addition of water to the resulting double bond,
- oxidation of the beta-hydroxyacyl-CoA to a ketone, and
- thiolytic cleavage by coenzyme A. (The identical reactions also occur in glyoxysomes)



One difference between the peroxisomal and mitochondrial pathways is in the chemistry of the first step.

In peroxisomes, the flavoprotein acyl-CoA oxidase that introduces the double bond passes electrons directly to O2, producing H2O2 (hydrogen peroxide). (Thus the name peroxisomes.)

This strong and potentially damaging oxidant is immediately cleaved to H2O and O2 by catalase.

Recall that in mitochondria, the electrons removed in the first oxidation step pass through the respiratory chain to O2 to produce H2O, and this process is accompanied by ATP synthesis.

In peroxisomes, the energy released in the first oxidative step of fatty acid breakdown is not conserved as ATP, but is dissipated as heat

A second important difference between mitochondrial and peroxisomal Beta oxidation in mammals is in the specificity for fatty acyl–CoAs; the peroxisomal system is much more active on very-long-chain fatty acids such as hexacosanoic acid (26:0) and on branched chain fatty acids such as phytanic acid and pristanic acid (see Fig. 17–18).

These less-common fatty acids are obtained in the diet from dairy products, the fat of ruminant animals, meat, and fish. Their catabolism in the peroxisome involves several auxiliary enzymes unique to this organelle.

The inability to oxidize these compounds is responsible for several serious human diseases.

Individuals with Zellweger syndrome are unable to make peroxisomes and therefore lack all the metabolism unique to that organelle.

In X-linked adrenoleukodystrophy (XALD), peroxisomes fail to oxidize very-long-chain fatty acids, apparently for lack of a functional transporter for these fatty acids in the peroxisomal membrane.

Both defects lead to accumulation in the blood of very-long-chain fatty acids, especially 26:0.

XALD affects young boys before the age of 10 years, causing loss of vision, behavioral disturbances, and death within a few years.
In mammals, high concentrations of fats in the diet result in increased synthesis of the enzymes of peroxisomal Beta oxidation in the liver.

Liver peroxisomes do not contain the enzymes of the citric acid cycle and cannot catalyze the oxidation of acetyl-CoA to CO2. Instead, long-chain or branched fatty acids are catabolized to shorter-chain products, such as hexanoyl-CoA, which are exported to mitochondria and completely oxidized.

TOPIC#121

Oxidation in peroxisomes -cont

Peroxisomes Also Carry Out β- Oxidation-cont..

Plant Peroxisomes and Glyoxysomes Use Acetyl-CoA from Oxidation as a Biosynthetic Precursor

In plants, fatty acid oxidation does not occur primarily in mitochondria but in the peroxisomes of leaf tissue and in the glyoxysomes of germinating seeds.

Plant peroxisomes and glyoxysomes are similar in structure and function; glyoxysomes, which occur only in germinating seeds, may be considered specialized peroxisomes.

The biological role of oxidation in these organelles is to use stored lipids primarily to provide biosynthetic precursors, not energy.

During seed germination, stored triacylglycerols are converted into glucose, sucrose, and a wide variety of essential metabolites (Fig. 17–15).

Fatty acids released from the triacylglycerols are first activated to their coenzyme A derivatives and oxidized in glyoxysomes by the same four-step process that takes place in peroxisomes (Fig. 17–14). The acetyl-CoA produced is converted via the glyoxylate cycle to four-carbon precursors for gluconeogenesis (see Fig. 16–24). Glyoxysomes, like peroxisomes, contain high concentrations of catalase, which converts the H2O2 produced by oxidation to H2O and O2.

The beta-Oxidation Enzymes of Different Organelles Have Diverged during Evolution

Although the beta-oxidation reactions in mitochondria are essentially the same as those in peroxisomes and glyoxysomes, the enzymes (isozymes) differ significantly between the two types of organelles.

The difference sapparently reflect an evolutionary divergence that occurred very early, with the separation of gram-positive and gram-negative bacteria

In mitochondria, the four -oxidation enzymes that act on short-chain fatty acyl–CoAs are separate, soluble proteins (as noted earlier), similar in structure to the analogous enzymes of gram-positive bacteria (Fig. 17–16a).

The gram-negative bacteria have four activities in three soluble subunits (Fig. 17–16b), and the eukaryotic enzyme system that acts on long-chain fatty acids—the trifunctional protein, TFP—has three enzyme activities in two subunits that are membrane-associated

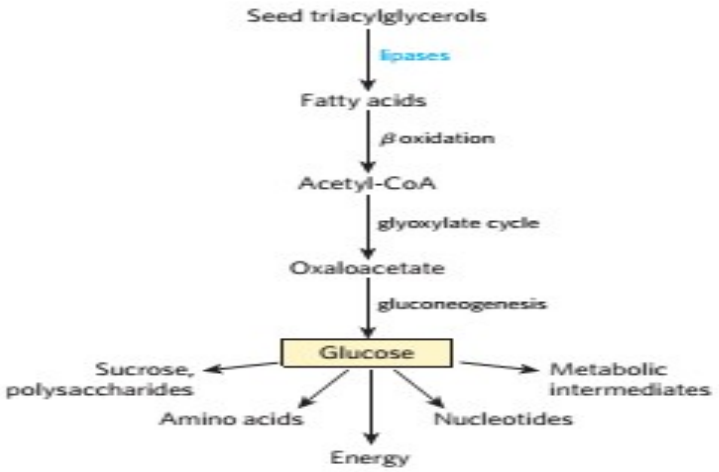


FIGURE 17–15 **Triacylglycerols as glucose source in seeds.** β Oxidation is one stage in a pathway that converts stored triacylglycerols to glucose in germinating seeds. For more detail, see Figure 16–24.

The beta-oxidation enzymes of plant peroxisomes and glyoxysomes, however, form a complex of proteins, one of which contains four enzymatic activities in a single polypeptide

The first enzyme, acyl-CoA oxidase, is a single polypeptide chain; the multifunctional protein (MFP) contains the second and third enzyme activities (enoyl-CoA hydratase and hydroxyacyl-CoA dehydrogenase) as well as two auxiliary activities needed for the oxidation of unsaturated fatty acids (D-3-hydroxyacyl-CoA epimerase and D3,D2-enoyl-CoA isomerase); the fourth enzyme, thiolase, is a separate, soluble polypeptide.

Interesting thing is that the enzymes that catalyze essentially the reversal of beta oxidation in the synthesis of fatty acids are also organized differently in bacteria and eukaryotes; in bacteria, the seven enzymes needed for fatty acid synthesis are separate polypeptides, but in mammals, all seven activities are part of a single, huge polypeptide chain.

One advantage to the cell in having several enzymes of the same pathway encoded in a single polypeptide chain is that this solves the problem of regulating the synthesis of enzymes that must interact functionally; regulation of the expression of one gene ensures production of the same number of active sites for all enzymes in the path.

When each enzyme activity is on a separate polypeptide, some mechanism is required to coordinate the synthesis of all the gene products.

The disadvantage of having several activities on the same polypeptide is that the longer the polypeptide chain, the greater is the probability of a mistake in its

synthesis: a single incorrect amino acid in the chain may make all the enzyme activities in that chain useless.

Comparison of the gene structures for these proteins in many species may shed light on the reasons for the selection of one or the other strategy in evolution.

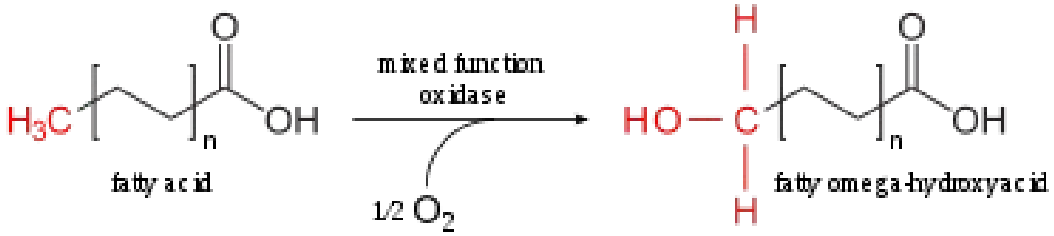
TOPIC#122

Omega Oxidation

The Omega-Oxidation of Fatty Acids

The Omega-Oxidation of Fatty Acids Occurs in the Endoplasmic Reticulum

Omega oxidation (ω-oxidation) is a process of fatty acid metabolism in some species of animals. It is an alternative pathway to beta oxidation that, instead of involving the β carbon, involves the oxidation of the ω carbon (the carbon most distant from the carboxyl group of the fatty acid).



In vertebrates, the enzymes for ω oxidation are located in the smooth ER of liver and kidney cells, instead of in the mitochondria as with β oxidation.

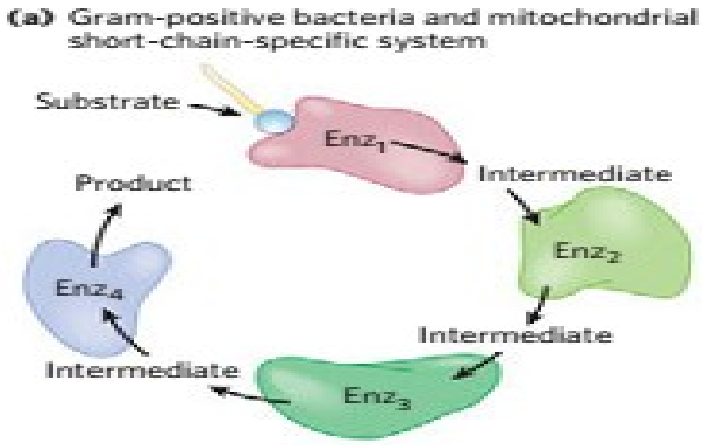
The Omega-Oxidation of Fatty Acids Occurs in the Endoplasmic Reticulum

Although mitochondrial Omega oxidation, in which enzymes act at the carboxyl end of a fatty acid, is by far the most important catabolic fate for fatty acids in animal cells, there is another pathway in some species, including vertebrates, that involves oxidation of the (omega) carbon—the carbon most distant from the carboxyl group.

The enzymes unique to Omega oxidation are located (in vertebrates) in the endoplasmic reticulum of liver and kidney, and the preferred substrates are fatty acids of 10 or 12 carbon atoms. In mammals Omega - oxidation is normally a minor pathway for fatty acid degradation, but when beta-oxidation is defective (because of mutation or a carnitine deficiency, for example) it becomes more important.

The first step introduces a hydroxyl group onto the Omega carbon

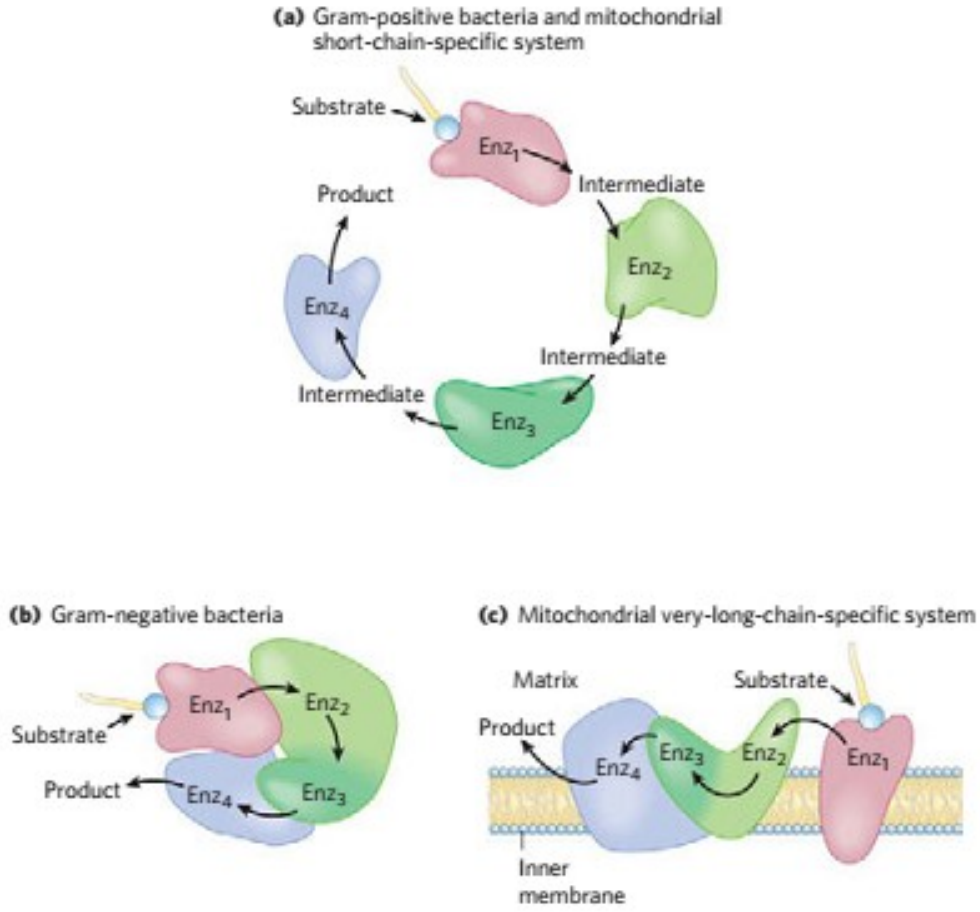
The oxygen for this group comes from molecular oxygen (O2) in a complex reaction that involves cytochrome P450 and the electron donor NADPH. Reactions of this type are catalyzed by mixed-function oxidases



The enzymes of oxidation.

In the picture Shown on next slides, are the different subunit structures of the enzymes of Beta oxidation in gram-positive and gram-negative bacteria, mitochondria, and plant peroxisomes and glyoxysomes.

Enz1 is acyl-CoA dehydrogenase; Enz2, enoyl-CoA hydratase; Enz3, L- Beta - hydroxyacyl-CoA dehydrogenase; Enz4, thiolase; Enz5, D-3-hydroxyacyl-CoA epimerase, and Enz6, D3,D2-enoyl-CoA isomerase.

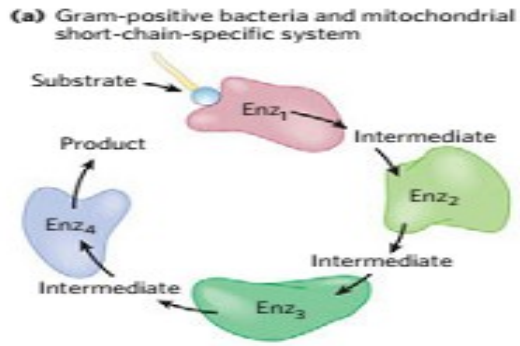


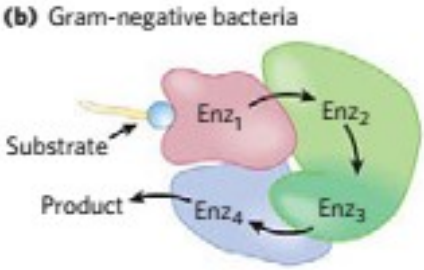
a. The four enzymes of beta oxidation in gram-positive bacteria are separate, soluble entities, as are those of the short-chain-specific system of mitochondria

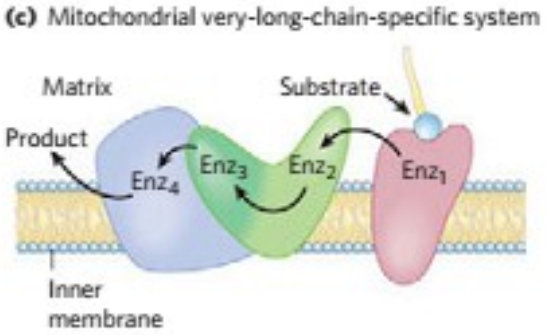
b. In gram negative bacteria, the four enzyme activities reside in three polypeptides; Enz2 and
Enz 3 are parts of a single polypeptide chain.

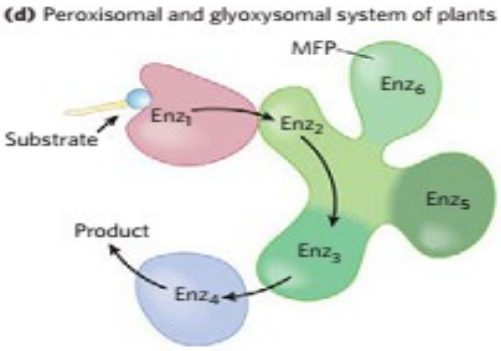
c. The very-long-chain-specific system of mitochondria is also composed of three polypeptides, one of which includes the activities of Enz2 and Enz3; in this case, the system is bound to the inner mitochondrial membrane.

d. In the peroxisomal and glyoxysomal beta oxidation systems of plants, Enz 1 and Enz4 are separate polypeptides, but Enz2 and Enz3, as well as two auxiliary enzymes (Enz5 and Enz6), are part of a single polypeptide chain: the multifunctional protein, MFP.


(a) Gram-positive bacteria and mitochondrial short-chain-specific system
Substrate
Enz1
Intermediate
Enz2
Intermediate
Enz3
Intermediate
Enz4
Product


(b) Gram-negative bacteria
Substrate
Enz1
Enz2
Enz3
Product
Enz4


(c) Mitochondrial very-long-chain-specific system
Matrix
Substrate
Product
Enz4
Enz3
Enz2
Enz1
Inner membrane


(d) Peroxisomal and glyoxysomal system of plants
MFP
Enz6
Enz1
Enz2
Substrate
Enz5
Product
Enz3
Enz4

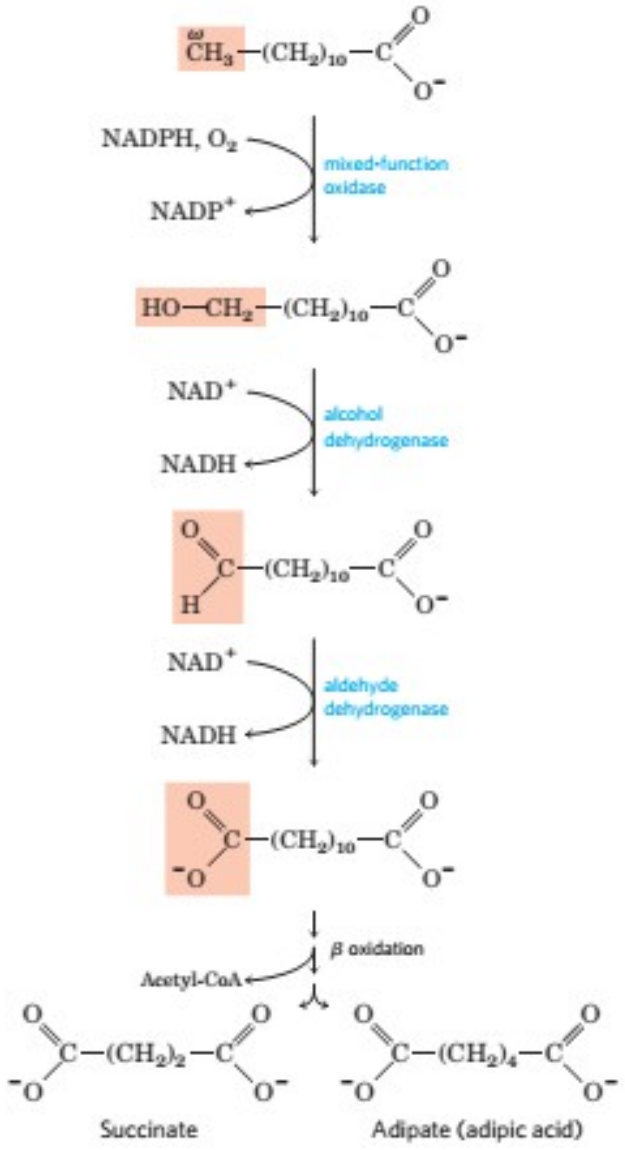


FIGURE 17–17 The ω oxidation of fatty acids in the endoplasmic reticulum. This alternative to β oxidation begins with oxidation of the carbon most distant from the β carbon—the ω (omega) carbon. The substrate is usually a medium-chain fatty acid; shown here is lauric acid (laurate). This pathway is generally not the major route for oxidative catabolism of fatty acids.

Two more enzymes now act on the Omega carbon :

1. alcohol dehydrogenase oxidizes the hydroxyl group to an aldehyde
2. aldehyde dehydrogenase oxidizes the aldehyde group to a carboxylic acid producing a fatty acid with a carboxyl group at each end.

At this point, either end can be attached to coenzyme A, and the molecule can enter the mitochondrion and undergo oxidation by the normal route.

In each pass through beta -oxidation pathway, the "double-ended" fatty acid yields dicarboxylic acids such as succinic acid, which can enter the citric acid cycle, and adipic acid.

TOPIC#123

Oxidation of phytanic acid

Phytanic Acid Undergoes Alpha-Oxidation in Peroxisomes

The presence of a methyl group on the carbon of a fatty acid makes beta-oxidation impossible, and these branched fatty acids are catabolized in peroxisomes of animal cells by Alpha-oxidation.

In the oxidation of phytanic acid, phytanoyl CoA is hydroxylated on its alpha carbon, in a reaction that involves molecular oxygen.

alpha carbon is decarboxylated to form an aldehyde, that is now having a carbon chain shorter of 1 carbon but has no functional group on beta carbon

Now it's oxidized at beta carbon and process continues in same way

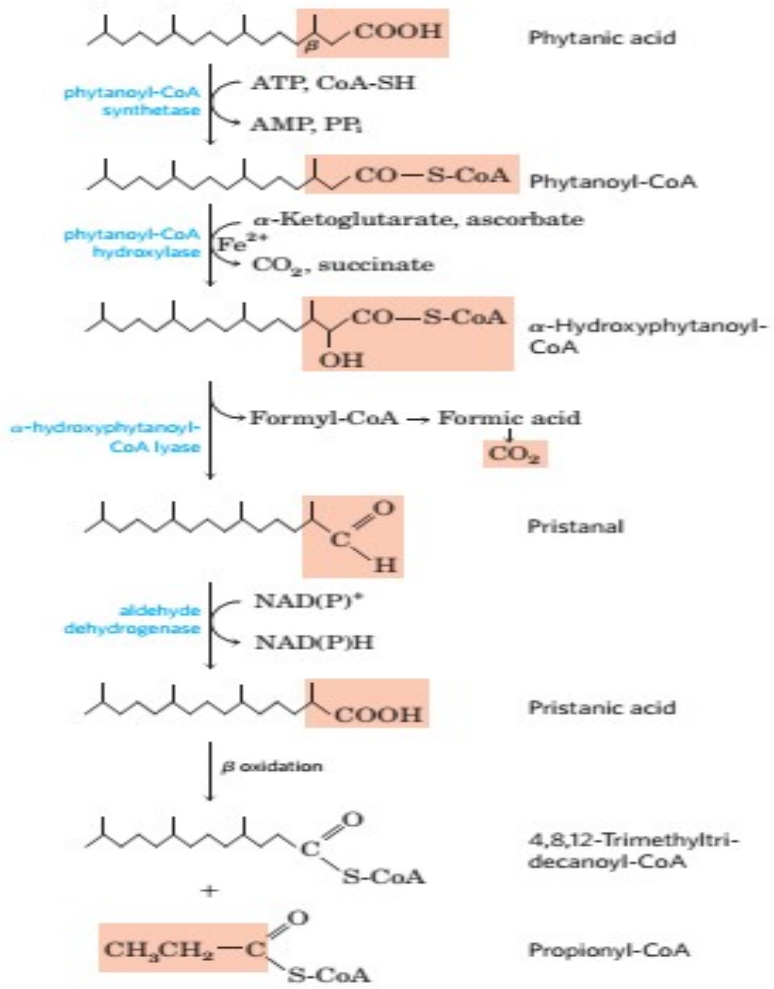


FIGURE 17–18 The α oxidation of a branched-chain fatty acid (phytanic acid) in peroxisomes. Phytanic acid has a methyl-substituted β carbon and therefore cannot undergo β oxidation. The combined action of the enzymes shown here removes the carboxyl carbon of phytanic acid to produce pristanic acid, in which the β carbon is unsubstituted, allowing β oxidation. Notice that β oxidation of pristanic acid releases propionyl-CoA, not acetyl-CoA. This is further catabolized as in Figure 17–12. (The details of the reaction that produces pristanal remain controversial.)

Phytanic Acid Undergoes Alpha-Oxidation in Peroxisomes

Refsum disease:

A disorder resulting from a genetic defect in phytanoyl-CoA hydroxylase, leads to very high blood levels of phytanic acid and severe neurological problems, including blindness and deafness.

TOPIC#124

Ketone bodies

Ketone Bodies

In humans and most other mammals, acetyl-CoA formed in the liver during oxidation of fatty acids can either enter the citric acid cycle (stage 2 of Fig. 17–7) or undergo conversion to the "ketone bodies,"

These include acetone, acetoacetate, and D-beta-hydroxybutyrate, for export to other tissues.

The term "bodies" is a historical artifact; the term is occasionally applied to insoluble particles, but these compounds are soluble in blood and urine.
Acetone, produced in smaller quantities than the other ketone bodies, is exhaled.

Acetoacetate and D-beta- hydroxybutyrate are transported by the blood to tissues other than the liver (extrahepatic tissues), where they
are converted to acetyl-CoA and oxidized in the citric acid cycle, providing much of the energy required by tissues such as skeletal and heart muscle and the renal cortex.

The brain, which preferentially uses glucose as fuel, can adapt to the use of acetoacetate or D-beta- hydroxybutyrate under starvation conditions, when glucose is unavailable. The production and export of ketone bodies from the liver to extrahepatic tissues allows continued oxidation of fatty acids in the liver when acetyl CoA is not being oxidized in the citric acid cycle.

Ketone Bodies, Formed in the Liver, Are Exported to Other Organs as Fuel

The first step in the formation of acetoacetate, occurring in the liver, is the enzymatic condensation of two molecules of acetyl-CoA, catalyzed by thiolase; this is simply the reversal of the last step of oxidation.

The acetoacetyl-CoA then condenses with acetyl-CoA to form Beta-hydroxy-Beta-methylglutaryl-CoA (HMG-CoA), which is cleaved to free acetoacetate and acetyl-CoA.

The acetoacetate is reversibly reduced by D-Beta-hydroxybutyrate dehydrogenase, a mitochondrial enzyme, to D-Beta-hydroxybutyrate.

This enzyme is specific for the D stereoisomer; it does not act on L Beta-hydroxyacyl-CoAs and is not to be confused with L-Beta-hydroxyacyl-CoA dehydrogenase of the Beta-oxidation pathway.

TOPIC#125

Ketone bodies-continued

Ketone Bodies

In healthy people, acetone is formed in very small amounts from acetoacetate, which is easily decarboxylated, either spontaneously or by the action of acetoacetate decarboxylase (Fig. 17–19).

Because individuals with untreated diabetes produce large quantities of acetoacetate, their blood contains significant amounts of acetone, which is toxic.

Acetone is volatile and imparts a characteristic odor to the breath, which is sometimes useful in diagnosing diabetes.

In extrahepatic tissues, D-Beta-hydroxybutyrate is oxidized to acetoacetate by D-Beta-hydroxybutyrate dehydrogenase (Fig. 17–20).

The acetoacetate is activated to its coenzyme A ester by transfer of CoA from succinylCoA, an intermediate of the citric acid cycle (see Fig.16–7), in a reaction catalyzed by Beta-ketoacyl-CoA transferase, also called thiophorase.

The acetoacetylCoA is then cleaved by thiolase to yield two acetylCoAs, which enter the citric acid cycle.

Thus the ketone bodies are used as fuels in all tissues except liver, which lacks thiophorase.

The liver is therefore a producer of ketone bodies for the other tissues, but not a consumer.
The production and export of ketone bodies by the liver allows continued oxidation of fatty acids with only minimal oxidation of acetyl-CoA.

When intermediates of the citric acid cycle are being siphoned off for glucose synthesis by gluconeogenesis, for example, oxidation of cycle intermediates slows—and so does acetyl-CoA oxidation.

Moreover, the liver contains only a limited amount of coenzyme A, and when most of it is tied up in acetyl-CoA, Beta-oxidation slows for want of the free coenzyme.

The production and export of ketone bodies frees coenzyme A, allowing continued fatty acid oxidation.

Ketone Bodies Are Overproduced in Diabetes and during Starvation

Starvation and untreated diabetes mellitus lead to overproduction of ketone bodies, with several associated medical problems. During starvation, gluconeogenesis depletes citric acid cycle intermediates, diverting acetyl-CoA to ketone body production (Fig. 17–21).

In untreated diabetes, when the insulin level is insufficient, extrahepatic tissues cannot take up glucose efficiently from the blood, either for fuel or for conversion to fat.

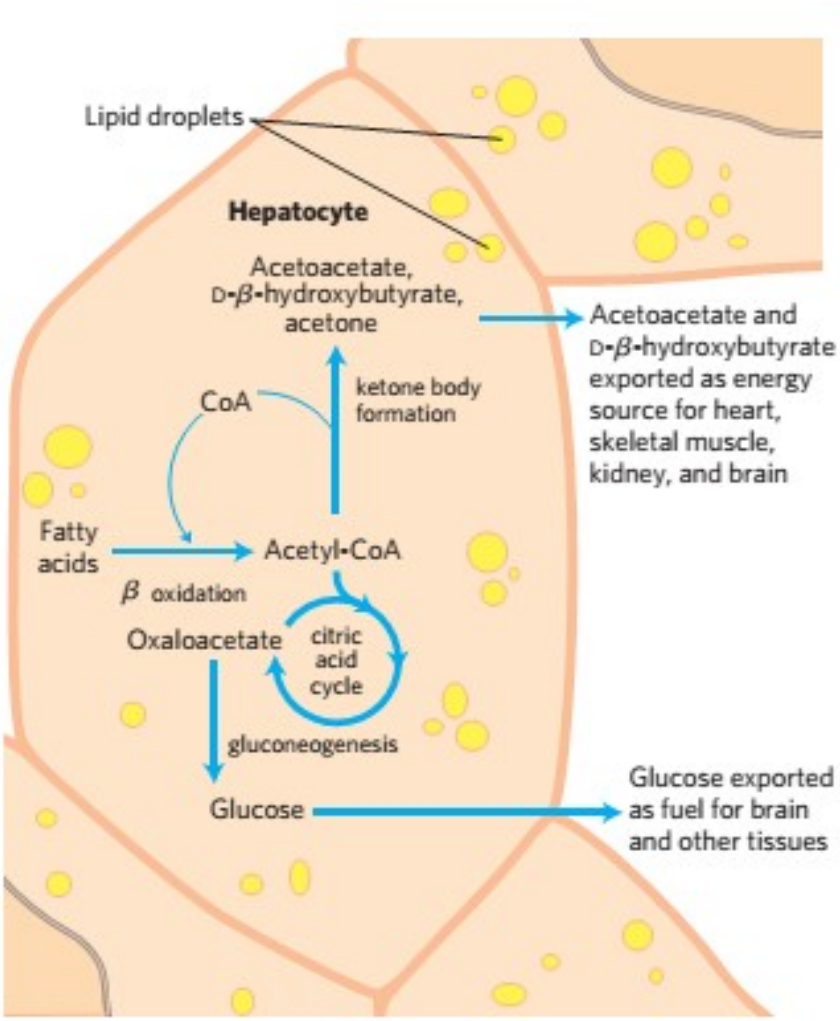
Under these conditions, levels of malonyl-CoA (the starting material for fatty acid synthesis) fall, inhibition of carnitine acyltransferase I is relieved, and fatty acids enter mitochondria to be degraded to acetylCoA—which cannot pass through the citric acid cycle because cycle intermediates have been drawn off for use as substrates in gluconeogenesis.

The resulting accumulation of acetyl-CoA accelerates the formation of ketone bodies beyond the capacity of extrahepatic tissues to oxidize them.

The increased blood levels of acetoacetate and D-betA-hydroxybutyrate lower the blood pH, causing the condition known as acidosis.

Extreme acidosis can lead to coma and in some cases death.
Ketone bodies in the blood and urine of individuals with untreated diabetes can reach extraordinary levels—a blood concentration of 90 mg/100 mL (compared with a normal level of ,3 mg/100 mL) and urinary excretion of 5,000 mg/24 hr (compared with a normal rate of #125 mg/24 hr). This condition is called ketosis.

Individuals on very low-calorie diets, using the fats stored in adipose tissue as their major energy source, also have increased levels of ketone bodies in their blood and urine. These levels must be monitored to avoid the dangers of acidosis and ketosis (ketoacidosis).



Ketone body formation and export from the liver.

Conditions that promote gluconeogenesis (untreated diabetes, severely reduced food intake) slow the citric acid cycle (by drawing off oxaloacetate) and enhance the conversion of acetyl-CoA to acetoacetate.

The released coenzyme A allows continued Beta oxidation of fatty acids.